MELANOMA

1600O Efficacy and safety of IMA203, a PRAME-directed T-cell receptor (TCR) T-cell therapy, in patients with previously treated advanced or metastatic uveal melanoma from a phase I trial

S.P. Patel[1], A.M. Tsimberidou[2], J.J. Luke[3], W. Alsdorf[4], A. Busse[5], S.C. DeVane[6], S. Hengler[7], N. Hilf[8], M.A. Kursunel[7], A. Mayer-Mokler[9], D. Pankov[10], C. Britten[11], M. Wermke[12], L. Hernandez-Aya[13]

[1]Medicine, University of Colorado Anschutz Medical Campus, Aurora, United States of America; [2]Investigational Cancer Therapeutics Dept., The University of Texas MD Anderson Cancer Center - Main Building, Houston, United States of America; [3]Immunotherapy and Drug Development Center, UPMC Hillman Cancer Center and University of Pittsburgh, Pittsburgh, United States of America; [4]Oncology, Hematology and Bone Marrow Transplantation, University Medical Center Hamburg-Eppendorf, Hamburg, Germany; [5]Medizinische Klinik III, Charité - Universitätsmedizin Berlin, Berlin, Germany; [6]Clinical Sciences, Immatics US., Inc., Houston, United States of America; [7]Translational Science, Immatics Biotechnologies GmbH, Munich, Germany; [8]Immatics Biotechnologies GmbH, Munich, Germany; [9]Clinical Development Department, Immatics Biotechnologies GmbH, Tuebingen, Germany; [10]Clinical Development, Immatics US., Inc., Houston, United States of America; [11]Immatics Biotechnologies GmbH, Munich, Germany; [12]Early Clinical Trial Dept., Dresden University, Dresden, Germany; [13]Oncology, University of Miami, Miami, United States of America

Background: Uveal melanoma (UM) represents 5% of all melanomas; metastatic disease develops in ∼50%. Currently, 1 systemic therapy is approved for UM, and metastatic disease prognosis is poor. PRAME is expressed in ∼90% of UM and is a negative predictive marker of outcomes. IMA203 is a PRAME-directed TCR T-cell therapy engineered to recognize intracellular PRAME-derived peptides presented by HLA and initiate a potent and specific anti-tumor response. Here, we present data on UM pts from an ongoing Ph1a/b trial (NCT03686124; EudraCT 2019-002370-31).

Methods: Pts were ≥18 y, HLA-A*02:01+, PRAME+, have recurrent and/or refractory solid tumors with no additional SOC treatments available, incl. UM with prior tebentafusp, measurable disease (RECIST 1.1) and ECOG PS 0-1. Following leukapheresis and IMA203 manufacture, pts underwent lymphodepletion (LD) with Cy (500 mg/m^2 x 4 d) and Flu (30 mg/m^2 x 4 d), IMA203 infusion followed by low-dose IL-2 (1 M IU QD x5 d, BID x5 d, SC).

Results: As of Apr 7, 2025, 74 heavily pretreated pts (median 3 prior lines) across tumor types were enrolled. Most common TEAEs were LD-related cytopenias in 99% (G3/4). Cytokine release syndrome occurred in 95%, was mostly low grade (G1: 37%; G2: 47%; G3: 11%; no ≥G4) and mostly resolved by day 14. ICANS was infrequent (G1/2: 9%; G3: 4%; no ≥G4). UM pts in Ph 1b (n=16) had a median of 2 prior lines of therapy (62.5% prior tebentafusp), 75% with the largest metastatic lesion measuring 3.1-8.0 cm in diameter (AJCC M1b) and 56% had elevated LDH. A confirmed objective response was achieved by 67% (10/15) (1 pt had ongoing unconfirmed PR); all pts (16/16) experienced tumor shrinkage. Median DOR was 11.0 mo (range 1.8+ to 31.6) with 5 of 10 confirmed responses ongoing at a median FU of 13.4 mo. Median PFS was 8.5 mo (range 1.4 to 32.9) and median OS was 16.2 mo (range 3.2+ to 34.2+). Rapid engraftment (median T_{max}: 3 days) and sustained persistence of IMA203 T cells (up to 903 days) were observed in pts with UM, indicating durable *in vivo* exposure.

Conclusions: IMA203 has a high degree of activity in pts with UM, with 67% ORR and promising survival outcomes. Tolerability was favorable. Further study in a larger UM cohort is warranted and planned.

Clinical trial identification: NCT03686124, EudraCT 2019-002370-31.

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A.M. Tsimberidou: Financial Interests, Institutional, Funding, No Financial Interest: OBI Pharma; Financial Interests, Institutional, Funding, No financial interest: Agenus, IMMATICS, Tachyon, Novocure, Orionis, Tempus, Tvardi; Financial Interests, Institutional, Funding, NO financial interest: Parker Institute for Cancer Immunotherapy; Financial Interests, Personal, Other, Advisory Role: Avstera Therapeutics, BrYet, Macrogenics, NEX-I, OBI Pharmaceuticals; Financial Interests, Institutional, Coordinating PI: Anaveon, AbbVie. J.J. Luke: Financial Interests, Personal, Advisory Role: Abbvie, Agenus, AstraZeneca, Bayer, Bristol Myers Squibb, Clasp, Curadev, Eisai, EMD Serono, Geneos, Gilead, HotSpot, Krystal, Janssen, Ikena, Immatics, Incyte, IO Biotech, iTeos, LegoChem, Lyvgen, Merck, Mersana, Novartis, Pfizer, Pioneering Medicines, R; Financial Interests, Personal, Advisory Board: BioCytics, Bright Peak, RefleXion, Xilio, Actym, Duke Street, Elipscience, Kanaph, NeoTx, Onc.AI, OncoNano, Pyxis, Saros, Tempest, Zola; Financial Interests, Institutional, Research Funding: AbbVie, Astellas, Astrazeneca, Bristol Myers Squibb, Corvus, Day One, EMD Serono, Fstar, Genmab, Hot Spot, Ikena, Immatics, Imugene, Incyte, Janux, Kadmon, KAHR, Macrogenics, Merck, Moderna, Nektar, Next Cure, Novartis, Numab, Palleon, Pfizer, Replimmune,. W. Alsdorf: Financial Interests, Personal, Advisory Role: AstraZeneca, Astellas, GSK, Johnson&Johnson; Financial Interests, Personal, Other: Biontech, Immatics, Johnson&Johnson; Financial Interests, Institutional, Research Funding: Biontech, Affimed. S.C. DeVane: Financial Interests, Full or part-time Employment: Immatics US; Financial Interests, Stocks or ownership: Immatics US; Financial Interests, Funding: Immatics; Financial Interests, Other: Immatics. S. Hengler: Financial Interests, Full or part-time Employment: Immatics Biotechnologies Gmbh; Financial Interests, Funding: Immatics Biotechnologies Gmbh. N. Hilf: Financial Interests, Personal, Full or part-time Employment: Immatics US; Financial Interests, Personal, Stocks or ownership: Immatics US; Financial Interests, Personal, Funding: Immatics; Financial Interests, Personal, Other: Immatics US. M..A. Kursunel: Financial Interests, Full or part-time Employment: Immatics Biotechnologies Gmbh; Financial Interests, Funding: Immatics Biotechnologies Gmbh; Financial Interests, Stocks or ownership: Immatics Biotechnologies Gmbh; Financial Interests, Leadership Role: Immatics Biotechnologies Gmbh; Financial Interests, Other: Immatics Biotechnologies Gmbh; Financial Interests, Royalties: Immatics Biotechnologies Gmbh. A. Mayer-Mokler: Financial Interests, Full or part-time Employment: Immatics Biotechnologies Gmbh; Financial Interests, Stocks or ownership: Immatics Biotechnologies Gmbh; Financial Interests, Royalties: Immatics Biotechnologies Gmbh. D. Pankov: Financial Interests, Full or part-time Employment: Immatics US. C. Britten: Financial Interests, Full or part-time Employment: Immatics Biotechnologies Gmbh; Financial Interests, Leadership Role: Immatics Biotechnologies Gmbh; Financial Interests, Stocks or ownership: Immatics Biotechnologies Gmbh. M. Wermke: Financial Interests, Personal, Advisory Board: Novartis, Boehringer Ingelheim, Bayer, Bristol Myers Squibb, ImCheck Therapeutics, Immatics, Pfizer, AstraZeneca, Tacalyx GmbH, Regeneron, Zymeworks, Daiichi Sankyo, PharmaMar, Iovance Biotherapeutics, Tknife, Genentech; Financial Interests, Personal, Invited Speaker: Boehringer-Ingelheim, Synlab GmbH, Janssen, EMD Merck Serono, GWT TUD GmbH, Amgen, Novartis, MJH Life Sciences, Bristol Myers Squibb International; Financial Interests, Personal, Other, Member of DSMC: ISA Therapeutics; Financial Interests, Institutional, Funding: Roche; Financial Interests, Institutional, Research Grant: Iovance Biotherapeutics; Non-Financial Interests, Personal, Other, Congress Travel Cost Support: AstraZeneca, EMD Merck Serono, Janssen, Boehringer Ingelheim, Immatics, Daiichi Sankyo, Iovance. L. Hernandez-Aya: Financial Interests, Personal, Advisory Board: Bristol Myers Squibb, Replimune; Financial Interests, Personal, Other: Castle Biosciences; Financial Interests, Institutional, Research Funding: Immatics, Replimune, Bristol Myers Squibb, Foghorn, AsherBio, Immunocore. All other authors have declared no conflicts of interest.

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1601O 3-year survival with neoadjuvant-adjuvant pembrolizumab from SWOG S1801

S.P. Patel[1], M. Othus[2], V. Prieto[3], M. Lowe[4], E. Buchbinder[5], Y. Chen[6], G.P. Wright[7], J.R. Hyngstrom[8], C.D. Lao[9], T-G. Truong[10], S. Chandra[11], K. Kendra[12], C. Devoe[13], A. Mangla[14], A. Ikeguchi[15], J.M. Kirkwood[16], L.A. Korde[17], J. Moon[18], V.K. Sondak[19], A. Ribas[20]

[1]Medicine, University of Colorado Anschutz Medical Campus, Aurora, United States of America; [2]Biostatistics, Fred Hutchinson Cancer Research Center, Seattle, United States of America; [3]Pathology, The University of Texas MD Anderson Cancer Center, Houston, United States of America; [4]Surgery Department, Emory University, Atlanta, United States of America; [5]Medical Oncology Department, Dana-Farber Cancer Institute, Boston, United States of America; [6]Medical Oncology Dept., Cancer and Hematology Centers of Western Michigan, Grand Rapids, United States of America; [7]Surgery, Corewell Health William Beaumont University Hospital, Royal Oak, United States of America; [8]Surgery, Rush MD Anderson Cancer Center, Chicago, United States of America; [9]Physician Engagement, Bristol Myers Squibb, New Jersey, United States of America; [10]Department of Haematology and Oncology, Taussig Cancer Center-Cleveland Clinic, Cleveland, United States of America; [11]Division of Hematology Oncology, Robert. H. Lurie Cancer Center of Northwestern University, Chicago, United States of America; [12]Medical Oncology, OSUCCC - The Ohio State University Comprehensive Cancer Center - James, Columbus, United States of America; [13]Medicine, Northwell Health Cancer Institute, Lake Success, United States of America; [14]Hematology and Oncology, Case Western Reserve University/University Hospitals, Cleveland, United States of America; [15]Melanoma Medical Oncology, University of Texas MD Anderson Cancer Center, Houston, United States of America; [16]Melanoma Program Department, University of Pittsburgh Medical Center Hillman Cancer Cente, Pittsburgh, United States of America; [17]Cancer Therapy Evaluation Program, National Cancer Institute, Bethesda, United States of America; [18]Biostatistics, Fred Hutchinson Cancer Center, University of Washington, Seattle, United States of America; [19]Surgery, Moffitt Cancer Center, Tampa, United States of America; [20]Department of Medicine, Jonsson Comprehensive Cancer Center at UCLA, Los Angeles, United States of America

Background: Data from two large randomized trials have shown neoadjuvant (Neo) immunotherapy for resectable melanoma (mel) improves event-free survival (EFS) over adjuvant (Adj) therapy. Adj therapy improves recurrence-free survival (RFS) over placebo in resectable mel but has no effect on overall survival (OS).

Methods: SWOG S1801 was a randomized phase 2 study in 345 pts of Adj pembrolizumab (pem) versus Neo-Adj pem for resectable, clinically detectable mel. Data were analyzed as of May 9, 2025 for updated endpoints.

Results: At a median follow-up of 44 months EFS benefit was maintained in the Neo group (HR 0.67, 95% CI 0.47 – 0.94). 3-year EFS was 68% in the Neo group (95% CI 62-72%) and 55% in the Adj group (95% CI 48-64%). With 32 death events in the Neo group and 47 deaths in the Adj group, 3-year OS was 84% versus 73% (HR 0.65, 95% CI 0.42 – 1.02). In an exploratory subgroup of 33 pts with resectable stage IV mel, EFS and OS demonstrate a trend toward improvement in the Neo-Adj group with only 1 event for EFS and 0 events for OS compared to 10 events for EFS and 5 deaths in the Adj group (HR for EFS 0.08 95% CI 0.01-0.61, HR for OS could not be estimated with 0 events in Neo-Adj group). EFS and OS were similar between treatment groups in pts harboring *BRAF* mutation. In the subgroup of Neo-Adj pts that did not undergo surgery (n=18) one pt did not undergo surgery due to toxicity and was still alive without recurrence at 35 months. 2 pts did not undergo surgery due to radiographical CR; 1 progressed at 5.2 months and died at 3.6 yr and 1 was alive without recurrence at 5 yr. 2 Two pts did not have surgery due to co-morbidities and/or COVID-related constraints at the hospital – both died in less than a year after going off protocol. The remaining 13 pts did not have surgery due to progression of mel rendering them unresectable - 6 have died, 3-year OS = 61%. With longer follow-up, grade 3/4 toxicity was noted in n=35 (20%) in Neo-Adj group and 29 (17%) in the Adj group (p=0.41). One grade 5 myocarditis was observed in the Adj group.

Conclusions: At a median follow-up of 3.7 years Neo-Adj pem maintains EFS benefit over Adj pem in resectable mel. Median OS remains immature but continues to show a trend toward benefit with Neo-Adj pem.

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1602O Enucleation prevention and vision preservation in primary uveal melanoma (UM): Preliminary results from a phase II study of neoadjuvant darovasertib

M.O. Butler[1], D. Reichstein[2], H. Krema[3], M. Mckean[4], A.M. Joshua[5], L-A. Lim[6], R. O'Day[7], M. Shackleton[8], S. Chandana[9], T. Aaberg[10], L.A. Dalvin[11], M. Angi[12], E. Rossi[13], Z. Correa[14], J. Lutzky[15], M.M. Chao[16], J.C. Sachdev[17], M. Orloff[18], D. Beaupre[19], C. Shields[20]

[1]Medical Oncology and Hematology Department, UHN - University Health Network - Princess Margaret Cancer Center, Toronto, Canada; [2]Ocular Oncology, Tennessee Retina, Nashville, United States of America; [3]Ocular Oncology, UHN - University Health Network - Princess Margaret Cancer Center, Toronto, Canada; [4]Drug Development Unit, SCRI - Sarah Cannon Research Institute, Nashville, United States of America; [5]Medical Oncology, The Kinghorn Cancer Centre, Sydney, Australia; [6]Ocular Oncology, Chatswood Private Hospital, Sidney, Australia; [7]Opthalmology, Royal Victorian Eye and Ear Hospital, Melbourne, Australia; [8]Oncology, Alfred Hospital, Melbourne, Australia; [9]Department of Medical Oncology, START Midwest, Grand Rapids, United States of America; [10]Opthalmology, Retina Specialists of Michigan, Grand Rapids, United States of America; [11]Department of Ophthalmology and Medical Oncology, Mayo Clinic - Rochester, Rochester, United States of America; [12]Surgical Oncology, Fondazione IRCCS - Istituto Nazionale dei Tumori, Milan, Italy; [13]Medical Oncology Dept., Fondazione Policlinico Universitario Agostino Gemelli IRCCS, Rome, Italy; [14]Ophthalmology, University of Miami Bascom Palmer Eye Institute, Miami, United States of America; [15]Cutaneous Oncology, University of Miami Sylvester Comprehensive Cancer Center, Miami, FL, United States of America; [16]Medical Oncology Dept., Ideaya Bio, S. San Francisco, United States of America; [17]Early Phase Clinical Trials, IDEAYA Biosciences, South San Francisco, United States of America; [18]Medical Oncology, Thomas Jefferson Univ Hospital, Philadelphia, United States of America; [19]Clinical Development, IDEAYA Biosciences, South San Francisco, CA, United States of America; [20]Ocular Oncology, Wills Eye Hospital, Philadelphia, United States of America

Background: Primary UM treatment (Rx) requires vision-compromising enucleation or plaque brachytherapy (PB). Approximately 95% of UMs harbor GNAQ/GNA11 mutations with activated oncogenic PKC signaling. Darovasertib (D), a selective PKC inhibitor that inhibits growth of UM tumors in vivo, could alter the treatment paradigm for primary UM.

Methods: IDE196-009 is a phase 2 study (NCT05907954) testing the safety & efficacy of D as neoadjuvant Rx in primary UM requiring enucleation (Cohort 1: Co1) or PB (Cohort 2: Co2). Primary endpoints: Enucleation Prevention (EP) for Co1 & radiation Dose Reduction (DR) for Co2. Tumor shrinkage by Product of Diameters (PoD: height (ht) * longest basal diameter {lbd}) (Co1 & 2) & best corrected visual acuity (BCVA) loss of $\geq$ 15 letters post PB (Co2) are secondary endpoints, while risk of 20/200 vision assessed by a prognostic tool is exploratory (Co2).

Results: Safety population: 90 patients {pts} (n=51 Co1, n=39 Co2) received D at 300 mg BID in 28-day cycles. Median age: 58 yrs (23-85); median tumor ht: Co1: 11mm (2-15), Co2: 7mm (3-14); lbd: Co1: 17 mm (10-22); Co2: 14 mm (7-21). All grade treatment related adverse events (TRAEs) in $\geq$20%; diarrhea (48%), nausea (43%) & fatigue (30%). The only Grade $\geq$3 TRAE in $\geq$ 5% pts was hypotension (6.7%). Drug interruption, modification, or discontinuation due to TRAEs was seen in 16 (18%), 7 (8%) & 4 (4%) pts, resp. Efficacy population: N= 62 (n= 32 Co1, n = 30 Co2) defined as pts with tumor assessment after $\geq$ 3 cycles of D (or progression/AE requiring early discontinuation). In this efficacy population, 59% (19/32) of pts in Co1 became eligible for EP; in Co2 for pts with available paired simulations, DR was seen in 14/17 (82%) pts with 59% (10/17) showing both $\geq$ 20% DR and a lower 3-year predicted risk of 20/200 vision. Tumor shrinkage $\geq \sim$ 20% in the PoD was noted in 62% (Co1) & 63% (Co2) of pts. Among pts who had EP or DR, the majority had this magnitude of tumor shrinkage.

Conclusions: Darovasertib is well-tolerated & is the first Rx to show tumor shrinkage in primary UM resulting in EP, radiation dose reduction & potential for vision preservation. Based on these promising results, a registrational study is planned.

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1603MO Analysis of second primary cutaneous squamous cell carcinoma (CSCC) tumors (SPTs) reported during the C-POST trial, a randomized phase III study of adjuvant cemiplimab vs placebo (pbo) for high-risk CSCC

D. Rischin[1], S. Porceddu[2], F. Day[3], D. Brungs[4], H. Christie[5], J.E. Jackson[6], B. Stein[7], Y.B. Su[8], R. Ladwa[9], G. Adams[10], S.E. Bowyer[11], Z. Otty[12], P. Bossi[13], A. Challapalli[14], A. Hauschild[15], A.M.L. Lim[16], F.A. Seebach[17], S-Y. Yoo[18], P. Goncalves[19], M. Fury[20]

[1]Medical Oncology Department, Peter MacCallum Cancer Centre, Melbourne, Australia; [2]Department of Radiation Oncology, Peter MacCallum Cancer Centre, Melbourne, Australia; [3]Medical Oncology, Calvary Mater Newcastle, Waratah, Australia; [4]Medical Oncology, The Wollongong Hospital, Wollongong, Australia; [5]Medical Oncology, Cancer Care Centre Hervey Bay, Urraween, Australia; [6]Radiation Oncology, Radiation Oncology Centers, Gold Coast, Australia; [7]Medical Oncology Dept., ICON Cancer Centre Adelaide (Kurralta Park), Adelaide, Australia; [8]Head and Neck Medical Oncology, Nebraska Cancer Specialists, Omaha, United States of America; [9]Medical Oncology, Princess Alexandra Hospital - Metro South Health, Woolloongabba, Australia; [10]Radiation Oncology, Genesis Care, Bundaberg, Australia; [11]Medical Oncology, Sir Charles Gairdner Hospital, Nedlands, Australia; [12]Townsville Cancer Centre, Townsville University Hospital, Douglas, Australia; [13]Medical Oncology Department, IRCCS Humanitas Research Hospital, Rozzano, Italy; [14]Department of Oncology, BHOC - Bristol Haematology and Oncology Centre, Bristol, United Kingdom; [15]Dermatology, Venereology and Allergology, UKSH - Universitätsklinikum Schleswig-Holstein - Campus Kiel, Kiel, Germany; [16]Medical Oncology, Peter MacCallum Cancer Centre, Melbourne, Australia; [17]Oncology and Translational Sciences, Regeneron Pharmaceuticals, Inc., Tarrytown, United States of America; [18]Biostatistics, Regeneron Pharmaceuticals, Inc., Tarrytown, United States of America; [19]Oncology, Regeneron Pharmaceuticals, Inc., Tarrytown, United States of America; [20]Oncology Clinical Sciences, Regeneron Pharmaceuticals, Inc., Tarrytown, United States of America

Background: The primary analysis of the phase 3 C-POST trial (NCT03969004) demonstrated that cemiplimab is the first systemic adjuvant therapy to improve disease-free survival (DFS; HR 0.32 vs pbo) in patients (pts) at high risk of CSCC recurrence after surgery and radiotherapy. We report results of CSCC SPTs that arose during the study period.

Methods: A secondary objective of C-POST was to compare the effect of adjuvant cemiplimab vs pbo on the cumulative incidence of SPTs. SPTs were defined as invasive CSCC lesions in the skin unrelated to the index high-risk CSCC. SPTs were managed by local modality therapy as part of routine clinical practice and were not DFS events per protocol. Pts could have multiple SPTs during the study.

Results: From Jun 2019 to Aug 2024, 415 pts were enrolled (209 cemiplimab; 206 pbo). There were 191 SPTs occurring in 68 pts during the study period (on-treatment plus follow-up; Table). Although the number (nbr) of pts with ≥1 SPT was similar in each arm (n=31 cemiplimab; n=37 pbo), the total nbr of SPTs and the annualized SPT rate were lower in the cemiplimab arm (68 SPTs; 0.63 [95% CI: 0.36−1.11]) vs the pbo arm (123 SPTs; 1.20 [0.59−2.44]). In a post-hoc analysis in which first SPTs were counted as DFS events, efficacy still favored cemiplimab (HR 0.43; 95% CI: 0.30−0.60).

Table: 1603MO

	Cemiplimab (n=209)	Pbo (n=206)
Total SPTs, n	68	123
Total pt-years followed	412.4	331.0
SPT rate		
Unadjusted†	0.2	0.4
Adjusted annualized (95% CI)‡	0.63 (0.36−1.11)	1.20 (0.59−2.44)
Pts with ≥1 SPT, n (%)	31 (15)	37 (18)
Pts by number of SPTs, n (%)		
1	17 (8)	25 (12)
2	6 (3)	4 (2)
3	2 (1)	1 (<1)
4	1 (<1)	2 (1)
5	4 (2)	0
6	0	1 (<1)
7	0	2 (1)
9	1 (<1)	0
16	0	1 (<1)
43	0	1 (<1)

†Total SPTs during the study period divided by total pt-years followed. ‡Via negative binomial model: response variable: total SPTs during the study period; covariates: treatment arm, anatomic region of resected high-risk tumor, geographic region; offset variable: log-transformed standardized observation duration.

Conclusions: Cemiplimab may be associated with a lower incidence and rate of SPTs in a subset of pts with high-risk CSCC. The robust efficacy signal with cemiplimab vs pbo is maintained in a post-hoc analysis in which first SPTs were included as DFS events.

Clinical trial identification: NCT03969004.

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1604MO Randomization to adjuvant nivolumab or ipilimumab + nivolumab based on pathological response to a single dose of neoadjuvant nivolumab in stage III melanoma

J. Miura[1], A.K.S. Salama[2], X. Xu[3], G.M. Beasley[4], R. Al-Rohil[5], R. Brody[6], S. Mitnick[7], J. Tabita-Martinez[7], L. Giles[7], S. McGettigan[7], K. Kreider[7], A. Nandi[7], G. Wang[7], G. Mo[7], L. Schuchter[7], R. Amaravadi[7], A. Huang[7], P. Gimotty[8], G. Karakousis[1], T.C. Mitchell[7]

[1]Division of Endocrine and Oncologic Surgery, Abramson Cancer Center - University of Pennsylvania, Philadelphia, United States of America; [2]Department of Medicine, Duke University, Durham, United States of America; [3]Department of Pathology and Laboratory Medicine, University of Pennsylvania, Philadelphia, United States of America; [4]Department of Surgery, Duke Cancer Institute, Durham, United States of America; [5]Department of Pathology, Duke Cancer Institute, currently University of North Carolina - Chapel Hill, Durham, United States of America; [6]Department of Otorhinolaryngology-Head and Neck Surgery, Abramson Cancer Center - University of Pennsylvania, Philadelphia, United States of America; [7]Division of Hematology/ Oncology, Abramson Cancer Center of the University of Pennsylvania, Philadelphia, United States of America; [8]Department of Biostatistics, Epidemiology and Informatics, Abramson Cancer Center - University of Pennsylvania, Philadelphia, United States of America

Background: Melanoma patients (pts) who do not have a pathological response to neoadjuvant anti-PD-1 have a poor prognosis; intensifying adjuvant therapy based on pathological response after a short course of neoadjuvant therapy may improve outcomes and reduce toxicity.

Methods: In this investigator-initiated study, pts with clinical stage III melanoma received one dose of nivolumab (480 mg) followed by surgery 4 weeks later. Pts with a major pathological response (MPR), including pathological complete (pCR) or near complete (<10% viable tumor; near pCR) response received adjuvant nivolumab for up to one year (Arm A). Patients with less than a MPR were randomized 1:2 to either adjuvant nivolumab (Arm B) or adjuvant ipilimumab (1mg/kg) plus nivolumab (3mg/kg) every 3 weeks for 4 doses followed by nivolumab (Arm C). The primary endpoint was 1-year recurrence-free survival (RFS). Secondary endpoints included overall survival (OS), MPR rate, and safety.

Results: 70 patients were enrolled from 2019-24. 66% men. 54% stage IIIC/D by baseline clinical nodes. All underwent resection. 23 pts had a MPR (33%; 20 pCR, 3 near pCR). 7 patients had pPR. At a median follow-up of 21.4 months, the estimated 1-year RFS rate was 100% for pts treated per arm A, 55.6% (22.6% − 77%, 95% CI) for Arm B and 61.5% (40.3% − 77.1%) for Arm C. The median OS was 21.4 months in all patients and was 22.9 in Arm A. 9 pts (37.5%) in Arms A and B, and 15 pts (62.5%) in Arm C had grade 3/4 treatment-related adverse events.

Conclusions: Patients who had a MPR after a single dose of neoadjuvant PD-1 blockade have a high RFS (100%) and OS (100%). MPR was prognostic of RFS (log-rank p=0.002) and OS (log-rank p=0.019). Pts without a MPR had a numerically higher RFS when treated with ipi/nivo than with nivo alone, which was not statistically significant. This study demonstrated that adaptive adjuvant therapy based on pathological response is feasible. Longer follow-up will confirm whether or not there may be any benefit of immunotherapy escalation in the adjuvant setting. Translational studies are under way to evaluate the effect of different adjuvant therapies on immunophenotype of melanoma-specific CD8 T cells and across cell types.

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Funding: BMS.

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1605MO Primary results from a randomized phase II trial of BNT111 in combination with cemiplimab with calibrator monotherapy arms in anti-PD-(L)1 relapsed/refractory melanoma

P.A. Ascierto[1], S. Grabbe[2], M. Guida[3], F. Carnevale Schianca[4], P. Rutkowski[5], A.M. Arance Fernandez[6], J.L. Markman[7], V. Chen[8], H. Yang[8], K. Cuk[9], M.A. Kaczorowska[10], I. Büttel[9], S. Liebscher[11], M. Wenger[10], I. Lowy[12], F.A. Seebach[13], P. Brück[10], Ö. Türeci[14], U. Sahin[14], G.V. Long[16]

[1]Melanoma, Cancer Immunotherapy & Developmental Therapeutics, Istituto Nazionale Tumori IRCCS - Fondazione G. Pascale, Naples, Italy; [2]Dermatology, JGU - Johannes Gutenberg University, Mainz, Germany; [3]Rare Tumors and Melanoma Unit, Istituto Tumori Bari Giovanni Paolo II - IRCCS, Bari, Italy; [4]Clinical Cell Therapy, IRCCS - Istituto di Candiolo - FPO, Candiolo, Italy; [5]Department of Soft Tissue/Bone Sarcoma and Melanoma, Maria Sklodowska-Curie National Research Institute of Oncology, Warsaw, Poland; [6]Department of Medical Oncology, H. Clinic Barcelona, Barcelona, Spain; [7]Translational Sciences Oncology, BioNTech US, Cambridge, United States of America; [8]Biostatistics, BioNTech US, Cambridge, United States of America; [9]Global Regulatory Affairs, BioNTech SE, Mainz, Germany; [10]Clinical Development, BioNTech SE, Mainz, Germany; [11]Biostatistics, BioNTech SE, Mainz, Germany; [12]Clinical Sciences Oncology, Regeneron Pharmaceuticals, Inc - Corporate Headquarters, Tarrytown, United States of America; [13]Global Clinical Development, Regeneron Pharmaceuticals, Inc., Tarrytown, United States of America; [14]BioNTech SE, Mainz, Germany; [15]Faculty of Medicine and Health, The University of Sydney, and Mater and Royal North Shore Hospitals, University of Sydney, Melanoma Institute Australia, Sydney, Australia

Background: BNT111 is an investigational uridine RNA-based lipoplex cancer immunotherapy targeting the non-mutated, tumor-associated antigens NY-ESO-1, MAGE-A3, Tyrosinase and TPTE. A previous phase 1 trial (Lipo-MERIT) established a tolerable dose range and indicated early signs of clinical activity of BNT111 monotherapy (mono) and in combination with PD-1 inhibition in heavily pre-treated melanoma. Here, we present data from the phase 2 BNT111-01 trial in patients (pts) with unresectable Stage III or IV non-acral cutaneous melanoma (NACM).

Methods: The open label, randomized, multi-site, interventional phase 2 BNT111-01 trial (NCT: 04526899) aims to determine efficacy of BNT111 + cemiplimab (cemi) in anti-PD-(L)1 refractory/relapsed pts with NACM. Pts were randomized 2:1:1 to Arm 1 (BNT111 + cemi) and calibrator arms 2 (BNT111 mono) and 3 (cemi mono). The primary endpoint is objective response rate (ORR) by blinded independent central review according to RECIST 1.1 in Arm 1 vs a hypothetical historical control of 10% (H0-hypothesis). Secondary endpoints include safety, progression-free survival (PFS) and overall survival (OS).

Results: 184 pts were randomized (94 in Arm 1, 46 in Arm 2, 44 in Arm 3), with comparable baseline characteristics between arms (median age 64.5 years; range 18 − 91). As of 14 FEB 2025, median follow-up was 15.6 months (mo). ORR of BNT111 + cemi was 18.1%, including 11 complete and 6 partial responses, resulting in rejection of the H0 hypothesis. Disease control rate (DCR) was 55.3%, median duration of response was not reached, median PFS was 3.1 mo and median OS 20.7 mo. ORR, DCR, median PFS and median OS for BNT111 mono were 17.4%, 58.7%, 2.8 mo and 13.7 mo; for cemi mono they were 13.6%, 47.7%, 3.2 mo and 22.3 mo, respectively. The safety profile in 181 treated pts was manageable, with typical cytokine release-related treatment-emergent adverse events in BNT111 treatment arms.

Conclusions: BNT111 + cemi in advanced NACM indicated statistically significant improvement against an assumed historic control ORR of 10%. BNT111 also indicated clinical activity as monotherapy. No meaningful differences in median PFS or OS were noted between combination and monotherapies.

Clinical trial identification: NCT04526899, EudraCT 2023-509513-36-00.

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1606P Use of a computational histologic artificial intelligence (CHAI) assay to predict overall survival in stage II and III cutaneous melanoma

L. Karapetyan[1], K.K. Tsai[2], V. Krishna[3], W.M. Abuzeid[4], S. Shingi[5], V. Krishna[5], K. Tsai[6], I. Aksit[1], R. Bansal[5], H. Zhang[3], A. Neema[3], T. Royce[4], L. Kiedrowski[7], A. Joshi[3], J.M. Kirkwood[8]

[1]Department of Cutaneous Oncology, Moffitt Cancer Center, Tampa, United States of America; [2]Clinical Medicine, Helen Diller Family Comprehensive Cancer Center, University of California San Francisco, San Francisco, United States of America; [3]Computational Oncology, Valar Labs Inc., Palo Alto, United States of America; [4]Clinical Development, Valar Labs Inc., Palo Alto, United States of America; [5]Computational Oncology, Valar Labs Inc., Palo Alto, United States of America; [6]Pathology, Moffitt Cancer Center, Tampa, United States of America; [7]Medical Affairs, Valar Labs Inc., Palo Alto, United States of America; [8]Melanoma Program Department, University of Pittsburgh Medical Center Hillman Cancer Cente, Pittsburgh, United States of America

Background: Most melanoma mortality occurs in patients diagnosed at earlier stages who then develop recurrent/advanced disease. This highlights the need for accurate prognostic tools to identify patients at high risk of mortality at the time of diagnosis who could benefit from increased surveillance or earlier interventions. We sought to determine if a Computational Histologic Artificial Intelligence (CHAI) assay, which uses deep learning to quantify features from digitized whole slide images (WSI) of diagnostic H&E specimens, could predict overall survival (OS).

Methods: Pathologic Stage II/III melanoma cases with WSI were extracted from The Cancer Genome Atlas and analyzed using the CHAI assay. Histologic features were correlated to OS using multivariate Cox proportional hazards models. Cases were stratified into "high" and "low" risk groups by median split of a continuous AI score and analyzed with Kaplan-Meier statistics and log-rank test in cross validation.

Results: 284 Stage II ($n=123$) or III ($n=161$) patients with available WSI were analyzed. There was no significant difference in clinicopathologic characteristics between patients identified as high or low risk. High risk cases were more likely to have worse OS versus low risk cases (HR 2.1, 95%CI 1.7-2.6, $p<0.0008$) with median 5-yr OS of 45% and 68%, respectively. Multivariate regression indicated that the CHAI biomarker prognosticated OS independently of age, stage, Breslow thickness, and ulceration (HR 2.4, 95%CI 1.9-3.1, $p<0.0007$).

Table: 1606P

Univariable analysis	Hazard ratio (95% confidence interval)	Regression coefficient	P-value
CHAI biomarker	2.08 (1.67, 2.59)	0.73	0.0008
Age	1.02 (1.01, 1.03)	0.02	0.01
Stage	1.26 (1.01, 1.56)	0.23	0.30
Breslow thickness	1.02 (1.01, 1.03)	0.02	0.05
Ulceration	2.04 (1.56, 2.67)	0.71	0.008
Multivariable analysis	**Hazard ratio (95% confidence interval)**	**Regression coefficient**	**P-value**
CHAI biomarker	2.43 (1.87, 3.15)	0.89	0.0006
Age	1.02 (1.01, 1.03)	0.02	0.02
Stage	2.23 (1.72, 2.88)	0.80	0.002
Breslow thickness	1.02 (1.01, 1.03)	0.02	0.07
Ulceration	2.07 (1.56, 2.75)	0.73	0.01

Conclusions: A deep learning-powered histologic assay that quantifies histologic features from diagnostic pathology slides predicted OS in Stage II and III melanoma patients independently of traditional clinicopathologic characteristics. With future work, including external validation, this histologic biomarker could facilitate personalized melanoma treatment by identifying patients at high risk of mortality based on their initial diagnostic biopsy.

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1607P Circulating tumor cell dynamics in disease relapse among patients with stage I-IV melanoma

J. Chen[1], V. Sarli[2], S. Meas[3], J. Upshaw[1], A. Lucci[4]

[1]Breast Surgical Oncology, The University of Texas MD Anderson Cancer Center, Houston, United States of America; [2]Breast Surgical Oncology, MD Anderson Cancer Center, Houston, United States of America; [3]Breast Surgery Research, MD Anderson Cancer Center, Houston, United States of America; [4]Breast Surgery Department, The University of Texas MD Anderson Cancer Center, Houston, United States of America

Background: Longitudinal monitoring of circulating tumor cells (CTC) may offer more prognostic value compared to assessments at a single timepoint. We characterized CTC dynamics and evaluated association with disease progression in melanoma patients.

Methods: Patients with $\geq$2 serial blood draws collected q6-12 months were included (2009-2024). CTCs were enumerated using CellSearch with $\geq$1 considered positive. CTC dynamics were categorized as persistently negative, last blood draw (BD) negative or positive, or persistently positive. Kaplan-Meier method and multivariable Cox proportional hazards model were used for statistical analysis.

Results: Of 403 patients, mean age was 64.1 years (SD14.2). Patients were 93.1% (375) non-Hispanic White, 64% (258) male, with predominantly stage III-IV disease (327, 81.1%). 32.3% (53) received neoadjuvant therapy, 90.6% (365) underwent primary tumor resection, and 89.5% underwent nodal dissection (118 CLND, 207 SLNB). In total, 1218 CTC assessments were performed with a median of 3 blood draws per patient (IQR2-4). 16.3% (229) had baseline blood draw within 3 months of diagnosis and 38.6% (141) had blood draw prior to surgery. Across serial CTC assessments, 22.6% (91) were persistently negative, 36.2% (146) ended on a negative CTC, 33.3% (134) ended on a positive CTC, and 7.9% (32) were persistently positive. There were no significant differences in age, gender, race, stage, and pathologic features by CTC dynamic categorization. At a median follow-up of 39 months (95% CI:36.5-41.5), 202 (50.1%) recurred and 69 (17.1%) died. Patients who remained persistently negative had significantly longer progression-free survival compared to those had a positive CTC on last BD or were persistently positive ($p=0.004$). On multivariable analysis adjusted for age and nodal burden, patients with last BD positive or persistently positive had significantly increased risk of disease progression (HR 2.4, 95% CI:1.4-4.0 and HR 3.0, 95% CI:1.5-6.0, both $p=0.001$, respectively).

Conclusions: Unfavorable CTC dynamics, those positive on last BD or persistently positive, had higher risk of melanoma disease relapse. Serial monitoring and characterization of CTCs are necessary for more accurate risk stratification among melanoma patients.

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1608P Fertility outcomes in young female patients treated with adjuvant anti-PD-1 therapy for resectable melanoma

S. Pedersen[1], E. Ellebaek[1], S.G. Kristensen[2], C.H. Ruhlmann[3], T.H. Borch[1], M. Donia[1], J. Fedder[4], L.B. Colmorn[2], K.L.T. Macklon[2], I.M. Svane[1]

[1]Department of Oncology, National Center for Cancer Immune Therapy (CCIT-DK), Herlev Hospital, Herlev, Denmark; [2]Department of Gynaecology, The Fertility Clinic, Rigshospitalet, Copenhagen, Denmark; [3]Department of Oncology, Odense University Hospital, Odense, Denmark; [4]Department of Reproductive Medicine, Center of Andrology & Fertility Clinic, Odense, Denmark

Background: Immune checkpoint inhibitors (ICIs) have transformed cancer treatment with a curative potential for some patients. As ICIs are increasingly used in the adjuvant and neoadjuvant setting for malignancies such as melanoma—often affecting younger adults—there is growing concern about long-term safety, including potential effects on fertility. To date, no studies have specifically evaluated the impact of ICIs on fertility in the adjuvant setting.

Methods: We conducted a national, multicenter, prospective observational study of female patients <45 years of age treated with adjuvant anti-PD-1 immunotherapy for resectable stage III—IV melanoma from 2018 to 2024. Fertility monitoring was performed at baseline and at 3, 6, 12, and 18 months after treatment initiation. Each visit included transvaginal ultrasound and serum analysis of anti-Müllerian hormone (AMH), estradiol, progesterone, follicle-stimulating hormone (FSH), luteinizing hormone (LH), and inhibin-B.

Results: Seventeen female patients were enrolled. The mean age was 28.9 years (range 17—38). Transvaginal ultrasound showed normal ovarian morphology in all patients at all time points. No patients had a decline in AMH during or after treatment. FSH, LH, estradiol, progesterone, and inhibin-B levels exhibited physiological fluctuations consistent with normal menstrual cycling. Five patients had at least one child following completion of adjuvant therapy. Two patients developed endocrine immune-related toxicity (type 1 diabetes and hypophysitis); both had children post-treatment. The median follow-up was 55 months.

Conclusions: This is the first study to report fertility outcomes during and after adjuvant anti-PD-1 immunotherapy in patients with melanoma. In this cohort, we found no evidence of ovarian insufficiency or impaired fertility, including in patients with endocrine toxicities. These findings are important for counseling young patients receiving adjuvant therapy who may wish to conceive in the future.

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1609P Final, 9-year results from the CheckMate 238 phase III trial of adjuvant nivolumab vs ipilimumab in resected stage IIIB—C or IV melanoma

P.A. Ascierto[1], M. Del Vecchio[2], B. Merelli[3], H. Gogas[4], A.M. Arance[5], S. Dalle[6], C.L. Cowey[7], M. Schenker[8], C. Gaudy Marqueste[9], J. Pigozzo[10], I. Marquez-Rodas[11], M.O. Butler[12], A.M. Di Giacomo[13], M.R. Middleton[14], O. Gligich[15], L. de la Cruz Merino[16], R. Soomro[17], D. Rathod[17], M. Askelson[17], J. Larkin[18]

[1]Melanoma, Cancer Immunotherapy & Developmental Therapeutics, Istituto Nazionale Tumori IRCCS Fondazione Pascale, Naples, Italy; [2]Medical Oncology Department, Fondazione IRCCS Istituto Nazionale dei Tumori, Milan, Italy; [3]Oncology Dept., ASST Papa Giovanni XXIII Bergamo, Bergamo, Italy; [4]First Department of Medicine, National and Kapodistrian University of Athens, Athens, Greece; [5]Medical Oncology Department, Hospital Clínic Barcelona, Barcelona, Spain; [6]Medical Oncology, Hospices Civils de Lyon, Pierre-Bénite, France; [7]Medical Oncology, Texas Oncology-Baylor Charles A. Sammons Cancer Center, Dallas, United States of America; [8]Medical Oncology Department, Oncology Center Sf Nectarie Ltd., Craiova, Romania; [9]Dermatology, Aix-Marseille University, APHM, Timone Hospital, Marseille, France; [10]Oncology, Veneto Institute of Oncology IOV-IRCCS, Padua, Italy; [11]Medical Oncology Department, Hospital General Universitario Gregorio Marañón, Madrid, Spain; [12]Medical Oncology and Hematology Department, Princess Margaret Cancer Centre, Toronto, Canada; [13]Department of Medical Oncology and Immunotherapy, University of Siena and Center for Immuno-Oncology, University Hospital, Siena, Italy; [14]Oncology Department, Churchill Hospital, Oxford, United Kingdom; [15]Hematology/ Oncology, Mount Sinai Comprehensive Cancer Center, Miami Beach, United States of America; [16]Clinical Oncology, Hospital Universitario Virgen Macarena, University of Seville, Seville, Spain; [17]Medical Oncology, Bristol Myers Squibb, Princeton, United States of America; [18]Medicine Department, The Royal Marsden Hospital, London, United Kingdom

Background: Nivolumab (NIVO) improved recurrence-free survival (RFS) vs ipilimumab (IPI) in pts with high-risk, resected stage IIIB—C or IV melanoma in CheckMate 238, which led to the first approval of a PD-1 inhibitor as adjuvant melanoma treatment. Final, 9-y results and new analyses are presented.

Methods: Pts aged > 15 y (stratified by disease stage and PD-L1 status) were randomized 1:1 to NIVO 3 mg/kg Q2W (n = 453) or IPI 10 mg/kg Q3W for 4 doses then Q12W (n = 453) for up to 1 y or until disease recurrence/unacceptable toxicity. Endpoints included RFS, overall survival (OS), distant metastasis-free survival (DMFS) in stage III pts, and safety. Other post hoc analyses included progression-free survival through next-line therapy (PFS2), melanoma-related mortality analyzed via melanoma-specific survival (MSS) and competing risks methodology, and the impact the timing of dose administration has on outcomes.

Results: At 9-y minimum follow-up, RFS with NIVO remained significantly improved vs IPI (HR 0.76 [95% CI 0.63—0.90), DMFS and PFS2 continued to favor NIVO; OS and MSS outcomes remained high (Table). Since the 7-y analysis, there were 8 new recurrence events for NIVO (2 regional, 2 local, 2 new primary, 2 death) and 2 for IPI (1 new primary, 1 death). For NIVO vs IPI respectively, there were a total of 7 and 10 new deaths (4 and 7 due to melanoma). Subsequent systemic therapy was less frequent with NIVO than with IPI (45% vs 51%), including subsequent immunotherapy in 26% vs 37% of pts. There were no new voluntarily reported late-emergent treatment-related adverse events since the 4-y database lock, and overall rates remain low.

Table: 1609P

	NIVO (n = 453)	9-y rates, % (95% CI) HR[a] (95% CI)	IPI (n = 453)
RFS	44 (39—49)	0.76 (0.63—0.90)	37 (32—42)
OS	69 (64—74)	0.88 (0.69—1.11)	65 (60—70)
MSS	74 (69—78)	0.87 (0.67—1.12)	70 (65—74)
DMFS[b]	54 (48—59)	0.81 (0.65—1.00)	48 (42—53)
PFS2	55 (49—59)	0.77 (0.63—0.93)	47 (42—52)

[a]NIVO vs IPI (stratified Cox proportional hazards model)
[b]In stage III patients (NIVO = 370; IPI = 366)

Conclusions: In this final analysis for CheckMate 238, NIVO continued to show superior long-term RFS vs IPI in pts with stage IIIB—C or IV resected melanoma. These data, which are the longest follow-up of a checkpoint inhibitor in any adjuvant solid tumor setting, show that DMFS and PFS2 favored NIVO. Moreover, OS and MSS rates are high, at least 70% in the NIVO arm.

Clinical trial identification: NCT02388906.

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Investigator: BMS, MSD, Regeneron. I. Marquez-Rodas: Financial Interests, Personal, Advisory Board: BMS, MSD, Novartis, Pierre Fabre, Roche, GSK, AstraZeneca, Celgene, Regeneron, Sanofi, Merck Serono, Highligth Therapeutics, Bioline Rx, Sun Pharma, Immunocore; Financial Interests, Personal and Institutional, Coordinating PI, GEM1802 clinical trial: Pierre Fabre; Financial Interests, Institutional, Coordinating PI, GEM1801 clinical study: Novartis, BMS, Roche, Pierre Fabre, Incyte, MSD; Financial Interests, Institutional, Coordinating PI, National coordinator of Spotlight 203 clinical trial: Highlight Therapeutics; Non-Financial Interests, Member of Board of Directors: Spanish Melanoma Group. M.O. Butler: Financial Interests, Personal, Advisory Board: BMS, Merck, Novartis, Adaptimmune, Iovance, GSK, Sanofi, La Roche-Possey, Pfizer, Medison, IDEAYA, Regeneron; Financial Interests, Personal, Invited Speaker: BMS, Merck, Novartis, Sanofi, Pfizer; Financial Interests, Personal, Advisory Board, Safety Review Committee: Adaptimmune; Financial Interests, Institutional, Other, Conduct Clinical Trial: TCR2, Novartis, Sanofi, Immunocore, GSK, Pfizer, Merck, Bristol Myers Squibb, Regeneron, AstraZeneca, Adaptimmune, IDEAYA Biosciences, Amgen, Instil Bio, Turnstone Biologics, Iovance, Ankara; Financial Interests, Institutional, Research Grant, support clinical trial: Merck; Financial Interests, Institutional, Funding, support clinical trial: Takara Bio; Financial Interests, Institutional, Funding, support quality improvement project: Novartis. A.M. Di Giacomo: Financial Interests, Personal, Advisory Board: Bristol Myers Squibb, MSD, Pierre Fabre, Novartis, Immunocore, Sunpharma; Financial Interests, Personal, Invited Speaker: Sanofi. M.R. Middleton: Financial Interests, Institutional, Coordinating PI, Fees for clinical trial conduct: Replimune; Financial Interests, Institutional, Local PI, Fees for trial conduct: BMS, Mural Oncology, MSD, Novartis, Bayer, Arcus Biosciences, AstraZeneca, Moderna; Financial Interests, Institutional, Funding, Fees for translational research: GSK; Financial Interests, Institutional, Funding, Educational grant towards IIS; fees for clinical trial conduct: Immunocore; Financial Interests, Institutional, Coordinating PI, Grant towards IIS: GRAIL; Non-Financial Interests, Other, Multiple roles as reviewer and committee member: Cancer Research UK; Non-Financial Interests, Member: AACR, ASCO, ACP. L. de la Cruz Merino: Financial Interests, Personal, Advisory Board: AstraZeneca, MSD; Financial Interests, Personal, Invited Speaker: Gilead, Roche, AbbVie, Novartis; Financial Interests, Institutional, Coordinating PI: Roche – Genentech; Financial Interests, Institutional, Local PI: Bristol Myers Squibb International, Merck, Novartis, AstraZeneca, Gilead, Regeneron Pharmaceuticals, Pierre Fabre. D. Rathod: Financial Interests, Personal, Full or part-time Employment: Bristol Myers Squibb International; Financial Interests, Personal, Stocks/Shares: Bristol Myers Squibb International. M. Askelson: Financial Interests, Personal, Full or part-time Employment: Bristol Myers Squibb; Financial Interests, Personal, Stocks/Shares: Bristol Myers Squibb. J. Larkin: Financial Interests, Personal, Other, Consultancy: Incyte, iOnctura, Apple Tree, Merck, BMS, Eisai, Debipharm, Pfizer, Novartis, MSD, Iovance Biotherapeutics, Boston Biomedical, YKT Global, Immunocore; Financial Interests, Personal, Other, Honorarium: touchIME, touchEXPERTS, Royal College of Physicians, Pfizer, Novartis, Incyte, Merck, Pfizer, Roche, iOnctura, Dynavax, CRUK, GSK, BMS; Financial Interests, Personal, Invited Speaker, Speaker Fee: BMS, Pfizer, Roche, Pierre Fabre, AstraZeneca, Novartis, EUSA Pharma, MSD, Merck, GSK, Ipsen, Aptitude, Eisai, Calithera, Ultimovacs, Seagen, eCancer; Financial Interests, Institutional, Funding: BMS, MSD, Novartis, Pfizer, Achilles, Roche, Nektar, Covance, Immunocore, Pharmacyclics, Aveo. All other authors have declared no conflicts of interest.

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1610P Adjuvant nivolumab vs placebo in resected stage IIB/C melanoma: 4-year update from CheckMate 76K

J.M. Kirkwood[1], M. Del Vecchio[2], C. Hoeller[3], P. Mohr[4], S. Grabbe[5], C. Dutriaux[6], J. Mackiewicz[7], P. Rutkowski[8], P. Arenberger[9], G. Quereux[10], J. Pigozzo[11], T. Meniawy[12], P.A. Ascierto[13], S. Dalle[14], J. Taube[15], A. Nassar[16], Q. duong[16], Z. Li[16], G.V. Long[17]

[1]Melanoma Program Department, UPMC Hillman Cancer Center, Pittsburgh, United States of America; [2]Medical Oncology Department, Fondazione IRCCS Istituto Nazionale dei Tumori, Milan, Italy; [3]Department of Dermatology and Dermatooncology, Medizinische Universität Wien, Vienna, Austria; [4]Dermato-Oncology department, Elbe Kliniken Buxtehude, Buxtehude, Germany; [5]Medical Oncology, University of Mainz Medical Center, Mainz, Germany; [6]Medical Oncology, Hôpital Saint-André, Bordeaux, France; [7]Medical Oncology, Institute of Oncology, Poznan University of Medical Sciences, Poznan, Poland; [8]Department of Soft Tissue/Bone Sarcoma and Melanoma, Maria Skłodowska-Curie National Research Institute of Oncology, Warsaw, Poland; [9]Medical Oncology, Charles University Third Faculty of Medicine and University Hospital of Královské Vinohrady, Prague, Czech Republic; [10]Dermatology, Nantes University Hospital, Nantes, France; [11]Oncology, Veneto Institute of Oncology-Istituto di Ricovero e Cura a Carattere Scientifico, Padua, Italy; [12]Medical Oncology, Sir Charles Gairdner Hospital, Subiaco, Australia; [13]Melanoma, Cancer Immunotherapy & Developmental Therapeutics, Istituto Nazionale Tumori IRCCS 'Fondazione G. Pascale', Naples, Italy; [14]Medical Oncology, Hospices Civils de Lyon, Cancer Research Center of Lyon, Pierre Benite, France; [15]Pathology, Johns Hopkins Medical Institutions, Baltimore, United States of America; [16]Medical Oncology, Bristol Myers Squibb, Princeton, United States of America; [17]Medical Oncology, Melanoma Institute Australia, University of Sydney, and Royal North Shore and Mater Hospitals, Sydney, Australia

Background: Patients (pts) with resected stage IIB or IIC melanoma are at high risk of recurrence following surgery. In CheckMate 76K (NCT04099251), treatment with nivolumab (NIVO) demonstrated a statistically significant improvement in recurrence-free survival vs placebo (PBO) in pts with completely resected stage IIB/C melanoma, leading to widespread regulatory approval. Long-term follow-up data are presented here.

Methods: Pts aged $\geq$ 12 y without evidence of residual disease following resection were stratified by tumor (T) category and randomized 2:1 to receive IV NIVO 480 mg or PBO Q4W for up to 1 y. The primary endpoint was recurrence-free survival (RFS); other endpoints were distant-metastasis free survival (DMFS), safety, and progression-free survival 2 (PFS2).

Results: With a minimum follow-up of 40.2 mo (last pt randomized to cut off) and median follow-ups of 46.8 (NIVO) and 46.4 (PBO) mo (median time from randomization date to death or last alive date), NIVO continued to demonstrate improved RFS and DMFS vs PBO (Table). Of 526 NIVO vs 264 PBO pts, recurrence events occurred in 131 (24.9%) vs 100 (37.9%) pts, respectively, including distant recurrence in 65 (12.4%) vs 52 (19.7%); distant recurrence sites included lung (50 [9.5%] vs 37 [14.0%]), lymph node (23 [4.4%] vs 23 [8.7%]), liver (18 [3.4%] vs 9 [3.4%]), and brain (10 [1.9%] vs 12 [4.5%]). Fewer NIVO pts received subsequent systemic therapy (15.0%) vs PBO (31.4%), and pts receiving adjuvant NIVO had sustained benefit vs PBO beyond first progression (PFS2 medians not yet reached, hazard ratio 0.76 [95% CI, 0.54–1.08]). There were no new safety signals. At this time, overall survival data are still not yet mature.

Table: 1610P

	NIVO (n = 526)		PBO (n = 264)
		4-y rates, % (95% CI) HR[a] (95% CI)	
RFS	66.1 (61.1–70.6)	0.64 (0.50–0.82)	55.3 (48.3–61.7)
DMFS	74.5 (69.7–78.6)	0.73 (0.54–0.98)	68.8 (62.0–74.6)

[a]NIVO vs PBO (stratified Cox proportional hazards model).

Conclusions: These long-term results at a 4-y follow-up continue to show a positive benefit–risk profile of adjuvant NIVO as a treatment option for pts with resected stage IIB/C melanoma. This long-term benefit confirms NIVO as an effective adjuvant treatment.

Clinical trial identification: NCT04099251.

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Kirkwood: Financial Interests, Personal, Advisory Role: Ankyra Therapeutics; Financial Interests, Personal, Advisory Board: Axio Research, LLC, Boxer Capital, Bristol Myers Squibb, CytomX Therapeutics, DermTech, Engage Health Media, iOnctura, Iovance Biotherapeutics, IQVIA, Istari Oncology, Jazz Pharmaceuticals, Inc., Lumira Capital Investment Management, Inc., Lytix Biopharma AS, Magnolia Innovation, LLC, Merck, Mural Oncology, Natera, Inc., Novartis Pharmaceuticals, OncoCyte Corporation, PathAI, Inc., Pfizer, Inc., Piper Sandler & Co., Regeneron Pharmaceuticals, Inc., Replimune, Inc., Scopus BioPharma, Inc., Takeda, Valar Labs, Inc., Zola Therapeutics, Applied Clinical Intelligence, LLC, Becker Pharmaceutical Consulting, Checkmate Pharmaceuticals, Fenix Group International, Harbour BioMed, Immunocore LLC, Oncosec Medical Inc., SR One Capital Management, LP; Financial Interests, Institutional, Research Grant: Amgen, Bristol Myers Squibb, Checkmate Pharmaceuticals, Harbour BioMed, Immvira Pharma Co., Immunocore, Ltd., Iovance Biotherapeutics, Lion Biotechnologies, Inc., Lytix Biopharma AS, Novartis Pharmaceuticals, Takeda, Verastem, Inc.; Financial Interests, Personal, Other, Board Member: AIM at Melanoma. M. Del Vecchio: Financial Interests, Personal, Advisory Role: Bristol Myers Squibb, Merck Sharp & Dohme, Novartis, Pierre Fabre, Immunocore, Regeneron. C. Hoeller: Financial Interests, Personal, Invited Speaker, Advisory Boards, Speaker: Amgen; Financial Interests, Personal, Advisory Board, Advisory Boards, Speaker: Bristol Myers Squibb, Merck Sharp & Dohme, Novartis, Pierre Fabre, Sanofi; Financial Interests, Personal, Advisory Board: Almirall, Immunocore, Ultimovacs; Financial Interests, Personal, Invited Speaker: Abbvie; Financial Interests, Institutional, Coordinating PI: Amgen, Bristol Myers Squibb, Pierre Fabre, Roche; Financial Interests, Institutional, Local PI: Merck Sharp & Dohme, Novartis, Regeneron, Immunocore, Nektar Therapeutics, Pfizer; Non-Financial Interests, Member, Board Member: European Association of Dermato-Oncology (EADO); Non-Financial Interests, Member: EORTC Melanoma Group. P. Mohr: Financial Interests, Personal, Advisory Board: Bristol Myers Squibb, Merck Sharp & Dohme, Novartis, GSK, Pierre Fabre, Sanofi, Delcath, Regeneron, Immunocore, Merck Serono; Financial Interests, Personal, Invited Speaker: Bristol Myers Squibb, Merck Sharp & Dohme, Novartis, Pierre Fabre, Sanofi, Sun Pharma, Delcath; Financial Interests, Institutional, Funding: Bristol Myers Squibb, Merck Sharp & Dohme, Novartis; Non-Financial Interests, Principal Investigator: Bristol Myers Squibb, Merck Sharp & Dohme, Regeneron, Sanofi, Novartis, Sun Pharma. S. Grabbe: Financial Interests, Personal, Advisory Board: BioNTech, BMS, Delcath, MSD, Novartis, SUN Pharma, Klinge Pharma ; Financial Interests, Personal, Other, travel support: Biontech, BMS, Delcath, MSD, Novartis, SUN Pharma, Klinge Pharma ; Financial Interests, Institutional, Research Grant: Novartis; Non-Financial Interests, Personal, Local PI: Anaveon AG, BioNTech SE, Parexel International (IRL) Limited, Agenus, Inc., HUYABIO International, LLC, I0 BIOTECH APS, Regeneron Pharmaceuticals, Inc, Novartis Pharma GmbH, Syneos Health UK Limited, Bristol Myers Squibb International Corporation, Genentech, Inc.. C. Dutriaux: Financial Interests, Personal, Other, Consulting fees: BMS, MSD, Novartis, Pierre Fabre, Sun Pharma, Sanofi; Financial Interests, Personal, Other, Honoraria: BMS, MSD, Novartis, Pierre Fabre, Sun Pharma, Sanofi; Financial Interests, Personal, Other, Support for attending meetings and/or travel: BMS, MSD, Novartis, Pierre Fabre, Sun Pharma, Sanofi. J. Mackiewicz: Financial Interests, Personal, Advisory Board: Bristol Myers-Squibb, MSD, Novartis; Financial Interests, Personal, Invited Speaker: Bristol Myers-Squibb, MSD, Novartis, Roche, Pierre Fabre; Other, Travel grant: Bristol Myers-Squibb, MSD, Novartis, Roche, Pierre Fabre. P. Rutkowski: Financial Interests, Personal, Invited Speaker, honoraria for lectures: MSD, BMS, Pierre Fabre; Financial Interests, Personal, Advisory Board: MSD, BMS, Pierre Fabre, Merck, Sanofi, Blueprint Medicines, Philogen, Medison Pharma, Genesis Pharma; Financial Interests, Personal, Invited Speaker: Merck, Sanofi, Novartis, AstraZeneca, Genesis Pharma; Financial Interests, Institutional, Research Grant, research grant for ISS: Pfzer; Financial Interests, Institutional, Funding, research grant for institution: BMS; Non-Financial Interests, Member of Board of Directors: Polish Society of Surgical Oncology; Non-Financial Interests, Member of Board of Directors, President: Polish Oncological Society. P. Arenberger: Financial Interests, Personal, Invited Speaker, Talk at their meeting: AbbVie, Pfizer, LaRoche Posay. G. Quereux: Financial Interests, Personal, Advisory Board: BMS, Pierre Fabre, MSD, Regeneron, Novartis, Sanofi; Non-Financial Interests, Principal Investigator: BMS, Regeneron, MSD, Novartis. J. Pigozzo: Financial Interests, Personal, Advisory Board: MSD; Non-Financial Interests, Principal Investigator: BMS, MSD, Regeneron. T. Meniawy: Financial Interests, Personal, Advisory Board: AstraZeneca, BMS, Eisai, MSD; Financial Interests, Personal, Invited Speaker: GSK; Non-Financial Interests, Institutional, Coordinating PI: AstraZeneca; Non-Financial Interests, Institutional, Local PI: BMS, GSK, Moderna, MSD. P.A. Ascierto: Financial Interests, Personal, Other, Consultant and Advisory Role: BMS, Roche Genentech, Novartis, Merck Serono, Sun Pharma, Sanofi, Sandoz, Immunocore, Boehringer Ingelheim, Regeneron; Financial Interests, Personal, Other, Consultant, Advisory Role and Travel support: MSD, Pierre Fabre; Financial Interests, Personal, Other, Consultant and Advisory Role.Travel support: Pfizer/Array; Financial Interests, Personal, Other, Consultant Role: Italfarmaco; Financial Interests, Personal, Other, Consultant Role: Pfizer, Nouscom, Lunaphore; Financial Interests, Personal, Other, Consultant role: Medicenna; Financial Interests, Personal, Other, Consultant role and travel support: Bio-AI Health; Financial Interests, Personal, Advisory Board, Consultant and Advisory role: ValoTx; Financial Interests, Personal, Advisory Board, Consultant and Advisor role.Travel support: Replimmune; Financial Interests, Personal, Advisory Board, Advisor role: Bayer; Financial Interests, Personal, Other, Consultant and Advisory: Erasca; Financial Interests, Personal, Other, Consultant: Philogen, Incyte; Financial Interests, Personal, Other, Consultant and Advisory role: BionTech, Genmab; Financial Interests, Personal, Advisory Board: Anaveon, Menarini; Financial Interests, Institutional, Funding, Clinical trial and translational research: BMS; Financial Interests, Institutional, Funding, Clinical Trial: Roche Genentech, Pfizer/Array, Sanofi; Non-Financial Interests, Leadership Role, President since 2010: Fondazione Melanoma Onlus Italy; Non-Financial Interests, Leadership Role, President since 2014: Campania Society of ImmunoTherapy of Cancer (SCITO) Italy; Non-Financial Interests, Other, Member of Steering Committee since 2016: Society for Melanoma Research (SMR); Non-Financial Interests, Member of Board of Directors, November 2017 - December 2021: Society for Immunotherapy of Cancer (SITC); Non-Financial Interests, Member: ASCO, SITC, EORTC Melanoma Cooperative Group, AIOM, SMR. S. Dalle: Financial Interests, Institutional, Advisory Board: MSD, BMS, Regeneron; Financial Interests, Personal, Other, Firt author: MSD; Financial Interests, Personal, Stocks/Shares, Spouse sanofi employee and stock owner: Sanofi; Financial Interests, Institutional, Research Grant: BMS, MSD, AstraZeneca, Pierre-Fabre. J. Taube: Financial Interests, Personal, Advisory Role: BMS, Merck &Co, AstraZeneca, Roche, Regeneron, Akoya Bio, Elephas, Compugen, Moderna, Next Point; Financial Interests, Personal, Stocks or ownership: Akoya Bio; Financial Interests, Institutional, Research Grant: BMS, Akoya Bio; Financial Interests, Personal, Leadership Role, previous lead of the Pathology Task Force: Society for Immunotherapy of Cancer. A. Nassar: Financial Interests, Personal, Full or part-time Employment, BMS is my employing organisation - I am employed full time as a Clinical Trial Physician: BMS. Z. Li: Financial Interests, Institutional, Full or part-time Employment: Bristol Myers Squibb; Financial Interests, Personal, Stocks/Shares: Bristol Myers Squibb. G.V. Long: Financial Interests, Personal, Other, Consultant Advisor: Agenus Inc, Amgen Inc, Array Biopharma Inc, AstraZeneca UK Limited, Bayer Healthcare Pharmaceuticals, BioNTech SE, Boehringer Ingelheim International GmbH, Bristol Myers Squibb, Evaxion Biotech A/S, Hexal AG, Highlight Therapeutics S. L., IOBiotech, Immunocore Ireland Limited, Innovent Bioilogics USA Inc, Merck Sharp & Dohme, Novartis Pharma AG, PHMR Limited, Pierre Fabre, Regeneron Pharmaceuticals Inc, Scancell Limited, SkylineDX B.V., Fortiva Biologics (USA); Financial Interests, Personal, Advisory Board, Consultant Advisor: GI Innovation; Non-Financial Interests, Principal Investigator, GL is PI on over 30 clinical trials: GL is PI on over 30 clinical trials. All other authors have declared no conflicts of interest.

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1611P Pembrolizumab versus placebo as adjuvant therapy for resected stage IIB or IIC melanoma: 5-year follow-up of the phase III KEYNOTE-716 study

J.J. Luke[1], G.V. Long[2], M.A. Khattak[3], P. Rutkowski[4], M. Del Vecchio[5], F. Spagnolo[6], J. Mackiewicz[7], L. de la Cruz Merino[8], J. Pigozzo[9], J.M. Kirkwood[10], C. Robert[11], M.S. Carlino[12], D. Schadendorf[13], F. De Galitiis[14], A.T. Odeleye-Ajakaye[15], T. Hoki[16], M. Kalabis[17], A.M.M. Eggermont[18], P.A. Ascierto[19]

[1]Immunotherapy and Drug Development Center, UPMC Hillman Cancer Center and University of Pittsburgh, Pittsburgh, United States of America; [2]Faculty of Medicine and Health, The University of Sydney, and Mater and Royal North Shore Hospitals, Melanoma Institute Australia, The University of Sydney, Royal North Shore and Mater Hospitals, Sydney, Australia; [3]Oncology, Fiona Stanley Hospital and Edith Cowan University, Perth, Australia; [4]Department of Soft Tissue/Bone Sarcoma and Melanoma, Maria Sklodowska-Curie National Research Institute of Oncology, Warsaw, Poland; [5]Medical Oncology, Fondazione IRCCS - Istituto Nazionale dei Tumori, Milan, Italy; [6]Medical Oncology, IRCCS Ospedale Policlinico San Martino, Genoa, Italy; [7]Department of Medical and Experimental Oncology, Poznan University of Medical Sciences Sciences and Greater Poland Cancer Center, Poznan, Poland; [8]Clinical Oncology, Instituto de Biomedicina de Sevilla (IBiS)/Hospital Universitario Virgen Macarena/Consejo Superior de Investigaciones Científicas (CSIC)/Universidad de Sevilla, Seville, Spain; [9]Medical Oncology, Veneto Institute of Oncology IOV-IRCCS, Padua, Italy; [10]Medicine, UPMC Hillman Cancer Center and University of Pittsburgh, Pittsburgh, United States of America; [11]Dermatology, Gustave Roussy, Villejuif and Paris-Saclay University, Villejuif, France; [12]Medical Oncology department, Melanoma Institute Australia, The University of Sydney, and Westmead and Blacktown Hospitals, Westmead, Australia; [13]Department of Dermatology - Hautklinik, University Hospital Essen and German Cancer Consortium Partner Site and National Center for Tumor Diseases (NCT)-West, Campus Essen, Research Alliance Ruhr, Research Center One Health, and University Duisburg-Essen, Essen, Germany; [14]Oncology, Dermopathic Institute of the Immaculate (IDI) IRCCS, Rome, Italy; [15]BARDS, Merck & Co., Inc., Rahway, United States of America; [16]Oncology, Merck & Co., Inc., Rahway, United States of America; [17]Global Clinical Development, Oncology, Merck & Co., Inc., Rahway, United States of America; [18]Cancer Medicine, University Medical Center Utrecht, Utrecht, Netherlands, and Comprehensive Cancer Center Munich of the Technical University Munich and the Ludwig Maximilian University, Utrecht, Netherlands; [19]Department of Skin Cancer, Cancer Immunotherapy and Development Therapeutics, Istituto Nazionale Tumori IRCCS Fondazione Pascale, Naples, Italy

Background: Adjuvant pembrolizumab (pembro) significantly prolonged RFS and DMFS vs placebo in participants (pts) with resected high-risk stage II melanoma in KEYNOTE-716 (NCT03553836). At the final DMFS analysis (median study follow-up, 39.4 mo), pembro continued to demonstrate a DMFS and RFS benefit. Here, we report 5-year follow-up data.

Methods: Pts aged $\geq$12 y with resected stage IIB or IIC melanoma were randomly assigned 1:1 to pembro 200 mg (2 mg/kg up to 200 mg for pts $<$18 y) or placebo IV Q3W for 17 cycles (part 1). Pts with recurrence after placebo or $>$6 mo after 17 cycles of pembro could crossover to or be rechallenged with pembro (part 2). RFS was the primary end point. DMFS and safety were secondary. Progression/recurrence-free survival 2 (PRFS2; time to PD beyond initial recurrence, second recurrence, or death) and time to subsequent therapy (TTST) were exploratory.

Results: A total of 976 pts were enrolled (pembro, n = 487; placebo, n = 489). Median study follow-up at data cutoff (Feb 13, 2025) was 64.7 mo (range, 51.4-76.7). 62 of 149 pts (41.6%) in the pembro arm and 83 of 218 (38.1%) in the placebo arm with an RFS event received subsequent therapy, most commonly combination anti—PD-1/anti—CTLA-4 (22.1% and 18.8%). Efficacy is presented in the table. Treatment-related AEs occurred in 399/483 pts (82.6%) in the pembro arm and 311/486 (64.0%) in the placebo arm; 84 (17.4%) and 25 (5.1%), respectively, were grade 3 or 4. Immune-mediated AEs and infusion reactions occurred in 185/483 pts (38.3%) in the pembro arm and 46/486 pts (9.5%) in the placebo arm; 53 (11.0%) and 6 (1.2%), respectively, were grade 3 or 4. No grade 5 immune-mediated AEs or infusion reactions occurred.

Table: 1611P

	Overall		Stage IIB		Stage IIC	
	Pembro n = 487	Placebo n = 489	Pembro n = 309	Placebo n = 316	Pembro n = 171	Placebo n = 169
RFS, HR (95% CI), mo	0.62 (0.50-0.76)		0.61 (0.46-0.80)		0.62 (0.44-0.86)	
5-y RFS, %	67.1	52.4	70.4	56.9	62.2	45.4
DMFS, HR (95% CI), mo	0.59 (0.46-0.77)		0.63 (0.45-0.89)		0.55 (0.36-0.82)	
5-y DMFS, %	78.7	66.6	80.4	70.7	76.0	60.1
PRFS2, HR (95% CI), mo	0.72 (0.55-0.94)		0.74 (0.51-1.07)		0.69 (0.45-1.04)	
5-y PRFS2, %	81.1	73.3	84.9	77.6	75.4	66.8
TTST, HR (95% CI), mo	0.65 (0.52-0.80)		0.64 (0.49-0.84)		0.62 (0.44-0.87)	
5-y TTST, %	67.5	55.2	70.2	58.7	64.3	49.5

Conclusions: After a median follow-up of $\geq$5 years, pembro continued to prolong RFS and DMFS. Of pts treated with pembro, 67.1% were without disease recurrence at 5 years vs 52.4% with placebo. PRFS2 and TTST results were promising. These findings support the use of adjuvant pembro for high-risk stage II melanoma.

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Luke: Non-Financial Interests, Personal, Advisory Board: BioCytics, Bright Peak, RefleXion, Xilio (stock) Actym, Duke Street Bio, Elipscience, Kanaph, NeoTx, Onc.AI, OncoNano, Pyxis, Saros, Tempest, Zola Therapeutics; Other, Institutional, Research Grant: AbbVie, Astellas, AstraZeneca, Bristol Myers Squibb, Corvus, Day One, EMD Serono, Fstar, Genmab, Hot Spot, Ikena, Immatics, Imugene, Incyte, Janux, Kadmon, KAHR, Macrogenics, Merck, Moderna, Nektar, Next Cure, Novartis, Numab, Palleon, Pfizer, Replimmune; Financial Interests, Personal, Other: (Patent) US-11638728 (Microbiome Biomarkers for Anti-PD-1/PD-L1 Responsiveness: Diagnostic, Prognostic and Therapeutic Uses Thereof); Financial Interests, Personal, Other, Data Safety Monitoring Board: Abbvie, Agenus, Evaxion, Immutep, Shionogi; Financial Interests, Personal, Other, Consultancy with compensation: AbbVie, Agenus, AstraZeneca, Bayer, Bristol Myers Squibb, Clasp, Curadev, Eisai, EMD Serono, Geneos, Gilead, HotSpot, Krystal, Janssen, Ikena, Immatics, Incyte, IO Biotech, iTeos, LegoChem, Lyvgen, Merck, Mersana, Novartis,. G.V. Long: Financial Interests, Personal, Other, Consultant Advisor: Agenus Inc, Amgen Inc, Array Biopharma Inc, AstraZeneca UK Limited, Bayer Healthcare Pharmaceuticals, BioNTech SE, Boehringer Ingelheim International GmbH, Bristol Myers Squibb, Evaxion Biotech A/S, Hexal AG, Highlight Therapeutics S.L., IOBiotech, Immunocore Ireland Limited, Innovent Bioilogics USA Inc, Merck Sharp & Dohme, Novartis Pharma AG, PHMR Limited, Pierre Fabre, Regeneron Pharmaceuticals Inc, Scancell Limited, SkylineDX B.V., Fortiva Biologics (USA); Financial Interests, Personal, Advisory Board, Consultant Advisor: GI Innovation; Non-Financial Interests, Principal Investigator, GL is PI on over 30 clinical trials: GL is PI on over 30 clinical trials. M.A. Khattak: Financial Interests, Personal, Advisory Board: Merck & Co., Inc.; Financial Interests, Personal, Invited Speaker: Merck & Co., Inc.; Financial Interests, Personal, Speaker's Bureau: Merck & Co., Inc.; Financial Interests, Personal, Principal Investigator: Merck & Co., Inc. P. Rutkowski: Financial Interests, Personal, Invited Speaker, honoraria for lectures: MSD, BMS, Pierre Fabre; Financial Interests, Personal, Advisory Board: MSD, BMS, Pierre Fabre, Merck, Sanofi, Blueprint Medicines, Philogen, Medison Pharma, Genesis Pharma; Financial Interests, Personal, Invited Speaker: Merck, Sanofi, Novartis, AstraZeneca, Genesis Pharma; Financial Interests, Institutional, Research Grant, research grant for ISS: Pfizer; Financial Interests, Institutional, Funding, research grant for institution: BMS; Non-Financial Interests, Member of Board of Directors: Polish Society of Surgical Oncology; Non-Financial Interests, Member of Board of Directors, President: Polish Oncological Society. M. Del Vecchio: Other, Personal, Advisory Board: Bristol Myers Squibb, Merck Sharp & Dohme, Novartis, Pierre Fabre, Immunocore, Regeneron; Other, Personal, Advisory Role: Bristol Myers Squibb, Merck Sharp & Dohme, Novartis, Pierre Fabre, Immunocore, Regeneron; Financial Interests, Personal, Invited Speaker: Bristol Myers Squibb; Other, Personal, Invited Speaker: Merck Sharp & Dohme, Novartis, Pierre Fabre, Immunocore, Regeneron. F. Spagnolo: Financial Interests, Personal, Invited Speaker: BMS, MSD, Novartis, Pierre Fabre, Sanofi, Merck & Co., Inc., Sunpharma, Regeneron; Financial Interests, Personal, Advisory Board: MSD, Regeneron. J. Mackiewicz: Financial Interests, Personal and Institutional, Invited Speaker: Bristol-Myers Squib, MSD, Pierre Fabre, Novartis; Financial Interests, Personal and Institutional, Advisory Role: Bristol-Myers Squib, MSD; Financial Interests, Personal and Institutional, Other, Travel grants: Bristol-Myers Squib, MSD, Pierre Fabre. L. de la Cruz Merino: Financial Interests, Personal and Institutional, Invited Speaker: Bristol Myers Squibb, Merck & Co., Inc., Roche; Financial Interests, Personal and Institutional, Speaker's Bureau: Bristol Myers Squibb, Merck & Co., Inc.; Financial Interests, Personal and Institutional, Funding: Bristol Myers Squibb, Merck & Co., Inc., ROCHE; Financial Interests, Personal and Institutional, Principal Investigator: Bristol Myers Squibb, Merck & Co., Inc., Roche; Financial Interests, Personal and Institutional, Advisory Board: Merck & Co., Inc.; Financial Interests, Personal and Institutional, Research Grant: Merck & Co., Inc., Roche; Financial Interests, Personal and Institutional, Project Lead: ROCHE; Financial Interests, Personal, Invited Speaker: GILEAD, ABBVIE; Financial Interests, Personal, Advisory Board: Gilead; Financial Interests, Personal, Principal Investigator: Gilead. J. Pigozzo: Other, Personal, Advisory Board: MSD; Other, Personal, Principal Investigator: MSD, BMS, Regeneron. J.M. Kirkwood: Financial Interests, Personal, Advisory Board: Ankyra Therapeutics, Bristol Myers Squibb, Iovance Biotherapeutics, Istari Oncology, Jazz Pharmaceuticals, Inc., Merck & Co., Inc., Mural Oncololgy, Natera, Inc., Novartis Pharmaceuticals, OncoCyte Corporation , Regeneron Pharmaceuticals, Inc., Replimune, Inc., Scopus BioPharma, Takeda; Financial Interests, Personal, Other, Data Safety Monitoring Board: Axio Research, LLC, IQVIA, Lytix Biopharma AS; Financial Interests, Personal, Advisory Role: Boxer Capital, CytomX Therapeutics, Engage Health Media, iOnctura, Lumira Capital Investment Management, Magnolia Innovation, LLC, PathAI, Inc., Pfizer, Inc., Piper Sandler & Co., PyrOjas Corporation, Valar Labs, Inc, Zola Therapeutics; Financial Interests, Personal, Other, Leader of Consensus Conference: DermTech; Financial Interests, Institutional, Principal Investigator: Amgen, Inc., Bristol Myers Squibb, Checkmate Pharmaceuticals, Harbour BioMed, Immvira Pharma Co, Immunocore, Ltd., Iovance Biotherapeutics, Lion Biotechnologies, Inc., Lytix Biopharma AS, Novartis Pharmaceuticals, Takeda, Verastem, Inc.. C. Robert: Financial Interests, Personal, Advisory Board: Pierre Fabre, Sanofi, BMS, MSD, Novaris, Merck, Roche, Pfizer, Sun Pharma, Ultimovacs, Regeneron, Egle, Philogen, Maat Pharma, Pierre Fabre; Financial Interests, Personal, Other, Steering Committee: Novartis, Regeneron, Pfizer, IO Biotech; Financial Interests, Personal, Invited Speaker: Sanofi, BMS, MSD, Novartis. M.S. Carlino: Financial Interests, Personal, Advisory Board: Amgen, Bristol Myers Squibb, Eisai, Ideaya, Merck Sharpe & Dohme, Medison, Moderna, Nektar, Novartis, Oncosec, Pierre Fabre, Qbiotic, Regeneron, Roche, Merck KGaA, Sanofi, Strand therapeutics. D. Schadendorf: Financial Interests, Personal, Invited Speaker: BMS, MSD, Merck Serono, Regeneron; Financial Interests, Personal, Advisory Board: BMS, Novartis, MSD, Immunocore, Pierre Fabre, Regeneron, Philogen, BioAlta, Daiichi Sanyko, BioNTech, Immatics, CureVac, Formycon, Replimune, SunPharma, Skyline Dx, IOVance, Ipsen, IOBioTech, Boehringer-Ingelheim; Financial Interests, Institutional, Coordinating PI: Novartis, BMS, MSD, Pierre Fabre, Immatics; Financial Interests, Institutional, Research Grant: BMS, MSD; Financial Interests, Personal, Steering Committee Member: MSD; Financial Interests, Institutional, Local PI: Philogen, Immunocore, Regeneron, Biontech Rna Pharmaceuticals GMBH, Replimune, IO Biotech; Non-Financial Interests, Member of Board of Directors: EORTC-MG; Non-Financial Interests, Leadership Role, Founding member and SC chair: European Melanoma Registry (EuMelaReg). F. De Galitiis: Financial Interests, Personal and Institutional, Invited Speaker: Bristol Myers Squibb, Novartis, Merck Sharp & Dohme; Financial Interests, Personal and Institutional, Advisory Board: Bristol Myers Squibb, Novartis, Merck Sharp & Dohme; Financial Interests, Personal and Institutional, Principal Investigator: Bristol Myers Squibb, Novartis, Merck Sharp & Dohme. A.T. Odeleye-Ajakaye: Financial Interests, Institutional, Full or part-time Employment: Merck & Co., Inc; Financial Interests, Personal, Stocks/Shares: Merck & Co., Inc. T. Hoki, M. Kalabis: Financial Interests, Personal, Full or part-time Employment: Merck & Co., Inc.; Financial Interests, Personal, Stocks or ownership: Merck & Co., Inc.: A.M.M. Eggermont: Financial Interests, Personal, Other, Data safety monitoring board: Boehringer Ingelheim, BioNTech, IQVIA, Pfizer, T3Pharma; Financial Interests, Personal, Advisory Board: Agenus, Boehringer Ingelheim, BioInvent, BioNTech, CatalYm, Egle, Eurobio, Imcheck, IO Biotech, IQVIA, Merck&Co, MSD, Pfizer, Oncolytics, Pierre Fabre, QBiotics, Regeneron, Sairopa, Secarna, SkylineDX, Thermosome, T3Pharma, Trained Immunity Discovery; Financial Interests, Personal, Speaker's Bureau: MSD; Financial Interests, Personal, Stocks or ownership: IO Biotech, Oncolytics, Sairopa, SkylineDX. P.A. Ascierto: Financial Interests, Personal, Other, Consultant and Advisory Role: BMS, Roche Genentech, Novartis, Merck Serono, Sun Pharma, Sanofi, Sandoz, Immunocore, Boehringer Ingelheim, Regeneron; Financial Interests, Personal, Other, Consultant, Advisory Role and Travel support: MSD, Pierre Fabre; Financial Interests, Personal, Other, Consultant and Advisory Role.Travel support: Pfizer/Array; Financial Interests, Personal, Other, Consultant Role: Italfarmaco; Financial Interests, Personal, Other, Consultant Role: Pfizer, Nouscom, Lunaphore; Financial Interests, Personal, Other, Consultant role: Medicenna; Financial Interests, Personal, Other, Consultant role and travel support: Bio-AI Health; Financial Interests, Personal, Advisory Board, Consultant and Advisory role: ValoTx; Financial Interests, Personal, Advisory Board, Consultant and Advisor role, Travel support: Replimmune; Financial Interests, Personal, Advisory Board, Advisor role: Bayer; Financial Interests, Personal, Other, Consultant and Advisory: Erasca; Financial Interests, Personal, Other, Consultant: Philogen, Incyte; Financial Interests, Personal, Other, Consultant and Advisory role: BionTech, Genmab; Financial Interests, Personal, Advisory Board: Anaveon, Menarini; Financial Interests, Institutional, Funding, Clinical trial and translational research: BMS; Financial Interests, Institutional, Funding, Clinical Trial: Roche Genentech, Pfizer/Array, Sanofi; Non-Financial Interests, Leadership Role, President since 2010: Fondazione Melanoma Onlus Italy; Non-Financial Interests, Leadership Role, President since 2014: Campania Society of ImmunoTherapy of Cancer (SCITO) Italy; Non-Financial Interests, Other, Member of Steering Committee since 2016: Society for Melanoma Research (SMR); Non-Financial Interests, Member of Board of Directors, November 2017 - December 2021: Society for Immunotherapy of Cancer (SITC); Non-Financial Interests, Member: ASCO, SITC, EORTC Melanoma Cooperative Group, AIOM, SMR.

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1612P Characterization of tumor and peripheral biomarkers in patients (pts) with resectable melanoma (MEL) treated with adjuvant nivolumab + relatlimab (NIVO + RELA) or NIVO alone in RELATIVITY-098

H.A. Tawbi[1], S. Dolfi[2], A. Yu[3], D. Paulucci[4], P.A. Ascierto[5], J. Guo[6], A.A. Tarhini[7], S. Chandra[8], E. Muñoz-Couselo[9], M. Del Vecchio[10], A.C. de Melo[11], M.O. Callahan[12], H. Gogas[13], R. Dummer[14], D. Schadendorf[15], A. Fisher[16], P. Djidel[17], M. Semaan[18], C. Garnett-Benson[19], G.V. Long[20]

[1]Melanoma Medical Oncology department, The University of Texas MD Anderson Cancer Center, Houston, United States of America; [2](Colead) Translational Medicine, Bristol Myers Squibb, Princeton, United States of America; [3]Translational Bioinformatics Late Development Oncology, Bristol Myers Squibb, Princeton, United States of America; [4]Data Science for Oncology Drug Development, Bristol Myers Squibb, Princeton, United States of America; [5]Melanoma, Cancer Immunotherapy & Developmental Therapeutics, Istituto Nazionale dei Tumori IRCCS "Fondazione G. Pascale", Naples, Italy; [6]Department of Renal Cancer and Melanoma, Peking University Cancer Hospital & Institute, Beijing, China; [7]Cutaneous Oncology and Immunology, H. Lee Moffitt Cancer Center & Research Institute, Tampa, United States of America; [8]Division of Hematology Oncology, Northwestern University, Chicago, United States of America; [9]Department of Medical Oncology, Vall d'Hebron Hospital, Barcelona, Spain; [10]Medical Oncology, Fondazione IRCCS Istituto Nazionale dei Tumori, Milan, Italy; [11]Clinical Research Division, Brazilian National Cancer Institute, Rio De Janeiro, Brazil; [12]Hematology and Oncology, Memorial Sloan Kettering Cancer Center, New York, United States of America; [13]First Department of Medicine, National and Kapodistrian University of Athens, Athens, Greece; [14]Dermatology Department, University of Zurich, Zurich, Switzerland; [15]Department of Dermatology - Hautklinik, University of Essen and the German Cancer Consortium, Essen, Germany; [16]Informatics & Predictive Sciences, Bristol Myers Squibb, Princeton, United States of America; [17]Biostatistics, Bristol Myers Squibb, Princeton, United States of America; [18]Clinical Science, Bristol Myers Squibb - Nassau Park, Princeton, United States of America; [19]Translational Medicine, Bristol Myers Squibb, Princeton, United States of America; [20]Medical Oncology, Melanoma Institute Australia, The University of Sydney, Royal North Shore and Mater Hospitals, Sydney, Australia

Background: RELA-098 (NCT05002569) evaluated adj NIVO + RELA fixed dose combination (FDC) vs NIVO in pts with resected stage III—IV MEL. NIVO + RELA did not significantly improve recurrence-free survival (RFS) vs NIVO alone. In advanced (adv) MEL, LAG-3+ CD8 T cells were primarily located in tumor tissue; on-treatment expansion of LAG-3+ CD8 T cells in blood was lower in adj vs adv MEL (Long, ASCO 2025). Here we characterized BM for pharmacodynamic (PD) effects of adj NIVO + RELA and NIVO alone and association of BM with RFS.

Methods: Pts received NIVO 480 mg + RELA 160 mg FDC or NIVO 480 mg. Baseline (BL) and post-treatment blood samples were analyzed for PD changes in immune cell populations by flow cytometry and cytokines by immunoassays. Tissue samples collected at BL resection and recurrence were analyzed by immunohistochemistry for tumor cell PD-L1, LAG-3, and CD8.

Results: Blood inflammatory cytokines (eg IFNγ, CXCL9, and CXCL10) were significantly increased in both arms, but to a greater degree by NIVO + RELA vs NIVO. Immunosuppressive cytokines (eg IL-10, CRP, and CCL8) were also elevated by NIVO + RELA. Blood LAG-3+ T cells were more significantly increased by adj NIVO + RELA vs NIVO. Higher PD-L1, LAG-3, and CD8 expression in BL resected tumors enriched for RFS benefit in both arms. 72% of resected tumors were LAG-3-positive but only 28% were PD-L1-positive despite most tumors being CD8 infiltrated ($>$95%). Pts with lower CD8 infiltration (lowest tertile) in resected tumors demonstrated a trend for RFS benefit of NIVO + RELA vs NIVO. BM expression in resected tumors compared with that in recurrent tumors is underway.

Conclusions: In resected stage III-IV MEL (RELA-098), NIVO + RELA induced similar PD changes as those reported for adv MEL (RELA-047) though smaller magnitude, and inflammation-related TME BM in resected tumors similarly enriched for efficacy to adj IO therapy. Despite the similarities, BM analyses, as previously reported, support that LAG-3+ tumor infiltrating lymphocytes at therapy initiation are likely needed for NIVO + RELA vs NIVO benefit. The significance of NIVO + RELA benefit in pts whose resected tumors have lower CD8 infiltration is being investigated.

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Genentech, Novartis, Merck Serono, Sun Pharma, Sanofi, Sandoz, Immunocore, Boehringer Ingelheim, Regeneron; Financial Interests, Personal, Other, Consultant, Advisory Role and Travel support: MSD, Pierre Fabre; Financial Interests, Personal, Other, Consultant and Advisory Role.Travel support: Pfizer/Array; Financial Interests, Personal, Other, Consultant Role: Italfarmaco; Financial Interests, Personal, Other, Consultant Role: Pfizer, Nouscom, Lunaphore; Financial Interests, Personal, Other, Consultant role: Medicenna; Financial Interests, Personal, Other, Consultant role and travel support: Bio-AI Health; Financial Interests, Personal, Advisory Board, Consultant and Advisory role: ValoTx; Financial Interests, Personal, Advisory Board, Consultant and Advisor role.Travel support: Replimmune; Financial Interests, Personal, Advisory Board, Advisor role: Bayer; Financial Interests, Personal, Other, Consultant and Advisory: Erasca; Financial Interests, Personal, Other, Consultant: Philogen, Incyte; Financial Interests, Personal, Other, Consultant and Advisory role: BionTech, Genmab; Financial Interests, Personal, Advisory Board: Anaveon, Menarini; Financial Interests, Institutional, Funding, Clinical trial and translational research: BMS; Financial Interests, Institutional, Funding, Clinical Trial: Roche Genentech, Pfizer/Array, Sanofi; Non-Financial Interests, Leadership Role, President since 2010: Fondazione Melanoma Onlus Italy; Non-Financial Interests, Leadership Role, President since 2014: Campania Society of ImmunoTherapy of Cancer (SCITO) Italy; Non-Financial Interests, Other, Member of Steering Committee since 2016: Society for Melanoma Research (SMR); Non-Financial Interests, Member of Board of Directors, November 2017 - December 2021: Society for Immunotherapy of Cancer (SITC); Non-Financial Interests, Member: ASCO, SITC, EORTC Melanoma Cooperative Group, AIOM, SMR. J. Guo: Financial Interests, Personal, Other, Honoraria: MSD, Roche, Pfizer, Bayer, Novartis, Simcere, Shanghai Junshi Bioscience, Oriengene. A.A. Tarhini: Financial Interests, Personal, Advisory Board: Bristol Myers Squib, Genentech/Roche, Regeneron, Sanofi-Genzyme, BioNTech, Merck / Moderna; Financial Interests, Personal, Advisory Board, Scientific Consultant: ConcertAI; Financial Interests, Personal and Institutional, Steering Committee Member: Bristol Myers Squib; Financial Interests, Institutional, Steering Committee Member: Genentech-Roche; Financial Interests, Institutional, Research Grant: Regeneron, Sanofi-Genzyme, Clinigen / IOVANCE, Acrotech, Pfizer, OncoSec; Financial Interests, Institutional, Trial Chair: Checkmate; Financial Interests, Institutional, Local PI: InflaRx, Scholar Rock, Agenus, Dragonfly, Werewolf, Merck, Moderna. S. Chandra: Financial Interests, Personal, Advisory Board: Bristol Myers Squibb, Novartis, Pfizer, Regeneron, Alkermes; Financial Interests, Personal, Leadership Role: NCCN Melanoma. E. Muñoz-Couselo: Financial Interests, Personal, Advisory Board: BMS, NOVARTIS, SANOFI, Pierre Fabre, Sun Pharma, MSD, Regeneron, Imunocore; Financial Interests, Personal, Invited Speaker: Novartis, Roche, BMS, Sanofi, MSD, Pierre Fabre; Non-Financial Interests, Member of Board of Directors, Vocal in the Society of Spanish Oncology group: SEOM; Non-Financial Interests, Member of Board of Directors, Member of the board diurectos in the Spanish Group of melanoma: GEM. M. Del Vecchio: Financial Interests, Personal, Advisory Role: Bristol Myers Squibb, Merck Sharp & Dohme, Novartis, Pierre Fabre, Immunocore, Regeneron. A.C. de Melo: Financial Interests, Personal, Advisory Board: AstraZeneca, Bristol Myers Squibb, F. Hoffmann-La Roche, GSK, Merck Sharp and Dohme; Financial Interests, Personal, Invited Speaker: AstraZeneca, Bristol Myers Squibb, F. Hoffmann-La Roche, GSK, Merck Sharp and Dohme, Novartis, Daiichi Sankyo; Financial Interests, Institutional, Local PI: Regeneron Pharmaceuticals, Novartis, Merck Sharp and Dohme, GSK, F. Hoffmann-La Roche, Clovis Oncology, Inc., Bristol Myers Squibb, AstraZeneca, Amgen. M.O. Callahan: Financial Interests, Personal, Advisory Board: Bristol Myers Squibb International, Epitopea; Financial Interests, Personal, Stocks/Shares, Family member with employment compensation at Merus: Merus; Financial Interests, Institutional, Research Grant: Bristol Myers Squibb International; Other, Family member employed at Merus with leadership position: Merus. H. Gogas: Financial Interests, Personal, Advisory Board: MSD, BMS, Pierre Fabre, Sanofi; Financial Interests, Personal, Invited Speaker: MSD, BMS, Novartis, Pierre Fabre, Sanofi; Financial Interests, Steering Committee Member: Amgen, Replimmune; Financial Interests, Institutional, Local PI: Amgen, MSD, BMS, Replimmune, Iovance, Bayer; Financial Interests, Institutional, Research Grant: BMS, Pfizer, Lilly, Pierre Fabre, AstraZeneca. R. Dummer: Financial Interests, Personal, Other, Consulting and/or advisory role: Novartis, Merck Sharp & Dohme (MSD), Bristol Myers Squibb (BMS), Roche, Amgen, Takeda, Pierre Fabre, Sun Pharma, Sanofi, Catalym, Second Genome, Regeneron, MaviVAX SA, T3 Pharma, Pfizer, Simcere; Financial Interests, Personal, Other, medical advisor: Oncobit. D. Schadendorf: Financial Interests, Personal, Invited Speaker: BMS, MSD, Merck Serono, Regeneron; Financial Interests, Personal, Advisory Board: BMS, Novartis, MSD, Immunocore, Pierre Fabre, Regeneron, Philogen, BioAlta, Daiichi Sanyko, BioNTech, Immatics, CureVac, Formycon, Replimune, SunPharma, Skyline Dx, IOVance, Ipsen, IOBioTech, Boehringer-Ingelheim; Financial Interests, Institutional, Coordinating PI: Novartis, BMS, MSD, Pierre Fabre, Immatics; Financial Interests, Institutional, Research Grant: BMS, MSD; Financial Interests, Personal, Steering Committee Member: MSD; Financial Interests, Institutional, Local PI: Philogen, Immunocore, Regeneron, Biontech Rna Pharmacuticals GMBH, Replimune, IO Biotech; Non-Financial Interests, Member of Board of Directors: EORTC-MG; Non-Financial Interests, Leadership Role, Founding member and SC chair: European Melanoma Registry (EuMelaReg). A. Fisher: Financial Interests, Personal, Full or part-time Employment, I've worked at BMS since 2020: Bristol Myers Squibb; Financial Interests, Personal, Other, Spouse's employment starting in 2025: Cullinan Therapeutics; Financial Interests, Personal, Other, Spouse's previous employment: Abata Therapeutics, TCR2 Therapeutics, Obsidian Therapeutics; Financial Interests, Personal, Other, Spouse previously consulted for: Clasp Therapeutics; Financial Interests, Personal, Stocks/Shares: Bristol Myers Squibb, Adaptimmune. P. Djidel: Financial Interests, Personal, Full or part-time Employment, I am employed at BMS as Senior Manager, Biostatistics since April 2022. The amount received corresponds to my salary: Bristol Myers Squibb International; Financial Interests, Personal, Stocks/Shares, I am employed at BMS as Senior Manager, Biostatistics since April 2022. The amount received corresponds to my salary. I also received stock options (around 20K$ since April 2022): Bristol Myers Squibb International; Other, I have recieved from BMS some compensation for travelling when attending conferences, such as ISCB 2023 and will receive compensation for attending the ISCB 2025: Bristol Myers Squibb International. M. Semaan: Financial Interests, Personal, Full or part-time Employment: BMS; Financial Interests, Personal, Stocks/Shares: BMS. C. Garnett-Benson: Financial Interests, Personal, Full or part-time Employment: Bristol Myers Squibb; Financial Interests, Personal, Stocks/Shares: Bristol Myers Squibb. G.V. Long: Financial Interests, Personal, Other, Consultant Advisor: Agenus Inc, Amgen Inc, Array Biopharma Inc, AstraZeneca UK Limited, Bayer Healthcare Pharmaceuticals, BioNTech SE, Boehringer Ingelheim International GmbH, Bristol Myers Squibb, Evaxion Biotech A/S, Hexal AG, Highlight Therapeutics S.L., IOBiotech, Immunocore Ireland Limited, Innovent Bioilogics USA Inc, Merck Sharp & Dohme, Novartis Pharma AG, PHMR Limited, Pierre Fabre, Regeneron Pharmaceuticals Inc, Scancell Limited, SkylineDX B.V., Fortiva Biologics (USA); Financial Interests, Personal, Advisory Board, Consultant Advisor: GI Innovation; Non-Financial Interests, Principal Investigator, GL is PI on over 30 clinical trials: GL is PI on over 30 clinical trials.

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1613P NEO-TIM: Predictive biomarkers and long-term outcomes in the NEO-TIM trial of neoadjuvant immunotherapy and targeted therapy in melanoma

C. Trojaniello[1], D. Mallardo[1], P. Quaglino[2], F. Spagnolo[3], M. Guidoboni[4], M. Del Vecchio[5], M. Guida[6], K. Peris[7], L. Fioretto[8], V. Caliendo[9], R. Marconcini[10], P. Queirolo[11], C. Caracò[12], S. Mocellin[13], M.G. Vitale[1], D. Giannarelli[14], M. Paone[1], D. Massi[15], L. Piccin[16], P.A. Ascierto[1]

[1]Melanoma, Cancer Immunotherapy & Developmental Therapeutics, Istituto Nazionale Tumori IRCCS - Fondazione G. Pascale, Naples, Italy; [2]Dermatology and Venereal Diseases, AOU Città della Salute e della Scienza Torino, Turin, Italy; [3]Medical Oncology, IRCCS Ospedale Policlinico San Martino, Genoa, Italy; [4]Oncology, Università degli Studi di Ferrara, Ferrara, Italy; [5]Medical Oncology Department, Fondazione IRCCS - Istituto Nazionale dei Tumori, Milan, Italy; [6]Oncology Department, Istituto Tumori Bari Giovanni Paolo II - IRCCS, Bari, Italy; [7]Dermatology, Università Cattolica del Sacro Cuore, Rome, Italy; [8]Oncologico, SOC Oncologia Medica Firenze e Dipartimento Oncologico Azienda Usl Toscana Centro, Florence, Italy; [9]Divisione di Oncologia Chirurgica Dermatologica dell'Ospedale San Lazzaro, Città della Salute e della Scienza di Torino - Sede Molinette, Turin, Italy; [10]Dipartimento di Oncologia Medica, AOU Pisana - Stabilimento di Santa Chiara, Pisa, Italy; [11]Division of Medical Oncology for Melanoma, Sarcoma and Rare Tumors, IEO - Istituto Europeo di Oncologia IRCCS, Milan, Italy; [12]SC Chirurgia Melanoma e dei Tumori Cutanei, Istituto Nazionale Tumori IRCCS - Fondazione G. Pascale, Naples, Italy; [13]UOC Chirurgia Oncologica dei Tessuti Molli, del Peritoneo e dei Melanomi, Università di Padova, Padua, Italy; [14]Facility of Epidemiology and Biostatistics, Fondazione Policlinico Universitario Agostino Gemelli IRCCS, Rome, Italy; [15]Department of Health Sciences, UniFI - Università degli Studi di Firenze, Florence, Italy; [16]DiSCOG Scienze Chirurgiche Oncologiche Gastroenterologiche, IOV - Istituto Oncologico Veneto IRCCS, Padua, Italy

Background: Neoadjuvant therapy (NAT) with immune checkpoint blockade (ICB) has shown strong correlation between major pathological response (MPR) and event-free survival (EFS). To investigate whether combining or sequencing targeted therapy (TT) with ICB affects pathological and survival outcomes, we conducted a randomized, non-comparative, phase II trial in high-risk resectable melanoma.

Methods: 95 patients (pts) with resectable stage IIIB/IV BRAF-mutant melanoma were randomized (1:1) to Arm A (vemurafenib+cobimetinib, V+C) or Arm B (V+C +atezolizumab, A) for 6 weeks. BRAF wild-type pts (Arm C) received NAT cobimetinib + A for 2 cycles. All pts subsequently underwent CLND followed by 17 cycles of A. Primary endpoint: pCR; secondary endpoints: RFS, EFS, OS, safety, and biomarker analyses. Baseline blood samples from 78 pts were utilized for gene expression and cytokine profiling.

Results: At September 2024 data-lock, 3-yrs OS: Arm A 91.4%, Arm B 72.7%, Arm C 82.4%; 3-yrs RFS: Arm A 58.2%, Arm B 56.3%, Arm C 61.9%. Pts achieving MPR had 3-year RFS/OS of 82.3%/92.4%. A 8 gene signature was evaluated for both Arm A and Arm B+C. In Arm A, a low 8-gene signature correlated with better RFS (HR 0.07, $p<0.0001$). MPR was observed in pts with high HLA-DQB1 and low ICOSLG expression. Higher baseline IL-2 levels were associated with MPR ($p=0.03$). In Arm B+C, a distinct low-8-gene profile was associated with RFS (HR 0.21, $p<0.0001$); MPR was linked to high OLR1 expression. Low serum CCL2 levels correlated with MPR ($p=0.01$), EFS ($p=0.002$), and RFS ($p=0.02$).

Table: 1613P

	Arm A	Arm B	Arm C
RFS			
At 1 year	72.4%	75.9%	71.4%
At 2 years	65.5%	61.4%	61.9%
At 3 years	58.2%	56.3%	61.9%
EFS			
At 1 year	92.1%	72.4%	63.0%
At 2 years	51.7%	55.2%	53.3%
At 3 years	46.0%	50.2%	53.3%
OS			
At 1 year	100%	96.6%	97.1%
At 2 years	91.4%	72.7%	82.4%
At 3 years	91.4%	72.7%	82.4%
	MPR	**pPR**	**pNR**
RFS			
At 1 year	94.4%	60.0%	65.5%
At 2 years	82.3%	60.0%	58.6%
At 3 years	82.3%	40.0%	54.1%
EFS			
At 1 year	86.1%	73.3%	65.5%
At 2 years	71.4%	46.7%	58.6%
At 3 years	71.4%	31.1%	54.1%
OS			
At 1 year	100%	100%	96.6%
At 2 years	92.4%	84.0%	77.5%
At 3 years	92.4%	84.0%	77.5%

Conclusions: As expected, the extent of pathological response emerged as the strongest predictor of long-term benefit. The addition of IO contributes to more durable disease control. Molecular profiling revealed that distinct low gene signatures were associated with improved RFS and MPR across all three arms. These findings support the role of molecular and immune markers in guiding treatment and predicting outcomes.

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1614P Long-term outcome of adding the histone deacetylase inhibitor (HDACi) domatinostat (DOM) to neoadjuvant (neoadj) nivolumab (NIVO) +/- ipilimumab (IPI) in interferon-gamma signature (IFNγ) high/low stage III melanoma patients (pts)

L. Hoeijmakers[1], I. Reijers[1], A.M. Menzies[2], P. Dimitriadis[3], M. Lopez-Yurda[3], S. Cornelissen[4], M.W. Wouters[5], R. Saw[6], M. Klop[7], T.E. Pennington[8], W. Van Houdt[5], L.J.W. Bosch[9], L. Grijpink-Ongering[10], R. Rawson[11], A.J. Spillane[12], R.A. Scolyer[13], B. van de Wiel[14], A.C.J. van Akkooi[15], G.V. Long[2], C.U. Blank[16]

[1]Medical Oncology Department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [2]Faculty of Medicine and Health, The University of Sydney, and Mater and Royal North Shore Hospitals, University of Sydney, Melanoma Institute Australia, Sydney, Australia; [3]Department of Biometrics, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [4]Core Facility Molecular Pathology & Biobanking, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [5]Surgical Oncology Department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [6]Department of Melanoma & Surgical Oncology, Royal Prince Alfred Hospital, Camperdown, Australia; [7]Head and Surgery and Oncology, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [8]Department of Surgery, Melanoma Institute Australia, Wollstonecraft, Australia; [9]Pathology Dept., Netherlands Cancer Institute, Amsterdam, Netherlands; [10]Department of biometrics, Netherlands Cancer Institute, Amsterdam, Netherlands; [11]Department of Pathology, Melanoma Institute Australia, Wollstonecraft, Australia; [12]Surgical Oncology, University of Sydney, Sydney, Australia; [13]Medical Dept., Melanoma Institute Australia, Wollstonecraft, Australia; [14]Pathology Department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [15]Surgical Oncology Dept., Melanoma Institute Australia, Wollstonecraft, Australia; [16]Medical Oncology Dept., NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands

Background: IFNγ at baseline is a predictive biomarker for response to neoadj IPI + NIVO. HDACi have shown to induce IFNγ response and reduce tumor burden in stage IV melanoma. Therefore, we hypothesized that addition of HDACi to neoadj checkpoint inhibition (CPI) in pts with a low IFNγ could improve the outcomes and conducted the phase Ib DONIMI trial. First analysis revealed no additional effects of DOM and toxicity lead to premature closure of the arm combining DOM twice daily (bid) and dual CPI. Here we report some unexpected results from long-term survival-analysis.

Methods: Baseline IFNγ was assessed using Nanostring nCounter Technologies, allocating 44 pts to IFNγ high or low cohorts. IFNγ high pts were randomized to receive 2x NIVO or 2x NIVO+6 weeks (w) DOM bid, and IFNγ low pts to 2x NIVO+6w DOM bid, 2x NIVO+IPI+6w DOM once daily, or 2x NIVO+IPI+6w DOM bid. All pts had surgery in w6 followed by adjuvant NIVO or dabrafenib + trametinib. The survival analysis was performed using the Kaplan Meijer method.

Results: At data cutoff (31-1-2025) the median follow-up was 48.7 months. 4-year survival-outcomes are displayed in the table. No new or ongoing grade $\geq$3 adverse events are reported. IFNγ high pts have an excellent outcome, independent of addition of DOM. In IFNγ low pts outcomes seem to be dose-dependent of DOM, and less on the CPI, as the IFNγ low pts treated with DOM bid have superior EFS. DOM once daily combined with IPI+NIVO increased the IFNγ score in 6/10 pts (in 5/10 from IFNγ low to high). DOM bid increased IFNγ in 9/10 pts combined with NIVO (in 6/10 from IFNγ low to high) and in 3/3 pts combined with IPI+NIVO (in 2/3 from IFNγ low to high).

Table: 1614P

		Estimated 4-year event-free survival (%)	Estimated 4-year overall survival (%)
All		73% (95%CI 61%-87%)	84% (95%CI 74%-96%)
IFNγ high	2x NIVO mono q3w (n=10)	90% (95%CI 73%-100%)	100%
	2x NIVO q3w + 6 weeks of DOM twice daily (n=10)	80% (95%CI 59%-100%)	100%
IFNγ low	2x NIVO + IPI q3w + 6 weeks of DOM once daily (n=10)	50% (95%CI 27%-93%)	50% (95%CI 27%-93%)
	2x NIVO q3w + 6 weeks of DOM twice daily (n=10)	70% (95%CI 47%-100%)	80% (95%CI 59%-100%)
	2x NIVO + IPI q3w + 6 weeks of DOM twice daily (n=4)	75% (95%CI 43%-100%)	100%

Conclusions: Addition of HDACi to neoadj CPI might improve the long-term outcomes of IFNγ low pts and warrants further investigation using alternative dosing schemes or novel HDACi to improve tolerability in combination with neoadj CPI.

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1615P Health-related quality of life in the phase III NADINA trial: Neoadjuvant vs adjuvant checkpoint blockade in stage III melanoma

M. Egeler[1], J. Lijnsvelt[2], M.W. Lucas[3], V.G. Atkinson[4], M.A. Khattak[5], M. Andrews[6], M. Boers-Sonderen[7], G. Hospers[8], M.S. Carlino[9], J.W. de Groot[10], M. Stevense-den Boer[11], E. Kapiteijn[12], K. Suijkerbuijk[13], P. Rutkowski[14], A.M. Menzies[15], S.K. Sandhu[16], A.A.M. van der Veldt[17], G.V. Long[18], C.U. Blank[19], L.V. Van De Poll-franse[20]

[1]PSOE, Netherlands Cancer Institute, Antoni Van Leeuwenhoek, Amsterdam, Netherlands; [2]Mod Department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [3]Medical Oncology Department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [4]Division of Cancer Services, Princess Alexandra Hospital - Metro South Health, Woolloongabba, Australia; [5]Oncology, Fiona Stanley Hospital, Perth, Australia; [6]Medical Oncology Department, Alfred Hospital, Melbourne, Australia; [7]Department of Medical Oncology, Radboud University Medical Center, Nijmegen, Nijmegen, Netherlands; [8]Medical Oncology, UMCG - University Medical Center Groningen, Groningen, Netherlands; [9]Department of Medical Oncology, Crown Princess Mary Cancer Centre Westmead, Westmead, Australia; [10]Oncology Center Department, Isala Hospital, Zwolle, Netherlands; [11]Medical Oncology Dept., Amphia Ziekenhuis-location Molengracht, Breda, Netherlands; [12]Medical Oncology Dept., Leiden University Medical Center (LUMC), Leiden, Netherlands; [13]Medical Oncology Dept, UMC-University Medical Center Utrecht, Utrecht, Netherlands; [14]Department of Soft Tissue/Bone Sarcoma and Melanoma, Maria Sklodowska-Curie National Research Institute of Oncology, Warsaw, Poland; [15]Medical Oncology department-Suite 5/6, Melanoma Institute Australia, Wollstonecraft, Australia; [16]Division of Cancer Medicine, Peter MacCallum Cancer Centre, Melbourne, Australia; [17]Medical Oncology department, Erasmus MC - University Medical Center, Rotterdam, Netherlands; [18]Faculty of Medicine and Health, The University of Sydney, and Mater and Royal North Shore Hospitals, University of Sydney, Melanoma Institute Australia, Sydney, Australia; [19]Medical Oncology Dept, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [20]Psychosocial Research and Epidemiology Department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands

Background: Neoadjuvant ipilimumab (IPI) plus nivolumab (NIVO) has demonstrated to improve event free and distant metastasis free survival compared to adjuvant NIVO for patients (pts) with clinically detectable stage III melanoma. However, this comes at cost of a doubling of the grade 3/4 toxicity frequency in this curative setting and it is currently unclear how this impacts health-related quality of life (HRQoL) over time. Here, we report the secondary HRQoL outcomes of the phase 3 NADINA trial.

Methods: Using an app based HRQoL collection, patients were asked to complete the EORTC QLQ-C30 at baseline and at week 6, 12,24,36,48,60, and every 24 weeks thereafter. Key HRQoL domains included all functioning scales: physical (PF), emotional (EF), role (RF), cognitive (CF), social (SF), and fatigue (FA). The analysis was truncated at week 60 due to falling compliance numbers.

Results: At the data cutoff of 14/04/2025, the 423 pts included in NADINA (NCT04949113), (neoadjuvant, n = 212; adjuvant, n = 211), questionnaire completion rates were more than 80% at baseline had remained above 60% through week 60 in both arms. Baseline functioning scores were high and comparable between the arms. At week 12, PF, RF, and SF were clinically and statistically significantly lower in the neoadjuvant arm than in the adjuvant arm (PF difference 7.9 points [95% CI 3.8−12.0]; $p < 0.01$; RF difference 11.9 points [95% CI 6.0−17.9]; $p < 0.01$; SF difference 11.8 points [95% CI 6.7−17.0]; $p < 0.01$), but scores converged by week 24 and remained similar thereafter. EF and CF did not differ between the two arms at any time point. Mean FA scores at week 12 were significantly higher in the neoadjuvant arm compared to the adjuvant arm (32.4 vs 26.7; FA difference 5.6 points [95% CI 0.9−10.3]; $p < 0.05$), with scores converging by week 24 and remaining similar thereafter.

Conclusions: Neoadjuvant IPI + NIVO results in more short-term deterioration in HRQoL compared to adjuvant NIVO, but HRQoL recovers shortly after and remains stable and comparable to those treated with adjuvant NIVO. Thus, despite higher toxicity rates associated with neoadjuvant IPI + NIVO versus adjuvant NIVO, HRQoL data support the tolerability of the neoadjuvant approach in stage III melanoma.

Clinical trial identification: NCT04949113.

Legal entity responsible for the study: The authors.

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Carlino: Financial Interests, Personal, Advisory Board, Consultant Advisor: MSD, BMS, Novartis, Amgen, Oncosec, Merck, Sanofi, Ideaya, Pierre-Fabre, Eisai, Nektar, Regeneron, Stand Therapeutics. E. Kapiteijn: Financial Interests, Institutional, Advisory Board: Delcath, Lilly; Financial Interests, Institutional, Invited Speaker: Immunocore; Financial Interests, Institutional, Coordinating PI: BMS, Pierre-Fabre, Delcath, Novartis. K. Suijkerbuijk: Financial Interests, Institutional, Advisory Board: BMS, AbbVie, Pierre Fabre; Financial Interests, Institutional, Other, honoraria for Safety Review Committee: Sairopa; Financial Interests, Institutional, Research Grant: Novartis, TigaTx, Bristol Myers Squibb, Philips, Genmab. P. Rutkowski: Financial Interests, Personal, Invited Speaker, honoraria for lectures: MSD, BMS, Pierre Fabre; Financial Interests, Personal, Advisory Board: MSD, BMS, Pierre Fabre, Merck, Sanofi, Blueprint Medicines, Philogen, Medison Pharma, Genesis Pharma; Financial Interests, Personal, Invited Speaker: Merck, Sanofi, Novartis, AstraZeneca, Genesis Pharma; Financial Interests, Institutional, Research Grant, research grant for ISS: Pfizer; Financial Interests, Institutional, Funding, research grant for institution: BMS; Non-Financial Interests, Member of Board of Directors: Polish Society of Surgical Oncology; Non-Financial Interests, Member of Board of Directors, President: Polish Oncological Society. A.M. Menzies: Financial Interests, Personal, Advisory Board, advisory board: BMS, MSD, Novartis, Roche, Pierre-Fabre, QBiotics. S.K. Sandhu: Financial Interests, Institutional, Advisory Board, I have served on advisory boards for MSD. The e contracts for my role are set up with the institution and the funds go to a research fund at Peter MacCallum Cancer Centre: Merck Sharp and Dohme; Financial Interests, Institutional, Advisory Board, I have served on advisory boards for AstraZeneca. The contracts for my role as an advisor are set up with the institution and the funds go to a research fund at Peter MacCallum Cancer Centre: AstraZeneca; Financial Interests, Institutional, Advisory Board, I have served on advisory boards for Novartis. The contracts for my role as an advisor are set up with the institution and the funds go to a research fund at Peter MacCallum Cancer Centre: Novartis; Financial Interests, Institutional, Advisory Board, I have served on an advisory board for Merck Serono. The contracts for my service are set up with the institution and funds go into a research fund at the Peter MacCallum Cancer Centre: Merck Serono; Financial Interests, Institutional, Advisory Board, I have served on an advisory board for SkylineDx. The contracts for my service are set up with the institution and funds go into a research fund at the Peter MacCallum Cancer Centre: SkylineDx; Financial Interests, Institutional, Advisory Board, I have served on an advisory board for AbbVie. The contracts for my role are with the institution and funds go into a research fund at the Peter MacCallum Cancer Centre: AbbVie; Financial Interests, Institutional, Invited Speaker, I have received honoraria for invited speaker role only which is paid to my institution. Funds go into a research fund at the Peter MacCallum Cancer Centre: Janssen; Financial Interests, Personal, Other, Equity (stock): AdvanCell; Financial Interests, Personal, Stocks/Shares: AdvanCell; Financial Interests, Institutional, Research Grant, My institution receives grant funding to run an investigator initiated trial that I lead: Novartis, Genentech, Amgen, AstraZeneca, Merck Serono, Merck Sharp and Dohme; Financial Interests, Institutional, Research Grant, Pfizer are providing funding to my institution for the conduct of an investigator initiated clinical trial: Pfizer; Financial Interests, Institutional, Research Grant, Senwha are providing funding to my institution for the conduct of an investigator initiated clinical trial: Senwha; Non-Financial Interests, Other, I serve on the Independent Safety and Data Monitoring committee for 2 of Novartis sponsored studies and don't receive any compensation for this: Novartis; Non-Financial Interests, Other, I serve on the steering committee for several Janssen sponsored trials and i do not receive compensation for this: Janssen; Non-Financial Interests, Other, I serve on the steering committee for one AstraZeneca sponsored trial and I do not receive compensation for this: AstraZeneca; Non-Financial Interests, Other, I serve on the steering committee for one Genentech sponsored trials and I do not receive compensation for this: Genentech; Non-Financial Interests, Other, I serve on the steering committee for a Bristol Myer Squibb sponsored trial and I do not receive renumeration for this: Bristol Myer Squibb; Non-Financial Interests, Principal Investigator, I am a principal investigator on several of Janssen sponsored studies and don't receive any renumeration for this: Janssen; Non-Financial Interests, Principal Investigator, I am a principal investigator on several Novartis sponsored studies and dont receive any renumeration for this: Novartis; Non-Financial Interests, Principal Investigator, I am a principal investigator on several of Genentech sponsored studies and don't receive any renumeration for this: Genentech; Non-Financial Interests, Principal Investigator, I am a principal investigator on several of Bristol Myer Squibb sponsored studies and don't receive any renumeration for this: Bristol Myer Squibb; Non-Financial Interests, Principal Investigator, I am the Priincipal Investigator for several AstraZeneca sponsored studies and am not renumerated for this: AstraZeneca; Non-Financial Interests, Principal Investigator, I am a principal investigator on a MacroGenics sponsored study and don't receive any renumeration for this: MacroGenics; Non-Financial Interests, Principal Investigator, I am a principal investigator on a Regeneron sponsored study and do not receive any renumeration for this: Regeneron; Non-Financial Interests, Principal Investigator, I am the Principal Investigator on several Merck Sharp and Dohme studies and I do not receive any renumeration for this: Merck Sharp and Dohme; Non-Financial Interests, Principal Investigator, I am a principal investigator on an IO Biotech study and I do not receive any renumeration for this: IO Biotech; Non-Financial Interests, Principal Investigator, I am principal investigator on an Evaxion sponsored study and I do not receive any renumeration for this: Evaxion. A.A.M. van der Veldt: Financial Interests, Institutional, Advisory Board: BMS, MSD, Sanofi, Ipsen, Novartis, Eisai, Pfizer, Roche, Pierre Fabre. G.V. Long: Financial Interests, Personal, Other, Consultant Advisor: Agenus Inc, Amgen Inc, Array Biopharma Inc, AstraZeneca UK Limited, Bayer Healthcare Pharmaceuticals, BioNTech SE, Boehringer Ingelheim International GmbH, Bristol Myers Squibb, Evaxion Biotech A/S, Hexal AG, Highlight Therapeutics S. L., IOBiotech, Immunocore Ireland Limited, Innovent Bioilogics USA Inc, Merck Sharp & Dohme, Novartis Pharma AG, PHMR Limited, Pierre Fabre, Regeneron Pharmaceuticals Inc, Scancell Limited, SkylineDX B.V., Fortiva Biologics (USA); Financial Interests, Personal, Advisory Board, Consultant Advisor: GI Innovation; Non-Financial Interests, Principal Investigator, GL is PI on over 30 clinical trials: GL is PI on over 30 clinical trials. C.U. 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1616P Neoadjuvant pembrolizumab in resectable stage III melanoma: Real-world data from the nationwide Dutch Melanoma Treatment Registry

C.D.M. Boreel[1], M. Bloem[2], A.J.M. van den Eertwegh[3], M. Aarts[4], F. Van den Berkmortel[5], C.U. Blank[6], W. Blokx[7], M. Boers-Sonderen[8], H. Bonenkamp[9], J.W. de Groot[10], J.B.A.G. Haanen[11], G. Hospers[12], D. Piersma[13], B. Rikhof[14], M. Stevense-den Boer[15], A.A.M. van der Veldt[16], G. Vreugdenhil[17], M.W. Wouters[18], E. Kapiteijn[19], K. Suijkerbuijk[20]

[1]Department of Medical Oncology, UMC - University Medical Center Utrecht, Utrecht, Netherlands; [2]Biomedical Data Sciences Dept., LUMC - Leiden University Medical Center, Leiden, Netherlands; [3]Medical Oncology, Amsterdam UMC, locatie VUmc, Amsterdam, Netherlands; [4]Medical Oncology, Maastricht University Medical Center (MUMC), Maastricht, Netherlands; [5]Medical Oncology, Zuyderland Medical Center, Heerlen, Netherlands; [6]Medical Oncology Dept, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [7]Department of Pathology, University Medical Centre Utrecht, Utrecht, Netherlands; [8]Department of Medical Oncology, Radboud University Medical Center, Nijmegen, Nijmegen, Netherlands; [9]Department of Surgery, Radboud University Medical Center, Nijmegen, Netherlands; [10]Oncology Center Department, Isala Hospital, Zwolle, Netherlands; [11]Department of Medical Oncology, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [12]Medical Oncology, UMCG - University Medical Center Groningen, Groningen, Netherlands; [13]Department of Internal Medicine, Medisch Spectrum Twente (MST) - Radiology, Enschede, Netherlands; [14]Internal medicine department, Frisius Medical Centre, Leeuwarden, Netherlands; [15]Medical Oncology Dept., Amphia Ziekenhuis-location Molengracht, Breda, Netherlands; [16]Medical Oncology department, Erasmus MC - University Medical Center, Rotterdam, Netherlands; [17]Internal Medicine Dept., Maxima Medisch Centrum -Veldhoven, Veldhoven, Netherlands; [18]Surgical Oncology Department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [19]Medical Oncology Dept., Leiden University Medical Center (LUMC), Leiden, Netherlands; [20]Medical Oncology Department, UMC - University Medical Center Utrecht, Utrecht, Netherlands

Background: The SWOG S1801 and NADINA trials have shown improved event-free survival of neoadjuvant immune checkpoint inhibition therapy (ICI) compared to adjuvant ICI in patients with resectable stage III melanoma. However, it is unclear how this translates to outcomes in the real-world setting.

Methods: Data from all patients with resectable stage III melanoma receiving neoadjuvant pembrolizumab in 2024 in the Netherlands were retrieved from the prospective nationwide Dutch Melanoma Treatment Registry. Patient and tumor characteristics were described, and an intention-to-treat analysis was conducted to evaluate pathological response. Major pathological response (MPR) included pathological complete response (pCR; 0% residual viable tumor (RVT)) or near complete response (pnCR; 1-10% RVT). Pathological response was also categorised into partial response (pPR; 11-50% RVT) and non-response (pNR; >50% RVT).

Results: A total of 175 patients were included. Median age was 66 years, 46.9% of the patients were female, and 47.6% of tested patients had a *BRAF* mutation. Of all patients, 80.0% had nodal-only disease, 6.9% had in-transit-only metastases, and 10.9% had both. Pathological response data were available for 153 patients (87.4%). Among these, 69 patients (45.1%) had MPR, including 59 patients (38.6%) with pCR and 10 patients (6.5%) with pnCR. pPR and NR were observed in 20 (13.1%) and 54 patients (35.3%), respectively (Table). Out of patients with NR, 4 patients did not undergo surgery due to progressive disease. Updated pathological response data and recurrence-free survival will be reported at the meeting.

Table: 1616P

Pathological response	(N = 175) *N (%)*
MPR (pCR+pnCR)	*69 (45.1%)*
pCR	*59 (38.6%)*
pnCR	*10 (6.5%)*
pPR	*20 (13.1%)*
pNR	*54 (35.3%)*
Not reported	*10 (6.5%)*
Total number of evaluated patients	*153 (100%)*
Not available	*22*

Conclusions: Neoadjuvant treatment with pembrolizumab in this real world stage III melanoma population resulted in a MPR rate of 45.1% and a pCR rate of 38.6%, suggesting benefit of neoadjuvant therapy for stage III melanoma beyond clinical trial setting.

Legal entity responsible for the study: The authors.

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1617P Leucine-rich 2 glycoprotein 1 (LRG1) as potential systemic biomarker for response to neoadjuvant immune checkpoint blockade in stage III melanoma

S. Chen[1], C.C.E. De Schrijver[1], H. Van Rossum[2], T. Linders[3], L. Hoeijmakers[1], M.W. Lucas[1], E.P. Hoefsmit[1], I. Reijers[1], P. DIMITRIADIS[4], B. van de Wiel[5], K. de Visser[6], A.A.M. van der Veldt[7], H. Eriksson[8], G. Hospers[9], R.A. Scolyer[10], K. Suijkerbuijk[11], E. Kapiteijn[12], A.M. Menzies[13], G.V. Long[14], C.U. Blank[1]

[1]Medical Oncology Department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [2]Clinical Chemistry department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [3]Clinical Laboratory department, Netherlands Cancer Institute, Amsterdam, Netherlands; [4]Biometrics Department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [5]Pathology Department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [6]Tumor Biology & Immunology Department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [7]Medical Oncology Department, Erasmus MC - University Medical Center, Rotterdam, Netherlands; [8]Oncology-Pathology Department, Karolinska University Hospital-Solna, Solna, Sweden; [9]Medical Oncology, UMCG - University Medical Center Groningen, Groningen, Netherlands; [10]Medical Dept., Melanoma Institute Australia, Wollstonecraft, Australia; [11]Medical Oncology Dept, UMC-University Medical Center Utrecht, Utrecht, Netherlands; [12]Medical Oncology Dept., Leiden University Medical Center (LUMC), Leiden, Netherlands; [13]Medical Oncology Department-Suite 5/6, Melanoma Institute Australia, Wollstonecraft, Australia; [14]Medical Oncology, University of Sydney, Sydney, Australia

Background: In a previous protomics analysis, we identified Leucine-rich 2 glycoprotein 1 (LRG1) as a potential biomarker in patient blood for predicting response to neoadjuvant immune checkpoint blockade (ICB) in stage III melanoma. LRG1 induces pathological neovascularisation in cancer, increasing metastasis risk, and is proposed to be induced by the IL-6/STAT3-pathway. Here, we aimed to analyse the potential of LRG1 as a novel (baseline) biomarker to subgroup patients, in addition to interferon-gamma signature (IFNγ) and PD-L1 expression.

Methods: Baseline plasma from patients with resectable stage III melanoma (n=32), treated with 2 cycles neoadjuvant ipilimumab 1 mg/kg + nivolumab 3 mg/kg in the PRADO trial, were analysed. LRG1 and IL-6 were measured using commercial ELISA kits (EH305RB & EH2IL6, Invitrogen). These data were correlated with previous proteomics data as a technical control, and with baseline PD-L1 expression (<1% low; ≥1% high), the 10-gene IFNγ signature, major pathological response (MPR) and 5 year survival.

Results: Low plasma LRG1 was associated with an improved 5y event free survival (EFS) rate versus a high LRG1 (p=0.07, see table). When high systemic LRG1 levels are combined with low intratumoral PD-L1 or low IFNγ scores, we identified a unfavourable patient population (5y event-free survival (EFS) of 50% and 33% respectively, compared to 83-100% and 71-80% in other situations). Similar observations were seen for MPR (MPR-rate of 30% and 0%, compared to 75-100% and 80-86%). Baseline systemic LRG1 levels showed a negative correlation with IL-6 (R=-0.5, p=0.015), suggesting that LRG1 is not IL-6-induced in stage III melanoma.

Table: 1617P			
	Number of patients	Proportion of MPR (%)	5y EFS (95% CI)
LRG1 low	14	79%	85% (69-100)
LRG1 high	17	52%	58% (40-88)
PD-L1 low − LRG1 low	6	83%	83% (58-100)
PD-L1 low − LRG1 high	10	30%	50% (27-93)
PD-L1 high − LRG1 low	3	100%	100%
PD-L1 high − LRG1 high	4	75%	100%
IFN-γ low − LRG1 low	5	80%	80% (52-100)
IFN-γ low − LRG1 high	6	0%	33% (11-100)
IFN-γ high − LRG1 low	6	83%	80% (52-100)
IFN-γ high − LRG1 high	7	86%	71% (45-100)

Conclusions: Pending confirmation in larger cohorts, baseline LRG1 in plasma might become a novel biomarker for neoadjuvant ICB in melanoma patients, especially in combination with PD-L1 expression. Since LRG1 does not correlate with IL-6, it emphasizes the potential of an LRG1-targeting therapy.

Clinical trial identification: NCT02977052 (PRADO extension of the OpACIN-NEO trial).

Legal entity responsible for the study: The Netherlands Cancer Institute.

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1618P Return to work in the phase III trial NADINA: Neoadjuvant versus adjuvant immunotherapy in stage III melanoma patients

J. Lijnsvelt[1], M. Egeler[2], M.W. Lucas[3], A.M. Menzies[4], R. Morton[5], A.A.M. van der Veldt[6], P. Rutkowski[7], E. Kapiteijn[8], S.K. Sandhu[9], K. Suijkerbuijk[10], J.W. de Groot[11], M.S. Carlino[12], G. Hospers[13], F. Van Den Eertwegh[14], M. Boers-Sonderen[15], M. Andrews[16], V.G. Atkinson[17], G.V. Long[18], C.U. Blank[19], L.V. Van De Poll-franse[20]

[1]Mod Department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [2]Psychosocial Oncology and Epidemiology Department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [3]Medical Oncology Department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [4]Medical Oncology Department-Suite 5/6, Melanoma Institute Australia, Wollstonecraft, Australia; [5]Clinical Trials Centre, NHMRC Clinical Trials Centre, University of Sydney, Camperdown, Australia; [6]Medical Oncology department, Erasmus MC - University Medical Center, Rotterdam, Netherlands; [7]Department of Soft Tissue/Bone Sarcoma and Melanoma, Maria Sklodowska-Curie National Research Institute of Oncology, Warsaw, Poland; [8]Medical Oncology Dept., Leiden University Medical Center (LUMC), Leiden, Netherlands; [9]Division of Cancer Medicine, Peter MacCallum Cancer Centre, Melbourne, Australia; [10]Medical Oncology Dept, UMC-University Medical Center Utrecht, Utrecht, Netherlands; [11]Oncology Center Department, Isala Hospital, Zwolle, Netherlands; [12]Department of Medical Oncology, Crown Princess Mary Cancer Centre Westmead, Westmead, Australia; [13]Medical Oncology, UMCG - University Medical Center Groningen, Groningen, Netherlands; [14]Medical oncology, Amsterdam UMC - Vrije University Medical Centre (VUmc), Amsterdam, Netherlands; [15]Department of Medical Oncology, Radboud University Medical Center, Nijmegen, Nijmegen, Netherlands; [16]Medical Oncology Department, Alfred Hospital, Melbourne, Australia; [17]Division of Cancer Services, Princess Alexandra Hospital - Metro South Health, Woolloongabba, Australia; [18]Medical Oncology department, University of Sydney, Sydney, Australia; [19]Medical Oncology Dept, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [20]Psychosocial Research and Epidemiology Department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands

Background: Neoadjuvant (Neoadj) ipilimumab + nivolumab in stage III melanoma improves event-free and distant metastasis free survival as compared to adjuvant (adj) therapy. Grade 3/4 toxicities were 30% versus 15%, while short-term quality of life was equal. Work participation after cancer therapy is considered a sensitive indicator of psychologiocal recovery, social integration, financial well-being and life normalization. Therefore, we compared both arms in regards of work activity.

Methods: The phase III NADINA trial radomized 423 patients with resectable, macroscopic stage III melanoma to either two cycles of neoadjuvant ipilimumab plus nivolumab followed by surgical resection plus adjuvant therapy in case of not achieving a major pathologic response (>10% vital tumor in the resection specimen), versus surgical resection followed by 12 cycles of adjuvant nivolumab. All participants were surveyed regarding their employment status at baseline (BL) and at week 60 (wk 60).

Results: Cut-off was 14 April 2025. At BL in the neoadjuvant arm (n = 173) 32.9% of patients worked full-time (mean 40.8 hrs/wk), 9.8% part-time (20.7 hrs/wk) and 9.3% were self-employed (33.9 hrs/wk), while in the adjuvant arm (n = 165), 27.3% worked full-time (39.6 hrs/wk), 17.6% part-time (24.3 hrs/wk) and 9.7% were self-employed (31.1 hrs/wk). At wk 60 among n=143 evaluable neoadjuvant patients, 26.6% remained full-time (41.7 hrs/wk), 13.3% part-time (20.7 hrs/wk) and 7.7% self-employed (37.5 hrs/wk), whereas of n=105 adjuvant patients, 26.7% worked full-time (37.5 hrs/wk), 16.2% part-time (21.3 hrs/wk) and 9.5% were self-employed (27.4 hrs/wk) at wk 60.

Conclusions: Overall, there were no substantial impairments in work participation for both, the neoadjuvant and adjuvant immunotherapy cohorts. Patients receiving neoadjuvant immunotherapy showed a slight decrease in full return to work at week 60, including those who were self-employed. This was compensated by a modest increase in the patients working partially. Among patients treated with adjuvant therapy, the changes were even less pronounced. Additional analyses adjusting for potential confounders, exploring subgroups, and analysing the timing of return to work are ongoing.

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Rutkowski: Financial Interests, Personal, Invited Speaker, honoraria for lectures: MSD, BMS, Pierre Fabre; Financial Interests, Personal, Advisory Board: MSD, BMS, Pierre Fabre, Merck, Sanofi, Blueprint Medicines, Philogen, Medison Pharma, Genesis Pharma; Financial Interests, Personal, Invited Speaker: Merck, Sanofi, Novartis, AstraZeneca, Genesis Pharma; Financial Interests, Institutional, Research Grant, research grant for ISS: Pfzer; Financial Interests, Institutional, Funding, research grant for institution: BMS; Non-Financial Interests, Member of Board of Directors: Polish Society of Surgical Oncology; Non-Financial Interests, Member of Board of Directors, President: Polish Oncological Society. E. Kapiteijn: Financial Interests, Institutional, Advisory Board: Delcath, Lilly; Financial Interests, Institutional, Invited Speaker: Immunocore; Financial Interests, Institutional, Coordinating PI: BMS, Pierre-Fabre, Delcath, Novartis. S.K. Sandhu: Financial Interests, Institutional, Advisory Board, I have served on advisory boards for MSD. The e contracts for my role are set up with the institution and the funds go to a research fund at Peter MacCallum Cancer Centre: Merck Sharp and Dohme; Financial Interests, Institutional, Advisory Board, I have served on advisory boards for AstraZeneca. The contracts for my role as an advisor are set up with the institution and the funds go to a research fund at Peter MacCallum Cancer Centre: AstraZeneca; Financial Interests, Institutional, Advisory Board, I have served on advisory boards for Novartis. The contracts for my role as an advisor are set up with the institution and the funds go to a research fund at Peter MacCallum Cancer Centre: Novartis; Financial Interests, Institutional, Advisory Board, I have served on an advisory board for Merck Serono. The contracts for my service are set up with the institution and funds go into a research fund at the Peter MacCallum Cancer Centre: Merck Serono; Financial Interests, Institutional, Advisory Board, I have served on an advisory board for SkylineDx. The contracts for my service are set up with the institution and funds go into a research fund at the Peter MacCallum Cancer Centre: SkylineDx; Financial Interests, Institutional, Advisory Board, I have served on an advisory board for AbbVie. The contracts for my role are with the institution and funds go into a research fund at the Peter MacCallum Cancer Centre: AbbVie; Financial Interests, Institutional, Invited Speaker, I have received honoraria for invited speaker role only which is paid to my institution. Funds go into a research fund at the Peter MacCallum Cancer Centre: Janssen; Financial Interests, Personal, Other, Equity (stock): AdvanCell; Financial Interests, Personal, Stocks/Shares: AdvanCell; Financial Interests, Institutional, Research Grant, My institution receives grant funding to run an investigator initiated trial that I lead: Novartis, Genentech, Amgen, AstraZeneca, Merck Serono, Merck Sharp and Dohme; Financial Interests, Institutional, Research Grant, Pfizer are providing funding to my institution for the conduct of an investigator initiated clinical trial: Pfizer; Financial Interests, Institutional, Research Grant, Senwha are providing funding to my institution for the conduct of an investigator initiated clinical trial: Senwha; Non-Financial Interests, Other, I serve on the Independent Safety and Data Monitoring committee for 2 of Novartis sponsored studies and dont receive any compensation for this: Novartis; Non-Financial Interests, Other, I serve on the steering committee for several Janssen sponsored trials and i do not receive compensation for this: Janssen; Non-Financial Interests, Other, I serve on the steering committee for one AstraZeneca sponsored trial and I do not receive compensation for this: AstraZeneca; Non-Financial Interests, Other, I serve on the steering committee for one Genentech sponsored trials and I do not receive compensation for this: Genentech; Non-Financial Interests, Other, I serve on the steering committee for a Bristol Myer Squibb sponsored trial and I do not receive renumeration for this: Bristol Myer Squibb; Non-Financial Interests, Principal Investigator, I am a principal investigator on several of Janssen sponsored studies and don't receive any renumeration for this: Janssen; Non-Financial Interests, Principal Investigator, I am a principal investigator on several Novartis sponsored studies and dont receive any renumeration for this: Novartis; Non-Financial Interests, Principal Investigator, I am a principal investigator on several of Genentech sponsored studies and don't receive any renumeration for this: Genentech; Non-Financial Interests, Principal Investigator, I am a principal investigator on several of Bristol Myer Squibb sponsored studies and don't receive any renumeration for this: Bristol Myer Squibb; Non-Financial Interests, Principal Investigator, I am the Priincipal Investigator for several AstraZeneca sponsored studies and am not remunerated for this: AstraZeneca; Non-Financial Interests, Principal Investigator, I am a principal investigator on a MacroGenics sponsored study and don't receive any renumeration for this: MacroGenics; Non-Financial Interests, Principal Investigator, I am a principal investigator on a Regeneron sponsored study and do not receive any renumeration for this: Regeneron; Non-Financial Interests, Principal Investigator, I am the Principal Investigator on several Merck Sharp and Dohme studies and I do not receive any renumeration for this: Merck Sharp and Dohme; Non-Financial Interests, Principal Investigator, I am a principal investigator on an IO Biotech study and I do not receive any renumeration for this: IO Biotech; Non-Financial Interests, Principal Investigator, I am principal investigator on an Evaxion sponsored study and I do not receive any renumeration for this: Evaxion. K. Suijkerbuijk: Financial Interests, Institutional, Advisory Board: BMS, AbbVie, Pierre Fabre; Financial Interests, Institutional, Other, honoraria for Safety Review Committee: Sairopa; Financial Interests, Institutional, Research Grant: Novartis, TigaTx, Bristol Myers Squibb, Philips, Genmab. M.S. Carlino: Financial Interests, Personal, Advisory Board, Consultant Advisor: MSD, BMS, Novartis, Amgen, Oncosec, Merck, Sanofi, Ideaya, Pierre Fabre, Eisai, Nektar, Regeneron, Stand Therapeutics. F. Van Den Eertwegh: Financial Interests, Institutional, Advisory Board: Bristol Myers Squibb International, MSD oncology, Ipsen, Pierre Fabre, Janssen-Cilag; Financial Interests, Institutional, Research Grant, Research grant for trial: IDERA. M. Andrews: Financial Interests, Institutional, Advisory Board: MSD Australia, BMS Australia; Financial Interests, Personal, Advisory Board: ImCheck; Financial Interests, Institutional, Local PI: Gilead Sciences, Merus, MSD Australia. V.G. Atkinson: Financial Interests, Personal, Advisory Board: BMS, MSD, Novartis, QBiotics; Financial Interests, Personal, Invited Speaker: BMS, MSD, Novartis, Pierre fabre, Limbic; Financial Interests, Personal, Other, Travel support: Pierre Fabre; Financial Interests, Personal, Other, Advisory Board: Immunocore. G.V. Long: Financial Interests, Personal, Other, Consultant Advisor: Agenus Inc, Amgen Inc, Array Biopharma Inc, AstraZeneca UK Limited, Bayer Healthcare Pharmaceuticals, BioNTech SE, Boehringer Ingelheim International GmbH, Bristol Myers Squibb, Evaxion Biotech A/S, Hexal AG, Highlight Therapeutics S. L., IOBiotech, Immunocore Ireland Limited, Innovent Bioilogics USA Inc, Merck Sharp & Dohme, Novartis Pharma AG, PHMR Limited, Pierre Fabre, Regeneron Pharmaceuticals Inc, Scancell Limited, SkylineDX B.V., Fortiva Biologics (USA); Financial Interests, Personal, Advisory Board, Consultant Advisor: GI Innovation; Non-Financial Interests, Principal Investigator, GL is PI on over 30 clinical trials: GL is PI on over 30 clinical trials. C.U. Blank: Financial Interests, Institutional, Advisory Board: BMS, MSD, Roche, Novartis, GSK, AZ, Pfizer, Lilly, GenMab, Pierre Fabre, Ventures, Senya; Financial Interests, Personal, Advisory Board: Third Rock; Financial Interests, Personal, Stocks/Shares: Flindr Tx; Financial Interests, Institutional, Other: PatentWO2021/177822. All other authors have declared no conflicts of interest.

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1619P Pooled outcomes with first-line nivolumab + relatlimab (NIVO + RELA) in patients (pts) with advanced melanoma (MEL)

H.A. Tawbi[1], P.A. Ascierto[2], E.J. Lipson[3], C. Gaudy Marqueste[4], S. Dalle[5], F.S. Hodi[6], D. Schadendorf[7], L.A. Matamala[8], E. Castillo Gutierrez[9], P. Rutkowski[10], K. Jalkanen[11], H. Gogas[12], L.A. Fecher[13], J. Menezes[14], I.M. Svane[15], R. Jiang[16], L. Sakkal[16], A. Selvan[17], M. Semaan[16], G.V. Long[18]

[1]Melanoma Medical Oncology Department, The University of Texas MD Anderson Cancer Center, Houston, United States of America; [2]Melanoma, Cancer Immunotherapy & Developmental Therapeutics, Istituto Nazionale dei Tumori IRCCS "Fondazione G. Pascale," Naples, Italy; [3]Oncology Department, Bloomberg-Kimmel Institute for Cancer Immunotherapy, Johns Hopkins Sidney Kimmel Comprehensive Cancer Center, Baltimore, United States of America; [4]Medical Oncology, Aix-Marseille University, APHM, CHU Timone, Marseille, France; [5]Unit of Dermatology, Hospices Civils de Lyon, Cancer Research Centre of Lyon, Pierre-Bénite, France; [6]Melanoma, Dana-Farber Cancer Institute, Boston, United States of America; [7]Department of Dermatology - Hautklinik, University Hospital Essen, German Cancer Consortium, Partner Site Essen and University Alliance Ruhr, Research Center One Health, Essen, Germany; [8]Department of Medical Oncology, Instituto Oncológico Fundación Arturo López Pérez and Department of Oncology, Instituto Nacional del Cáncer, Santiago, Chile; [9]Medical Oncology Department, FAICIC Clinical Research, Veracruz, Mexico; [10]Department of Soft Tissue/Bone Sarcoma and Melanoma, Maria Skłodowska-Curie National Research Institute of Oncology, Warsaw, Poland; [11]Clinical Trial Unit, Helsinki University Hospital Cancer Center, Helsinki, Finland; [12]First Department of Medicine, National and Kapodistrian University of Athens, Athens, Greece; [13]Internal Medicine, University of Michigan, Rogel Cancer Center, Ann Arbor, United States of America; [14]Oncology Department, Hospital Nossa Senhora da Conceição, Porto Alegre, Brazil; [15]Department of Oncology, Copenhagen University Hospital, Herlev, Denmark; [16]Medical Oncology, Bristol Myers Squibb, Princeton, United States of America; [17]Melanoma, Oncology, Bristol Myers Squibb, Princeton, United States of America; [18]Medical Oncology, Melanoma Institute Australia, The University of Sydney, Royal North Shore and Mater Hospitals, Sydney, Australia

Background: NIVO + RELA as a fixed dose combination (FDC) is approved for advanced MEL based on results from the phase 2/3 RELATIVITY-047 (RELA-047) trial, which showed a statistically significant progression-free survival (PFS) benefit with first-line (1L) NIVO + RELA vs NIVO alone. RELATIVITY-020 (RELA-020) is a phase 1/2a trial of RELA ± NIVO in advanced solid tumors that included pts in the 1L setting. Both included pts with acral and mucosal MEL, subtypes with typically poor prognosis. Here, we present a large, pooled analysis of efficacy and safety in pts from RELA-047 and RELA-020.

Methods: Pts with previously untreated advanced MEL who received NIVO 480 mg + RELA 160 mg FDC Q4W from RELA-047 and NIVO 240 mg + RELA 80 mg Q2W (Part C) and coadministered NIVO 480 mg + RELA 160 mg Q4W (Part E) from RELA-020 were included. Objective response rate (ORR), PFS, overall survival (OS), MEL-specific survival (MSS), and safety were assessed.

Results: The minimum follow-up was 45 mo for RELA-047 and 76 mo (Part C) and 33 mo (Part E) for RELA-020. NIVO + RELA showed generally similar response and survival outcomes for the pooled intent-to-treat (ITT) population (n = 498) vs RELA-047 and RELA-020; disease control was observed in pts with acral and mucosal MEL (Table). MSS was comparable; 2-y MSS rates were 68% for the pooled ITT population and 43% for acral, 44% for mucosal, and 73% for cutaneous nonacral/other MEL subgroups. The rate of grade 3/4 treatment-related adverse events was 24% in the pooled ITT population. Results in pts with PFS ≥ 3 y will be presented.

Conclusions: In this pooled analysis of the largest prospective dataset to date for 1L NIVO + RELA, efficacy and safety were generally consistent across pts with advanced MEL. While benefit was most pronounced in cutaneous nonacral/other MEL, meaningful activity was observed in acral and mucosal MEL, suggesting NIVO + RELA may have clinical utility across the advanced MEL disease spectrum. These results support NIVO + RELA as 1L treatment for advanced MEL.

Table: 1619P

	Pooled				047 (n = 355)	020	
	ITT (n = 498)	Acral (n = 50)	Mucosal (n = 37)	Cutaneous nonacral/other (n = 411)		C (n = 66)	E (n = 77)
ORR, % (95% CI)	45.2 (40.7−49.7)	28.0 (16.2−42.5)	32.4 (18.0−49.8)	48.4 (43.5−53.4)	43.9 (38.7−49.3)	47.0 (34.6−59.7)	49.4 (37.8−61.0)
PFS							
2-y, % (95% CI)	41 (37−46)	18 (8−30)	24 (10−40)	46 (40−50)	40 (34−45)	34 (22−46)	56 (44−67)
Median, mo (95% CI)	12.2 (8.3−17.0)	3.3 (2.8−6.5)	10.0 (2.8−16.6)	15.7 (11.0−25.0)	10.2 (6.5−15.7)	11.6 (3.7−16.6)	33.4 (11.0−NR)
OS							
2-y, % (95% CI)	63 (58−67)	37 (24−50)	36 (21−52)	68 (64−72)	62 (57−67)	56 (43−67)	73 (62−82)
Median, mo (95% CI)	55.8 (38.6−NR)	12.6 (7.0−21.4)	18.9 (9.1−24.3)	NR (55.8−NR)	53.3 (34.0−NR)	34.6 (18.9−NR)	NR (40.0−NR)

NR, not reached.

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Ascierto: Financial Interests, Personal, Other, Consultant and Advisory Role: BMS, Roche Genentech, Novartis, Merck Serono, Sun Pharma, Sanofi, Sandoz, Immunocore, Boehringer Ingelheim, Regeneron; Financial Interests, Personal, Other, Consultant, Advisory Role and Travel support: MSD, Pierre Fabre; Financial Interests, Personal, Other, Consultant and Advisory Role.Travel support: Pfizer/Array; Financial Interests, Personal, Other, Consultant Role: Italfarmaco; Financial Interests, Personal, Other, Consultant Role: Pfizer, Nouscom, Lunaphore; Financial Interests, Personal, Other, Consultant role: Medicenna; Financial Interests, Personal, Other, Consultant role and travel support: Bio-AI Health; Financial Interests, Personal, Advisory Board, Consultant and Advisory role: ValoTx; Financial Interests, Personal, Advisory Board, Consultant and Advisor role, Travel support: Replimmune; Financial Interests, Personal, Advisory Board, Advisor role: Bayer; Financial Interests, Personal, Other, Consultant and Advisory: Erasca; Financial Interests, Personal, Other, Consultant: Philogen, Incyte; Financial Interests, Personal, Other, Consultant and Advisory role: BionTech, Genmab; Financial Interests, Personal, Advisory Board: Anaveon, Menarini; Financial Interests, Institutional, Funding, Clinical trial and translational research: BMS; Financial Interests, Institutional, Funding, Clinical Trial: Roche Genentech, Pfizer/Array, Sanofi; Non-Financial Interests, Leadership Role, President since 2010: Fondazione Melanoma Onlus Italy; Non-Financial Interests, Leadership Role, President since 2014: Campania Society of ImmunoTherapy of Cancer (SCITO) Italy; Non-Financial Interests, Other, Member of Steering Committee since 2016: Society for Melanoma Research (SMR); Non-Financial Interests, Member of Board of Directors, November 2017 - December 2021: Society for Immunotherapy of Cancer (SITC); Non-Financial Interests, Member: ASCO, SITC, EORTC Melanoma Cooperative Group, AIOM, SMR. E.J. Lipson: Financial Interests, Personal, Other, Consultant: Bristol Myers Squibb International, Merck, Natera, Nektar Therapeutics, Pfizer, Regeneron, Replimune, IO Biotech, Huya Bioscience, Novartis, Oncosec, Merck kgaa; Financial Interests, Personal, Advisory Board: Caredx, Genentech, Instil Bio, Sun Pharma; Financial Interests, Personal, Stocks/Shares: Iovance; Financial Interests, Institutional, Coordinating PI: Bristol Myers Squibb; Financial Interests, Institutional, Local PI: Regeneron. C. Gaudy Marqueste: Financial Interests, Personal, Advisory Board: BMS, Pierre Fabre, MSD, Sun Pharma, Regeneron, Iovance; Financial Interests, Personal, Invited Speaker: BMS, Pierre Fabre, UCB; Financial Interests, Institutional, Local PI: BMS, MSD, Pfizer, Amgen, GSK, Roche, Novartis, Pierre Fabre, Replimune, Jansen, Kartos, IFX, Huyabio, Regeneron, IO Biotech, Philogen, Sotio, Kinnate, AstraZeneca, Day One biopharmaceuticals, Sairopa, Catalyst, Sanofi, Immunocore, Daiichi, Dragonfly, ERASCA, Microbiotica; Other, Travel expenses: BMS, Pierre Fabre, MSD, Janssen. S. Dalle: Financial Interests, Institutional, Advisory Board: MSD, BMS, Regeneron; Financial Interests, Personal, Other, First author: MSD; Financial Interests, Personal, Stocks/Shares, Spouse sanofi employee and stock owner: Sanofi; Financial Interests, Institutional, Research Grant: BMS, MSD, AstraZeneca, Pierre-Fabre. F.S. Hodi: Financial Interests, Personal, Leadership Role: Bristol Myers Squibb, Merck, Novartis, Compass Therapeutics, Apricity, Pionyr, 7 Hills Pharma, Bicara, Checkpoint Therapeutics, Bioentre, Gossamer, Iovance, Catalym, Immunocore, Kairos, Rheos, Bayer, Zumotor, Corner Therapeutics, Puretech, Curis, AstraZeneca, Pliant, Solu Therapeutics, Vir Biotechnology, 92 Bio; Financial Interests, Personal, Stocks or ownership: Apricity, Bicara, Checkpoint Therapeutics, Corner Therapeutics, Solu Therapeutics; Financial Interests, Institutional, Licencing Fees or royalty for IP: Bristol Myers Squibb, Novartis; Financial Interests, Institutional, Research Funding: Bristol Myers Squibb, Novartis; Financial Interests, Institutional, Research Grant: Bristol Myers Squibb, Novartis, Merck, Genentech, Seagen, Merus, Pfizer; Non-Financial Interests, Personal, Other, protocol scientific steering committee: Bristol Myers Squibb . D. Schadendorf: Financial Interests, Personal, Invited Speaker: BMS, MSD, Merck Serono, Regeneron; Financial Interests, Personal, Advisory Board: BMS, Novartis, MSD, Immunocore, Pierre Fabre, Regeneron, Philogen, BioAlta, Daiichi Sankyo, BioNTech, Immatics, CureVac, Formycon, Replimune, SunPharma, Skyline Dx, IOVance, Ipsen, IOBioTech, Boehringer-Ingelheim; Financial Interests, Institutional, Coordinating PI: Novartis, BMS, MSD, Pierre Fabre, Immatics; Financial Interests, Institutional, Research Grant: BMS, MSD; Financial Interests, Personal, Steering Committee Member: MSD; Financial Interests, Institutional, Local PI: Philogen, Immunocore, Regeneron, Biontech Rna Pharmacuticals GMBH, Replimune, IO Biotech; Non-Financial Interests, Member of Board of Directors: EORTC-MG; Non-Financial Interests, Leadership Role, Founding member and SC chair: European Melanoma Registry (EuMelaReg). P. Rutkowski: Financial Interests, Personal, Invited Speaker, honoraria for lectures: MSD, BMS, Pierre Fabre; Financial Interests, Personal, Advisory Board: MSD, BMS, Pierre Fabre, Merck, Sanofi, Blueprint Medicines, Philogen, Medison Pharma, Genesis Pharma; Financial Interests, Personal, Invited Speaker: Merck, Sanofi, Novartis, AstraZeneca, Genesis Pharma; Financial Interests, Institutional, Research Grant, research grant for ISS: Pfizer; Financial Interests, Institutional, Funding, research grant for institution: BMS; Non-Financial Interests, Member of Board of Directors: Polish Society of Surgical Oncology; Non-Financial Interests, Member of Board of Directors, President: Polish Oncological Society. K. Jalkanen: Financial Interests, Personal, Advisory Board: MSD, Ipsen, Roche, BMS, Pfizer, Lilly, Novartis, Bayer; Financial Interests, Personal, Stocks/Shares: Faron Pharmaceuticals, Curify Labs; Financial Interests, Institutional, Local PI, Conduct of sponsored clinical trial: Novartis; Financial Interests, Institutional, Local PI, Sponsored clinical trial: Exelixis; Financial Interests, Institutional, Local PI, Several clinical trials: BMS, MSD, Roche; Financial Interests, Institutional, Local PI, clinical trials: Incyte; Financial Interests, Institutional, Local PI, Conduct of clinical trials: Pfizer; Financial Interests, Institutional, Local PI, Conduct of clinical trial: Bayer; Financial Interests, Institutional, Local PI, Conduct sponsored trial: Orion Pharma. H. Gogas: Financial Interests, Personal, Advisory Board: MSD, BMS, Pierre Fabre, Sanofi; Financial Interests, Personal, Invited Speaker: MSD, BMS, Novartis, Pierre Fabre, Sanofi; Financial Interests, Steering Committee Member: Amgen, Replimmune; Financial Interests, Institutional, Local PI: Amgen, MSD, BMS, Replimmune, Iovance, Bayer; Financial Interests, Institutional, Research Grant: BMS, Pfizer, Lilly, Pierre Fabre, AstraZeneca. L.A. Fecher: Financial Interests, Personal, Other, Co-Chair, Melanoma Committee for Via Oncology pathway: Elsevier; Financial Interests, Institutional, Local PI: Bristol Myers Squibb International, Linnaeus Therapeutics, Merck, moderna, EMD Serono, Kartos Therapeutics; Financial Interests, Institutional, Trial Chair, ECOG-ACRIN Study Chair with support/funding from Pfizer: Pfizer; Non-Financial Interests, Other, Immunotherapy Toxicity Guidelines Member: ASCO; Non-Financial Interests, Other, Member, Palliative Care Guidelines Panel: NCCN; Non-Financial Interests, Principal Investigator, EA6183 clinical trial: ECOG-ACRIN. I.M. Svane: Financial Interests, Institutional, Other, Conference participation: MSD; Financial Interests, Personal, Invited Speaker: BMS, Janssen; Financial Interests, Personal, Advisory Board: Mendus, Genmab; Financial Interests, Personal, Stocks/Shares, Cofounder and Founder warrents: IO Biotech; Financial Interests, Institutional, Research Grant: Adaptimmune, Enara Bio, Lytix Biopharma, TILT Biotherapeutics, Asgard Therapeutics, IO Biotech; Financial Interests, Institutional, Funding: Evaxion; Financial Interests, Institutional, Other, drug for investigator driven trial: BMS; Non-Financial Interests, Principal Investigator: BMS, Roche, TILT Biotherapeutics, Lytix Biopharma, Novartis, Immunocore, MSD. R. Jiang, A. Selvan: Financial Interests, Personal, Full or part-time Employment: Bristol Myers Squibb; Financial Interests, Personal, Stocks or ownership: Bristol Myers Squibb. L. Sakkal, M. Semaan: Financial Interests, Personal, Full or part-time Employment: Bristol Myers Squibb; Financial Interests, Personal, Stocks/Shares: Bristol Myers Squibb. G.V. Long: Financial Interests, Personal, Other, Consultant Advisor: Agenus Inc, Amgen Inc, Array Biopharma Inc, AstraZeneca UK Limited, Bayer Healthcare Pharmaceuticals, BioNTech SE, Boehringer Ingelheim International GmbH, Bristol Myers Squibb, Evaxion Biotech A/S, Hexal AG, Highlight Therapeutics S. L., IOBiotech, Immunocore Ireland Limited, Innovent Bioilogics USA Inc, Merck Sharp & Dohme, Novartis Pharma AG, PHMR Limited, Pierre Fabre, Regeneron Pharmaceuticals Inc, Scancell Limited, SkylineDX B.V., Fortiva Biologics (USA); Financial Interests, Personal, Advisory Board, Consultant Advisor: GI Innovation; Non-Financial Interests, Principal Investigator, GL is PI on over 30 clinical trials: GL is PI on over 30 clinical trials. All other authors have declared no conflicts of interest.

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1620P Evaluation of two dosing regimens, NIVO3+IPI1 or NIVO1 +IPI3 in 399 patients with advanced melanoma

H. Helgadottir[1], K. Björkström.[2], C. Liu[2], A. Fager[3], L.L. Liu Burström[4], L. Ny[5]

[1]Department of Oncology-Pathology, Karolinska Institutet - Cancer Center Karolinska (CCK), Solna, Sweden; [2]Department of Oncology-Pathology, Karolinska Institute, Stockholm, Sweden; [3]Department of Oncology, Sahlgrenska Academy at University of Gothenburg, Gothenburg, Sweden; [4]Department of Oncology, Karolinska University Hospital, Stockholm, Sweden; [5]Department of Oncology, Sahlgrenska University Hospital, Gothenburg, Sweden

Background: Nivolumab 1 mg/kg plus ipilimumab 3 mg/kg (NIVO1+IPI3) is an effective treatment for advanced melanoma that was EMA approved in 2016. After the publication of the CheckMate 511 trial in 2019, the NIVO3+IPI1 regimen was also implemented in Sweden. In this study the efficacy and safety of the two dosing regimens was compared.

Methods: All patients with advanced inoperable melanoma treated with NIVO+IPI at the Karolinska and Sahlgrenska university hospitals were included in 2016-2023. The melanoma oncology is centralized to these clinics, and the study is hence a population-based study from the Stockholm and Gothenburg regions with altogether nearly 4 million inhabitants.

Results: In total, 399 patients were included, 209 had NIVO3+IPI1 and 190 NIVO1 +IPI3 (Table). The median follow-up in the two cohorts was 41 (range 20-86) and 53 months (range 19-130). The response rate was 48.8% in NIVO3+IPI1 and 36.8% in NIVO1+IPI3 treated (P=0.018). Adjusted hazard ratios (HR) for progression-free survival (PFS) was 0.73 (95% CI 0.57-0.94, P=0.017) and for overall survival (OS), 0.65 (95% CI 0.48-0.86, P=0.003) and in all but one of the studied subgroup HR was also <1, in favor of the NIVO3+IPI1 regimen (Table). The incidence of treatment-related grade 3-5 AEs was 30.6% with NIVO3+IPI1 vs. 50.5% with NIVO1+IPI3 (P<0.001). Of the patients treated with NIVO3+IPI1, 57.4% received all four NIVO+IPI doses, compared to 33.8% with NIVO1+IPI3 (P<0.001).

Table: 1620P PFS and OS in patients receiving N3+I1 or N1+I3

	N3+I1 (n)	N1+I3 (n)	PFS: N3+I1 vs. N1+I3 HR (95% CI)	OS: N3+I1 vs N1+I3 HR (95% CI)
All patients	209	190	0.73*	0.65*
Male	130	110	0.80	0.66*
Female	79	80	0.69	0.79
<60 years	83	95	0.69	0.67
≥60 years	126	95	0.78	0.70
Cutaneous	200	178	0.74*	0.66*
Mucosal	9	12	0.20*	0.19
V600E/K mut	98	105	0.74	0.67
V600E/K wt	111	85	0.70*	0.63*
Stage III, M1a-b	50	29	1.03	0.86
M1c	77	61	0.74	0.63
M1d	44	95	0.57*	0.57*
Normal LDH	116	92	0.78	0.62*
Elevated LDH	93	98	0.72	0.69*
ECOG 0	146	132	0.81	0.71
ECOG≥1	63	58	0.52*	0.44*
No previous treatment	91	95	0.65*	0.54*
Previously treated	118	95	0.77*	0.86

*Statistically significant

HR: Hazard ratio, adjusted for age, ECOG, LDH and stage. Nivo=N, Ipi=I

Conclusions: In the CheckMate 511 trial (n=360), no significant PFS and OS differences were seen at a median follow-up of 18.7 months. The CheckMate 511 trial was however not designed to demonstrate non-inferiority for efficacy, and the NIVO3 +IPI1 regimen has not been approved for melanoma by regulatory authorities. The presents study however showed that in a real-world setting the NIVO3+IPI1 had superior efficacy, possibly related to its improved tolerability and more received doses. The results warrant continued evaluation of this regimen in advanced melanoma.

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1621P First-line pembrolizumab alone or with investigational agents for advanced melanoma: Updated results from the phase I/II KEYMAKER-U02 substudy 02B

H. Gogas[1], R. Dummer[2], R. Shapira-Frommer[3], J. Pigozzo[4], P.A. Ascierto[5], C. Gaudy Marqueste[6], D. Portnoy[7], M. Del Vecchio[8], G. Bar-Sela[9], J. Cohen[10], A.M. Di Giacomo[11], A. Ribas[12], C. Dutriaux[13], C. Belli[14], G.L. Cohen[15], R. Dokhan[16], N. Jing[17], R. Lal[18], G.V. Long[19]

[1]First Department of Medicine, National and Kapodistrian University of Athens - School of Medicine, Athens, Greece; [2]Dermatology Department, USZ - University Hospital Zürich, Zurich, Switzerland; [3]Oncology Department, The Ella Institute for Treatment and Research of Melanoma and Skin Cancer - Sheba Medical Center, Ramat Gan, Israel; [4]Oncology, Veneto Institute of Oncology IOV - IRCCS, Padua, Italy; [5]Melanoma, Cancer Immunotherapy & Developmental Therapeutics, Istituto Nazionale Tumori IRCCS Fondazione Pascale, Naples, Italy; [6]CEPCM, Dermatology and skin cancer, Aix-Marseille Université, Marseille, France; [7]Medical Oncology and Hematology, West Cancer Center & Research Institute, Germantown, United States of America; [8]Medical Oncology Department, Fondazione IRCCS - Istituto Nazionale dei Tumori, Milan, Italy; [9]Oncology, Emek Medical Center, Afula, Israel; [10]Sharett Institute of Oncology, Hadassah Hebrew-University Medical Center, Jerusalem, Israel; [11]Oncology, University Hospital of Siena, Siena, Italy; [12]Department of Medicine, UCLA, Los Angeles, United States of America; [13]Dermatology Department, CHU Bordeaux - Hopital St. André, Bordeaux, France; [14]Department of Oncology, European Institute of Oncology, IRCCS, Milan, Italy; [15]Medical Oncology, Mary Potter Oncology Centre, Pretoria, South Africa; [16]Oncology, Rambam Health Care Campus, Haifa, Israel; [17]Clinical, Merck & Co., Inc. - Rahway, Rahway, United States of America; [18]Clinical Development Dept., MSD - Merck Sharp & Dohme Limited, Hoddesdon, United Kingdom; [19]Medical Oncologist, University of Sydney, Sydney, Australia

Background: Prior analysis of KEYMAKER-U02 substudy 02B (NCT04305054) showed promising activity for first-line vibostolimab (vibo) + pembrolizumab (pembro) and coformulated quavonlimab (qmab)/pembro in advanced melanoma. Manageable safety was observed across the first 4 arms (vibo + pembro, pembro alone, qmab/pembro; qmab/pembro + lenvatinib [len]). Updated results are presented, including from 2 previously unreported arms: coformulated favezelimab (fave; anti—LAG-3)/pembro and fave/pembro + all-trans retinoic acid (ATRA).

Methods: Participants (pts) with untreated, unresectable stage III or IV melanoma were randomized to open arms: vibo 200 mg + pembro 200 mg IV Q3W (arm 1), pembro 400 mg IV Q6W (arm 2), qmab 25 mg/pembro 400 mg IV Q6W (arm 3), qmab 25 mg/pembro 400 mg IV Q6W + len 20 mg PO QD (arm 4), fave 800 mg/pembro 200 mg IV Q3W (arm 5), or fave 800 mg/pembro 200 mg IV Q3W + ATRA 75 mg/m^2 PO BID for 3 d around each fave/pembro infusion (arm 6). Treatment continued for $\sim$2 y. Primary end points were safety and ORR per RECIST v1.1 by BICR. DOR per RECIST v1.1 by BICR was secondary. PFS per RECIST v1.1 by BICR and OS were exploratory.

Results: A total of 329 pts were treated: 90 in arm 1, 24 in arm 2, 31 in arm 3, 87 in arm 4, 49 in arm 5, and 48 in arm 6. Median (range) study follow-up at data cutoff (Nov 11, 2024) was 29.1 mo (19.6-51.2), 44.3 mo (27.1-51.9), 36.1 mo (27.1-42.5), 23.7 mo (15.9-44.8), 6.6 mo (4.1-13.7), and 6.6 mo (3.9-13.8), respectively. Treatment-related AEs occurred in 89% of pts in arm 1, 79% in arm 2, 97% in arm 3, 97% in arm 4, 76% in arm 5, and 92% in arm 6; grade 3-5 treatment-related AEs in 34%, 29%, 42%, 64%, 27%, and 27%. 3 pts (3%) in arm 1 died due to treatment-related AEs (myasthenia gravis, rhabdomyolysis, encephalitis). Efficacy data are reported in the table.

Conclusions: Safety was manageable across arms. Moderate antitumor activity was observed with fave/pembro (arm 5); no additional benefit was observed with combining ATRA with fave/pembro (arm 6).

Clinical trial identification: NCT04305054.

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Shapira-Frommer: Financial Interests, Personal and Institutional, Advisory Board: MSD; Financial Interests, Personal, Invited Speaker: BMS, AstraZeneca, Novartis ; Financial Interests, Institutional, Invited Speaker: Sanofi, Merck Serono; Financial Interests, Personal and Institutional, Research Funding: MSD; Non-Financial Interests, Personal, Invited Speaker: AstraZeneca; Financial Interests, Personal, Advisory Board: Novartis . J. Pigozzo: Other, Personal, Advisory Board: MSD; Other, Personal, Principal Investigator: MSD, BMS, Regeneron. P.A. Ascierto: Financial Interests, Personal, Other, Consultant and Advisory Role: BMS, Roche Genentech, Novartis, Merck Serono, Sun Pharma, Sanofi, Sandoz, Immunocore, Boehringer Ingelheim, Regeneron; Financial Interests, Personal, Other, Consultant, Advisory Role and Travel support: MSD, Pierre Fabre; Financial Interests, Personal, Other, Consultant and Advisory Role, Travel support: Pfizer/Array; Financial Interests, Personal, Other, Consultant Role: Italfarmaco; Financial Interests, Personal, Other, Consultant Role: Pfizer, Nouscom, Lunaphore; Financial Interests, Personal, Other, Consultant role: Medicenna; Financial Interests, Personal, Other, Consultant role and travel support: Bio-AI Health; Financial Interests, Personal, Advisory Board, Consultant and Advisory role: ValoTx; Financial Interests, Personal, Advisory Board, Consultant and Advisor role, Travel support: Replimmune; Financial Interests, Personal, Advisory Board, Advisor role: Bayer; Financial Interests, Personal, Other, Consultant and Advisory: Erasca; Financial Interests, Personal, Other, Consultant: Philogen, Incyte; Financial Interests, Personal, Other, Consultant and Advisory role: BioNTech, Genmab; Financial Interests, Personal, Advisory Board: Anaveon, Menarini; Financial Interests, Institutional, Funding, Clinical trial and translational research: BMS; Financial Interests, Institutional, Funding, Clinical Trial: Roche Genentech, Pfizer/Array, Sanofi; Non-Financial Interests, Leadership Role, President since 2010: Fondazione Melanoma Onlus Italy; Non-Financial Interests, Leadership Role, President since 2014: Campania Society of ImmunoTherapy of Cancer (SCITO) Italy; Non-Financial Interests, Other, Member of Steering Committee since 2016: Society for Melanoma Research (SMR); Non-Financial Interests, Member of Board of Directors, November 2017 - December 2021: Society for Immunotherapy of Cancer (SITC); Non-Financial Interests, Member: ASCO, SITC, EORTC Melanoma Cooperative Group, AIOM, SMR. C. Gaudy Marqueste: Financial Interests, Personal, Advisory Board: BMS, Pierre Fabre, MSD, Sun Pharma, Regeneron, Iovance; Financial Interests, Personal, Invited Speaker: BMS, Pierre Fabre, UCB; Financial Interests, Institutional, Local PI: BMS, MSD, Pfizer, Amgen, GSK, Roche, Novartis, Pierre Fabre, Replimune, Jansen, Kartos, IFX, Huyabio, Regeneron, IO Biotech, Philogen, Sotio, Kinnate, AstraZeneca, Day One biopharmaceuticals, Sairopa, Catalyst, Sanofi, Immunocore, Daiichi, Dragonfly, Erasca, Microbiotica; Other, Travel expenses: BMS, Pierre Fabre, MSD, Janssen. M. Del Vecchio: Other, Personal, Advisory Board: Bristol Myers Squibb , Merck Sharp & Dohme, Novartis, Immunocore, Regeneron; Other, Personal, Invited Speaker: Bristol Myers Squibb , Merck Sharp & Dohme, Novartis, Immunocore; Other, Personal, Principal Investigator: Bristol Myers Squibb , Merck Sharp & Dohme, Novartis, Immunocore; Other, Personal, Advisory Role: Bristol Myers Squibb , Merck Sharp & Dohme, Novartis, Immunocore, Regeneron. J. Cohen: Financial Interests, Personal, Invited Speaker: AstraZeneca, Roche, Medison Pharma, Merck Serono, Bristol Myers Squibb Israel, MSD Oncology, Ideya biosciences; Financial Interests, Personal, Principal Investigator: Roche, Merck Serono, Beyond Air, MSD Oncology, AbbVie, Ideya biosciences, Ideaya biosciences, Sanofi, Pfizer, Nectin therapeutics, Beyond biologics, Regeneron; Financial Interests, Personal, Other, Travel & accommodations: Roche, Medison Pharma; Financial Interests, Personal, Advisory Board: Beyond Air. A.M. Di Giacomo: Financial Interests, Personal, Advisory Board: Bristol Myers Squibb, MSD, Immunocore, SunPharma. A. Ribas: Financial Interests, Personal, Advisory Role: Amgen, Merck, Appia, Arcus, Compugen, Larkspur, Lutris, Lyell, MapKure, Merus, Sastra, Synthekine, Tango, Apricity, Appia, Arcus, Dispatch, Compugen, Highlight, ImaginAb, Inspirna, Larkspur, Lutris, Lyell, MapKure, Merus, Synthekine, Tango; Financial Interests, Personal, Stocks/Shares: Apricity, Appia, Arcus, Dispatch,

Table: 1621P

	Arm 1 Vibo + pembro n = 90	Arm 2 Pembro alone n = 24	Arm 3 Qmab/pembro n = 31	Arm 4 Qmab/pembro + len n = 87	Arm 5 Fave/pembro n = 49	Arm 6 Fave/pembro + ATRA n = 48
ORR, % (95% CI)	46 (35-6)	38 (19-9)	68 (49-83)	52 (41-63)	33 (20-48)	33 (20-48)
DOR, median (range), mo	NR (2.1+ to 47.2)	NR (2.0+ to 47.0+)	NR (2.1+ to 38.9+)	NR (2.0+ to 42.0+)	NR (1.7+ to 11.0+)	NR (1.4+ to 8.6+)
PFS, median (95% CI), mo	13.1 (7.4-24.0)	10.4 (2.2-43.5)	NR (16.2-NR)	32.5 (7.4-NR)	NR (2.3-NR)	7.6 (4.3-NR)
24-mo PFS, %	39	42	60	51	NE	NE
OS, median (95% CI), mo	NR (27.0-NR)	29.5 (14.5-NR)	NR (NR-NR)	NR (28.5 to NR)	NR (NR-NR)	NR (NR-NR)
24-mo OS, %	66	57	87	66	NE	NE

"+" indicates no PD by time of last disease assessment.
NE, not estimable; NR, not reached.

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1622P Response and survival in in-transit metastatic unresectable melanoma according to treatment strategy

C.D.M. Boreel[1], M. Bloem[2], M. Aarts[3], F. Van den Berkmortel[4], C.U. Blank[5], M. Boers-Sonderen[6], H. Bonenkamp[7], J.W. de Groot[8], J.B.A.G. Haanen[5], G. Hospers[9], D. Piersma[10], B. Rikhof[11], M. Stevense-den Boer[12], A.A.M. van der Veldt[13], G. Vreugdenhil[14], M.W. Wouters[15], E. Kapiteijn[16], F. Van Den Eertwegh[17], K. Suijkerbuijk[18]

[1]Oncology Dept., UMC - University Medical Center Utrecht, Utrecht, Netherlands; [2]Biomedical Data Sciences Dept., LUMC - Leiden University Medical Center, Leiden, Netherlands; [3]Medical Oncology, Maastricht University Medical Center (MUMC), Maastricht, Netherlands; [4]Medical Oncology, Zuyderland Medical Center, Heerlen, Netherlands; [5]Medical Oncology Dept., NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [6]Department of Medical Oncology, Radboud University Medical Center, Nijmegen, Nijmegen, Netherlands; [7]Department of Surgery, Radboud University Medical Center, Nijmegen, Netherlands; [8]Oncology Center Department, Isala Hospital, Zwolle, Netherlands; [9]Medical Oncology, UMCG - University Medical Center Groningen, Groningen, Netherlands; [10]Medical Oncology, Medisch Spectrum Twente (MST) - Radiology, Enschede, Netherlands; [11]Internal Medicine Department, Frisius Medical Centre, Leeuwarden, Netherlands; [12]Department of Internal Medicine, Amphia Ziekenhuis-location Molengracht, Breda, Netherlands; [13]Medical Oncology department, Erasmus MC - University Medical Center, Rotterdam, Netherlands; [14]Internal Medicine Dept., Maxima Medisch Centrum -Veldhoven, Veldhoven, Netherlands; [15]Surgical Oncology Department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [16]Medical Oncology Dept., Leiden University Medical Center (LUMC), Leiden, Netherlands; [17]Medical Oncology, Amsterdam UMC - Vrije University Medical Centre (VUmc), Amsterdam, Netherlands; [18]Medical Oncology Dept, UMC-University Medical Center Utrecht, Utrecht, Netherlands

Background: There is no consensus on the optimal treatment for patients with unresectable melanoma in-transit metastases (ITM). This study describes response and survival outcomes of different treatment strategies for ITM.

Methods: Patients diagnosed with unresectable ITM with or without lymph node metastases (stage III or IV M1a) from January 2013 to June 2025, were included from the nationwide Dutch Melanoma Treatment Registry (DMTR). We analysed objective response rate (ORR), progression-free survival (PFS), and overall survival (OS). A multivariable Cox proportional hazards model was performed to adjust for confounding factors: age, sex, WHO performance status, LDH level, and nodal disease.

Results: A total of 458 patients were included: 212 received first-line anti-PD1 (aPD1), 179 T-VEC, and 67 BRAF(/MEK) inhibition (BMi). Patients treated with T-VEC less often had stage IV-M1a disease and more often had ITMs without nodal involvement. ORR was 55.2% for aPD1, 54.7% for T-VEC, and 67.2% for BMi. Median PFS was 7.8 months (95%CI 5.7-13.8) for aPD1, 5.9 months (95%CI 4.3-7.5) for T-VEC, and 8.5 months (95%CI 6.9-14.3) for BMi. Median OS was 38.7 months (95%CI 32.4-64.7) for aPD1, not reached for T-VEC, and 24.7 months (95%CI 17.0-60.0) for BMi. When separately analyzing patients without nodal disease in all 3 treatment groups, OS for T-VEC remained significantly better than for BMi ($p<0.001$) or aPD1 ($p=0.001$). However, multivariable Cox regression for PFS demonstrated no significant difference in the hazard of progression or death for BMi (HR_{adj}: 1.00, 95%CI: 0.62−1.60, $p=0.992$) or aPD1 (HR_{adj}: 1.00, 95%CI: 0.66−1.51, $p=0.996$) compared to T-VEC. Similarly, multivariable Cox regression for OS did not find a significant difference in hazard of death for BMi (HR_{adj}: 1.84, 95%CI: 0.97-3.50, $p<0.200$) or aPD1 (HR_{adj}: 1.49, 95%CI: 0.81-2.73, $p=0.062$) compared to T-VEC.

Conclusions: For ITM unresectable melanoma, aPD1, T-VEC, and BMi-based treatments can induce clinically meaningful responses. After adjusting for potential confounding factors PFS and OS were comparable across patients receiving aPD1, TVEC or BMi.

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1623P Impact of switching from BRAF/MEK inhibition to immune checkpoint inhibition before secondary resistance in metastatic melanoma: A EUMelaReg real-world study

M. Weichenthal[1], I. Gavrilova[2], J. Mangana[3], U. Leiter-Stoppke[4], E. Ellebaek[5], C.H. Ruhlmann[6], S. Ugurel-Becker[7], N. Asher[8], L. Kandolf Sekulovic[9], A. Gerasimov[10], R. Dummer[11], D.C. Ziogas[12], J.B.A.G. Haanen[13], I.M. Svane[5], E. Espinosa[14], P. Rutkowski[15], P. Mohr[16], H. Gogas[12], L. Bastholt[6], D. Schadendorf[17]

[1]Dermatology Department, Christian-Albrechts-University Kiel, Kiel, Germany; [2]Oncodermatology department, USBAL Oncology EAD Sofia, Sofia, Bulgaria; [3]Dermatologie department, USZ - Universitätsspital Zürich - Klinik für Dermatologie, Zurich, Switzerland; [4]Dermatology, Universitaets-Hautklinik Tuebingen, Tuebingen, Germany; [5]Department of Oncology, National Center for Cancer Immune Therapy (CCIT-DK), Herlev Hospital, Herlev, Denmark; [6]Department of Oncology, OUH - Odense University Hospital, Odense, Denmark; [7]Department of Dermatology, Venereology and Allergology, University Hospital Essen, Essen, Germany; [8]Department of Oncology, Skin Cancer and Melanoma Center at Davidoff Cancer Center, Rabin, Israel; [9]Dermatology and Venerology Department, Military Medical Academy VMA, Belgrade, Serbia; [10]Department of Medical Oncology, UMBAL Saint Ekaterina, Sofia, Bulgaria; [11]Dermatology Department, University Hospital Zurich, Zurich, Switzerland; [12]First Department of Medicine, National and Kapodistrian University of Athens, Athens, Greece; [13]Division of Medical Oncology, Netherlands Cancer Institute, Amsterdam, Netherlands; [14]Medical Oncology Department, Hospital Universitario La Paz, Madrid, Spain; [15]Department of Soft Tissue/Bone Sarcoma and Melanoma, Maria Skłodowska-Curie National Research Institute of Oncology, Warsaw, Poland; [16]Department of Dermatology, Elbe Hospital Buxtehude, Buxtehude, Germany; [17]Department of Dermatology, Venereology and Allergologyy, University Hospital Essen, Essen, Germany

Background: Recent studies have shown that immune checkpoint inhibition (ICI) is a preferable treatment of choice in first-line (1L) in BRAF mutated metastatic melanoma. For a variety of reasons, such as poor performance status, elevated LDH levels and aggressive disease, some patients might still benefit from 1L BRAF/MEK inhibitors (BRAF/MEKi). However, these patients still have limited progression-free survival (PFS) and poor long-term outcomes, and it is unclear whether a switch to ICI should be considered before secondary resistance develops.

Methods: This retrospective study evaluated 1,591 patients from the EUMelaReg registry with $BRAF^{V600}$ mutated metastatic melanoma who achieved tumor control (complete/partial response or stable disease) from 1L BRAF/MEKi and either received ICI in second line (2L) or no 2L therapy. We compared those who switched to 2L ICI without having progressed (switch cohort) to the remaining cohort (control cohort). To adjust for guaranteed time bias and baseline imbalances a 1:2 matching of the populations was performed on several prognostic covariates. The main outcome was overall survival from 1L treatment (OS), calculated by Kaplan-Meier analysis. As secondary outcomes, 2L ICI was evaluated for response rates and PFS, calculated from the start of 2L treatment (2L PFS).

Results: We identified 100 patients (6%) in the switch cohort who were matched to 200 patients in the control cohort. Median OS from start of 1L treatment (42.1 [25.6-73.8] months vs 23 [18.2-28.8] months; $p=0.017$) and median 2L PFS (5.6 [4.1-10.5] months vs 3.4 [2.2-5.9] months; $p=0.01$) were significantly longer in the switch cohort than in the control group (95% CI for both). ORR after 2L treatment was higher in the switch group than in the control group (34% vs 24%), although not statistically significant ($p=0.13$).

Conclusions: Our results indicate that in patients achieving tumor control from BRAF/MEKi, switching to ICI before progression might improve clinical outcomes including OS. This should be considered in the current guideline recommendations in 1L for patients who are not suitable for ICI upfront.

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Gavrilova: Financial Interests, Personal, Advisory Role: Merck Sharp & Dohme, Bristol Myers and Squibb, Pierre Fabre, Medison; Financial Interests, Personal, Other, Honoraria: Roche, Novartis, Merck Sharp & Dohme, Bristol Myers and Squibb, Sanofi, Merck, Pierre Fabre, Medison; Financial Interests, Speaker's Bureau: Roche , Novartis, Merck Sharp & Dohme, Bristol Myers and Squibb, Sanofi, Merck, Pierre Fabre, Medison; Financial Interests, Personal, Other, Travel/accommodations/expenses: Meck Sharp & Dohme, Bristol Myers and Squibb, Pierre Fabre. J. Mangana: Financial Interests, Personal, Advisory Board: Merck Sharp & Dohme; Financial Interests, Personal, Advisory Board, immune-related Adverse Events Management (ESMO): Bristol Myers and Squibb; Financial Interests, Personal, Other, Travel grant: Ultrasun, L'oreal, Merck Sharp & Dohme, Pierre Fabre; Financial Interests, Personal, Invited Speaker: Johnson and Johnson; Financial Interests, Personal, Other, Travel Grant: Sun pharmaceutical; Financial Interests, Institutional, Research Grant, The melanoma registry database of the Department of Dermatology has been partially supported by an unrestricted grant to the University of Zurich from Amgen, Novartis, BMS, MSD and Pierre Fabre: Novartis, BMS, Pierre Fabre, Amgen, MSD; Non-Financial Interests, Leadership Role, President WG Melanoma Group: SAKK; Non-Financial Interests, Leadership Role, president WG Melanoma: SAKK; Other, immune-related AEs podcast /paid personal: Bristol Myers Squibb International. U. Leiter-Stoppke: Financial Interests, Personal, Advisory Board, and speaker's honoraria: Sun Pharma, Regeneron, Sanofi; Financial Interests, Personal, Advisory Board: Pierre Fabre, Novartis, Almirall Hermal; Financial Interests, Institutional, Research Grant: MSD. E. Ellebaek: Financial Interests, Personal, Invited Speaker: Pierre Fabre, BMS, Novartis, MSD; Other, Travel and conference expenses: MSD, Pierre Fabre. C.H. Ruhlmann: Financial Interests, Personal, Invited Speaker: Bristol Myers Squibb, Helsinn Healthcare, MSD; Financial Interests, Personal, Other, Consultant: Astellas Pharma; Financial Interests, Institutional, Coordinating PI, Funding for a clinical trial: Helsinn Healthcare, Novo Nordisk Foundation; Non-Financial Interests, Principal Investigator, PI at Odense University Hospital for the V940-001 study: MSD. S. Ugurel-Becker: Financial Interests, Personal, Advisory Board: Bristol Myers Squibb, Merck Sharp and Dohme, Merck Serono, Novartis; Financial Interests, Personal, Other, Provide travel cost for conventions / scientific meetings: Sun Pharma; Financial Interests, Institutional, Coordinating PI: Bristol Myers Squibb, Merck Serono; Non-Financial Interests, Leadership Role: Dermatologic Cooperative Oncology Group (DeCOG). N. Asher: Financial Interests, Personal, Advisory Role: Novartis, Merck Sharp & Dohme, Bristol Myers and Squibb, Medison, Taro, Sanofi; Financial Interests, Institutional, Research Funding: Bristol Myers and Squibb. L. Kandolf Sekulovic: Financial Interests, Personal, Advisory Role: Merck Sharp & Dohme, Bristol Myers and Squibb, Novartis, Immunocore; Financial Interests, Personal, Speaker's Bureau: Merck Sharp & Dohme, Bristol Myers and Squibb, Roche , Novartis, Janssen, Immunocore, AbbVie; Financial Interests, Other, Honoraria: Merck Sharp & Dohme, Bristol Myers and Squibb, Roche, Novartis, Janssen, Immunocore, AbbVie; Financial Interests, Other, Travel/accommodations/expenses: Merck Sharp & Dohme, Bristol Myers and Squibb, Janssen. R. Dummer: Financial Interests, Personal, Other, Consulting and/or advisory role: Novartis, Merck Sharp & Dohme (MSD), Bristol Myers Squibb (BMS), Roche, Amgen, Takeda, Pierre Fabre, Sun Pharma, Sanofi, Catalym, Second Genome, Regeneron, MaviVAX SA, T3 Pharma, Pfizer, Simcere; Financial Interests, Personal, Other, medical advisor: Oncobit. D.C. Ziogas: Financial Interests, Personal, Invited Speaker: AstraZeneca, MSD, BMS, Ipsen, Pierre Fabre, Gilead, Amgen, Genesis; Financial Interests, Personal, Advisory Board: Roche, Winmedica. J.B.A.G. Haanen: Financial Interests, Institutional, Advisory Board: Bristol Myers Squipp, Achilles Therapeutics, Immunocore, Gadeta, Ipsen, Merck Sharpe & Dohme, Merck Serono, Pfizer, Molecular Partners, Novartis, Roche, Sanofi, Instil Bio, Iovance Biotherapeutics, AstraZeneca; Financial Interests, Institutional, Advisory Board, SAB member: BioNTech, PokeAcel, T-Knife; Financial Interests, Personal, Advisory Board, SAB member: Neogene Therapeutics, Sastra Cell Therapy; Financial Interests, Personal, Advisory Board: Third Rock Venture, Scenic, CureVac, Imcyse; Financial Interests, Personal, Stocks/Shares: Neogene Therapeutics; Financial Interests, Institutional, Research Grant: Bristol Myers Squibb, BioNTech US, Merck Sharpe & Dohme, Amgen, Novartis, Asher Bio, Sastra Cell Therapy; Non-Financial Interests, Member: ASCO, AACR, SITC; Other, Editor-in-Chief IOTECH: ESMO; Other, Editorial Board ESMO Open: ESMO; Other, Editorial Board: Kidney Cancer. I.M. Svane: Financial Interests, Institutional, Other, Conference participation: MSD; Financial Interests, Personal, Invited Speaker: BMS, Janssen; Financial Interests, Personal, Advisory Board: Mendus, Genmab; Financial Interests, Personal, Stocks/Shares, Cofounder and Founder warrents: IO Biotech; Financial Interests, Institutional, Research Grant: Adaptimmune, Enara Bio, Lytix Biopharma, TILT Biotherapeutics, Asgard Therapeutics, IO Biotech; Financial Interests, Institutional, Funding: Evaxion; Financial Interests, Institutional, Other, drug for investigator driven trial: BMS; Non-Financial Interests, Principal Investigator: BMS, Roche, TILT Biotherapeutics, Lytix Biopharma, Novartis, Immunocore, MSD. E. Espinosa: Financial Interests, Personal, Advisory Role: Merck Sharp & Dohme; Financial Interests, Other, Honoraria payments: Merck Sharp & Dohme, Bristol Myers and Squibb, Immunocore, Pierre Fabre, Novartis; Financial Interests, Personal, Other, Travel/accommodations/expenses: Pierre Fabre. P. Rutkowski: Financial Interests, Personal, Invited Speaker, honoraria for lectures: MSD, BMS, Pierre Fabre; Financial Interests, Personal, Advisory Board: MSD, BMS, Pierre Fabre, Merck, Sanofi, Blueprint Medicines, Philogen, Medison Pharma, Genesis Pharma; Financial Interests, Personal, Invited Speaker: Merck, Sanofi, Novartis, AstraZeneca, Genesis Pharma; Financial Interests, Institutional, Research Grant, research grant for ISS: Pfzer; Financial Interests, Institutional, Funding, research grant for institution: BMS; Non-Financial Interests, Member of Board of Directors: Polish Society of Surgical Oncology; Non-Financial Interests, Member of Board of Directors, President: Polish Oncological Society. P. Mohr: Financial Interests, Personal, Advisory Board: Bristol Myers Squibb, Merck Sharp & Dohme, Novartis, GSK, Pierre Fabre, Sanofi, Delcath, Regeneron, Immunocore, Merck Serono; Financial Interests, Personal, Invited Speaker: Bristol Myers Squibb, Merck Sharp & Dohme, Novartis, Pierre Fabre, Sanofi, Sun Pharma, Delcath; Financial Interests, Institutional, Funding: Bristol Myers Squibb, Merck Sharp & Dohme, Novartis; Non-Financial Interests, Principal Investigator: Bristol Myers Squibb, Merck Sharp & Dohme, Regeneron, Sanofi, Novartis, Sun Pharma. H. Gogas: Financial Interests, Personal, Advisory Board: MSD, BMS, Pierre Fabre, Sanofi; Financial Interests, Personal, Invited Speaker: MSD, BMS, Novartis, Pierre Fabre, Sanofi; Financial Interests, Steering Committee Member: Amgen, Replimmune; Financial Interests, Institutional, Local PI: Amgen, MSD, BMS, Replimmune, Iovance, Bayer; Financial Interests, Institutional, Research Grant: BMS, Pfizer, Lilly, Pierre Fabre, AstraZeneca. L. Bastholt: Non-Financial Interests, Advisory Role, Scientific committee under Danish Medicines Agency regarding new treatments of melanoma, skin cancer and thyroid cancer: Danish Medicines Agency. D. Schadendorf: Financial Interests, Personal, Invited Speaker: BMS, MSD, Merck Serono, Regeneron; Financial Interests, Personal, Advisory Board: BMS, Novartis, MSD, Immunocore, Pierre Fabre, Regeneron, Philogen, BioAlta, Daiichi Sankyo, BioNTech, Immatics, CureVac, Formycon, Replimune, SunPharma, Skyline Dx, IOVance, Ipsen, IOBioTech, Boehringer-Ingelheim; Financial Interests, Institutional, Coordinating PI: Novartis, BMS, MSD, Pierre Fabre, Immatics; Financial Interests, Institutional, Research Grant: BMS, MSD; Financial Interests, Personal, Steering Committee Member: MSD; Financial Interests, Institutional, Local PI: Philogen, Immunocore, Regeneron, BioNTech Rna Pharmacuticals GMBH, Replimune, IO Biotech; Non-Financial Interests, Member of Board of Directors: EORTC-MG; Non-Financial Interests, Leadership Role, Founding member and SC chair: European Melanoma Registry (EuMelaReg). All other authors have declared no conflicts of interest.

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1624P Comparison of patients' clinical outcomes treated with flat dose or weight-adjusted dose of pembrolizumab in advances melanoma, a multicenter observational study

I. Campo le Brun[1], S. Dalle[2], L. Mortier[3], O. Dereure[4], S. Dalac Rat[5], C. Dutriaux[6], M-T. Leccia[7], D. Legoupil[8], H. Montaudie[9], J. De Quatrebarbes[10], C. Gaudy Marqueste[11], E. Maubec[12], P. Saiag[13], C. Pages[14], F. Brunet Possenti[15], F. Granel Brocard[16], R. Porcher[17], W. Lefevre[18], C. Lebbe[19], E. Kempf[20]

[1]Medical Oncology, Henri Mondor University Hospital Center, Créteil, France; [2]Dermatology Dept., Lyon Sud Hospital Center, Pierre Benite, France; [3]Dermatology Dept., CHRU de Lille - Hôpital Roger Salengro, Lille, France; [4]Dermatology Dept., CHU de Montpellier - Hopital Saint-Eloi, Montpellier, France; [5]Dermatology Dept., CHU Dijon, Dijon, France; [6]Dermatology Dept., CHU Bordeaux - Hopital St. André, Bordeaux, France; [7]Dermatology Dept., CHU Grenoble Alpes - Site Nord (La Tronche), La Tronche, France; [8]Dermatology Dept., CHRU Brest - Hôpital Morvan, Brest, France; [9]Dermatology Dept., Hopital del l'Archet, Nice, France; [10]Dermatology Dept., Le Centre Hospitalier Annecy Genevois, Metz-Tessy, France; [11]Oncodermatology, AP-HM - CHU La Timone Enfants, Marseille, France; [12]Dermatology Department, Hopital Avicenne AP-HP, Bobigny, France; [13]Dermatology and Oncology Department, Hopital Ambroise Pare AP-HP, Boulogne-Billancourt, France; [14]Dermatology Dept., IUCT - Institut Universitaire du Cancer de Toulouse - Oncopole, Toulouse, France; [15]Onco-dermatology, Hopital Bichat - Claude-Bernard AP-HP, Paris, France; [16]Onco Dermatology, CHRU Nancy, Nancy, France; [17]Center for Research in Epidemiology and StatisticS, Université de Paris - Faculté de Médecine, Paris, France; [18]Dermatology Dept., Assistance Publique - Hopitaux de Paris, Paris, France; [19]Dermatology Dept., Hopital Saint Louis APHP, Paris, France; [20]Medical Oncology, Centre Hospitalier Universitaire Henri-Mondor AP-HP, Creteil, France

Background: Pembrolizumab obtained approval in advanced melanoma (AM) with weight-adjusted dose (WAD) administration (2mg/kg/3weeks). In 2018, the dosage regimen was changed to flat dose (FD) administration (200mg/3 weeks or 400mg/6 weeks) based on a modeling study, without clinical data. This multicenter real-world study aimed to compare clinical outcomes of patients with AM treated by FD or WAD.

Methods: AM patients treated at 26 French centers have been prospectively included in the MelBase database since 2013. First-line patients treated with pembrolizumab monotherapy were included in the WAD or FD groups of this study. The primary end point was the incidence of grade $\geq$3 immune-related adverse events (irAEs). Secondary end points were incidence of any grade irAEs, and overall survival (OS) and progression-free survival (PFS). Inverse probability of treatment weighting was used to balance groups on their baseline characteristics.

Results: Between 2015 and 2023, 606 patients were included: 418 in the WAD and 188 in the FD groups. The median follow-up was 26.3 months in the WAD and 13.6 months in the FD groups. In the FD group, 155 (87.3%) patients weighed $<$95 kg and 9 (5.1%) $>$105 kg. Grade $\geq$3 immune-related adverse events occurred in 58 (13.8%) patients in the WAD group and 21 (11.1%) patients in the FD group ($p = 0.4$). After weighting, the median progression free survival was 4 and 6.7 months (HR, 0.88; 95% CI, 0.71−1.09), and the median overall survival was 32.5 months and not reached (HR, 0.75; 95% CI, 0.56-1.01) in the WAD and FD groups, respectively.

Conclusions: This is the first real-world study comparing WAD and FD pembrolizumab in AM patients. There were no statistically significant differences between AM patients treated with WAD or FD pembrolizumab in terms of safety and efficacy, though a significant proportion of patients in the FD group received a higher dose.

Clinical trial identification: NCT02828202, 2013-02.

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1625P Incidence of immune related adverse events in malignant melanoma patients treated with concurrent radiotherapy or stereotactic radiosurgery

F. van Someren[1], T.J. Carter[2], H.M. Shaw[3]

[1]Internal Medicine, Mount Vernon Cancer Centre, Northwood, United Kingdom; [2]Department of Medical Oncology, Mount Vernon Cancer Centre, Northwood, United Kingdom; [3]Department of Medical Oncology, UCLH - University College London Hospitals NHS Foundation Trust, London, United Kingdom

Background: Immune Checkpoint Inhibitors (ICI) are first-line therapy in metastatic melanoma (MM) but are associated with treatment-limiting immune related adverse events (IRAE). Patients may also receive concurrent radiation therapy (RT) to treat symptomatic extracranial metastases or stereotactic radiosurgery (SRS) to treat intracranial metastases. It is poorly understood whether exposure to radiation therapy increases the risk of IRAE in melanoma patients treated with ICI.

Methods: This retrospective cohort study identified 321 patients with MM treated at a single tertiary oncology centre between 2019 -2023. Data was collated on ICI treatments, incidence of any-grade IRAE, and concurrent RT or SRS.

Results: Of 321 patients, 149 received pembrolizumab (67 with concurrent RT/SRS) and 148 received ipilimumab/nivolumab (62 with concurrent RT/SRS). 72.3 % of all patients had any-grade IRAE with rates of IRAE higher in patients receiving both RT (84.51% p = 0.005, OR = 3.2 (1.63-6.77)) and SRS (86.36% p = 0.001, OR = 3.71 (1.59-10.14)). Significantly higher IRAE rates were observed for patients receiving either pembrolizumab + SRS/RT (81.97%, p = <0.001, OR = 4.88 (2.30-11.04)) or ipilimumab/nivolumab + SRS/RT (92.45%, p = <0.001, OR = 13.15 (4.83-46.38)), compared to either ICI treatment alone (48.24% and 79.31% respectively). Mean IRAE grade was similar in patients receiving either ICI + RT/SRS, or ICI alone (2.21 *vs.* 2.08, p = 0.35). Timing of radiation treatment also appeared to be relevant, with patients receiving RT/SRS prior to ICI having higher rates of IRAE than those who received RT/SRS after first ICI exposure.

Conclusions: To our knowledge, increased rates of IRAE have not previously been described in patients receiving RT/SRS alongside ICI for MM. Whilst this study is limited in terms of patient numbers and confounding factors associated with retrospective data collection, these preliminary findings warrant further investigation and consideration in clinical practice.

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1626P DNV3 plus toripalimab and chemotherapy in advanced melanoma: An update on an open-label investigator-initiated trial

J. Lin[1], L. Chen[2], L. Chen[1], D. Wang[3], P. Chen[1], Y. Lu[1], H. Zhang[3], W. Yan[1], Z. Xiao[4], Y. Chen[1]

[1]Department of Medical Oncology, Clinical Oncology School of Fujian Medical University, Fujian Cancer Hospital, Fuzhou, China; [2]Clinical Oncology Department, Fujian Cancer Hospital and Fujian Medical University Cancer Hospital, Fuzhou, China; [3]Department of Medical Oncology, Fujian Cancer Hospital and Fujian Medical University Cancer Hospital, Fuzhou, China, [3]Phase I Clinical Trial Ward, Clinical Oncology School of Fujian Medical University & Fujian Cancer Hospital, Fuzhou, China; [4]Clinical Research Center, Zhejiang Shimai Pharmaceutical Co., Ltd., Hangzhou, China

Background: DNV3, a recombinant anti-LAG-3 fully human monoclonal antibody, is a first-in-class biological anti-tumor drug. This investigator-initiated trial evaluated the novel combination of DNV3 (anti-LAG-3), toripalimab (anti-PD-1), and chemotherapy in advanced melanoma, building upon the established efficacy of dual checkpoint inhibition.

Methods: Eligible patients (ECOG PS 0-1) received nab-paclitaxel (initial dose 260 mg/m^2, reduced to 200 mg/m^2), cisplatin (25 mg/m^2), DNV3 (3 mg/kg), and toripalimab (240 mg) every 3 weeks for 6 cycles, followed by maintenance immunotherapy. The primary endpoint was objective response rate (ORR); secondary endpoints included safety, progression-free survival (PFS), duration of response (DOR), disease control rate (DCR), and overall survival (OS).

Results: 27 patients were enrolled (13 mucosal, 6 acral, 5 cutaneous, 3 unknown primary melanomas); 77.8% had prior anti-PD-(L)1 therapy. Initial nab-paclitaxel 260 mg/m^2 caused grade$\geq$3 TRAEs in 55.6% (1 fatal), leading to dose reduction (200 mg/m^2) which reduced grade$\geq$3 TRAEs to 22.2%. irAEs occurred in 55.6% (22.2% grade$\geq$3). The regimen showed 44.4% ORR (85.2% DCR) in 27 patients. Median PFS was 7.29 months (6-/12-month: 60.0%/15.1%), with 87.8% 12-month OS. Mucosal melanoma showed 40.3% 6-month PFS (26.9% at 9 months) with sustained 100% OS. Among anti-PD-(L)1-pretreated patients (n=21), the ORR was 44.4%, DCR 85.2% . Subtype ORRs: mucosal 28.6%, cutaneous 100%, acral 16.7%; DCRs: 71.4%, 100%, 100%. The median DOR was 8.67 months overall, shorter with liver metastases (4.57 months). The median PFS was 7.29 months overall; mucosal: 8.44 months, cutaneous: NE, acral: 6.14 months. The PFS rates were 60.0% (6-month), 30.3% (9-month), and 15.1% (12-month). The OS rates were 95.8% (6-month) and 87.8% (9-/12-month); mucosal 12-month OS was 100%. Treatment-naïve mucosal patients (n=6) achieved 50.0% ORR with 5.78-month median PFS.

Conclusions: This study demonstrates preliminary tolerability and efficacy of DNV3 combined with toripalimab and chemotherapy in advanced melanoma, particularly in mucosal melanoma and liver metastases.

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1627P Toripalimab plus bevacizumab and chemoradiotherapy as treatment in patients with advanced melanoma: The IRAC phase II nonrandomized clinical trial

Z. Zou, Y. Ren, L. Zhao, Y. Yang

The Comprehensive Cancer Center, The Affiliated Drum Tower Hospital of Nanjing University Medical School, Nanjing, China

Background: Malignant melanoma, characterized by its aggressiveness and early metastasis, poses significant challenges in terms of treatment efficacy. Combining nab-paclitaxel, bevacizumab (bev), and immunotherapy has shown promising efficacy across various cancer types. The IRAC phase 2 nonrandomized clinical trial investigated the efficacy and safety of this triple regimens combined with radiotherapy in Chinese patients (pts) with advanced melanoma.

Methods: Inoperable stage III-IV melanoma pts with ECOG PS 0-1 were eligible to receive nab-paclitaxel (200mg/m^2) plus bev (7.5mg/kg) and toripalimab (tor) (240mg) (Q3W). Maintenance therapy with bev and tor was given after 6 cycles until disease progression or intolerable toxicity. Individualized hypofractionated radiotherapy was administered to multiple lesions over multiple cycles. In principle, the prescribed dose was 5-8 Gy per fraction for 3 fractions. The primary endpoint was ORR and DCR according to the RECIST V1.1.

Results: Between July 2022 and May 2025, a total of 37 pts were enrolled (Table). As of May 2025, 36 pts underwent at least once efficacy evaluation and the median follow-up was 10.9 months. The ORR was 63.9% (95% CI 46.2-79.2) and DCR was 97.2% (95% CI 85.5-99.9). The median PFS was 13.3 months (95% CI 10.4-16.2). Subgroup analysis showed that the ORR was 65.0% (95% CI 40.8-84.6) in the acral subtype and 66.7% (95% CI 29.9-92.5) in the mucosal subtype. The mPFS was 13.3 months (95% CI 10.4-16.1) and 9.5 months (95% CI 1.8-17.2), respectively. Grade$\geq$3 TRAEs occurred in 13 (35.1%) pts, included leukopenia (5 [13.5%]), neutropenia (10 [27.0%]) and hypertension (2 [5.4%]). No TRAE led to death.

Table: 1627P Baseline characteristics

Characteristics		Patients (n=37)
Age (years), median (range)		61 (43-78)
Sex (male/female)		18/19
Current therapy line (1/$\geq$2)		22/15
Stage (III/IV)		8/29
Primary site (acral/mucosal/cutaneous/unknown)		21/9/6/1
Metastatic site	Liver/no	8/29
	Lung/no	14/23
	Lymph node/no	29/8
	Bone/no	8/29
Gene mutation (BRAF/C-KIT/NRAS/unknown)		4/5/7/6

Conclusions: Toripalimab plus bevacizumab and chemoradiotherapy showed encouraging efficacy with advanced melanoma, especially in acral and mucosal subtype, more common in Chinese pts, with an acceptable safety profile. Follow-up for survival outcomes is ongoing.

Clinical trial identification: ChiCTR2200062880.

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1628P Anti-PD1 plus lenvatinib as a treatment option in pretreated advanced melanoma patients: A retrospective multicenter study

L.J. Albrecht[1], J. Garnier[1], L. Brauch[2], F. Brunnert[3], R. Rauschenberg[4], A. Forschner[5], X. Garza Vazquez[6], J. Mangana[7], M. Erdmann[8], Y. Angela[9], C. Franklin[10], J.K. Tietze[11], S. Ugurel-Becker[12], F. Meier[4], P. Mohr[13], J.C. Hassel[2], D. Schadendorf[1], E. Livingstone[1], L. Zimmer[1]

[1]Department of Dermatology, University Hospital Essen, Essen, Germany; [2]Department of Dermatology and National Center for Tumor Diseases (NCT), University Hospital Heidelberg, Heidelberg, Germany; [3]Department of Dermatology, Elbe Hospital Buxtehude, Buxtehude, Germany; [4]Department of Dermatology and NCT/UCC, University Hospital Carl Gustav Carus, Technische Universität Dresden, Dresden, Germany; [5]Dermato-Oncology, Universitaets-Hautklinik Tuebingen, Tuebingen, Germany; [6]Department of Dermatology and Allergology, LMU University Hospital Munich, Ludwig-Maximilian University Munich, Munich, Germany; [7]Dermatology Department, USZ - Universitätsspital Zürich - Klinik für Dermatologie, Zurich, Switzerland; [8]Department of Dermatology, Friedrich-Alexander University Erlangen-Nürnberg, Erlangen, Germany; [9]Department of Dermatology, Johannes Weseling Klinikum, Minden, Germany; [10]Department of Dermatology - Skin Cancer Center at the Center of Integrated Oncol, Universitätsklinikum Köln (AöR), Cologne, Germany; [11]Clinic and Policlinic for Dermatology and Venereology, University Medical Center Rostock, Rostock, Germany; [12]University Clinic for Dermatology, Allergology und Venerology, Klinikum Bielefeld, Bielefeld, Germany; [13]Dermato-Oncology department, Dermatologic Center Buxtehude, Buxtehude, Germany

Background: Treatment options for melanoma patients (pts) after progression on BRAF/MEK and immune checkpoint inhibitors (ICI) are still limited, thus the demand for new therapeutic options remains. A phase 2 study (LEAP004) evaluating the safety and efficacy of pembrolizumab plus lenvatinib (PEM/LENVA) in ICI refractory melanoma pts showed an ORR of 21%.[1] Although no significant benefit was seen in the 1[st] line setting[2], it has become a valuable option when available treatment options have been exhausted. To date, there is only limited *real-world* data on the efficacy of PEM/LENVA in melanoma pts after failure of standard therapies.

Methods: This retrospective, multicenter study included pts with advanced melanoma who were treated with anti-PD-1 plus lenvatinib (PD1/LENVA) after failure of standard therapies in 11 major skin cancer centers between Oct 2020 - Mar 2025. Demographics, treatment characteristics, and outcome were evaluated.

Results: 120 pts were identified; med age was 59 yrs (range, 26-85), 84 pts (70%) were male, 46 pts (38%) had ECOG PS$\geq$1, 83 pts (69%) had $\geq$3 metastatic sites, 50 pts (42%) were M1d (14% symptomatic), and 69 (58%) had elevated LDH levels prior PD1/LENVA. Med follow-up was 13.4 months (IQR, 6.6-22.9). Med number of prior treatment lines was 2 (range, 1-10); 56 pts (47%) had $\geq$3 prior treatment lines. 74 pts (62%) had primary resistance, and 46 pts (38%) had secondary resistance to previous ICI therapy. Most pts received pembrolizumab 200 mg (n=115; 96%) plus lenvatinib 20mg (n=109; 91%). PD1/LENVA resulted in an ORR of 23% (n=27), a DCR of 46% (n=55) and a med DOR of 5 months (range, 2.8-7.3). Med PFS was 3.5 months (95% CI, 2.9-4.2), with a 12-months PFS of 28% (95% CI, 21-37). Med OS was 9.7 months (95% CI, 7.2-12.3 months), with a 12-months OS of 41% (95% CI, 32-50). LENVA was discontinued due to PD (n=80, 67%), death (n=6, 5%), treatment-related adverse events (TRAE) (n=11, 9%), and patient decision (n=1, 1%). Grade $\geq$3 TRAE occurred in 19 pts (16%) (n=9; 8% related to PD1, n=16 only; 13% related to LENVA only).

Conclusions: Treatment with PD1/LENVA showed promising efficacy in heavily pretreated pts with advanced melanoma, representing a valuable treatment option after failure of standard therapy.

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Mangana: Financial Interests, Personal, Advisory Board: Merck Sharp & Dohme; Financial Interests, Personal, Advisory Board, immune-related Adverse Events Management (ESMO): Bristol Myers and Squibb; Financial Interests, Personal, Other, Travel grant: Ultrasun, L'oreal, Merck Sharp & Dohme, Pierre Fabre; Financial Interests, Personal, Invited Speaker: Johnson and Johnson; Financial Interests, Personal, Other, Travel Grant: Sun pharmaceutical; Financial Interests, Institutional, Research Grant, The melanoma registry database of the Department of Dermatology has been partially supported by an unrestricted grant to the University of Zurich from Amgen, Novartis, BMS, MSD and Pierre Fabre: Novartis, BMS, Pierre Fabre, Amgen, MSD; Non-Financial Interests, Leadership Role, President WG Melanoma Group: SAKK; Non-Financial Interests, Leadership Role, president WG Melanoma: SAKK; Other, immune-related AEs podcast /paid personal: Bristol Myers Squibb International. M. Erdmann: Financial Interests, Personal, Advisory Board: Immunocore; Financial Interests, Personal, Invited Speaker: Novartis, Pierre Fabre; Financial Interests, Personal, Other, travel expenses: Sanofi. Y. Angela: Financial Interests, Personal, Advisory Board: Roche, BMS, MSD, Novartis, Amgen, Merch Serono, Almirall Hermal, SUN, Sanofi, Pierre Fabre; Financial Interests, Personal, Other: Novartis, Pierre Fabre. C. Franklin: Financial Interests, Personal, Advisory Board: Novartis, BMS, MSD, Immucore; Financial Interests, Personal, Other: BMS, Pierre Fabre. S. Ugurel-Becker: Financial Interests, Personal, Advisory Board: Bristol Myers Squibb, Merck Sharp and Dohme, Merck Serono, Novartis; Financial Interests, Personal, Other, Provide travel cost for conventions / scientific meetings: Sun Pharma; Financial Interests, Institutional, Coordinating PI: Bristol Myers Squibb, Merck Serono; Non-Financial Interests, Leadership Role: Dermatologic Cooperative Oncology Group (DeCOG). F. Meier: Financial Interests, Personal, Advisory Board: Bristol Myers Squibb, Novartis, MSD, Sanofi, Philogen, Immunocore; Financial Interests, Personal, Invited Speaker: Sanofi, Immunocore; Financial Interests, Personal, Advisory Board, Advisory Board: Regeneron. P. Mohr: Financial Interests, Institutional, Research Grant: MSD, Novartis, BMS; Financial Interests, Personal, Expert Testimony: MSD, Novartis, Pierre Fabre, Sanofi, Roche Pharma, Beiersdorf, Amgen, Allmiral Hermal, Sunpharma, BMS, Regeneron, Delcath, Immunocore; Financial Interests, Personal, Other, Travel support: Pierre Fabre, BMS, MSD, Regeneron, Delcath, Novartis; Financial Interests, Personal, Advisory Board: Novartis, BMS, Pierre Fabre, Sanofi Genzyme, Beiersdorf, MSD, Biotech, Regeneron, Sun Pharma; Non-Financial Interests, Personal, Leadership Role, unpaid: MID melanoma info Deutschland, HKND Hautkrebsnetzwerk Deutschland. J.C. Hassel: Financial Interests, Personal, Invited Speaker: BMS, Novartis, Sanofi, MSD, Sunpharma, Pierre Fabre, Immunocore, Delcath; Financial Interests, Institutional, Advisory Board: MSD, Pierre Fabre, Novartis, BMS, Immunocore, Philogen, Sanofi; Financial Interests, Personal, Advisory Board: Sunpharma, Onkowissen; Financial Interests, Institutional, Local PI: Philogen, Genentech, 4SC, BioNTech, Iovance, Pierre Fabre, Regeneron, Sanofi, Replimune, Immatics, MSD, Pfizer; Financial Interests, Institutional, Coordinating PI: BMS, Immunocore, Novartis, Genmab, Seagan; Financial Interests, Personal, Steering Committee Member: Immunocore, IO Biotech; Financial Interests, Institutional, Research Grant: Sanofi; Non-Financial Interests, Leadership Role: DeCOG; Non-Financial Interests, Member: ASCO. D. Schadendorf: Financial Interests, Personal, Invited Speaker: BMS, MSD, Merck Serono, Regeneron; Financial Interests, Personal, Advisory Board: BMS, Novartis, MSD, Immunocore, Pierre Fabre, Regeneron, Philogen, BioAlta, Daiichi Sanyko, BioNTech, Immatics, CureVac, Formycon, Replimune, SunPharma, Skyline Dx, IOVance, Ipsen, IOBioTech, Boehringer-Ingelheim; Financial Interests, Institutional, Coordinating PI: Novartis, BMS, MSD, Pierre Fabre, Immatics; Financial Interests, Institutional, Research Grant: BMS, MSD; Financial Interests, Personal, Steering Committee Member: MSD; Financial Interests, Institutional, Local PI: Philogen, Immunocore, Regeneron, Biontech Rna Pharmacuticals GMBH, Replimune, IO Biotech; Non-Financial Interests, Member of Board of Directors: EORTC-MG; Non-Financial Interests, Leadership Role, Founding member and SC chair: European Melanoma Registry (EuMelaReg). E. Livingstone: Financial Interests, Personal, Advisory Board: Novartis, BMS, Sanofi, Sun Pharma, Recordati, MSD; Financial Interests, Personal, Invited Speaker: Novartis, BMS; Financial Interests, Personal, Other, Innvator's Circle: Pierre Fabre; Non-Financial Interests, Principal Investigator: BMS, Novartis, Regeneron, MSD, Philogen, 4SC, Biontech, Immatics, Immunocore; Non-Financial Interests, Member: DeCOG, DKG; Other, Travel expenses: Pierre Fabre, Sun Pharma. L. Zimmer: Financial Interests, Personal, Advisory Board: Bristol Myers Squibb, Novartis, MSD, Sunpharma, Pierre Fabre, Sanofi; Financial Interests, Personal, Other, Travel Support: Bristol Myers Squibb, MSD, Sunpharma, Pierre Fabre, Sanofi, Novartis; Financial Interests, Personal, Invited Speaker: Novartis, MSD, Pierre Fabre; Financial Interests, Personal and Institutional, Research Grant: Novartis; Non-Financial Interests, Member: German Cancer Society, Dermatologic Cooperative Oncology Group. 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1629P Real-world effectiveness and safety of pembrolizumab biosimilar BCD-201 in metastatic or unresectable melanoma

L.Y. Martínez Espinosa[1], E. Ibañez Carrillo[2], I. Caballero Aguirrechu[3], L.S. Pomar Durruthy[4], R.M. EspinCoto[5], R. Rodriguez Barrios[6], A. Potente Hernández[7], M. Nazario Doltz[8], J.C. Avila Perez[9], D. Mariño Puente[10], M. Betancourt Cabeza[11], R.D.J. Ropero[12], O.N. Rodríguez Carrera[13], E.A. Gracia Medina[1]

[1]Medical Oncology, INOR - Institute of Oncology and Radiobiology, Havana, Cuba; [2]Medical Oncology, Hospital Universitario Oncológico Regional Celestino Hernández Robau, Santa Clara, Cuba; [3]Medical Oncology, HHA - Hospital Clinico Quirurgico Hermanos Ameijeiras, Havana, Cuba; [4]Medical Oncology, INOR - Insitutte of Oncology and Radiobiology, Havana, Cuba; [5]Medical Oncology, Hospital Oncológico-Universitario Territorial "Dr. Celestino Hdez Robau", Santa Clara, Cuba; [6]Oncology Department, HHA - Hospital Clinico Quirurgico Hermanos Ameijeiras, Havana, Cuba; [7]Radiology Department, INOR - Instituto de Oncologia y Radiobiologia, Havana, Cuba; [8]Pathology Department, INOR - Instituto de Oncologia y Radiobiologia, Havana, Cuba; [9]Clinical Trials, INOR - Instituto Nacional de Oncologia y Radiobiologia, Havana, Cuba; [10]Medical Oncology, HHA - Hospital Clinico Quirurgico Hermanos Ameijeiras, Havana, Cuba; [11]Clinical Reseach, INOR - Instituto Nacional de Oncologia y Radiobiologia, Havana, Cuba; [12]Clinical Research Unit, INOR - Instituto of Oncology and Radiobiology, Havana, Cuba; [13]Clinical Research, INOR - Instituto de Oncologia y Radiobiologia, Havana, Cuba

Background: Pembrolizumab is a standard therapy for metastatic/unresectable melanoma. BCD-201, a pembrolizumab biosimilar, has shown comparable pharmacokinetics, efficacy, and safety in prior studies. This prospective, multicenter observational study assessed BCD-201 in routine clinical practice across three Cuban centers.

Methods: Between January 1 and July 7, 2024, 47 patients with metastatic or unresectable melanoma were enrolled. Eligible patients had ECOG $\leq$2 and no systemic cancer treatment within 6 weeks. Immunosuppressive therapy was excluded for 14 days prior to enrollment. BCD-201 was administered at 200 mg IV over 30 minutes Q3W until progression or unacceptable toxicity. Tumor response was evaluated every 12 weeks per RECIST 1.1. Adverse events (AEs) were graded using CTCAE v4.03.

Results: Median age was 58 years (95% CI: 47–69); 59.6% were male and had metastatic disease. BRAF status was tested in 72.3% and mutations found in 25.5% of pts. Objective response rate (ORR) was 34.0% (CR: 10.6%; PR: 23.4%), and stable disease was achieved in 42.5%. Two patients with PR underwent surgery and acheived CR. At 12.2 mo median follow-up, 12-mo PFS and OS were 65.4% and 73.5%, respectively. Relapses mainly involved brain (8.5%) and lung (6.4%). Most AEs were grade 1–2. Common treatment-related AEs included pruritus (37.8%), fatigue

(35.1%), headache (27.0%), pain (27.0%), constipation (27.0%), and rash (24.3%), consistent with known pembrolizumab profiles.

Conclusions: BCD-201 demonstrated promising real-world efficacy and tolerability, supporting its use as an alternative to reference pembrolizumab in advanced melanoma.

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1630P HBI-8000 and nivolumab combination in advanced melanoma patients with brain metastases: Analysis of HBI-8000-303 open-label cohort

N.I. Khushalani[1], R.P. Pereira[2], T.D. Nakakogue[3], J. Segalla[4], V.T. Liutti[5], F.J.S. Melo Cruz[6], G.L. Mcadam[7], R. Marais-Nieman[8], M. Mekebeb[9], B.L. Rapoport[10], A.M. Arance[11], M. Gillings[12], P. Stiegler[13], H. Onogi[14], G.T. Lee[14], P.A. Ascierto[15]

[1]Cutaneous Oncology, H. Lee Moffitt Cancer Center, Tampa, United States of America; [2]Clinical Oncology Department, Hospital de Clinicas de Porto Alegre, Porto Alegre, Brazil; [3]Medical Oncology, Hospital Erasto Gaertner, Curitiba, Brazil; [4]Oncology, Fundacao Amaral Carvalho, Sao Paulo, Brazil; [5]Medical Oncology Department, HCL - Hospital do Cancer de Londrina, Londrina, Brazil; [6]Medical Oncology, Instituto Brasileiro de Controle do Câncer, Sao Paulo, Brazil; [7]Oncology, Rondebosch Medical Centre, Rondebosch Cape Town, South Africa; [8]Radiation Oncology Dept., Curo Oncology, Pretoria, South Africa; [9]Oncology, Excellentis Clinical Trial Consultants, George, Western Cape, South Africa; [10]Department of Medical Oncology, The Medical Oncology Centre of Rosebank, Johannesburg, South Africa; [11]Medical Oncology Department, Hospital Clinic y Provincial de Barcelona, Barcelona, Spain; [12]HUYABIO International, San Diego, United States of America; [13]Clinical Operations Oncology, HUYABIO International, San Diego, United States of America; [14]Research and Development, HUYABIO International, San Diego, New York, United States of America; [15]Melanoma, Cancer Immunotherapy & Developmental Therapeutics, Istituto Nazionale Tumori IRCCS - Fondazione G. Pascale, Naples, Italy

Background: HBI-8000, an oral selective histone deacetylase inhibitor in combination with nivolumab demonstrated response rate of 66% in a phase 1b/2 study in anti-PD (L)1-naive advanced melanoma (Khushalani, SITC 2024). A global randomized phase 3 trial of this combination compared to nivolumab in patients (pts) with advanced melanoma not previously treated with immune checkpoint inhibitors for metastatic disease has completed accrual (NCT04674683). Within this, an independent, embedded, non-randomized open-label cohort of HBI-8000 plus nivolumab in melanoma patients with asymptomatic brain metastases was enrolled. We report descriptive results from the open label cohort.

Methods: Key eligibility included advanced melanoma (non-uveal) with asymptomatic brain metastases not requiring high-dose steroids, maximum intra-cranial (IC) metastatic lesion size 3cm, ECOG PS 0-1, and adequate major organ function. 45 pts were enrolled from Apr22 to Feb23 and received HBI-8000 30 mg orally twice weekly + nivolumab 480 mg IV on Day 1 of each 28-day cycle until progression, intolerable toxicity or completion of 24 months. Overall tumor response was assessed by Blinded Independent Central Review (BICR).

Results: Median age was 63 years (range,30-85); 58% male; 64% ECOG PS 0; 56% with ↑ LDH. All pts had extra-cranial (EC) metastases. The median exposure to HBI-8000 and nivolumab was 6 and 7 cycles (range 1 — 26 each) respectively. Of the 45 pts treated in the safety population, 38 had IC lesions confirmed by BICR and 37 were included in the efficacy analysis; 31 had post-treatment imaging studies available. The overall (IC + EC) response rate was 37.8%, 95% CI [22.5, 55.2], including 13 partial and 1 complete response. With median follow-up of 16.4 months (range 1—29), the median PFS was 8.7 months (95% CI [2.3, 18.4]). Any grade treatment-related adverse events (AEs) occurred in 43/45 (96%) pts. Most common AEs were diarrhea (53%), nausea (47%), and thrombocytopenia (42%; 7% Grade 3).

Conclusions: The combination of HBI-8000 and nivolumab has promising efficacy and good tolerability in advanced melanoma patients with asymptomatic brain metastases. Further investigation in this setting is warranted.

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Nakakogue: Financial Interests, Personal, Invited Speaker: Adium, Astellas Pharma, AstraZeneca, Bayer, Janssen, Merck; Financial Interests, Institutional, Local PI: Daiichi Sankya, Genentech Roche, Gilead Sciences, HUYABIO International, Regeneron Pharmaceuticals, Seagen. V.T. Liutti: Financial Interests, Personal, Invited Speaker: AstraZeneca, BMS, Janssen, MSD, Novartis, Pfizer, Roche, Takeda; Financial Interests, Personal, Expert Testimony: Daiichi Sankyo; Financial Interests, Personal, Advisory Board: Daiichi Sankyo, Novartis, Roche; Financial Interests, Institutional, Local PI: AstraZeneca, BeiGene, Daiichi Sankyo, GSK, Incyte, Janssen, Lilly, MSD, Novartis, Pfizer, Roche, Seagen. F.J.S. Melo Cruz: Financial Interests, Personal, Invited Speaker: Astellas, Bayer, MSD; Financial Interests, Personal, Advisory Board: Bayer, Jansen; Financial Interests, Institutional, Local PI: AstraZeneca, Bayer, HUYABIO International, MSD, Novartis; Financial Interests, Personal, Local PI: Roche. B.L. Rapoport: Financial Interests, Personal, Speaker, Consultant, Advisor: MSD, Roche, AstraZeneca; Financial Interests, Institutional, Research Funding: HUYABIO International, Eikon, Regeneron, Amgen, MSD, Celltrion. A.M. Arance: Financial Interests, Personal, Advisory Board, Consulting or Advisory Role, Speakers' Bureau: Pierre Fabre; Financial Interests, Personal, Advisory Board, Consulting or Advisory Role, Speakers' Bureau: BMS, MSD; Financial Interests, Personal, Advisory Board, Consulting or Advisory Role: BioNTech, Almirall, Genmab; Financial Interests, Institutional, Local PI, Research Funding: Novartis, BioNTech, BMS, MSD, Pierre Fabre, Replimune, IO Biotech, HUYABIO International, Iovance Biotherapeutics, Immunocore, Sairopa, Mural Oncology; Other, Travel, Accomodations, Expenses: MSD, Pierre Fabre. M. Gillings: Financial Interests, Personal, Leadership Role, Executive Chairman and CEO: HUYABIO ; Financial Interests, Personal, Member of Board of Directors: Sente. P. Stiegler: Financial Interests, Personal, Leadership Role, VP Clinical Operations Oncology: HUYABIO International. H. Onogi: Financial Interests, Personal, Leadership Role, Executive Vice President: HUYABIO International. G.T. Lee: Financial Interests, Personal, Leadership Role, Chief Medical Officer: HUYABIO International. P.A. Ascierto: Financial Interests, Personal, Other, Consultant and Advisory Role: BMS, Roche Genentech, Novartis, Merck Serono, Sun Pharma, Sanofi, Sandoz, Immunocore, Boehringer Ingelheim, Regeneron; Financial Interests, Personal, Other, Consultant, Advisory Role and Travel support: MSD, Pierre Fabre; Financial Interests, Personal, Other, Consultant and Advisory Role, Travel support: Pfizer/Array; Financial Interests, Personal, Other, Consultant Role: Italfarmaco; Financial Interests, Personal, Other, Consultant Role: Pfizer, Nouscom, Lunaphore; Financial Interests, Personal, Other, Consultant role: Medicenna; Financial Interests, Personal, Other, Consultant role and travel support: Bio-AI Health; Financial Interests, Personal, Advisory Board, Consultant and Advisory role: ValoTx; Financial Interests, Personal, Advisory Board, Consultant and Advisor role, Travel support: Replimmune; Financial Interests, Personal, Advisory Board, Advisor role: Bayer; Financial Interests, Personal, Other, Consultant and Advisory: Erasca; Financial Interests, Personal, Other, Consultant: Philogen, Incyte; Financial Interests, Personal, Other, Consultant and Advisory role: BioNTech, Genmab; Financial Interests, Personal, Advisory Board: Anaveon, Menarini, ImCheck therapeutics; Financial Interests, Institutional, Funding, Clinical trial and translational research: BMS; Financial Interests, Institutional, Funding, Clinical Trial: Roche Genentech, Pfizer/Array, Sanofi; Non-Financial Interests, Leadership Role, President since 2010: Fondazione Melanoma Onlus Italy; Non-Financial Interests, Leadership Role, President since 2014: Campania Society of ImmunoTherapy of Cancer (SCITO) Italy; Non-Financial Interests, Other, Member of Steering Committee since 2016: Society for Melanoma Research (SMR); Non-Financial Interests, Member of Board of Directors, November 2017 - December 2021: Society for Immunotherapy of Cancer (SITC); Non-Financial Interests, Member: ASCO, SITC, EORTC Melanoma Cooperative Group, AIOM, SMR. 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1631P Clinical outcomes in patients with melanoma brain metastases: First-line treatment options

J. Mangana[1], N. Asher[2], L. Zimmer[3], B. Özdemir[4], S. Pedersen[5], I. von Wasielewski[6], I. Gavrilova[7], M-Á. Berciano-Guerrero[8], H. Schmidt[9], A.A. Luczak[10], D. Herceg[11], S.K. Petersen[12], P. Cerezuela-Fuentes[13], C. Gebhardt[14], H. Gogas[15], P.A. Ascierto[16], L. Kandolf Sekulovic[17], E. Ellebaek[18], M. Weichenthal[19]

[1]Dermatologie Department, USZ - Universitätsspital Zürich - Klinik für Dermatologie, Zurich, Switzerland; [2]Department of Oncology, Skin Cancer and Melanoma Center at Davidoff Cancer Center, Rabin, Israel; [3]Dermatology, Universitätsklinikum Essen, Essen, Germany; [4]Oncology Department, Inselspital - Universitatsklinik fur Medizinische Onkologie, Bern, Switzerland; [5]Oncology Department, Herlev Hospital - National Center for Cancer Immune Therapy (CCIT-DK), Herlev, Denmark; [6]Department of Dermatology and Allergy, MHH - Medizinische Hochschule Hannover, Hannover, Germany; [7]Oncodermatology Department, USBAL Oncology EAD Sofia, Sofia, Bulgaria; [8]Medical Oncology Department, Hospital Regional Universitario (HRU). Instituto de Investigaciones Biomédicas de Málaga (IBIMA), Málaga, Spain; [9]Department of Oncology, Aarhus University Hospital, Aarhus, Denmark; [10]Oncology Department, Aalborg University Hospital, Aalborg, Denmark; [11]Department Of Oncology, KBC - University Hospital Centre Zagreb, Zagreb, Croatia; [12]Department of Oncology, OUH - Odense University Hospital, Odense, Denmark; [13]Medical Oncology Department, Hospital Clínico Universitario Virgen de la Arrixaca, Instituto Murciano de Investigación Biosanitaria (IMIB)-Arrixaca, Ciudad de Murcia, Spain; [14]Dermatology and Venereology, University Medical Center Hamburg-Eppendorf, Hamburg, Germany; [15]First Department of Medicine, National and Kapodistrian University of Athens - School of Medicine, Athens, Greece; [16]Melanoma, Cancer Immunotherapy & Developmental Therapeutics, Istituto Nazionale Tumori IRCCS - Fondazione G. Pascale, Naples, Italy; [17]Dermatology and Venerology Department, Military Medical Academy VMA, Belgrade, Serbia; [18]Department of Oncology, National Center for Cancer Immune Therapy (CCIT-DK), Herlev Hospital, Herlev, Denmark; [19]Dermatology Department, Christian-Albrechts-University Kiel, Kiel, Germany

Background: Combination immunotherapy with ipilimumab and nivolumab (Ipi/Nivo) is the current standard of care in patients with asymptomatic melanoma brain metastasis (MBM). However, optimal strategies for managing symptomatic MBM remain unclear and treatment with BRAF/MEK inhibitors (BRAF/MEKi) is still in common use for symptomatic BRAF V600-mutated MBM.

Methods: Patients with MBM who received first-line (1L) systemic treatment within the EUMelaReg registry were included. Symptomatic patients were defined as those requiring corticosteroids at the initiation of 1L treatment. Concurrent use of local therapies such as surgery and/or radiotherapy were recorded. Primary endpoints were overall survival (OS) and melanoma-specific survival (MSS) stratified by 1L treatment. Secondary endpoints included progression-free survival (PFS), response rates and stratified analyses for BRAF mutational status and various prognostic factors.

Results: Among a total of 1,515 patients with MBM, 385 were symptomatic at start of 1L therapy, of whom 250 (64.9%) were BRAF V600 mutated. Symptomatic patients had worse baseline prognostic factors, including higher ECOG score, high LDH, and greater intercranial tumor burden. Median PFS, OS, and MSS were significantly shorter in symptomatic patients (4.1 [3.5-4.5] months, 9.4 [8.0-11.9] months, and 9.7 [8.1-12.0] months, respectively) compared to asymptomatic patients (6.3 [5.8−6.9] months, 18.9 [16.3-21.3] months, and 19.4 [16.8-22.3] months, respectively) (95% confidence interval for all). In BRAF mutated patients with symptomatic MBM, 148 (59.2%) received 1L BRAF/MEKi with an ORR of 60.7%, while 50 (25.0%) received Ipi/Nivo with an ORR of 32.0%. Median PFS and OS were 5.3 (4.5-5.8) and 8.8 (7.5-11.8) months with BRAF/MEKi versus 2.7 (1.9-4.4) and 17.3 (10.5-23.9) months, with Ipi/Nivo.

Conclusions: Symptomatic patients with MBM show markedly inferior survival. Despite lower ORR and shorter PFS, symptomatic patients treated with Ipi/Nivo had notably longer OS than those treated with BRAF/MEKi, supporting the use of 1L immunotherapy even in this high-risk group.

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Zimmer: Financial Interests, Personal, Advisory Board: Bristol Myers Squibb, Novartis, MSD, Sunpharma, Pierre Fabre, Sanofi; Financial Interests, Personal, Other, Travel Support: Bristol Myers Squibb, MSD, Sunpharma, Pierre Fabre, Sanofi, Novartis; Financial Interests, Personal, Invited Speaker: Novartis, MSD, Pierre Fabre; Financial Interests, Personal and Institutional, Research Grant: Novartis; Non-Financial Interests, Member: German Cancer Society, Dermatologic Cooperative Oncology Group. B. Özdemir: Financial Interests, Institutional, Advisory Board: Roche, Ipsen, MSD, Pfizer, Sanofi, Immunocore; Financial Interests, Institutional, Invited Speaker: Merck, Novartis; Financial Interests, Institutional, Other, Reduction for registration fee for ASCO Gu: Janssen; Financial Interests, Institutional, Local PI: Parexel, Seattle Genetics, MSD, BMS; Financial Interests, Institutional, Research Grant: Novartis. I. von Wasielewski: Financial Interests, Advisory Role: BMS, MSD, Sanofi; Financial Interests, Other, Honoraria payment: Novartis, BMS, MSD, Sanofi, Stemline, Kyowa Kirin; Financial Interests, Other, Travel/accommodations/expenses: Novartis, BMS, MSD, Sanofi, Stemline, Kyowa Kirin. I. Gavrilova: Financial Interests, Advisory Role: Merck Sharp & Dohme, Bristol Myers and Squibb, Pierre Fabre, Medison; Financial Interests, Other, Honoraria: Roche, Novartis, Merck Sharp & Dohme, Bristol Myers and Squibb, Sanofi, Merck, Pierre Fabre, Medison; Financial Interests, Speaker's Bureau: Roche, Novartis, Merck Sharp & Dohme, Bristol Myers and Squibb, Sanofi, Merck, Pierre Fabre, Medison; Financial Interests, Other, Travel/accommodations/expenses: Merck Sharp & Dohme, Bristol Myers and Squibb, Pierre Fabre. M. Berciano-Guerrero: Financial Interests, Advisory Role: BMS, MSD, Pierre Fabre, Novartis, Eisei, Ipsen, Lilly; Financial Interests, Speaker's Bureau: BMS, MSD, Pierre Fabre, Novartis, Eisei, Novartis, Ipsen, Lilly; Financial Interests, Research Funding: Novartis. A.A. Luczak: Financial Interests, Advisory Role: Member of Danish Medicines Council. D. Herceg: Financial Interests, Other, congress sponsorship and sponsored lecture: MSD; Financial Interests, Other, sponsored lecture: Novartis. P. Cerezuela-Fuentes: Financial Interests, Advisory Role: Novartis, Bristol Myers and Squibb, MSD, Pierre Fabre, Roche, Sanofi, SUN Pharma; Financial Interests, Other, attendance at conferences: Novartis, MSD, Pierre Fabre. C. Gebhardt: Financial Interests, Personal, Advisory Board: Almirall, BMS, Beiersdorf, BioNTech, Delcath, ImCheck, Immatics, Immunocore, Iovance Biotherapeutics, Janssen, MSD, Novartis, Pierre Fabre, Regeneron, Replimune, Roche, Sanofi Aventis, Skyline DX, Sun Pharmaceuticals, Sysmex; Financial Interests, Personal, Invited Speaker: Almirall, BMS, Beiersdorf, Delcath, Immunocore, MSD, Novartis, Pierre Fabre, Regeneron, Replimune, Sun Pharmaceuticals; Financial Interests, Personal, Ownership Interest, Co-Founder/Shareholder: Dermagnostix, Dermagnostix R&D; Financial Interests, Institutional, Coordinating PI: Almirall, BMS, BioNTech, Genentech, Immatics, Immunocore, Iovance Biotherapeutics, Iovance Biotherapeutics, MSD, Novartis, Pierre Fabre, Regeneron, Replimune, Sanofi, Sun Pharmaceuticals; Financial Interests, Institutional, Research Grant: Delcath, Novartis, Regeneron; Financial Interests, Personal, Steering Committee Member: MSD; Non-Financial Interests, Member of Board of Directors: DeCOG / ADO - German Dermatooncology Group, European Association of Dermato-Oncology (EADO), German Skin Cancer Foundation - Fleur Hiege, Roggenbuck Foundation; Non-Financial Interests, Member of Board of Directors, Member of the Scientific Board: Melanoma World Society (MWS); Other, Section Editor / Editor: The Journal of European Association of Dermatology and Venereology (EADV); Other, Editor: The Journal of the German Dermatological Society (JDDG). H. Gogas: Financial Interests, Personal, Advisory Board: MSD, BMS, Pierre Fabre, Sanofi; Financial Interests, Personal, Invited Speaker: MSD, BMS, Novartis, Pierre Fabre, Sanofi; Financial Interests, Steering Committee Member: Amgen, Replimmune; Financial Interests, Institutional, Local PI: Amgen, MSD, BMS, Replimmune, Iovance, Bayer; Financial Interests, Institutional, Research Grant: BMS, Pfizer, Lilly, Pierre Fabre, AstraZeneca. P.A. Ascierto: Financial Interests, Personal, Other, Consultant and Advisory Role: BMS, Roche Genentech, Novartis, Merck Serono, Sun Pharma, Sanofi, Sandoz, Immunocore, Boehringer Ingelheim, Regeneron; Financial Interests, Personal, Other, Consultant, Advisory Role and Travel support: MSD, Pierre Fabre; Financial Interests, Personal, Other, Consultant and Advisory Role.Travel support: Pfizer/Array; Financial Interests, Personal, Other, Consultant Role: Italfarmaco; Financial Interests, Personal, Other, Consultant Role: Pfizer, Nouscom, Lunaphore; Financial Interests, Personal, Other, Consultant role: Medicenna; Financial Interests, Personal, Other, Consultant role and travel support: Bio-AI Health; Financial Interests, Personal, Advisory Board, Consultant and Advisory role: ValoTx; Financial Interests, Personal, Advisory Board, Consultant and Advisor role, Travel support: Replimmune; Financial Interests, Personal, Advisory Board, Advisor role: Bayer; Financial Interests, Personal, Other, Consultant and Advisory: Erasca; Financial Interests, Personal, Other, Consultant: Philogen, Incyte; Financial Interests, Personal, Other, Consultant and Advisory role: BioNTech, Genmab; Financial Interests, Personal, Advisory Board: Anaveon, Menarini, ImCheck therapeutics; Financial Interests, Institutional, Funding, Clinical trial and translational research: BMS; Financial Interests, Institutional, Funding, Clinical Trial: Roche Genentech, Pfizer/Array, Sanofi; Non-Financial Interests, Leadership Role, President since 2010: Fondazione Melanoma Onlus Italy; Non-Financial Interests, Leadership Role, President since 2014: Campania Society of ImmunoTherapy of Cancer (SCITO) Italy; Non-Financial Interests, Other, Member of Steering Committee since 2016: Society for Melanoma Research (SMR); Non-Financial Interests, Member of Board of Directors, November 2017 - December 2021: Society for Immunotherapy of Cancer (SITC); Non-Financial Interests, Member: ASCO, SITC, EORTC Melanoma Cooperative Group, AIOM, SMR. L. Kandolf Sekulovic: Financial Interests, Advisory Role: Merck Sharp & Dohme, Bristol Myers and Squibb, Novartis, Immunocore; Financial Interests, Speaker's Bureau: Merck Sharp & Dohme, Bristol Myers and Squibb, Roche, Novartis, Janssen, Immunocore, Abbvie; Financial Interests, Other, Honoraria: Merck Sharp & Dohme, Bristol Myers and Squibb, Roche, Novartis, Janssen, Immunocore, AbbVie; Financial Interests, Other, Travel/accommodations/expenses: Merck Sharp & Dohme, Bristol Myers and Squibb, Janssen. E. Ellebaek: Financial Interests, Personal, Invited Speaker: Pierre Fabre, BMS, Novartis, MSD; Other, Travel and conference expenses: MSD, Pierre Fabre. M. Weichenthal: Financial Interests, Personal, Advisory Board: Merck Sharp & Dohme, Bristol Myers and Squibb, Sun Pharma, Sanofi, Pierre Fabre; Financial Interests, Institutional, Local PI: Millennium, Merck Sharp & Dohme, 4SC, Innate Pharma; Financial Interests, Institutional, Research Grant: Bristol Myers and Squibb, Novartis; Non-Financial Interests, Leadership Role: EuMelaReg; Non-Financial Interests, Member of Board of Directors: DeCOG. All other authors have declared no conflicts of interest.

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1632P Predicting real-world overall survival in advanced melanoma using machine learning

F. Wang[1], D. Hathi[2], L. Chi[2], N. Wu[2], E. Nkhoma[2], J. Harnett[2], M.E. Salvati[2], K. Lewis[2], Y. Li[2]

[1]Department of Population Health Sciences, Weill Cornell Medicine, New York, United States of America; [2]Health Economics and Outcomes Research, Regeneron Pharmaceuticals, Inc., Tarrytown, United States of America

Background: This study used machine learning (ML) to predict and identify predictors of real-world overall survival (OS) among advanced melanoma patients (aMel pts) receiving 1L treatment.

Methods: 1L aMel pts were identified from the ConcertAI EHR database (Jan 2016 − Oct 2023). Index date was the start of 1L therapy. Data was split 80:20 into training and test sets, respectively. A gradient-boosted survival ML model (XGBoost) was trained to predict all-cause OS using baseline clinical and patient characteristics. Model performance was evaluated using the concordance index (c-index). Top predictors were visualized in a cox proportional hazard nomogram of OS at 6 months, 1 yr, and 2 yrs. Pts with ≥median of predicted risk scores were classified as high-risk.

Results: The study cohort had 2,038 pts (mean age, 65.1 yrs; 64.8% male) with median OS of 22.9 months (1,091 events). The XGBoost achieved a c-index of 0.71. Higher risk of death was associated with ECOG-PS $\geq$ 2, presence of brain or liver metastases, elevated lactate dehydrogenase (LDH) and white blood cell (WBC) counts, and age $\geq$ 70 yrs at 1L initiation (Table). ECOG-PS $\geq$ 2 increased the risk of death by 119%, brain and liver metastases by 98% and 45%, respectively, and elevated LDH by 29%. Median probability of 6-month, 1-yr, and 2-yr OS for the high-risk group was 71.3%, 55.8%, and 37.8% vs. 85.3%, 75.9%, and 63.1% for lower-risk group, respectively.

Conclusions: We developed a melanoma-specific model to predict OS using a large real-world dataset derived from diverse oncology practices across the US that may have implications across drug development and clinical practice decisions in the future.

Table: 1632P Summary of top predictors at baseline for all-cause OS

	Mean HR (95% CI)
ECOG-PS ($\geq$ vs $<$ 2)	2.19 (1.88−2.56)
Brain metastasis (present vs no brain metastasis)	1.98 (1.64−2.40)
Liver metastasis (present vs no liver metastasis)	1.45 (1.17−1.80)
Other metastasis (vs no other metastasis)	1.24 (1.04−1.48)
Elevated LDH ($\geq$1×ULN vs normal)	1.29 (1.17−1.43)
Age at 1L initiation ($\geq$ vs $<$ 70) yrs	1.44 (1.27−1.62)
WBC ($\geq$ vs $<6.6 \times 10^3$ cells/µL)	1.15 (1.09−1.22)
Chloride ($\geq$ vs $<$ 104 mM)	0.91 (0.86−0.96)
% Lymphocytes ($\geq$ vs $<$ 24.2%)	0.89 (0.84−0.94)

Metastatic sites are mutually exclusive

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1633P Onset and progression of atherosclerosis in patients with melanoma treated with immune checkpoint inhibitors

T.J.J. Uyl[1], D.V. Dorst[2], A.A.M. van der Veldt[2], A. Joosse[2], E. Oomen-de Hoop[2], J. Danser[3], R.H. Mathijssen[2], D. Bos[4], J. Versmissen[3]

[1]Department of Hospital Pharmacy/Medical Oncology, Erasmus MC - Erasmus University Rotterdam, Rotterdam, Netherlands; [2]Department of Medical Oncology, Erasmus MC - Erasmus University Rotterdam, Rotterdam, Netherlands; [3]Department of Vascular Medicine, Erasmus MC - Erasmus University Rotterdam, Rotterdam, Netherlands; [4]Department of Epidemiology, Erasmus MC - Erasmus University Rotterdam, Rotterdam, Netherlands

Background: Immune checkpoint inhibitors (ICIs) are effective anti-cancer agents but significantly increase cardiovascular risk. This could be due to its potential to induce or worsen atherosclerosis. We evaluated the onset and progression of atherosclerosis during ICI treatment in patients with melanoma in five segments of the arterial tree and investigated risk factors for substantial (>10%/year) atherosclerotic plaque growth in the descending thoracic aorta.

Methods: Onset and yearly progression of atherosclerosis were assessed in the aortic arch, descending thoracic aorta, abdominal aorta, left and right iliac arteries via CT scans performed prior to and one year (+/-three months) after ICI therapy initiation in patients with melanoma in the adjuvant and advanced disease setting. The primary outcome was the yearly progression of maximal plaque thickness in each arterial segment. Secondary outcomes were changes in the number of plaques, incidence of arterial thrombosis (ATE), and factors associated with substantial plaque growth in the descending thoracic aorta.

Results: In total, 244 patients were included. Plaque thickness increased significantly in all aortic segments, ranging from +3.0-8.0% per year (Table; all $P<0.001$). In 75% of included patients, substantial plaque growth in $\geq$1 segment occurred. Number of plaques remained identical in 64-86% of arterial segments. Three patients (1.2%) developed ATE within 1 year after ICI initiation. ICI combination therapy demonstrated a trend towards increased risk of substantial plaque growth compared to monotherapy (OR 2.10 [0.95-4.66; $P=0.068$]), whereas antihypertensive drug usage associated with a lower risk (OR 0.47 [0.26-0.86; $P=0.014$]).

Table: 1633P

Segment	Total			Growth ≥10%		
	N (total)	Median growth (%)	*P*-value	N (% of total)	Median growth (%)	*P*-value
Aortic arch	165	8.0 (18)	**<0.001**	72 (44)	19.0 (21)	**<0.001**
Descending thoracic aorta	191	5.2 (19)	**<0.001**	74 (39)	20.0 (16)	**<0.001**
Abdominal aorta	208	3.0 (17)	**<0.001**	64 (31)	19.0 (13)	**<0.001**
Left iliac artery	171	6.5 (25)	**<0.001**	77 (45)	21.7 (17)	**<0.001**
Right iliac artery	189	5.6 (19)	**<0.001**	76 (40)	20.6 (24)	**<0.001**

Conclusions: The majority of melanoma patients experience substantial atherosclerotic plaque growth during ICI therapy. Possible preventive strategies should be investigated to minimize cardiovascular events while maintaining the beneficial effects of ICI therapy.

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1634P Quantifying the invisible: Circulating tumour fraction and ctDNA dynamics predict clinical outcome in the OPTIMEL trial

A. Harlé[1], A. Lhuillier[2], F. Granel Brocard[3], A. Schoeffler[4], H. Martin[4], P. Muller[5], A. Hervieu[6], M. Rouyer[7], M. Husson[7], D. Bechet[8], M. Meyer-Lefebvre[7], T. Vetterli[9], J. Behr[9], A. Rapin[9], J. Salleron[7], J.L. Merlin[10], L. Geoffrois[11]

[1]Service de Médecine de Précision, Institut de Cancérologie de Lorraine - Alexis Vautrin, Vandoeuvre-lès-Nancy, France; [2]Medical Oncology, Institut de Cancérologie de Lorraine - Alexis Vautrin, Vandoeuvre-lès-Nancy, France; [3]Onco Dermatology, CHRU Nancy, Nancy, France; [4]Dermatology, CHR Metz-Thionville - Hopital Bel-Air, Thionville, France; [5]Dermatology, Hopital de Mercy-CHR Metz-Thionville, Metz-Tessy, France; [6]Médical Oncology, Centre Georges-François Leclerc (Dijon), Dijon, France; [7]Biopathologie, Institut de Cancérologie de Lorraine - Alexis Vautrin, Vandoeuvre-lès-Nancy, France; [8]Centre de Promotion des Essais Cliniques, Institut de Cancérologie de Lorraine - Alexis Vautrin, Vandoeuvre-lès-Nancy, France; [9]Bioinformatics, HederaDx, Epalinges, Switzerland; [10]Department of Biopathology, Institut de Cancérologie de Lorraine - Alexis Vautrin, Vandoeuvre-lès-Nancy, France; [11]Medical Oncology Department, Institut de Cancérologie de Lorraine - Alexis Vautrin, Vandoeuvre-lès-Nancy, France

Background: Despite initial responses to BRAF/MEK inhibition in metastatic melanoma, reliable early and long-term outcome biomarkers remain limited. The OPTIMEL trial (NCT03416933) evaluated liquid biopsy to dynamically predict outcomes in BRAF V600-mutant melanoma treated with dabrafenib and trametinib.

Methods: 35 patients with histologically confirmed advanced or metastatic cutaneous melanoma harbouring BRAF V600 mutations were prospectively enrolled and received first-line dabrafenib + trametinib. Matched tumour and plasma were collected at baseline (D0), and plasma was sampled again at D15, 30, 90, 180, 270, 365, or at progression. DNA from FFPE tumour samples was sequenced using a 519-gene panel (Nonacus, UK). ctDNA from plasma was analysed with a 32-gene panel (HederaDx, Switzerland). Circulating tumour fraction (TF) was also quantified using a proprietary algorithm based on variant allele frequencies and cfDNA features. Progression-free survival (PFS) and tumour burden (via imaging) were assessed in the intention-to-treat population. Associations between ctDNA dynamics, TF, and clinical outcomes were explored using survival and correlation analyses.

Results: Baseline concordance between tissue and cfDNA was 63%. No BRAF mutation was detected in blood for 7 patients (20%) and 6 patients (17.1%) had no detectable ctDNA during the entire follow-up, suggesting a “non-shedder” profile. Non-shedders showed significantly improved PFS ($p=0.001$). Tumour burden correlated strongly with liquid TF ($p=0.0033$). Patients with TF >20% had significantly poorer outcomes (PFS at 12 months: 0%) compared to those with TF <20% (PFS at 12 months: 45.4%).

Table: 1634P

	n	n event	6 months-PFS [95% CI]	9 months-PFS [95% CI]	12 months-PFS [95%CI]	p-value
BRAF detection at baseline	28	21	66.2 [45;80.8]	32.3 [15;51]	9.2 [0.9;30.1]	0.001
No BRAF detection at baseline	7	1	100 [100;100]	100 [100;100]	85.7 [33.4;97.9]	
ctDNA at baseline	29	22	67.4 [46.7;81.6]	35 [17.4;53.3]	8.8 [0.8;28.9]	0.001
No ctDNA at baseline	6	0	100	100	100	
Tumour Fraction > 20%	10	9	36 [9;64.8]	0	0	<0.001
Tumour Fraction < 20%	21	9	95 [69.5;99.3]	68.1 [42.1;84.3]	45.4 [18.7;68.9]	

Abbreviation: n number of patients; n event n number of patients with progression or death; 95%CI 95% confidence interval

Conclusions: ctDNA analysis offers a powerful, non-invasive tool to stratify patients with BRAF V600-mutant melanoma. Absence of ctDNA—either at baseline or across treatment—was associated with markedly improved outcomes, suggesting a potential role for liquid biopsy in treatment monitoring and re-staging.

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1635P Real-world outcomes after immunotherapy discontinuation in advanced melanoma: Insights from the GEM1801 study

E. Muñoz-Couselo[1], M-Á. Berciano-Guerrero[2], G. Crespo[3], A. García Castaño[4], J.L. Manzano[5], P. Ayala de Miguel[6], C. Aguado de la Rosa[7], P. Cerezuela-Fuentes[8], T. Puértolas[9], L. Gutiérrez Sanz[10], J. Medina[11], A. Cunquero-Tomás[12], M. Majem[13], L.A. Fernández[14], E. Espinosa[15], L. Bellido[16], A. Soria[17], A. Bujosa[18], S. Martin Algarra[19], I. Marquez-Rodas[20]

[1]Medical Oncology Department, Hospital Universitari Vall d'Hebron y Vall d'Hebron Institute of Oncology (VHIO), Barcelona, Spain; [2]Medical Oncology Department, Hospital Regional Universitario (HRU). Instituto de Investigaciones Biomédicas de Málaga (IBIMA), Málaga, Spain; [3]Medical Oncology Department, Hospital Universitario de Burgos, Burgos, Spain; [4]Medical Oncology Department, Hospital Universitario Marqués de Valdecilla, Santander, Spain; [5]Medical Oncology Department, Instituto Catalán de Oncología, ICO-Badalona, H. Germans Trias i Pujol, Badalona, Spain; [6]Medical Oncology Department, Hospital Universitario San Pedro de Alcántara, Cáceres, Spain; [7]Medical Oncology Department, Hospital Universitario Clínico San Carlos, Madrid, Spain; [8]Medical Oncology Department, Hospital Clínico Universitario Virgen de la Arrixaca, Instituto Murciano de Investigación Biosanitaria (IMIB)-Arrixaca, Ciudad de Murcia, Spain; [9]Medical Oncology Department, Hospital Universitario Miguel Servet, Zaragoza, Spain; [10]Medical Oncology Department, Hospital Universitario Puerta de Hierro, Madrid, Spain; [11]Medical Oncology Department, Hospital Universitario Toledo, Toledo, Spain; [12]Medical Oncology Department, Hospital General Universitario Valencia, Valencia, Spain; [13]Medical Oncology Department, Hospital de la Santa Creu i Sant Pau, Barcelona, Spain; [14]Medical Oncology Department, Hospital Parc Taulí, Sabadell, Spain; [15]Medical Oncology Department, Hospital Universitario La Paz, CIBERONC, Madrid, Spain; [16]Medical Oncology Department, Complejo Asistencial Universitario de Salamanca. Instituto de Investigación Biomédica de Salamanca (IBSAL), Salamanca, Spain; [17]Medical Oncology Department, Hospital Universitario Ramón y Cajal, Madrid, Spain; [18]Medical Oncology Department, Hospital Son Llàtzer, Palma de Mallorca, Spain; [19]Medical Oncology Department, Clínica Universidad de Navarra, Pamplona, Spain; [20]Medical Oncology Department, Hospital General Universitario Gregorio Marañón, Madrid, Spain

Background: Immunotherapy (IT) has transformed the prognosis of patients (pts) with advanced melanoma. However, the optimal treatment duration remains undefined. While pivotal clinical trials often allow treatment discontinuation after 2 years in responding pts, real-world patterns and outcomes are less clear.

Methods: GEM1801 is a prospective, multicenter, real-world cohort study conducted across Spain. We identified pts with advanced melanoma treated with first-line IT who discontinued for reasons other than disease progression or death. Patients were grouped based on treatment duration: $<$2 years (C1) and $\geq$2 years (C2). We analyzed baseline characteristics, treatment-free interval (TFI), treatment-free survival (TFS), progression-free survival (PFS), and overall survival (OS).

Results: Of 1302 pts included between 2018 and 2024, 162 (12.4%) discontinued first-line IT in the absence of progression or death. Among them, 139 (85.8%) received $<$2 years of treatment. Reasons for discontinuation were: maximum benefit (38.1% C1, 47.8% C2), toxicity (40.3% C1, 8.7% C2), investigator decision (16.5% C1, 30.4% C2), patient decision (2.9% C1, 8.7% C2), and unknown (2.2% C1, 4.3% C2). Median age was 71 (C1) vs. 67 (C2), and most were male (61% C1, 78% C2). Complete response was achieved in 48.2% of C1 and 60.9% of C2. After a median follow-up of 36 months (95% CI: 33−39), 4-year outcomes were as follows:

Table: 1635P

Outcome	C1 (<2 years) % (95% CI)	C2 (≥2 years) % (95% CI)	p-value
TFI	73.9 (64.0-85.3)	75.0 (42.6-100)	0.120
TFS	49.6 (39.6-62.1)	75.0 (42.6-100)	0.003
PFS	62.2 (53.8-72.0)	100 (100-100)	0.001
OS	72.9 (64.9-82.0)	100 (100-100)	0.001

Conclusions: In this large real-world cohort, pts with advanced melanoma who discontinued IT in the absence of progression showed durable benefit. A treatment duration $\geq$2 years was associated with better long-term outcomes, including lower risk of relapse and death, supporting treatment individualization and reinforcing the importance of balancing efficacy, toxicity, and patient preferences when determining immunotherapy duration. However, intrinsic biases in prognostic basal factors among the two groups cannot be ruled out.

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1636P Gonadal toxicity in men receiving immune checkpoint inhibitors for melanoma

W. Cui[1], A. Winship[2], L.R. Alesi[2], K.J. Hutt[2], Z.X. Lu[3], A. Pasam[4], A. Rao[1], L. Spain[5], G. Rozen[6], C. Stern[7], S.R. Catford[6], R.A. Anderson[8], K-A. Phillips[1], S.K. Sandhu[1]

[1]Department of Medical Oncology; The Sir Peter MacCallum Department of Oncology, Peter MacCallum Cancer Centre; University of Melbourne, Melbourne, Australia; [2]Development and Stem Cell Program and Dept of Anatomy and Developmental Biology, Monash University, Clayton, Australia; [3]Medicine, Monash University, Clayton, Australia; [4]Department of Translational Research, Peter MacCallum Cancer Center, Melbourne, Australia; [5]Department of Medical Oncology, Peter MacCallum Cancer Center, Melbourne, Australia; [6]Reproductive Services, The Royal Women's Hospital, Parkville, Australia; [7]Reproductive Services, Royal Women's Hospital, Parkville, Australia; [8]12. Centre for Reproductive Health, University of Edinburgh - MRC Institute of Genetics and Molecular Medicine, Edinburgh, United Kingdom

Background: It is uncertain if immune checkpoint inhibitors (ICI) affect gonadal function. This is critical information for patients (pts). Elevated cytokines were observed in ICI treated mice with ovarian damage. We aimed to assess inflammatory and gonadal responses over time in melanoma pts receiving ICI.

Methods: We leveraged a longitudinal prospective biomarker study in melanoma pts receiving ICI and analysed gonadal markers and cytokines on serum taken before and at 12, 24 and 54 weeks (wk) after ICI commencement in men aged 18-60 years. The relationship between gonadal measures and cytokines was assessed by Pearson correlation; the significance of changes in gonadal markers and cytokines over time was assessed by Kruskal Wallis test. Analyses were performed using STATA 18.

Results: 150 pts were enrolled in the melanoma biomarker study; 18 men met the criteria for this analysis. Reasons for exclusion included advanced age, did not consent to future research, baseline serum unavailable. Median age was 48 years (range 24-59), 11 had stage IV melanoma. Pts received combined anti-CTLA4 and anti-PD1 (12/18), anti-PD1 alone (4/18) or clinical trial with $\geq$1 ICI (2/18). 16 men experienced immune adverse events, 13 received steroids. Half had stopped ICI before wk 54. Serum follicle stimulating hormone (FSH), luteinising hormone (LH) and total testosterone (T) were stable during ICI (Table). Cytokine levels were variable but appeared to transiently increase in some men (Table). Exploratory analysis found moderate positive correlation between post-treatment T and IL-6 ($r = 0.54$, $p < 0.001$), post-treatment T and granzyme A ($r = 0.52$, $p < 0.001$) and post-treatment T and granzyme B ($r = 0.50$, $p < 0.001$). No significant correlation was found between pre treatment T and IL-6 ($r = 0.43$, $p = 0.8$), granzyme A ($r = 0.41$, $p = 0.09$) and granzyme B ($r = 0.39$, $p = 0.10$).

Table: 1636P

	Baseline		Wk 12		Wk 24		Wk 54		
No. of pts with serum tested	18		16		17		10		
No. of tested pts receiving ICI	0		12		8		5		
	Median	IQR	Median	IQR	Median	IQR	Median	IQR	P Value
FSH IU/L	5.6	3.0-7.3	6.8	4.8-9.0	8.2	4.7-10.6	6.8	3.8-13.2	0.41
LH IU/L	3.4	2.4-6.3	4.6	2.4-8.3	3.6	2.6-7.2	4.8	2.8-10.1	0.82
T nmol/L	11.6	9.8-18.1	11.8	9.1-17.8	13.7	12.0-16.1	14.7	11.3-20.6	0.38
IFNγ pg/mL	22.1	1.6-47.8	41.5	12.5-93.8	32.3	12.2-133	15.9	1.6-29.6	0.27
Granzyme A pg/mL	421	0-9306	1003	0-5834	1841	0-6483	888	0-1768	0.92
Granzyme B pg/mL	28.0	4.8-439	81.0	3.1-295	181	2.4-571	39.0	3.8-78.0	0.93
TNF α pg/mL	14.9	12.8-20.4	15.0	11.4-20.3	15.7	11.3-20.2	11.0	9.2-13.9	0.08
IL 6 pg/mL	16.0	2.9-107	11.9	2.1-76.1	30.9	2.2-82.6	16.0	3.9-32.3	0.98

Conclusions: This study supports the limited data that testicular function likely remains stable and some inflammatory cytokines transiently increase on ICI but larger studies are needed.

Legal entity responsible for the study: The authors.

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The e contracts for my role are set up with the institution and the funds go to a research fund at Peter MacCallum Cancer Centre: Merck Sharp and Dohme; Financial Interests, Institutional, Advisory Board, I have served on advisory boards for AstraZeneca. The contracts for my role as an advisor are set up with the institution and the funds go to a research fund at Peter MacCallum Cancer Centre: AstraZeneca; Financial Interests, Institutional, Advisory Board, I have served on advisory boards for Novartis. The contracts for my role as an advisor are set up with the institution and the funds go to a research fund at Peter MacCallum Cancer Centre: Novartis; Financial Interests, Institutional, Advisory Board, I have served on an advisory board for Merck Serono. The contracts for my service are set up with the institution and funds go into a research fund at the Peter MacCallum Cancer Centre: Merck Serono; Financial Interests, Institutional, Advisory Board, I have served on an advisory board for SkylineDx. The contracts for my service are set up with the institution and funds go into a research fund at the Peter MacCallum Cancer Centre: SkylineDx; Financial Interests, Institutional, Advisory Board, I have served on an advisory board for AbbVie. The contracts for my role are with the institution and funds go into a research fund at the Peter MacCallum Cancer Centre: AbbVie; Financial Interests, Institutional, Invited Speaker, I have received honoraria for invited speaker role only which is paid to my institution. Funds go into a research fund at the Peter MacCallum Cancer Centre: Janssen; Financial Interests, Personal, Other, Equity (stock): AdvanCell; Financial Interests, Personal, Stocks/Shares: AdvanCell; Financial Interests, Institutional, Research Grant, My institution receives grant funding to run an investigator initiated trial that I lead: Novartis, Genentech, Amgen, AstraZeneca, Merck Serono, Merck Sharp and Dohme; Financial Interests, Institutional, Research Grant, Pfizer are providing funding to my institution for the conduct of an investigator initiated clinical trial: Pfizer; Financial Interests, Institutional, Research Grant, Senwha are providing funding to my institution for the conduct of an investigator initiated clinical trial: Senwha; Non-Financial Interests, Other, I serve on the Independent Safety and Data Monitoring committee for 2 of Novartis sponsored studies and dont receive any compensation for this: Novartis; Non-Financial Interests, Other, I serve on the steering committee for several Janssen sponsored trials and i do not receive compensation for this: Janssen; Non-Financial Interests, Other, I serve on the steering committee for one AstraZeneca sponsored trial and I do not receive compensation for this: AstraZeneca; Non-Financial Interests, Other, I serve on the steering committee for one Genentech sponsored trials and I do not receive compensation for this: Genentech; Non-Financial Interests, Other, I serve on the steering committee for a Bristol Myer Squibb sponsored trial and I do not receive renumeration for this: Bristol Myer Squibb; Non-Financial Interests, Principal Investigator, I am a principal investigator on several of Janssen sponsored studies and don't receive any renumeration for this: Janssen; Non-Financial Interests, Principal Investigator, I am a principal investigator on several Novartis sponsored studies and dont receive any renumeration for this: Novartis; Non-Financial Interests, Principal Investigator, I am a principal investigator on several of Genentech sponsored studies and don't receive any renumeration for this: Genentech; Non-Financial Interests, Principal Investigator, I am a principal investigator on several of Bristol Myer Squibb sponsored studies and don't receive any renumeration for this: Bristol Myer Squibb; Non-Financial Interests, Principal Investigator, I am the Priincipal Investigator for several AstraZeneca sponsored studies and am not remunerated for this: AstraZeneca; Non-Financial Interests, Principal Investigator, I am a principal investigator on a MacroGenics sponsored study and don't receive any renumeration for this: MacroGenics; Non-Financial Interests, Principal Investigator, I am a principal investigator on a Regeneron sponsored study and do not receive any renumeration for this: Regeneron; Non-Financial Interests, Principal Investigator, I am the Principal Investigator on several Merck Sharp and Dohme studies and I do not receive any renumeration for this: Merck Sharp and Dohme; Non-Financial Interests, Principal Investigator, I am a principal investigator on an IO Biotech study and I do not receive any renumeration for this: IO Biotech; Non-Financial Interests, Principal Investigator, I am principal investigator on an Evaxion sponsored study and I do not receive any renumeration for this: Evaxion. 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1637P FDG PET/CT response as a predictive biomarker of long-term outcomes and patterns of recurrence to immune checkpoint inhibitor (ICI) therapy in unresectable cutaneous melanoma

R. Wallace[1], M. Jayawardana[2], S. Rajapaksa[2], S. Williams[3], T. Lee[3], R. Alipour[4], L. Spain[3], S.K. Sandhu[5], G. McArthur[6], A. Rao[7]

[1]Medical Oncology Department, Peter MacCallum Cancer Centre, Melbourne, Australia; [2]Office of Cancer Research (Research Computing Facility), Peter MacCallum Cancer Center, Melbourne, Australia; [3]Department of Medical Oncology, Peter MacCallum Cancer Center, Melbourne, Australia; [4]Cancer Imaging, Peter MacCallum Cancer Center, Melbourne, Australia; [5]Division of Cancer Medicine, Peter MacCallum Cancer Centre, Melbourne, Australia; [6]Victorian Comprehensive Cancer Center, Peter MacCallum Cancer Centre, Melbourne, Australia; [7]Medical Oncology, Peter MacCallum Cancer Centre, East Melbourne, Australia

Background: ICI results in sustained remission and improved overall survival (OS) in patients (pts) with advanced cutaneous melanoma. No clinical biomarkers exist to predict long-term responses. FDG PET/CT scans are used to define responses to ICI. We investigated the clinical impact of attaining a complete metabolic response (CMR), partial metabolic response (PMR) or stable metabolic disease (SMD) to predict risk of progression and characterised patterns of recurrence in pts who progress after a CMR.

Methods: This single centre, retrospective study included all pts with unresectable melanoma who received ICI from 1/1/2010 to 6/8/2020 at the Peter MacCallum Cancer Centre and had a baseline and follow up FDG PET/CT scan within 4 months of commencing ICI. 'Best' FDG PET/CT response was determined by nuclear medical specialists' reports adhering to PERCIST criteria: CMR, PMR, SMD and progressive metabolic disease (PMD). OS was calculated from date of first ICI to date of death or last follow up.

Results: 426 out of 893 pts were included. 205/426 had a CMR, 53 PMR, 29 SMD and 139 PMD. The OS was significantly higher for pts achieving a CMR, (HR 0.09, 95% CI 0.06-0.13, $P<0.001$), adjusted for line of therapy, ECOG and stage (Table). The time to achieve a CMR $\leq$6 months (78%), 6-12 months (15%) or $\geq$12 months (7%) did not clearly affect OS. 86/205 (42%) pts with CMR progressed - 49/86 (57%) extracranially, 30/86 (35%) intracranially and 7/86 (8%) intra and extracranally. 64/86 (74%) pts progressed with $\leq$ 3 sites (oligometastatic), all received salvage local therapy. 21/64 (33%) pts treated for oligo-progression died from melanoma.

Table: 1637P

Response	Median OS
CMR	NR
PMR	27.5 (23.7 - 43.9)
SMD	31.2 (20.3 - NR)
PMD	9.1 (7.6 - 12.4)

Conclusions: A CMR on FDG PET/CT from ICI treatment predicts long-term response and OS. Recurrence occurred in 42% of pts who achieved a CMR. Mostly, this was oligometastatic allowing local treatment. Further investigation is required to determine pts at risk of recurrence despite attaining a CMR.

Clinical trial identification: HREC/55187/PMCC-2019.

Legal entity responsible for the study: R. Wallace.

Funding: Has not received any funding.

Disclosure: L. Spain: Financial Interests, Institutional, Advisory Board, Renal Cell Ad Board: Ipsen; Financial Interests, Personal, Invited Speaker, Renal Cell dinner meeting 08/2023: BMS; Financial Interests, Personal, Other, Engagement in steering committee of non-interventional study (ongoing from 2022); also invited chair at local meeting 08/2024: MSD; Financial Interests, Personal, Stocks/Shares, Stock in Impedimed: Impedimed; Financial Interests, Institutional, Local PI, PI for TILVANCE-301 trial: IOVANCE; Financial Interests, Institutional, Local PI, PI for PRISM-MEL study at Peter Mac: Immunocore. S.K. Sandhu: Financial Interests, Institutional, Advisory Board, I have served on advisory boards for MSD. The e contracts for my role are set up with the institution and the funds go to a research fund at Peter MacCallum Cancer Centre: Merck Sharp and Dohme; Financial Interests, Institutional, Advisory Board, I have served on advisory boards for AstraZeneca. 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The contracts for my service are set up with the institution and funds go into a research fund at the Peter MacCallum Cancer Centre: SkylineDx; Financial Interests, Institutional, Advisory Board, I have served on an advisory board for Abbvie. The contracts for my role are with the institution and funds go into a research fund at the Peter MacCallum Cancer Centre: Abbvie; Financial Interests, Institutional, Invited Speaker, I have received honoraria for invited speaker role only which is paid to my institution. 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G. McArthur: Financial Interests, Institutional, Coordinating PI, Financial interest is reimbursed costs for activities on clinical trials: Genentech Roche; Non-Financial Interests, Other, Advisory Board Member: Canthera; Non-Financial Interests, Institutional, Product Samples, IMP for investigator initiated and funded trial: Pierre-Fabre, Pfizer; Non-Financial Interests, Other, Director: Melanoma World Society; Non-Financial Interests, Other, Board of Directors: National Breast Cancer Foundation (Australia); Non-Financial Interests, Leadership Role, Board of Directors: Children's Cancer CoLab. A. Rao: Financial Interests, Institutional, Research Grant, Outside submitted work: Peter MacCallum Cancer Centre. All other authors have declared no conflicts of interest.

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1638P Dissecting the relationship between response to ICB and CMV status in melanoma

G.B. Milotay[1], M. Little[1], R. Watson[2], S. Mackay[1], D. Muldoon[1], O. Tong[2], B.P. Fairfax[1]

[1]Department of Oncology, University of Oxford, Oxford, United Kingdom; [2]Department of Oncology, The MRC Weatherall Institute of Molecular Medicine Oxford, Oxford, United Kingdom

Background: We previously demonstrated seropositivity for Cytomegalovirus (CMV) to be associated with improved overall survival (OS) in metastatic melanoma (MM) patients treated with anti-PD-1 monotherapy, but not combination anti-PD-1/anti-CTLA-4 Immune Checkpoint Blockade (ICB). Understanding how CMV enhances anti-PD-1 responses may provide novel therapeutic insights and assist in patient stratification. Here we demonstrate the presence of distinct molecular drivers and cellular determinants of ICB response according to CMV serostatus.

Methods: Using bulk $CD8^+$ T-cell RNA-seq and a single-cell RNA/V(D)J-seq atlas from MM samples within the OxCITE cohort (n=326) we characterise T cell clonal dynamics and molecular interactions that underpin effector function, controlling for covariates and assessing for interaction with CMV IgG seropositivity and ICB regimen. We integrate immuno-transcriptomic findings with clinical outcomes with a view to stratify patients by treatment.

Results: We find CMV seropositivity was associated with improved OS (HR_{adj}=0.44, p=0.019) following treatment with anti-PD-1 monotherapy. CMV+ patients showed specific induction of a cytotoxic geneset that was independently predictive of improved OS across all MM patients (HR_{adj}=0.37, p<0.01). Notably, elevated pre-treatment cytotoxicity has strong immunological correlates, with increased numbers of effector T-cells, gamma-delta T-cells, and reduced Treg activation. We identify distinct cell-cell interactions in single-cell analysis upon ICB treatment, specifically related to CMV status.

Conclusions: Our findings highlight a crucial role for chronic viral infection in cancer immunotherapy response. CMV seropositivity is associated with a cytotoxic T-cell milieu composed of expanded effector and gamma-delta subsets, with corresponding Treg depletion. Conversely, CMV- patients have reduced T-cell cytotoxicity coupled to increased Treg activation. We propose CMV serostatus to be a novel peripheral biomarker for predicting ICB response that has translational implications for patient stratification in MM, with CMV+ patients having excellent outcomes post PD-1 monotherapy, while CMV- patients may preferentially benefit from anti-CTLA-4 driven Treg depletion.

Legal entity responsible for the study: The authors.

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1639P Characterizing the immune landscape in melanoma upon αPD-1 and αTIGIT mAbs

E. Ramelyte[1], R. Dätwyler[2], C. Panetti[2], M. Teufer[1], F. Sella[1], P. Turko[1], M.P. Levesque[1], R. Dummer[1], L. Kraehenbuehl[3], N. Joller[2]

[1]Dermatology Department, University Hospital Zurich, Zurich, Switzerland; [2]Department of Quantitative Biomedicine, University of Zurich, Zurich, Switzerland; [3]Dermatology Department, KSA am Bahnhof - Kantonsspital Aarau, Aarau, Switzerland

Background: TIGIT emerged as a promising target in cancer immunotherapy, particularly in combination strategies with anti-PD-1/PD-L1. However, studies combining pembrolizumab (anti-PD-1, pembro) and vibostolimab (anti-TIGIT, vibo) have not met their primary endpoints: the adjuvant trial (NCT05665595) was terminated early due to futility, while in the neoadjuvant setting, the efficacy of pembro + vibo combination was similar to pembro alone (Dummer et al. 2025). Here we report distinct immune activation during pembro + vibo vs. pembro alone as a potential explanation for the observed clinical outcome.

Methods: Baseline (BL) and on treatment (oT) PBMCs were collected from melanoma patients who received pembro + vibo or pembro alone in an adjuvant (n=12, 7), neoadjuvant (n=3, 4), or metastatic (n=3, 9) setting, respectively. PBMCs were characterized by spectral flow cytometry to assess immune cell composition and function.

Results: BL and oT samples from patients who received pembro + vibo (n=18) or pembro alone (n=20) were analyzed. Median time between BL and oT sample collection was 91d (range, 32-106) and 120d (range, 48-171), respectively. Compared to pembro alone, pembro + vibo downregulated the proinflammatory cytokines IFNγ and TNFα in $CD8^+$ (p=.005, .0003) and $CD4^+$ (p=.015, .077) T cells and upregulated the regulatory cytokine IL-10 in $CD8^+$ (p=.010), $CD4^+$ (p=.009) and regulatory T cells (Tregs, p=.003). The IFNγ/IL-10 and TNFα/IL-10 ratios were reduced in $CD8^+$ (p=.003, .002) and $CD4^+$ (p=.003, .003) T cells, indicating a shift towards an anti-inflammatory cytokine profile upon pembro + vibo. No differences in BL immune cell composition were found in the two groups. However, TIGIT-expressing T cells decreased upon pembro + vibo treatment ($CD8^+$ p=.003, Tregs p=.001).

Conclusions: The observed decrease in proinflammatory immune response upon pembro + vibo contrasts pre-clinical studies (Schorer et al. 2020) and may explain the lack of added clinical benefit. The decrease of TIGIT-expressing cells following treatment could suggest depletion of recently activated cells but further analyses are needed to fully elucidate the mechanism of action and its impact on melanoma outcomes to improve future treatment approaches.

Legal entity responsible for the study: The authors.

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1640P ADU-1604, a differentiated CTLA-4 blocking antibody shows benchmark clinical efficacy with a milder safety profile in PD1 relapse/refractory melanoma patients

G.A. De Velasco Oria[1], P. Rutkowski[2], A.M. Di Giacomo[3], A.M. Arance[4], C. Robert[5], V. Chiarion Sileni[6], I. Marquez-Rodas[7], J. Kundapur[8], J. Villaudy[8], H. van Eenennaam[8], L. Lassouw-Polman[9], A.M.M. Eggermont[10]

[1]Medical Oncology Department, Hospital Universitario 12 de Octubre, Madrid, Spain; [2]Department of Soft Tissue/Bone Sarcoma and Melanoma, Maria Sklodowska-Curie National Research Institute of Oncology, Warsaw, Poland; [3]Department of Medical Oncology and Immunotherapy, University of Siena, Siena, Italy; [4]Medical Oncology, Hospital Clinic de Barcelona, Barcelona, Spain; [5]Dermatology, Gustave Roussy, Villejuif and Paris-Saclay University, Villejuif, France; [6]Clinical and Experimental Oncology Dept., Synlab Euganea Medica, Albignasego, Italy; [7]Medical Oncology Department, Hospital General Universitario Gregorio Marañon - Fundación Investigación Biomedica, Madrid, Spain; [8]Research, Sairopa B.V., Rotterdam, Netherlands; [9]Clinical, Sairopa B.V., Rotterdam, Netherlands; [10]Cancer Medicine, University Medical Center Utrecht, Utrecht, Netherlands, and Comprehensive Cancer Center Munich of the Technical University Munich and the Ludwig Maximilian University, Utrecht, Netherlands

Background: Cytotoxic T-lymphocyte-associated protein 4 (CTLA-4) is a negative regulator of T-cell responses, also known as an immune checkpoint. The clinical relevance of CTLA-4 blockade was demonstrated by the approval of ipilimumab for the treatment of melanoma. However, current CTLA-4 blocking antibodies come with significant adverse events. ADU-1604 is a humanized hIgG1 CTLA-4 blocking antibody that binds a unique epitope on CTLA-4 and demonstrates pH-sensitive binding to CTLA-4.

Methods: This is a phase 1, first-in-human (FIH), two-part, open-label clinical trial of intravenous (IV) administration of ADU-1604 given as monotherapy in subjects with advanced-stage, relapsed/refractory melanoma who relapsed or were refractory to a prior anti-PD-1/PD-L1 therapy. Main endpoints are safety, pharmacokinetics, pharmacodynamics and clinical efficacy. 20 subjects received escalating doses of ADU-1604 IV (25, 75, 225, 450 mg flat dose) Q3W. In the dose expansion part 21 additional patients were treated with 225 mg dose to confirm recommended phase 2 dose (RP2D).

Results: ADU-1604 has similar exposure as was observed for ipilimumab[1] and dose-dependently increased absolute lymphocyte count (ALC) and proliferating CD4 and CD8 T cells. Of the 27 patients treated at 225 mg dose level (RP2D), one patient experienced a treatment-related Grade 4 event (3.7%) and six patients (22%) experienced treatment-related Grade 3 events. Most notably, no patients experienced immune-related enterocolitis or immune-related diarrhea. Three patients demonstrated partial response and twelve patients exhibited stable disease as best response (of which two >10 months).

Conclusions: ADU-1604 at RP2D (225 mg dose) shows clinical activity in line with what has been observed with other CTLA-4 blocking antibodies. Importantly, no DLTs and limited adverse events were reported during the study suggesting a milder safety profile than observed with ipilimumab.

Clinical trial identification: The clinical trial is registered with EudraCT number 2021-002623-38.

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1641P Activity of ulixertinib plus palbociclib in patients (Pts) with NRASQ61-mutant (mut) PD1 inhibitor refractory (PD1iR metastatic melanoma (MM))

S.J. Moschos[1], F. Collichio[1], M. Flores[2], D.L. Knoerzer[3], H. Sanoff[1]

[1]Medicine, The University of North Carolina at Chapel Hill, Chapel Hill, United States of America; [2]Clinical Trials Unit, UNC - Lineberger Comprehensive Cancer Center, Chapel Hill, United States of America; [3]Translational Sciences Department, BioMed Valley Discoveries, Inc., Kansas City, United States of America

Background: Pts with *NRASQ61*-mut MM exhibit frequent activation of the MAPK pathway. However, MEK inhibitor monotherapy does not provide substantially greater benefit than cytotoxic chemotherapy in *NRAS*-mut MM. Ulixertinib, an ERK1/2 inhibitor, has demonstrated single-agent activity in *NRAS*-mut MM. Genetic alterations in cell cycle-associated genes can occur in >50% of *NRAS*-mut MM. We hypothesized that concurrent administration of ulixertinib plus palbociclib, a CDK4/6 inhibitor, would significantly benefit pts with PD1iR MM. The phase I study, previously completed by our group, determined that the RP2D for ulixertinib in advanced solid tumors is 450 mg BID and for palbociclib is 125 mg QD, 21 days-on, 7 days-off (Raybould ASCO 2021 abstr 3103).

Methods: Pts with *N(K,H)RAS* G12/G13/Q61-mut and *NF1* loss-of-function mutant MM who have progressed from PD1i-based treatments are eligible in an exploratory (up to 15 pt) MM-expansion cohort that would assess safety and tolerability (CTCAE v4.03), antitumor response rate (RECIST v1.1), progression-free (PFS), and overall survival (OS).

Results: 9 pts (6 males; median age 65, range 23-82; 5 with M1c; all with *NRASQ61*-mut; 4 with cell cycle-associated genetic aberrations; median prior systemic treatments 3, range 1-4) were enrolled and treated in the study so far. The treatment regimen was well-tolerated (median regimen-related adverse events (RRAE) were 8, range 4-22). RRAE of any grade occurring in >=4 pts were rash, serum creatinine increase, fatigue, diarrhea, anemia, vomiting, thrombocytopenia, and neutropenia, resulting in dose reduction in 7 pts. Serious (grade >=3) RRAE occurring in >=3 patients were neutropenia and rash. 1 pt had a partial response, and 3 more pts had prolonged stable disease (all with minor response). The median PFS was 2.0 months (range 0.8-9.2) and the median OS was 8.3 months (range 3.3-22.2+).

Conclusions: In this preliminary analysis, ulixertinib plus palbociclib had a moderate clinical benefit not significantly higher than ulixertinib monotherapy; palbociclib may impair lymphocytic proliferation and offset its direct antitumor effect. Nevertheless, the regimen was well-tolerated in this heavily pretreated pt population.

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1642P The role of BRAF/MEKi rechallenge in BRAFV600 mutated melanoma patients: Insights from a EUMelaReg real-world study

I. Gavrilova[1], M. Weichenthal[2], N. Asher[3], J. Ulrich[4], E. Ellebaek[5], A. Kreuter[6], A. Popovic[7], I. Stojkovski[8], S. Levari[9], B. Özdemir[10], A. García Castaño[11], J.B.A.G. Haanen[12], I.M. Svane[5], P. Mohr[13], P.A. Ascierto[14], P. Rutkowski[15], H. Gogas[16], J. Mangana[17], L. Bastholt[18], D. Schadendorf[19]

[1]Oncodermatology Department, USBAL Oncology EAD Sofia, Sofia, Bulgaria; [2]Dermatology Department, Christian-Albrechts-University Kiel, Kiel, Germany; [3]Department of Oncology, Skin Cancer and Melanoma Center at Davidoff Cancer Center, Rabin, Israel; [4]Dept. of Dermatology and Skin Cancer Center, Harzklinikum Dorothea Christiane Erxleben GmbH-Wernigerode, Wernigerode, Germany; [5]Department of Oncology, National Center for Cancer Immune Therapy (CCIT-DK), Herlev Hospital, Herlev, Denmark; [6]Department of Oncology, Helios St. Elisabeth Klinik Oberhausen, Oberhausen, Germany; [7]Oncology Department, Clinical Center Nis, Nis, Serbia; [8]CNS, Skin and Bone Tumors Dept., UCRO - University Clinic of Radiotherapy and Oncology, Skopje, North Macedonia; [9]Melanoma and Skin Cancer, Sheba Tel Hashomer Medical Center, Ramat Gan, Israel; [10]Oncology Department, Inselspital - Universitatsklinik fur Medizinische Onkologie, Bern, Switzerland; [11]Department of Oncology, Hospital Universitario Marqués de Valdecilla, Santander, Spain; [12]Department of Medical Oncology, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [13]Department of Dermatology, Elbe Hospital Buxtehude, Buxtehude, Germany; [14]Melanoma, Cancer Immunotherapy & Developmental Therapeutics, Istituto Nazionale Tumori IRCCS Fondazione Pascale, Naples, Italy; [15]Department of Soft Tissue/Bone Sarcoma and Melanoma, Maria Skłodowska-Curie National Research Institute of Oncology, Warsaw, Poland; [16]First Department of Medicine, National and Kapodistrian University of Athens - School of Medicine, Athens, Greece; [17]Dermatologie Department, USZ - Universitätsspital Zürich - Klinik für Dermatologie, Zurich, Switzerland; [18]Dept. of Oncology, OUH - Odense University Hospital, Odense, Denmark; [19]Department of Dermatology, Venereology and Allergology, University Hospital Essen, Essen, Germany

Background: Rechallenging patients with BRAF/MEK inhibitors (BRAF/MEKi) in later lines can be successful in patients with $BRAF^{V600}$ mutated melanoma who have progressed after other treatments.

Methods: This retrospective study evaluated 194 patients from the EUMelaReg registry with $BRAF^{V600}$ mutated metastatic melanoma who were rechallenged with BRAF/MEKi. These patients had all failed immune checkpoint inhibitors (ICI) following prior treatment with either adjuvant BRAF/MEKi (adjuvant cohort) or non-adjuvant BRAF/MEKi (non-adjuvant cohort) treatment. Sensitivity analyses compared rechallenged patients with BRAF/MEKi-naive patients treated with non-adjuvant BRAF/MEKi after ICI failure (control cohort). To adjust for baseline imbalances, a 1:1 covariate-based matching of the rechallenge and control populations was performed on several prognostic factors. Objectives focused on overall response rates (ORR), progression-free survival (PFS), and overall survival (OS) from start of rechallenge BRAF/MEKi treatment.

Results: Rechallenge with BRAF/MEKi resulted in an ORR of 29.4%. Median PFS was 5.6 (4.6-6.7) months, and median OS from start of rechallenge BRAF/MEKi was 9.6 (8.0-12.2) months (95% CI for both). We identified 42 (21.6%) patients in the adjuvant and 152 (78.4%) in the non-adjuvant cohort. Kaplan-Meier estimates showed a significantly longer median OS (13.8 [9.6-not reached] months vs 8.6 [6.9-11.1] months; $p=0.03$) for patients in the adjuvant compared to the non-adjuvant cohort (95% CI for both). Median PFS, and ORR were equally distributed between the two analyzed groups. These results were compared with the 1:1 matched BRAF/MEKi-naive population. The outcomes were significantly better in BRAF/MEKi-naive patients compared to the rechallenge cohort with ORR 54.6% ($p<0.0001$), median OS (16.9 [13.2-19.2] months, $p=0.0026$) and median PFS (7.6 [5.9-8.6] months; $p=0.012$) (95% CI for both).

Conclusions: Rechallenging patients with $BRAF^{V600}$ mutated melanoma with BRAF/MEKi lead to clinically meaningful benefit in terms of ORR and survival outcomes, and represents a viable treatment option, although efficacy is inferior compared to BRAF/MEKi naive patients.

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Ulrich: Financial Interests, Personal, Other, Honoraria: BMS, MSD, Novartis, Sanofi, SUN Pharma; Financial Interests, Institutional, Other, Honoraria: Astra-Zeneca, Regeneron, Roche , Gilead, Kyowa-Kirin; Financial Interests, Other, Support for attending meetings and/or travel: Pierre Fabre; Financial Interests, Advisory Board: BMS, Sanofi. E. Ellebaek: Financial Interests, Personal, Invited Speaker: Pierre Fabre, BMS, Novartis, MSD; Other, Travel and conference expenses: MSD, Pierre Fabre. A. Kreuter: Financial Interests, Speaker, Consultant, Advisor: Merck Sharp & Dohme, AbbVie, Boehringer Ingelheim, Janssen, Sanofi. A. Popovic: Financial Interests, Other, Honoraria payments: MSD, BMS, Roche, El Pharma, Novartis; Financial Interests, Other, Travel/accommodations/expenses: MSD, BMS, Roche. I. Stojkovski: Financial Interests, Personal, Advisory Board: Immunocore, Novartis; Financial Interests, Personal, Invited Speaker: MSD, Roche. S. Levari: Financial Interests, Other, Travel/accommodations/expenses: BMS, MSD, Medison; Financial Interests, Research Funding: BMS. B. Özdemir: Financial Interests, Institutional, Advisory Board: Roche, Ipsen, MSD, Pfizer, Sanofi, Immunocore; Financial Interests, Institutional, Invited Speaker: Merck, Novartis; Financial Interests, Institutional, Other, Reduction for registration fee for ASCO Gu: Janssen; Financial Interests, Institutional, Local PI: Parexel, Seattle Genetics, MSD, BMS; Financial Interests, Institutional, Research Grant: Novartis. J.B.A.G. Haanen: Financial Interests, Institutional, Advisory Board: Bristol Myers Squibb, Achilles Therapeutics, Immunocore, Gadeta, Ipsen, Merck Sharpe & Dohme, Merck Serono, Pfizer, Molecular Partners, Novartis, Roche, Sanofi, Instil Bio, Iovance Biotherapeutics, AstraZeneca; Financial Interests, Institutional, Advisory Board, SAB member: BioNTech, PokeAcel, T-Knife; Financial Interests, Personal, Advisory Board, SAB member: Neogene Therapeutics, Sastra Cell Therapy; Financial Interests, Personal, Advisory Board: Third Rock Venture, Scenic, CureVac, Imcyse; Financial Interests, Personal, Stocks/Shares: Neogene Therapeutics; Financial Interests, Institutional, Research Grant: Bristol Myers Squibb, BioNTech US, Merck Sharpe & Dohme, Amgen, Novartis, Asher Bio, Sastra Cell Therapy; Non-Financial Interests, Member: ASCO, AACR, SITC; Other, Editor-in-Chief IOTECH: ESMO; Other, Editorial Board ESMO Open: ESMO; Other, Editorial Board: Kidney Cancer. I.M. Svane: Financial Interests, Institutional, Other, Conference participation: MSD; Financial Interests, Personal, Invited Speaker: BMS, Janssen; Financial Interests, Personal, Advisory Board: Mendus, Genmab; Financial Interests, Personal, Stocks/Shares, Cofounder and Founder warrents: IO Biotech; Financial Interests, Institutional, Research Grant: Adaptimmune, Enara Bio, Lytix Biopharma, TILT Biotherapeutics, Asgard Therapeutics, IO Biotech; Financial Interests, Institutional, Funding: Evaxion; Financial Interests, Institutional, Other, drug for investigator driven trial: BMS; Non-Financial Interests, Principal Investigator: BMS, Roche, TILT Biotherapeutics, Lytix Biopharma, Novartis, Immunocore, MSD. P. Mohr: Financial Interests, Personal, Advisory Board: Bristol Myers Squibb, Merck Sharp & Dohme, Novartis, GSK, Pierre Fabre, Sanofi, Delcath, Regeneron, Immunocore, Merck Serono; Financial Interests, Personal, Invited Speaker: Bristol Myers Squibb, Merck Sharp & Dohme, Novartis, Pierre Fabre, Sanofi, Sun Pharma, Delcath; Financial Interests, Institutional, Funding: Bristol Myers Squibb, Merck Sharp & Dohme, Novartis; Non-Financial Interests, Principal Investigator: Bristol Myers Squibb, Merck Sharp & Dohme, Regeneron, Sanofi, Novartis, Sun Pharma. P.A. Ascierto: Financial Interests, Personal, Other, Consultant and Advisory Role: BMS, Roche Genentech, Novartis, Merck Serono, Sun Pharma, Sanofi, Sandoz, Immunocore, Boehringer Ingelheim, Regeneron; Financial Interests, Personal, Other, Consultant, Advisory Role and Travel support: MSD, Pierre Fabre; Financial Interests, Personal, Other, Consultant and Advisory Role.Travel support: Pfizer/Array; Financial Interests, Personal, Other, Consultant Role: Italfarmaco; Financial Interests, Personal, Other, Consultant Role: Pfizer, Nouscom, Lunaphore; Financial Interests, Personal, Other, Consultant role: Medicenna; Financial Interests, Personal, Other, Consultant role and travel support: Bio-AI Health; Financial Interests, Personal, Advisory Board, Consultant and Advisory role: ValoTx; Financial Interests, Personal, Advisory Board, Consultant and Advisor role.Travel support: Replimmune; Financial Interests, Personal, Advisory Board, Advisor role: Bayer; Financial Interests, Personal, Other, Consultant and Advisory: Erasca; Financial Interests, Personal, Other, Consultant: Philogen, Incyte; Financial Interests, Personal, Other, Consultant and Advisory role: BionTech, Genmab; Financial Interests, Personal, Advisory Board: Anaveon, Menarini, ImCheck therapeutics; Financial Interests, Institutional, Funding, Clinical trial and translational research: BMS; Financial Interests, Institutional, Funding, Clinical Trial: Roche Genentech, Pfizer/Array, Sanofi; Non-Financial Interests, Leadership Role, President since 2010: Fondazione Melanoma Onlus Italy; Non-Financial Interests, Leadership Role, President since 2014: Campania Society of ImmunoTherapy of Cancer (SCITO) Italy; Non-Financial Interests, Other, Member of Steering Committee since 2016: Society for Melanoma Research (SMR); Non-Financial Interests, Member of Board of Directors, November 2017 - December 2021: Society for Immunotherapy of Cancer (SITC); Non-Financial Interests, Member: ASCO, SITC, EORTC Melanoma Cooperative Group, AIOM, SMR. 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Gogas: Financial Interests, Personal, Advisory Board: MSD, BMS, Pierre Fabre, Sanofi; Financial Interests, Personal, Invited Speaker: MSD, BMS, Novartis, Pierre Fabre, Sanofi; Financial Interests, Steering Committee Member: Amgen, Replimmune; Financial Interests, Institutional, Local PI: Amgen, MSD, BMS, Replimmune, Iovance, Bayer; Financial Interests, Institutional, Research Grant: BMS, Pfizer, Lilly, Pierre Fabre, AstraZeneca. J. Mangana: Financial Interests, Personal, Advisory Board: Merck Sharp & Dohme; Financial Interests, Personal, Advisory Board, immune-related Adverse Events Management (ESMO): Bristol Myers and Squibb; Financial Interests, Personal, Other, Travel grant: Ultrasun, L'oreal, Merck Sharp & Dohme, Pierre Fabre; Financial Interests, Personal, Invited Speaker: Johnson and Johnson; Financial Interests, Personal, Other, Travel Grant: Sun pharmaceutical; Financial Interests, Institutional, Research Grant, The melanoma registry database of the Department of Dermatology has been partially supported by an unrestricted grant to the University of Zurich from Amgen, Novartis, BMS, MSD and Pierre Fabre: Novartis, BMS, Pierre Fabre, Amgen, MSD; Non-Financial Interests, Leadership Role, President WG Melanoma Group: SAKK; Non-Financial Interests, Leadership Role, president WG Melanoma: SAKK; Other, immune-related AEs podcast /paid personal: Bristol Myers Squibb International. L. Bastholt: Non-Financial Interests, Advisory Role, Scientific committee under Danish Medicines Agency regarding new treatments of melanoma, skin cancer and thyroid cancer: Danish Medicines Agency. D. Schadendorf: Financial Interests, Personal, Invited Speaker: BMS, MSD, Merck Serono, Regeneron; Financial Interests, Personal, Advisory Board: BMS, Novartis, MSD, Immunocore, Pierre Fabre, Regeneron, Philogen, BioAlta, Daiichi Sanyko, BioNTech, Immatics, CureVac, Formycon, Replimune, SunPharma, Skyline Dx, IOVance, Ipsen, IOBioTech, Boehringer-Ingelheim; Financial Interests, Institutional, Coordinating PI: Novartis, BMS, MSD, Pierre Fabre, Immatics; Financial Interests, Institutional, Research Grant: BMS, MSD; Financial Interests, Personal, Steering Committee Member: MSD; Financial Interests, Institutional, Local PI: Philogen, Immunocore, Regeneron, BioNTech Rna Pharmacuticals GMBH, Replimune, IO Biotech; Non-Financial Interests, Member of Board of Directors: EORTC-MG; Non-Financial Interests, Leadership Role, Founding member and SC chair: European Melanoma Registry (EuMelaReg). All other authors have declared no conflicts of interest.

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1643P A phase II clinical trial of regorafenib (REGO) combined with BRAF-/MEK-inhibitors in advanced pretreated BRAFV600-mutant (mut) melanoma (RegoMel)

I. Dirven[1], C. Bertels[1], M. Vounckx[1], J.I. Kessels[1], N. Vanlaer[1], B.A. Von Kemp[2], G. Fasolino[3], G. Awada[1], B. Neyns[1]

[1]Department of Medical Oncology, Vrije Universiteit Brussel (VUB)/Universitair Ziekenhuis Brussel (UZ Brussel), Brussel, Belgium; [2]Department of Cardiology, Vrije Universiteit Brussel (VUB)/Universitair Ziekenhuis Brussel (UZ Brussel), Brussel, Belgium; [3]Department of Ophthalmology, Vrije Universiteit Brussel (VUB)/Universitair Ziekenhuis Brussel (UZ Brussel), Brussel, Belgium

Background: Most *BRAF*^{V600}mut melanoma patients (pts) will progress despite immune checkpoint- and BRAF-/MEK-inhibitors (ICI and BRAF/MEKi). Preclinical data suggest that addition of a RAF-dimer inhibitor, such as REGO, can overcome adaptive resistance to BRAF/MEKi.

Methods: In cohort B of this multicohort, single-center, prospective, phase 2 clinical trial, pts with advanced *BRAF*^{V600}mut melanoma, progressive (incl. brain metastases) on ICI and BRAF/MEKi, added REGO (40-80 mg uninterrupted once daily (OD) oral dosing) to the BRAF/MEKi they were receiving ($\leq$4 weeks interruption allowed). Objective response rate (ORR, per RECIST v1.1) served as primary endpoint. The sample size (total n=16) was determined according to a Simon's two-stage design. Secondary endpoints were safety, median progression free (mPFS) and overall survival (mOS). (Database lock Apr 1st, 2025).

Results: Between Oct 2023 and Jan 2025, 18 pts were enrolled (9 male, med age 60y; AJCC stage IIIC: 1; M1c: 5, M1d: 12). All pts initiated REGO 40 mg OD, 12 pts (67%) escalated to 80 mg OD. REGO was added to encorafenib/binimetinib (E/B) and dabrafenib/trametinib (D/T) in 10 and 8 pts respectively. 15 pts experienced at least one treatment-related adverse event (TRAE). The most common TRAE were diarrhea (56%; 6% gr3) and acneiform dermatitis (33%; 6% gr3). Eight pts needed temporary treatment interruptions due to TRAE, but all were reversible. There were no grade 5 TRAE. To mitigate drug-specific TRAE, E/B was switched to D/T in 5 pts and vice versa in 1 pt. There was 1 complete and 1 partial response (both M1d; duration of response 6 and 24 weeks), 4 pts had stable disease. 4 pts experienced rapid intracranial progression prior to the first response evaluation. The ORR and disease control rate in response evaluable patients were 14% and 43% (intent-to-treat 11% and 33%). 17 pts have progressed, mPFS and mOS were 5.9 (95% CI 5-6) and 28.3 weeks (95% CI 16-40).

Conclusions: This phase 2 trial of REGO + BRAF/MEKi in *BRAF*^{V600}mut melanoma pts progressing on BRAF/MEKi failed to meet its predefined primary efficacy endpoint but demonstrated activity (incl. in the brain) in a subset of patients, with manageable toxicity.

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1644P Efficacy and safety of RP1 plus nivolumab in patients with advanced anti−PD-1-failed acral melanoma

C. Robert[1], J.J. Sacco[2], E. Muñoz-Couselo[3], D. Schadendorf[4], G.K. In[5], G.M. Beasley[6], J.J. Niu[7], B. Chmielowski[8], T.M. Wise-draper[9], M.M. Milhem[10], T.L. Bowles[11], K.K. Tsai[12], C. Lebbe[13], C. Gaudy Marqueste[14], A. Samson[15], J. Zhu[16], M. Viana[17], J.W. Hou[17], M.K. Wong[18]

[1]Department of Dermato-Oncology, Gustave Roussy and Paris-Saclay University, Villejuif, France; [2]Department of Medical Oncology, The Clatterbridge Cancer Centre and University of Liverpool, Bebington, United Kingdom; [3]Department of Medical Oncology, Vall d'Hebron Institute of Oncology (VHIO) and Vall d'Hebron Hospital, Barcelona, Spain; [4]Department of Dermatology, West German Cancer Center, University Hospital Essen, Essen & National Center for Tumor Diseases (NCT-West), Campus Essen & University Alliance Ruhr, Research Alliance Ruhr, Research Center One Health, University Duisburg-Essen, Campus, Essen, Germany; [5]Departments of Medicine and Dermatology, University of Southern California Norris Comprehensive Cancer Center, Los Angeles, United States of America; [6]Department of Surgery, Duke Cancer Institute, Duke University, Durham, United States of America; [7]Department of Medical Oncology, Banner MD Anderson Cancer Center, Gilbert, United States of America; [8]Division of Hematology and Medical Oncology, Jonsson Comprehensive Cancer Center, University of California Los Angeles, Los Angeles, United States of America; [9]Division of Hematology/Oncology, University of Cincinnati Cancer Center, Cincinnati, United States of America; [10]Department of Internal Medicine, Medical Oncology, Holden Comprehensive Cancer Center, University of Iowa, Iowa City, United States of America; [11]Department of Surgical Oncology, Intermountain Medical Center, Murray, United States of America; [12]Division of Hematology/Oncology, Department of Medicine, Helen Diller Family Comprehensive Cancer Center, University of California San Francisco, San Francisco, United States of America; [13]Service Dermatologie, Université Paris Cité, AP-HP Dermato-Oncology and CIC, Cancer Institute APHP. Nord−Université Paris Cité, INSERM U976, Saint Louis Hospital, Paris, France; [14]Dermatology and Skin Cancer Department, Aix-Marseille Université, APHM, Centre de Recherche en Cancérologie de Marseille (CRCM), INSERM, U1068, CNRS, UMR7258, UM105, Hôpital Timone, CEPCM, Marseille, France; [15]Department of Medical Oncology, Leeds Institute of Medical Research at St. James's, University of Leeds, Leeds, United Kingdom; [16]Biometrics, Replimune, Inc., Woburn, United States of America; [17]Clinical Development, Replimune, Inc., Woburn, United States of America; [18]Department of Medicine, Roswell Park Comprehensive Cancer Center, Buffalo, United States of America

Background: Acral melanoma is a rare and aggressive type of melanoma (2%−3% of all cases) that often has poor outcomes. Acral melanoma responds poorly to available therapies, such as immune checkpoint inhibitors, particularly after progression on first-line treatment. RP1 (vusolimogene oderparepvec) is an oncolytic immunotherapy expressing human granulocyte-macrophage colony-stimulating factor and a fusogenic glycoprotein (GALV-GP-R^{-}). RP1 + nivolumab (nivo) demonstrated an ORR of 32.9% by blinded independent central review (BICR) using RECIST 1.1 in patients (pts) with anti−PD-1−failed advanced melanoma from the IGNYTE trial (NCT03767348). Here we report an ad hoc analysis of RP1 + nivo efficacy and safety in pts with acral melanoma.

Methods: The IGNYTE phase 2 registrational cohort enrolled pts with stage IIIB−IV cutaneous melanoma and confirmed progression on anti−PD-1 $\pm$ anti−CTLA-4 for $\geq$8 weeks as the last prior treatment (N = 140). RP1 was administered intratumorally at 1×10^6 plaque-forming units (PFU)/mL initially, then at 1×10^7 PFU/mL Q2W ($\leq$7 doses) with intravenous nivo.

Results: Of 140 pts with anti−PD-1−failed cutaneous melanoma, 18 (12.9%) had acral melanoma. Of these 18 pts, 50.0% (9/18) had stage IVM1b−d disease, 94.4% (17/18) had *BRAF* wild-type tumors, and 72.2% (13/18) had PD-L1−negative ($<$1%) tumors. Most pts (61.1% [11/18]) had both anti−PD-1 and anti−CTLA-4 prior treatment, and 72.2% (13/18) had primary resistance to anti−PD-1. Following treatment with RP1 + nivo, the confirmed ORR was 44.4% (8/18) by BICR using RECIST 1.1, including 16.7% (3/18) complete response and 27.8% (5/18) partial response. Median (95% CI) duration of response was 11.9 (3.6−NR) months (ongoing). Most treatment-related adverse events (TRAEs) were grade 1/2; the most common TRAEs (any grade) were chills, pyrexia, fatigue, injection-site pain, and nausea. Grade $\geq$3 TRAEs were reported in 2 (11.1%) pts.

Conclusions: RP1 + nivo demonstrated deep and durable efficacy and was well tolerated in pts with advanced anti−PD-1−failed acral melanoma. RP1 + nivo represents a promising treatment approach for this rare, aggressive melanoma subtype.

Clinical trial identification: NCT03767348.

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Sacco: Financial Interests, Institutional, Research Funding: AstraZeneca, Bristol Myers Squibb; Financial Interests, Institutional, Research Funding, Research funding: Replimune, Inc., Immunocore; Financial Interests, Institutional, Speaker, Consultant, Advisor, Consulting fees: Sun Pharmaceuticals, Immunocore, Replimune, Inc.; Financial Interests, Personal, Speaker, Consultant, Advisor: Replimune, Inc.; Financial Interests, Personal, Other, Travel support: Merck Sharp & Dohme, Replimune, Inc.; Financial Interests, Institutional, Advisory Board: Immunocore, Replimune, Inc. E. Muñoz-Couselo: Financial Interests, Personal, Advisory Board: BMS, Novartis, Sanofi, Pierre Fabre, Sun Pharma, MSD, Regeneron, Imunocore; Financial Interests, Personal, Invited Speaker: Novartis, Roche, BMS, Sanofi, MSD, Pierre Fabre; Non-Financial Interests, Member of Board of Directors, Vocal in the Society of Spanish Oncology group: SEOM; Non-Financial Interests, Member of Board of Directors, Member of the board diurectos in the Spanish Group of melanoma: GEM. D. Schadendorf: Financial Interests, Personal, Invited Speaker: BMS, MSD, Merck Serono, Regeneron; Financial Interests, Personal, Advisory Board: BMS, Novartis, MSD, Immunocore, Pierre Fabre, Regeneron, Philogen, BioAlta, Daiichi Sanyko, BioNTech, Immatics, CureVac, Formycon, Replimune, SunPharma, Skyline Dx, IOVance, Ipsen, IOBioTech, Boehringer-Ingelheim; Financial Interests, Institutional, Coordinating PI: Novartis, BMS, MSD, Pierre Fabre, Immatics; Financial Interests, Institutional, Research Grant: BMS, MSD; Financial Interests, Personal, Steering Committee Member: MSD; Financial Interests, Institutional, Local PI: Philogen, Immunocore, Regeneron, Biontech Rna Pharmaceuticals GMBH, Replimune, IO Biotech; Non-Financial Interests, Member of Board of Directors: EORTC-MG; Non-Financial Interests, Leadership Role, Founding member and SC chair: European Melanoma Registry (EuMelaReg). G.K. In: Financial Interests, Personal, Invited Speaker, Speaker for Merck from 2018-2020: Merck; Financial Interests, Personal, Advisory Board: Regeneron Pharmaceuticals, Replimune, Obsidian Therapeutics, Pfizer, Bristol Myers Squibb International, Novartis; Financial Interests, Institutional, Coordinating PI, PI on an investigator-initiated study: Regeneron Pharmaceuticals; Financial Interests, Institutional, Local PI: Regeneron Pharmaceuticals, Replimune, Iovance Biotherapeutics, Xencor, Checkpoint Therapeutics, Obsidian Therapeutics, Georgiamune, Bicara Therapeutics, Immunocore. G.M. Beasley: Financial Interests, Institutional, Research Funding: Replimune, Inc., Delcath, Philogen, Skyline; Financial Interests, Personal, Speaker, Consultant, Advisor, Payment/honoraria: Bristol Myers Squibb. J.J. Niu: Financial Interests, Personal, Advisory Board: AstraZeneca, Bristol Myers Squibb, Daiichi Sankyo, pfizer, sanofi, EMD Serono. B. Chmielowski: Financial Interests, Institutional, Research Funding: Bristol Myers Squibb, Macrogenics, Infinity Pharmaceuticals, Advenchen Laboratories, Xencore, Compugen, Iovance, RAPT Therapeutics, IDEAYA Biosciences, Ascentage, Atreca, Replimune, Inc., InstilBio, Adagene, Trisalus Life Sciences, Kinnate , PTC Therapeutics, Xilio Therapeutics, Kezar Life Sciences, Immunocore, AskGene Pharma, Krystal Bio, Nested Therapeutics, Pierre Fabre, Georgiamune, Immatics; Financial Interests, Personal, Advisory Board, Data monitoring board: Servier, SpringWorks Therapeutics; Financial Interests, Personal, Advisory Board: Regeneron, Treeline Biosciences, Atreca. T.M. Wise-draper: Financial Interests, Personal, Advisory Board: Merck & Co, Adaptimmune, EMD Serono, Genmab, Replimmune; Financial Interests, Personal, Other, Consultant for Molecular Tumor Board: Caris Life Sciences; Financial Interests, Personal, Other, Consultant: NEED INC; Financial Interests, Personal, Invited Speaker: Medscape; Financial Interests, Personal, Ownership Interest, Funded 10K as initial investment: High Enroll; Financial Interests, Personal, Research Grant: BMS, Merck & Co, Tesaro/GSK, AstraZeneca, Janssen; Financial Interests, Personal and Institutional, Local PI: BMS, EMD Serono, Replimune, AstraZeneca, Alkermes, Exicure, Shattuck Labs, Boston Medical, Debio Pharm, Eli Lilly, Epizyme, Iovance, Agenus, SQZbiotech, AdlaiNortye, Yingli, Vyriad, Ideaya, MINK, Sinogen, Takeda, duality Bio, Daiichi Sankyo, Coherus, Xencor, Bicara. M.M. Milhem: Financial Interests, Institutional, Research Funding: Amgen, Novartis, Merck, Pfizer, E.R. Squibb Sons, LLC, Prometheus. T.L. Bowles: Financial Interests, Institutional, Research Funding: Amgen Inc., Genentech, Natera, Iovance; Non-Financial Interests, Personal, Leadership Role: Society of Surgical Oncology, Melanoma Research Foundation, AIM at Melanoma. K.K. Tsai: Financial Interests, Institutional, Research Funding: ABM Therapeutics, AstraZeneca, BioAtla, Bristol Myers Squibb, Genentech, Georgiamune, Ideaya, Innovent, OnKure, Pfizer, Regeneraon, Replimune, Inc.; Financial Interests, Personal, Speaker, Consultant, Advisor, Consulting fees: Bristol Myers Squibb. C. Lebbe: Financial Interests, Personal, Advisory Board: Bristol Myers Squibb, MSD, Novartis, Roche, Pierre Fabre, Regeneron Pharmaceuticals; Financial Interests, Personal, Funding: Bristol Myers Squibb; Financial Interests, Coordinating PI: MSD; Non-Financial Interests, Advisory Role: Bristol Myers Squibb, MSD, Novartis, Pierre Fabre; Other, Honoraria, Speaker's bureau, Travel, Accommodations, Expenses, Board: MSD; Other, Honoraria, Travel, Accommodations, Expenses, Board: Pierre Fabre. C. Gaudy Marqueste: Financial Interests, Personal, Advisory Board: BMS, Pierre Fabre, MSD, Sun Pharma, Regeneron, Iovance; Financial Interests, Personal, Invited Speaker: BMS, Pierre Fabre, UCB; Financial Interests, Institutional, Local PI: BMS, MSD, Pfizer, Amgen, GSK, Roche, Novartis, Pierre Fabre, Replimune, Jansen, Kartos, IFX, Huyabio, Regeneron, IO Biotech, Philogen, Sotio, Kinnate, Astra Zenecca, Day One biopharmaceuticals, Sairopa, Catalyst, Sanofi, Immunocore, Daiichi, Dragonfly, Erasca, Microbiotica; Other, Travel expenses: BMS, Pierre Fabre, MSD, Janssen. A. Samson: Financial Interests, Institutional, Research Funding: CRUK, MRC, Histosonics, YCR; Financial Interests, Personal, Speaker, Consultant, Advisor, Consulting fees: Roche, Chugai; Financial Interests, Personal, Expert Testimony: Moore-Barlow LLP, McCollum Consultants; Financial Interests, Personal, Advisory Board: CUBIC; Other, Personal, Leadership Role: Leeds Cancer Research Centre. J. Zhu, J.W. Hou: Financial Interests, Personal, Full or part-time Employment: Replimune, Inc.; Financial Interests, Personal, Stocks or ownership: Replimune, Inc. M. Viana: Financial Interests, Personal, Full or part-time Employment: Replimune, Inc.; Financial Interests, Personal, Stocks or ownership: Replimune, Inc.; Financial Interests, Personal, Advisory Board: Replimune, Inc.; Other, Personal, Other, Patent: Zentalis Pharmaceuticals. M.K. Wong: Financial Interests, Personal, Advisory Board: Regeneron, Castle Biosciences, Replimune, Inc., Incyte, Sun Pharmaceuticals.

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1645P A phase Ib study to evaluate the safety, tolerability and efficacy of the combination of encorafenib, binimetinib and palbociclib in patients with BRAF-mutant metastatic melanoma (CELEBRATE)

J. Chan[1], J.R. Dixon Douglas[1], J. Xie[2], C. Khoo[1], J. Soon[1], A. Weppler[1], P. Parente[3], M. Shackleton[4], D. Kee[5], G. McArthur[1]

[1]*Department of Medical Oncology, Peter MacCallum Cancer Centre, Melbourne, Australia;* [2]*Centre for Biostatistics and Clinical Trials (BaCT), Peter MacCallum Cancer Center, Melbourne, Australia;* [3]*Department of Medical Oncology, Eastern Health - Box Hill Hospital - Building B, Box Hill, Australia;* [4]*Oncology, Alfred Hospital, Melbourne, Australia;* [5]*Medical Oncology, Austin Health - Austin Hospital, Heidelberg, Australia*

Background: Despite high initial response rates to BRAF/MEK inhibitors (BRAF/MEKi), most patients (pts) with *BRAFV600*-mutant (*BRAFV600*m) metastatic melanoma acquire secondary resistance. The cyclin D pathway is dysregulated in 90% of melanomas and converges with BRAF/ERK signalling at the CDK4/6-cyclinD1 complex. Increased cyclin D signalling promotes resistance to BRAF/MEKi and modulates anti-tumour immunity. Concurrent CDK4/6 and BRAF/MEK inhibition may overcome resistance pathways and improve outcomes.

Methods: CELEBRATE is an open-label, phase 1b, non-randomised dose escalation study evaluating the safety, tolerability and efficacy of encorafenib, binimetinib and palbociclib in pts with *BRAF V600m* melanoma. Eligible pts had unresectable stage III/IV melanoma; *BRAF V600E/K* and ECOG $\leq$ 2. Prior BRAF/MEKi and/or checkpoint inhibitors was allowed. Pts were enrolled using a 3 + 3 design with 75mg, 100mg and 125mg palbociclib daily day 1-21 every 28 days with encorafenib 450mg once daily plus binimetinib 45mg twice daily. Primary endpoints were maximum tolerated dose (MTD), dose-limiting toxicities (DLT) and recommended phase 2 dose (RP2D). Secondary endpoints included objective response rate (ORR), progression free survival (PFS) and overall survival (OS).

Results: A total of 23 pts were enrolled, 17 (74%) pts had prior BRAF/MEKi and 18 (78%) had M1c/d disease. One DLT occurred at each palbociclib dose level with G3 hyponatraemia at 75mg, G3 alkaline phosphatase rise at 100mg and G3 hepatic failure at 125mg. MTD and RP2D for palbociclib was 125mg. Common treatment related adverse events (TRAE all grades) were platelet count decreased (52%), fatigue (43%) and diarrhoea (39%). ORR was 65% [95% CI: 43 − 84), 65% with prior BRAF/MEKi [38 − 86]. Median follow up was 12.7 months (m), median PFS was 9.3m [95% CI: 3.6 − 12.7], 9.2m with prior BRAF/MEKi [3.6 − 11.2], and median OS was 16.1m [95% CI: 7.3 − 22.3], 14.4m with prior BRAF/MEKi [5.3-22.3].

Conclusions: Concurrent encorafenib, binimetinib and palbociclib has a manageable safety profile and promising anti-tumour activity against *BRAFV600*m metastatic melanoma, including pts who had prior BRAF/MEKi.

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1646P Efficiency of second-line BRAF/MEK inhibitors in BRAFV600 mutated metastatic melanoma patients after first-line immunotherapy failure: A EUMelaReg real-world study

E. Ellebaek[1], M. Weichenthal[2], I. Gavrilova[3], N. Asher[4], F. Meier[5], I. von Wasielewski[6], B. Dujović[7], V. Šabarić[8], T. Duvancic[9], A. Soria Rivas[10], L. Gutierrez Sanz[11], J.B.A.G. Haanen[12], I.M. Svane[1], P. Mohr[13], P.A. Ascierto[14], J. Mangana[15], P. Rutkowski[16], H. Gogas[17], L. Bastholt[18], D. Schadendorf[19]

[1]*Department of Oncology, National Center for Cancer Immune Therapy (CCIT-DK), Herlev Hospital, Herlev, Denmark;* [2]*Dermatology Department, Christian-Albrechts-University Kiel, Kiel, Germany;* [3]*Oncodermatology Department, USBAL Oncology EAD Sofia, Sofia, Bulgaria;* [4]*Department of Oncology, Skin Cancer and Melanoma Center at Davidoff Cancer Center, Rabin, Israel;* [5]*Skin Cancer Center, National Center for Tumor Diseases, Dept. of Dermatology, Un, Universitaetsklinikum Carl Gustav Carus Dresden, Dresden, Germany;* [6]*Department of Dermatology and Allergy, MHH - Medizinische Hochschule Hannover, Hannover, Germany;* [7]*Department of Oncology, Military Medical Academy VMA, Belgrade, Serbia;* [8]*Department of Oncology, KBC - University Hospital Centre Zagreb, Zagreb, Croatia;* [9]*Department of Dermatology and Venereology, Sestre Milosrdnice University Hospital Center, Zagreb, Croatia;* [10]*Medical Oncology Department, Hospital Universitario Ramon y Cajal, Madrid, Spain;* [11]*Medical Oncology Department, Hospital Universitario Puerta de Hierro, Madrid, Spain;* [12]*Division of Medical Oncology, Netherlands Cancer Institute, Amsterdam, Netherlands;* [13]*Department of Dermatology, Elbe Hospital Buxtehude, Buxtehude, Germany;* [14]*Melanoma, Cancer Immunotherapy & Developmental Therapeutics, Istituto Nazionale Tumori IRCCS Fondazione Pascale, Naples, Italy;* [15]*Dermatologie Department, USZ - Universitätsspital Zürich - Klinik für Dermatologie, Zurich, Switzerland;* [16]*Department of Soft Tissue/Bone Sarcoma and Melanoma, Maria Skłodowska-Curie National Research Institute of Oncology, Warsaw, Poland;* [17]*First Department of Medicine, National and Kapodistrian University of Athens, Athens, Greece;* [18]*Dept. of Oncology, OUH - Odense University Hospital, Odense, Denmark;* [19]*Department of Dermatology, Venereology and Allergology, University Hospital Essen, Essen, Germany*

Background: The use of BRAF/MEK inhibitors (BRAF/MEKi) in the first-line (1L) treatment has recently been deprioritized due to inferior outcomes compared to upfront immune checkpoint inhibitor (ICI) therapy. However, BRAF/MEKi are still the treatment of choice for BRAF mutated melanoma patients who have failed from 1L ICI. Our study aimed to evaluate the effectiveness of BRAF/MEKi in second-line (2L) in one of the largest real-world populations available.

Methods: This retrospective study evaluated 2,343 patients from the EUMelaReg registry diagnosed with BRAF^{V600} mutated non-resectable or advanced cutaneous melanoma who received BRAF/MEKi in 2L after failure of 1L ICI versus those who received BRAF/MEKi in 1L. Patients pre-treated with adjuvant ICI or BRAF/MEKi were excluded. For adjustment of imbalances of prognostic factors between 1L and 2L patients, a 1:1 inverse propensity score-based matching for several prognostic factors was performed. Key outcomes focused on overall survival (OS), progression-free survival (PFS), overall response rate (ORR), and time on treatment (TOT).

Results: 654 patients (28%) treated with 2L BRAF/MEKi were matched 1:1 to patients wo received BRAF/MEKi in 1L. Kaplan-Meier estimates showed a longer median PFS (8.4 [7.8-9.8] months vs 7.7 [6.9-8.3] months; p=0.01) and longer median TOT (7.8 [6.6-8.6] months vs 6.2 [5.8-6.9] months; p=0.002) for patients treated with 2L BRAF/MEKi compared to 1L BRAF/MEKi (95% CI for both). Median OS from start of BRAF/MEKi (17.2 [14.8-19.4] months vs 16.0 [13.2-19.1] months; p=0.73) and ORR (56.4% vs 53.5%; p=0.32) were similar in both groups.

Conclusions: In conclusion, our results support the preferred sequencing of ICI in 1L and BRAF/MEKi in 2L for BRAF V600 mutated melanoma patients by showing that the use of BRAF/MEKi in 2L does not compromise key outcomes, affirming their value in this sequencing approach.

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Gavrilova: Financial Interests, Personal, Advisory Role: Merck Sharp & Dohme, Bristol Myers and Squibb, Pierre Fabre, Medison; Financial Interests, Personal, Other, Honoraria: Roche, Novartis, Merck Sharp & Dohme, Bristol Myers Squibb, Sanofi, Merck, Pierre Fabre, Medison; Financial Interests, Speaker's Bureau: Roche, Novartis, Merck Sharp & Dohme, Bristol Myers and Squibb, Sanofi, Merck, Pierre Fabre, Medison; Financial Interests, Personal, Other, Travel/accommodations/expenses: Merck Sharp & Dohme, Bristol Myers and Squibb, Pierre Fabre. F. Meier: Financial Interests, Personal, Advisory Board: Bristol Myers Squibb, Novartis, MSD, Sanofi, Philogen, Immunocore; Financial Interests, Personal, Invited Speaker: Sanofi, Immunocore; Financial Interests, Personal, Advisory Board, Advisory Board: Regeneron. I. von Wasielewski: Financial Interests, Advisory Role: BMS, MSD, Sanofi; Financial Interests, Other, Honoraria payment: BMS, MSD, Sanofi, Novartis, Stemline, Kyowa Kirin; Financial Interests, Other, Travel/accommodations/expenses: BMS, MSD, Sanofi, Novartis, Stemline, Kyowa Kirin. V. Šabarić: Financial Interests, Other, Honoraria: Novartis, Bristol Myers and Squibb; Financial Interests, Other, Travel/Accommodation/Expenses: Novartis, MSD. A. Soria Rivas: Financial Interests, Personal, Speaker, Consultant, Advisor: Novartis, Bristol Myers and Squibb, MSD, Pierre Fabre, Regeneron, Pfizer. J.B.A.G. Haanen: Financial Interests, Institutional, Advisory Board: Bristol Myers Squibb, Achilles Therapeutics, Immunocore, Gadeta, Ipsen, Merck Sharpe & Dohme, Merck Serono, Pfizer, Molecular Partners, Novartis, Roche, Sanofi, Instil Bio, Iovance Biotherapeutics, AstraZeneca; Financial Interests, Institutional, Advisory Board, SAB member: BioNTech, PokeAcel, T-Knife; Financial Interests, Personal, Advisory Board, SAB member: Neogene Therapeutics, Sastra Cell Therapy; Financial Interests, Personal, Advisory Board: Third Rock Venture, Scenic, CureVac, Imcyse; Financial Interests, Personal, Stocks/Shares: Neogene Therapeutics; Financial Interests, Institutional, Research Grant: Bristol Myers Squibb, BioNTech US, Merck Sharpe & Dohme, Amgen, Novartis, Asher Bio, Sastra Cell Therapy; Non-Financial Interests, Member: ASCO, AACR, SITC; Other, Editor-in-Chief IOTECH: ESMO; Other, Editorial Board ESMO Open: ESMO; Other, Editorial Board: Kidney Cancer. I.M. Svane: Financial Interests, Institutional, Other, Conference participation: MSD; Financial Interests, Personal, Invited Speaker: BMS, Janssen; Financial Interests, Personal, Advisory Board: Mendus, Genmab; Financial Interests, Personal, Stocks/Shares, Cofounder and Founder warrents: IO Biotech; Financial Interests, Institutional, Research Grant: Adaptimmune, Enara Bio, Lytix Biopharma, TILT Biotherapeutics, Asgard Therapeutics, IO Biotech; Financial Interests, Institutional, Funding: Evaxion; Financial Interests, Institutional, Other, drug for investigator driven trial: BMS; Non-Financial Interests, Principal Investigator: BMS, Roche, TILT Biotherapeutics, Lytix Biopharma, Novartis, Immunocore, MSD. P. Mohr: Financial Interests, Personal, Advisory Board: Bristol Myers SquibbMerck Sharp & Dohme, Merck Sharp & Dohme, Novartis, GSK, Pierre Fabre, Sanofi, Delcath, Regeneron, Immunocore, Merck Serono; Financial Interests, Personal, Invited Speaker: Bristol Myers SquibbMerck Sharp & Dohme, Merck Sharp & Dohme, Novartis, Pierre Fabre, Sanofi, Sun Pharma, Delcath; Financial Interests, Institutional, Funding: Bristol Myers Squibb, Merck Sharp & Dohme, Novartis; Non-Financial Interests, Principal Investigator: Bristol Myers Squibb, Merck Sharp & Dohme, Regeneron, Sanofi, Novartis, Sun Pharma. P.A. Ascierto: Financial Interests, Personal, Other, Consultant and Advisory Role: BMS, Roche Genentech, Novartis, Merck Serono, Sun Pharma, Sanofi, Sandoz, Immunocore, Boehringer Ingelheim, Regeneron; Financial Interests, Personal, Other, Consultant, Advisory Role and Travel support: MSD, Pierre Fabre; Financial Interests, Personal, Other, Consultant and Advisory Role, Travel support: Pfizer/Array; Financial Interests, Personal, Other, Consultant Role: Italfarmaco; Financial Interests, Personal, Other, Consultant Role: Pfizer, Nouscom, Lunaphore; Financial Interests, Personal, Other, Consultant role: Medicenna; Financial Interests, Personal, Other, Consultant role and travel support: Bio-AI Health; Financial Interests, Personal, Advisory Board, Consultant and Advisory role: ValoTx; Financial Interests, Personal, Advisory Board, Consultant and Advisor role, Travel support: Replimmune; Financial Interests, Personal, Advisory Board, Advisor role: Bayer; Financial Interests, Personal, Other, Consultant and Advisory: Erasca; Financial Interests, Personal, Other, Consultant: Philogen, Incyte; Financial Interests, Personal, Other, Consultant and Advisory role: BionTech, Genmab; Financial Interests, Personal, Advisory Board: Anaveon, Menarini, ImCheck therapeutics; Financial Interests, Institutional, Funding, Clinical trial and translational research: BMS; Financial Interests, Institutional, Funding, Clinical Trial: Roche Genentech, Pfizer/Array, Sanofi; Non-Financial Interests, Leadership Role, President since 2010: Fondazione Melanoma Onlus Italy; Non-Financial Interests, Leadership Role, President since 2014: Campania Society of Immuno-Therapy of Cancer (SCITO) Italy; Non-Financial Interests, Other, Member of Steering Committee since 2016: Society for Melanoma Research (SMR); Non-Financial Interests, Member of Board of Directors, November 2017 - December 2021: Society for Immunotherapy of Cancer (SITC); Non-Financial Interests, Member: ASCO, SITC, EORTC Melanoma Cooperative Group, AIOM, SMR. J. Mangana: Financial Interests, Personal, Advisory Board: Merck Sharp & Dohme; Financial Interests, Personal, Advisory Board, immune-related Adverse Events Management (ESMO): Bristol Myers and Squibb; Financial Interests, Personal, Other, Travel grant: Ultrasun, L'oreal, Merck Sharp & Dohme, Pierre Fabre; Financial Interests, Personal, Invited Speaker: Johnson and Johnson; Financial Interests, Personal, Other, Travel Grant: Sun pharmaceutical; Financial Interests, Institutional, Research Grant, The melanoma registry database of the Department of Dermatology has been partially supported by an unrestricted grant to the University of Zurich from Amgen, Novartis, BMS, MSD and Pierre Fabre: Novartis, BMS, Pierre Fabre, Amgen, MSD; Non-Financial Interests, Leadership Role, President WG Melanoma Group: SAKK; Non-Financial Interests, Leadership Role, president WG Melanoma: SAKK; Other, immune-related AEs podcast /paid personal: Bristol Myers Squibb International. P. Rutkowski: Financial Interests, Personal, Invited Speaker, honoraria for lectures: MSD, BMS, Pierre Fabre; Financial Interests, Personal, Advisory Board: MSD, BMS, Pierre Fabre, Merck, Sanofi, Blueprint Medicines, Philogen, Medison Pharma, Genesis Pharma; Financial Interests, Personal, Invited Speaker: Merck, Sanofi, Novartis, AstraZeneca, Genesis Pharma; Financial Interests, Institutional, Research Grant, research grant for ISS: Pfzer; Financial Interests, Institutional, Funding, research grant for institution: BMS; Non-Financial Interests, Member of Board of Directors: Polish Society of Surgical Oncology; Non-Financial Interests, Member of Board of Directors, President: Polish Oncological Society. H. Gogas: Financial Interests, Personal, Advisory Board: MSD, BMS, Pierre Fabre, Sanofi; Financial Interests, Personal, Invited Speaker: MSD, BMS, Novartis, Pierre Fabre, Sanofi; Financial Interests, Steering Committee Member: Amgen, Replimmune; Financial Interests, Institutional, Local PI: Amgen, MSD, BMS, Replimmune, Iovance, Bayer; Financial Interests, Institutional, Research Grant: BMS, Pfizer, Lilly, Pierre Fabre, AstraZeneca. L. Bastholt: Non-Financial Interests, Advisory Role, Scientific committee under Danish Medicines Agency regarding new treatments of melanoma, skin cancer and thyroid cancer: Danish Medicines Agency. D. Schadendorf: Financial Interests, Personal, Invited Speaker: BMS, MSD, Merck Serono, Regeneron; Financial Interests, Personal, Advisory Board: BMS, Novartis, MSD, Immunocore, Pierre Fabre, Regeneron, Philogen, BioAlta, Daiichi Sanyko, BioNTech, Immatics, CureVac, Formycon, Replimune, SunPharma, Skyline Dx, IOVance, Ipsen, IOBioTech, Boehringer-Ingelheim; Financial Interests, Institutional, Coordinating PI: Novartis, BMS, MSD, Pierre Fabre, Immatics; Financial Interests, Institutional, Research Grant: BMS, MSD; Financial Interests, Personal, Steering Committee Member: MSD; Financial Interests, Institutional, Local PI: Philogen, Immunocore, Regeneron, BioNTech Rna Pharmacuticals GMBH, Replimune, IO Biotech; Non-Financial Interests, Member of Board of Directors: EORTC-MG; Non-Financial Interests, Leadership Role, Founding member and SC chair: European Melanoma Registry (EuMelaReg). All other authors have declared no conflicts of interest.

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1647P Dabrafenib/trametinib versus encorafenib/binimetinib in BRAF-mutant metastatic melanoma: A real-world propensity score matched survival analysis

Y. Garitaonaindía Díaz[1], S. Khan[1], R.B. Friis[2], S.K. Petersen[3], T.H. Borch[1], C.H. Ruhlmann[3], L.M. Guldbrandt[2], A.A. Luczak[4], H. Schmidt[2], L. Bastholt[3], I.M. Svane[1], M. Donia[1], E. Ellebaek[1]

[1]Department of Medical Oncology, Herlev Hospital - National Center for Cancer Immune Therapy (CCIT-DK), Herlev, Denmark; [2]Oncology Department, Institute of Clinical Medicine, Aarhus University Hospital Skejby, Aarhus, Denmark; [3]Department of Oncology, OUH - Odense University Hospital, Odense, Denmark; [4]Oncology Department, Aalborg University Hospital, Aalborg, Denmark

Background: Encorafenib is a BRAF inhibitor with a pharmacodynamic profile distinct from that of dabrafenib, featuring a longer dissociation half-life that may enable more sustained BRAF inhibition. It has been hypothesized that this could translate into superior clinical efficacy. We aimed to determine whether encorafenib/binimetinib (E/B) is superior to dabrafenib/trametinib (D/T) in terms of clinical activity and outcomes.

Methods: Patients were identified through the Danish Metastatic Melanoma Database (DAMMED), a national registry collecting prospective data on all patients undergoing systemic treatments. We retrospectively collected baseline, treatment, and clinical outcome information for patients with metastatic melanoma harboring BRAF mutations treated either with E/B or D/T from 2017 to 2023. Both unmatched and propensity score matched (PSM) analyses were conducted to account for potential residual confounding (despite initial balance).

Results: A total of 751 patients were included (422 D/T, 329 E/B). Baseline characteristics were balanced between groups: median age 64.5 vs. 65.0 years; ECOG 0−1 in 68.2% vs. 74.8%; elevated LDH at diagnosis in 64.4% vs. 64.6%; and liver metastases in 36.7% vs. 37.7%. Median follow-up was 28.8 months (D/T) and 45.8 months (E/B). In the unmatched cohort, no significant differences were observed between D/T and E/B in progression-free survival (PFS: median 7.94 vs. 8.02 months; HR 0.99, p=0.889), overall survival (OS: 15.48 vs. 15.36 months; HR 0.91, p=0.302), or melanoma-specific survival (MSS: 16.20 vs. 15.72 months; HR 0.94, p=0.498). PSM analyses confirmed these findings (PFS: HR 1.05, p=0.596; OS: HR 0.922, p=0.407; MSS: HR 0.97, p=0.740). Post-hoc power analysis for detecting a three months survival difference was adequate for PFS (97.5%) but limited for OS (56.3%) and MSS (52.1%).

Conclusions: We found no evidence that E/B is superior to D/T in the treatment of metastatic melanoma. These findings suggest that the choice between these combinations should be guided by tolerability profiles and economic considerations rather than efficacy.

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1648P Evaluating curative potential of lifileucel in previously treated advanced melanoma: Analyses from C-144-01 trial

J. Larkin[1], P. Lorigan[2], E. Dlotko[3], M. Kurt[4], L. Cox[5], D. Freigofaite[3], J. Wang[6], J. Jiang[6], X. Wu[7], P. Mohr[8]

[1]Medicine Department, The Royal Marsden Hospital - Chelsea, London, United Kingdom; [2]Medical Oncology Dept., The Christie NHS Foundation Trust, Manchester, United Kingdom; [3]Health Economics and Outcomes Research, Access, FIECON Limited, London, United Kingdom; [4]Global Health Economics & Outcomes Research, Iovance Biotherapeutics, Inc. - Headquarters, San Carlos, United States of America; [5]Health Economics & Outcomes Research, Access, Fiecon Limited, London, United Kingdom; [6]Global Patient Operations, Value and Access, Iovance Biotherapeutics, Inc. - Headquarters, San Carlos, United States of America; [7]Biostatistics, Iovance Biotherapeutics, Inc. - Headquarters, San Carlos, United States of America; [8]Dermato-Oncology Department, Dermatologic Center Buxtehude, Buxtehude, Germany

Background: In the single-arm phase 2, C-144-01 trial, lifileucel, an autologous tumor-infiltrating lymphocyte cell therapy, showed durable survival and response benefits for unresectable or metastatic melanoma patients who are relapsed or refractory to immune checkpoint inhibitors. The trial exhibited survival plateaus that can be attributed to potential long-term survivors (LTS). This study investigated the underlying fractions of LTS in the trial.

Methods: Mixture cure models (MCMs) were applied to both overall survival (OS) and progression-free survival (PFS) outcomes using patient-level data (n=106) from the trial with 47.2-months of median follow-up. In the MCMs, patients were classified in two mutually exclusive latent subgroups as LTS and non-LTS. In both PFS and OS analysis, LTS were subject to only non-melanoma related mortality and non-LTS were subject to all-cause mortality. In the PFS analysis, non-LTS were also subject to risk of disease progression. The survival trend of LTS was assumed to follow that of age- and sex-adjusted general population, and derived from US lifetables. PFS and OS for non-LTS were modeled by parametric survival functions and estimated along with the corresponding proportions of LTS via maximum likelihood methods. Candidate models were examined based on statistical goodness-of-fit, clinical plausibility and visual comparison to reported survival data and underlying hazard trends from the trial.

Results: Exponential MCM and log-normal MCM provided the best fit to the observed OS and PFS data, respectively, and estimated the proportion of LTS as 25.7% [95% CI : 16.7% - 35.7%] from the OS data and as 17.0% [95%CI: 10.0% - 25.5%] from the PFS data. Corresponding 10-year restricted mean OS and PFS were 39.6 and 24.3 months, respectively. For both PFS and OS, smoothed hazard rates in the observed data showed convergence towards age- and sex-adjusted general population mortality rates around year 4 indicating maturity of data for analysis with MCMs.

Conclusions: Survival plateaus and hazard trends in C-144-01 trial can be adequately captured by MCMs. Estimated fractions of LTS in the trial highlight lifileucel's potential in fulfilling the unmet need in previously treated unresectable or metastatic melanoma.

Clinical trial identification: Single arm, phase II, C-144-01 trial. NCT02360579.

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1649P First real-world data for TIL therapy in patients with metastatic cutaneous melanoma

W.S. Tak[1], I. Jedema[2], T. Seijkens[3], B. Nuijen[4], S. Kouwen[5], C. Nijenhuis[6], M. Klous[7], E. van den Haak[8], I. Dirven[9], M. Delfos[10], A. Daletzakis[11], W. Van Houdt[12], A. Kuijpers[13], K.J. Hartemink[14], C.U. Blank[3], S. Wilgenhof[3], A. Meerveld - Eggink[9], H.V. van Thienen[9], J.B.A.G. Haanen[3], S. Klobuch[3]

[1]Division of Molecular Oncology and Immunology, Netherlands Cancer Institute, Antoni Van Leeuwenhoek, Amsterdam, Netherlands; [2]Molecular Oncology & Immunology Dept., Netherlands Cancer Institute, Amsterdam, Netherlands; [3]Medical Oncology Department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [4]Division of Pharmacy and Pharmacology, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [5]Biotherapeutics Unit, Netherlands Cancer Institute, Amsterdam, Netherlands; [6]Biotherapeutics Unit, Netherlands Cancer Institute, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [7]Division of Pharmacy and Pharmacology, Netherlands Cancer Institute, Amsterdam, Netherlands; [8]Division of Molecular Oncology and Immunology, Netherlands Cancer Institute, Amsterdam, Netherlands; [9]Division of Medical Oncology, Netherlands Cancer Institute, Amsterdam, Netherlands; [10]Division of Medical Oncology, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [11]Division of Biometrics, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [12]Surgical Oncology Department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [13]Division of Surgical Oncology, Netherlands Cancer Institute, Amsterdam, Netherlands; [14]Surgery Department, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands

Background: Patients with advanced melanoma face limited treatment options after progression on immune checkpoint inhibitors (ICI). Tumor-Infiltrating Lymphocyte (TIL) therapy has shown promising efficacy in this setting, with response rates of 49% in a randomized phase III trial. Based on this trial, hospital exemption (HE) was granted to the Netherlands Cancer Institute (NKI) in January 2023, allowing treatment of advanced melanoma patients with the TIL product TM001. Here, we present real-world data on TM001 therapy in advanced melanoma patients treated within the ongoing HE program.

Methods: Clinical data was retrospectively collected from advanced melanoma patients who received TM001 infusion followed by high-dose interleukin-2 (IL-2), after non-myeloablative lymphocyte-depleting chemotherapy (data cut-off 6 November 2024). We analyzed progression-free survival (PFS), overall survival (OS), objective response rate (ORR) according to RECIST v.1.1 and treatment emergent adverse events (TEAEs).

Results: Fifty patients with advanced melanoma received TM001 therapy within the HE program. All patients experienced disease progression after anti-programmed death-1 (PD-1) containing ICI treatment. Median follow-up was 9.4 months (95% confidence interval = 6.7 – 16.4). Median PFS was 3.5 months after TM001 infusion, ORR was 34% with an 8% complete response rate, and the median OS was 13.9 months. Patients pretreated with anti-PD-1 monotherapy showed an ORR of 39% and patients pretreated with anti-PD-1/anti-CTLA-4 also responded to TM001 therapy, with an ORR of 24% and a similar PFS. Grade $\geq$3 TEAE were observed for all patients. One serious adverse event, a grade 4 ventricular tachycardia related to IL-2 infusion, was recorded. All patients experienced chemotherapy-related TEAEs, 90% experienced IL-2 related TEAEs and 96% experienced any TEAE related to the TM001 infusion.

Conclusions: Real-world data from advanced melanoma patients treated with TM001 therapy within an HE program, following prior anti-PD-1 containing ICI treatment, confirm efficacy and safety profiles from the previous randomized controlled trial. Notably, patients pretreated with anti-PD-1/anti-CTLA-4 also demonstrate responses to TM001 therapy.

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1650P Real-world outcomes of tumor-infiltrating lymphocyte (TIL) therapy for metastatic melanoma patients in 2025

S. Grynberg[1], O. Itzhaki[2], D. Zikich[2], Y. Eshet[3], S. Lev Ari[4], R. Harel-Mutai[4], E. Mor[5], M. Bar[4], J. Schachter[4], R. Shapira-Frommer[6]

[1]Ella Lemelbaum Institute for Immuno-Oncology, Sheba Tel Hashomer Medical Center, Ramat Gan, Israel; [2]Cell Therapy, The Ella Institute for Treatment and Research of Melanoma and Skin Cancer - Sheba Medical Center, Ramat Gan, Israel; [3]Nuclear Medicine, Sheba Medical Center, Ramat Gan, Israel; [4]Oncology, The Ella Institute for Treatment and Research of Melanoma and Skin Cancer - Sheba Medical Center, Ramat Gan, Israel; [5]Surgical Oncology, Sheba Medical Center, Ramat Gan, Israel; [6]Oncology Department, The Ella Institute for Treatment and Research of Melanoma and Skin Cancer - Sheba Medical Center, Ramat Gan, Israel

Background: Immune checkpoint inhibitors (ICIs) have transformed metastatic melanoma care, yet response rates (RRs) to even the best combinations rarely exceed 60%, and many patients eventually progress. Adoptive cell therapy with tumor-infiltrating lymphocytes (ACT-TIL) has shown promise in refractory melanoma, with durable responses in a subset. However, real-world data following modern ICI regimens, including anti−PD-1 + anti−CTLA-4 and in non-cutaneous melanoma, remain limited. To address this, we analyzed outcomes in a cohort of heavily pretreated melanoma patients treated with ACT-TIL from 2019−2025.

Methods: Eligible patients (age 18−70, ECOG 0−1) with metastatic melanoma and prior ICI failure were treated under IRB protocol 3566-16. Inclusion required adequate organ function; stable or small brain metastases ($<$1 cm) were permitted. Treatment included lymphodepleting chemotherapy, TIL infusion, high-dose IL-2, and, from 2021, one nivolumab dose pre-/post-infusion. Follow-up included clinical/lab assessments and imaging at defined intervals. Primary endpoints: (1) Efficacy: $\geq$30% clinical benefit rate (CBR) (CR + PR + SD $>$12 weeks); (2) Safety: treatment-related toxicity. Secondary endpoints: PFS, OS, and QoL (EORTC QLQ-C30 v3.0).

Results: Thirty-four patients (median age 48; 50% female) received ACT-TIL. Twenty-five had cutaneous melanoma (incl. 2 acral, 1 conjunctival); 8 BRAF V600 mutant; 4 had brain metastases. Patients had a median of 2 prior lines; 50% had primary progression to ICI. Median ICI exposure was 11 months, with a 4-month median interval to TIL. Median IL-2 doses: 3; infused cells: 6.18×10^{10}. Twelve patients received further therapy post-TIL without notable AEs. At 3 months, RR was 20.6% (7/34) overall and 28% (7/25) in cutaneous melanoma. PFS/OS data are maturing. Toxicity was consistent with prior TIL data.

Conclusions: TIL therapy offers potential for durable responses in metastatic melanoma, but outcomes were modest in this heavily pre-treated cohort, especially post anti−PD-1 + anti−CTLA-4. Non-cutaneous melanoma showed no benefit. These findings support earlier use of TIL, though optimal sequencing with ICIs and BRAF inhibitors remains unresolved.

Clinical trial identification: NCT03166397.

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1652P A phase I clinical trial of autologous tumor-infiltrating lymphocytes FAST-TIL (HS-IT101) with low-dose IL-2 for the treatment of advanced solid tumors: Subgroup analysis of melanoma

Y. Jiang[1], Y. Fang[2], Z. Guo[3], J. Liu[1], Y. Deng[1], J. Lin[4], S. Lan[3], S. Wang[2], Z. Sun[5], L. Jiang[5], W. Li[5], W. Liu[5], Z. Li[5], P. Wang[5], X. Zhang[5], Y. Zhao[5], Y. Chen[4], D. Wu[3], N. Li[2]

[1]Medical Oncology Dept., West China School of Medicine/West China Hospital of Sichuan University, Chengdu, China; [2]Clinical Trials Center, National Cancer Center/National Clinical Research Center for Cancer/Cancer Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China; [3]Cancer Center, The First Hospital of Jilin University, Changchun, China; [4]Department of Medical Oncology, Fujian Cancer Hospital and Fujian Medical University Cancer Hospital, Fuzhou, China; [5]Research and Development Dept., Qingdao Sino-Cell Biomedicine Co., Ltd., Qingdao, China

Background: Adoptive cell therapy (ACT) based on tumor-infiltrating lymphocytes (TIL) has shown significant efficacy in immunotherapy checkpoint inhibitor (ICI)-resistant melanoma. Yet challenges such as long manufacturing time , high-dose non-myeloablative conditioning (HD-NMA) and high-dose IL-2 infusion were linked to severe treatment-related adverse events (TRAEs).We developed FAST-TIL (HS-IT101), an innovative TIL-ACT product with a 14-day manufacturing time. FAST-TIL is now in phase I clinical trial with optimized lymphodepletion conditioning and low-dose IL-2 infusion, to evaluate its safety and efficacy.

Methods: The phase I trial of HS-IT101 is an open-label, single-arm, multi-center trial (NCT06342336) in patients(pts) with advanced solid tumors. The primary endpoint is safety, with efficacy and pharmacokinetics as the secondary endpoint.

Results: As of Feb 2025, 8 pts with advanced melanoma were enrolled (5 males and 3 females), including 1 cutaneous melanoma, 5 acral melanoma, and 2 mucosal melanoma. All pts were resistant to ICI treatment. 2 pts received low-dose non-myeloablative conditioning(LD-NMA) (Day -5 to -3: Cyclophosphamide (Cy) 300 mg/m^2 qd; Fludarabine (Flu) 30 mg/m^2 qd), and 6 pts received middle-dose non-myeloablative conditioning(MD-NMA) (Day -4 to -2: Cy: 750 mg/m^2, qd; Day -4 to -1: Flu: 30 mg/m^2, qd). After TIL infusion, pts received subcutaneous IL-2 injections at 2 MIU/m^2, qd for 3 days. Fifty percent (4/8) of the pts experienced grade 3 non-hematologic toxicity, with no grade 4-5 events reported; all adverse reactions responded quickly to treatment with little complications. The overall objective response rate (ORR) was 50% (4/8). The ORR in the MD-NMA group was 66.7% (4/6), with 1 patient achieved complete response (CR) and 3 pts achieved partial responses (PR). TCR clonal analysis showed that clones of TIL infusion product persisted long-term in the peripheral blood.

Conclusions: HS-IT101 injection demonstrated a favorable safety profile and promising efficacy in patients with advanced cutaneous and non-cutaneous melanoma.

Clinical trial identification: HS-IT101ST01-I 2024-03-19.

Legal entity responsible for the study: Qingdao Sino-Cell Biomedicine Co., Ltd., Qingdao, China.

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1653P CRISPR-PD1 modified tumor infiltrating lymphocytes for adoptive therapy for patients with metastatic melanoma

J.E. Sohlin, M.D. Crowther, S.M. Hammer Mannering, S.E. Weis-Banke, Ö. Met, I.M. Svane

Department of Oncology, Herlev Hospital - National Center for Cancer Immune Therapy (CCIT-DK), Herlev, Denmark

Background: Immune checkpoint inhibitors and adoptive cell therapy using tumor-infiltrating lymphocytes (TILs) have improved outcomes in metastatic melanoma (MM), but durable responses remain limited. Gene engineering strategies, such as PD-1 knockout, may enhance TIL efficacy by overcoming tumor-induced immunosuppression. We previously demonstrated that non-viral CRISPR-Cas9 delivery can efficiently disrupt PD-1 in TILs (Chamberlain et al, 2022). Here, we report early feasibility data from the first-in-human trial testing non-viral CRISPR-Cas9 PD-1 knockout in TILs (CRISPR-TILs) in advanced MM.

Methods: Patients with metastatic or inoperable cutaneous melanoma progressing after first-line checkpoint inhibitors were enrolled (NCT06783270). PD-1 knockout was integrated into the in-house GMP TIL manufacturing process using non-viral CRISPR-Cas9. Following lymphodepletion (cyclophosphamide, fludarabine), patients received $\geq5\times10^9$ CRISPR-TILs and up to six doses of high-dose IL-2. Infusion products were assessed for PD-1 knockout efficiency, cellular composition, memory phenotype, and tumor-reactivity by flow cytometry. Clinical responses were evaluated per RECIST 1.1, and safety was monitored throughout.

Results: Three patients have received CRISPR-TILs. Infusion products showed efficient PD-1 disruption, preserved cellular composition and effector memory phenotype. Tumor recognition assays confirmed antigen-specific reactivity and preserved cytokine production post-editing. No unexpected adverse events or increased severity of known toxicities were observed. At six weeks post-infusion, two patients achieved objective responses per RECIST 1.1.

Conclusions: These initial feasibility data from a first-in-human clinical trial demonstrate that non-viral CRISPR-Cas9−mediated PD-1 knockout can be efficiently integrated into TIL manufacturing workflow and safely administered to patients with metastatic melanoma, showing promising early antitumor activity. Recruitment is ongoing, and continued follow-up will provide further insights into the therapeutic potential and durability of CRISPR-engineered TIL therapy.

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1654P Impact of tumor infiltrating lymphocyte therapy on the outcome of BRAF and MEK inhibitor treatment for advanced melanoma

S. Pedersen[1], E. van den Haak[2], M. Presti[1], M. Donia[1], S. Klobuch[2], J.B.A.G. Haanen[2], I.M. Svane[1], T.H. Borch[1]

[1]Department of Oncology, National Center for Cancer Immune Therapy, Herlev Hospital, Herlev, Denmark; [2]Department of Medical Oncology, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands

Background: Adoptive cell therapy with tumor-infiltrating lymphocytes (TILs) has documented clinical efficacy for patients with advanced melanoma and was FDA approved in patients failing anti-PD-1 based immunotherapy and, if BRAF V600 mutant, BRAF/MEK inhibitor (BRAF/MEKi) therapy. Treatment with TILs can induce long-term responses and potentially cure patients with complete response. However, a recent study suggests that BRAF/MEKi prior to TIL therapy significantly reduces response rate to TILs. Thus, the optimal sequencing of TIL therapy is uncertain. To further elucidate this question, we evaluated treatment responses to BRAF/MEKi after progression on TIL therapy.

Methods: We conducted a multinational retrospective study of patients included in clinical trials (NCT00937625, NCT02354690, NCT02278887, and NCT02379195) who received BRAF/MEKi after progression on TIL therapy in Denmark and the Netherlands. Data were extracted from electronic patient records. Outcomes were compared to a propensity-matched cohort (matched on treatment line, performance status, LDH level, and brain metastases) of anti-PD-1 refractory patients from the Danish Metastatic Melanoma Database (DAMMED) who received BRAF/MEKi without prior TIL therapy.

Results: A total of 41 patients were included. Median number of prior treatment lines was three. The overall response rate to BRAF/MEKi was 75%. Median progression free survival (PFS) and median overall survival (OS) were 8.9 and 13.8 months, respectively. Five patients were still alive at three years (12%). Propensity matching showed similar PFS and OS curves for patients receiving BRAF/MEKi after TIL therapy compared to BRAF/MEKi without prior TIL therapy.

Conclusions: BRAF/MEKi treatment after failure of TIL therapy appears at least as effective as when given after progression on anti-PD-1 alone. Given the curative potential of TIL therapy and possible reduced efficacy following BRAF/MEKi, these findings support the use of TILs before progression on BRAF/MEKi in patients with advanced melanoma.

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1655P Translational data validate OBX-115 mechanism of action: Impact of dosing on clinical outcome in advanced (adv) melanoma

G.K. In[1], J.A. Chesney[2], A.N. Shoushtari[3], J.T. Moyers[4], Y. Samhouri[5], T. Johnson[6], R. Amaria[7], G. Ramsingh[8], C. Renard[8], B.A. Aksoy[9], R. Burga[10], M. Reuter[11], P. Prabhakar[11], L. Mclaughlin[11], A. Betof Warner[12]

[1]Medicine Dept., USC Norris Comprehensive Cancer Center, Los Angeles, United States of America; [2]Department of Medicine, UofL Health - Brown Cancer Center, Louisville, United States of America; [3]Melanoma Oncology and Cell Therapy, Memorial Sloan Kettering Cancer Center, New York, United States of America; [4]Clinical Translational Research and Immunotherapeutics, The Angeles Clinic & Research Institute, A Cedars-Sinai Affiliate, Santa Monica, United States of America; [5]Division of Hematology and Cellular Therapy, Allegheny Health Network Cancer Institute, Pittsburgh, United States of America; [6]Department of Hematology and Oncology, Orlando Health Cancer Institute, Lake Mary, United States of America; [7]Department of Melanoma Medical Oncology, The University of Texas MD Anderson Cancer Center, Houston, United States of America; [8]Clinical Development, Obsidian Therapeutics, Inc., Cambridge, United States of America; [9]Computational Biology, Obsidian Therapeutics, Inc., Cambridge, United States of America; [10]Cell Therapy, Obsidian Therapeutics, Inc., Cambridge, United States of America; [11]Clinical Operations, Obsidian Therapeutics, Inc., Cambridge, United States of America; [12]Medicine Department, Stanford University School of Medicine, Stanford, United States of America

Background: OBX-115 TIL express membrane-bound IL15 (mbIL15) regulated by the FDA-approved small-molecule drug acetazolamide (ACZ), abrogating the need for IL2 and resulting in differentiated safety and promising efficacy (Amaria ASCO 2024, Chesney ASCO 2025). We report translational data from pts with adv melanoma treated with escalating dose levels (DL) of cryopreserved OBX-115 and ACZ in the multicenter Agni-01 study to further validate OBX-115 mechanism of action.

Methods: This single-arm open-label ph 1/2 study (NCT06060613) assesses the OBX-115 regimen in pts with adv melanoma and NSCLC. After lymphodepletion (LD), OBX-115 is infused at doses of $\leq 30 \times 10^9$ (DL1) or $\leq 100 \times 10^9$ (DL2, DL3) cells. Oral ACZ is administered daily ($\leq$14 d) after OBX-115 infusion (Day 0 [D0]) at 250 (DL1, DL2) or 500 (DL3; D0−6, D14−20) mg and can be redosed as an outpatient ($\leq$7 d) Q6W starting ∼D35. OBX-115 infusion product and Baseline (BL) and post-infusion PBMC and tumor (as available) were analyzed.

Results: Among the first 11 pts with melanoma, 10 (91%) received low-dose LD (Cy 750 mg/m^2/d $\times$3, Flu 30 mg/m^2/d $\times$4). ORR was 67% (4/6, 1 CR) at DL3 (RP2D); 3 responses were ongoing past Wk 24. Initial OBX-115 expansion was associated with cumulative ACZ dose. PBMC TCR seq showed dose-dependent persistence and repertoire remodeling by OBX-115−derived clonotypes. In 3 pts at RP2D with BL and post-infusion tumor biopsies, TCR seq showed tumor clonotype remodeling; RNA FISH confirmed OBX-115 TIL infiltration and persistence. Spectral flow cytometry demonstrated peripheral expansion of infused CD8+ cells and a surge of endogenous NK and NK-like T cells during lymphocyte recovery from LD.

Table: 1655P Pre- and post-infusion immune profile				
Characteristic, median	N	Pre-infusion (N=11)	Post-infusion DL1 / DL2 (n=5)	Post-infusion DL3 (RP2D) (n=6)
Cumulative ACZ dose received by D28, mg	11	NA	2500	7000
OBX-115 peak in PBMC first 28 d (ddPCR), cell/μL	10*	NA	175.4	506.1
OBX-115 TCR clonotypes in PBMC at D42, %	7*	NA	10.9	33.1
OBX-115 TCR clonotypes in tumor at D42, %	3	NA	21.9	82.2
CD8+ peak (of CD3+ in PBMC), %	10	25.7	77.7	76.6
CD3-CD56+ peak (of PBMC), %	10	4.9	26.3	27.1
CD3+CD56+ peak (of PBMC), %	10	1.1	4.9	2.2

*Excluding 1 asplenic patient.

Conclusions: ACZ dose-dependent mbIL15 regulation supports expansion, persistence, and infiltration of OBX-115 into tumors, remodeling of PBMC and tumor TCR repertoire, and cis- and trans-immune cell activation. This regimen allows for low-dose LD and no IL2, which may expand pt eligibility. Ph 2 enrollment is ongoing.

Clinical trial identification: NCT06060613.

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Disclosure: G.K. In: Financial Interests, Institutional, Research Grant: Regeneron, Array, Idera, Replimune, Xencor, Instil Bio, Pfizer, Checkmate Pharmaceuticals; Financial Interests, Personal, Advisory Role: Sanofi, Pfizer; Financial Interests, Personal, Advisory Board: BMS, Merck, Regeneron, Array, Castle Biosciences, Sanofi, Replimune, Pfizer; Financial Interests, Personal, Leadership Role: Replimune. A.N. Shoushtari: Financial Interests, Institutional, Research Grant: Bristol Myers Squibb, Novartis, Immunocore, Targovax, Pfizer, Alkermes, Checkmate Pharmaceuticals, Foghorn Therapeutics, Linnaeus Therapeutics, Iovance Biotherapeutics, Prelude Therapeutics; Financial Interests, Personal, Advisory Role: Bristol Myers Squibb, Novartis, Erasca. J.T. Moyers: Financial Interests, Personal, Speaker's Bureau: Clinical Care Options; Financial Interests, Personal, Speaker, Consultant, Advisor: Medical Oncology Association of Southern California, Scripps Oncology Review; Financial Interests, Personal, Advisory Board: Replimune; Financial Interests, Personal, Other, DSMB: Houston Methodist. T. Johnson: Financial Interests, Personal, Speaker, Consultant, Advisor: ASCO Direct™ Hawaii Conference; Financial Interests, Personal, Leadership Role: Florida Association of Clinical Oncology H&N Disease Management Team Leader. R. Amaria: Financial Interests, Institutional, Coordinating PI: Merck, Bristol Myers Squibb, OnKure, Erasca, KSQ, Regeneron Pharmaceuticals; Financial Interests, Institutional, Local PI: Roche; Financial Interests, Institutional, Trial Chair: Obsidian. G. Ramsingh, C. Renard, B.A. Aksoy, R. Burga, M. Reuter, P. Prabhakar, L. Mclaughlin: Financial Interests, Personal, Full or part-time Employment: Obsidian Therapeutics; Financial Interests, Personal, Stocks/Shares: Obsidian Therapeutics. A. Betof Warner: Financial Interests, Personal, Advisory Board: Bristol Myers Squibb, Immatics, Instil Bio, Iovance Biotherapeutics, Lyell Immunopharma, Novartis, Merck, Pfizer, cTRL Therapeutics, Genmab, IO Biotech, Replimune; Financial Interests, Personal, Other, Consultant: Adaptimmune; Financial Interests, Institutional, Coordinating PI: Bristol Myers Squibb International; Financial Interests, Institutional, Local PI: Lyell Immunopharma, Obsidian Therapeutics, Replimune; Financial Interests, Institutional, Steering Committee Member: Replimune; Non-Financial Interests, Member: ASCO, AACR; Non-Financial Interests, Leadership Role: SITC. All other authors have declared no conflicts of interest.

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1656P Clinical and molecular outcomes of tebentafusp in metastatic uveal melanoma (MUM): A retrospective cohort of 168 patients

R. Sanchez[1], T. Ramtohul[2], V. Servois[2], G. Pierron[3], P. Mariani[4], N. Cassoux[5], A. Matet[5], S. Piperno-Neumann[1], M.J. Rodrigues[1]

[1]Department of Medical Oncology, PSL Research University, Institut Curie, Paris, France; [2]Radiology Department, PSL Research University, Institut Curie, Paris, France; [3]Somatic Genetic Unit, Department of Genetics, PSL Research University, Institut Curie, Paris, France; [4]Department of Surgical Oncology, PSL Research University, Institut Curie, Paris, France; [5]Ophtalmology Department, Institut Curie, Paris, France

Background: Tebentafusp, a bispecific T-cell receptor targeting gp100, improved overall survival (OS) in HLA-A*02:01-positive patients (pts) with MUM in the IMCgp100-202 phase 3 trial (median OS: 21.6 months). However, predictive factors of efficacy remain unclear. This study evaluates clinical outcomes in Tebentafusp-treated patients at Institut Curie, Paris, France, with subgroup analyses by mutational and chromosomal profiles.

Methods: Retrospective collection of clinical and genomic data from MUM pts treated between 2018 and 2024. Outcomes included objective response rate (ORR) per RECIST 1.1, progression-free survival (PFS), and OS. Survival was analyzed using Kaplan−Meier estimates and the log-rank test.

Results: Among 168 patients (18 from IMCgp100-202, 150 from early access program), Tebentafusp was first-line treatment in 82% pts (n=138). Median follow-up was 20 months, median PFS was 3 months, and the 20-month OS rate was 67%. ORR was 8.9% (n=15) and 46% pts (n=78) had stable disease. Pts with extra-hepatic metastases only (n=4) had the best outcomes (median PFS and OS not reached), while those with hepatic disease only (n=130) or hepatic/extra-hepatic (mixed) disease (n=34) had shorter PFS (3 months in both cases) and OS (28 months for hepatic, 18 for mixed). *EIF1AX* mutations were associated with sustained disease control and no deaths. *BAP1* and *SF3B1* mutations were associated with short PFS (median 5 and 8 months, respectively) and OS (median not reached and 27 months, respectively). Disomy 3 tended to be correlated with longer PFS (8 months) and OS (30 months) than monosomy 3 (PFS: 3 months; OS: 28 months).

Conclusions: These findings support the efficacy of tebentafusp in MUM patients with outcomes in line with the IMC-gp100-202 trial and suggest that genomic and metastatic features may help refine prognostic stratification. Further prospective studies are warranted to disentangle prognostic from predictive biomarkers in this setting.

Legal entity responsible for the study: Institut Curie.

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1657P De-squamate study: Event-free survival outcomes in clinical or pathological complete responders

R. Ladwa[1], B.G.M. Hughes[2], B. Panizza[3], C. Cooper[4], H. Liu[1], J. Bowman[3], R. Gupta[5], C. Cuscaden[6], M.K. Nottage[2], J. Clarke[7], M. McGrath[1], S.V. Porceddu[8], J. Lee[9]

[1]Cancer Services, Princess Alexandra Hospital - Metro South Health, Woolloongabba, Australia; [2]Cancer Care Services, Royal Brisbane and Women's Hospital, Herston, Australia; [3]ENT, Princess Alexandra Hospital - Metro South Health, Woolloongabba, Australia; [4]Pathology, Princess Alexandra Hospital - Metro South Health, Woolloongabba, Australia; [5]Pathology, Chris O'Brien Lifehouse, Camperdown, Australia; [6]Radiology, Princess Alexandra Hospital - Metro South Health, Woolloongabba, Australia; [7]Surgical Oncology, Chris O'Brien Lifehouse, Camperdown, Australia; [8]Radiation Oncology, Peter McCallum Cancer Centre, Melbourne, Australia; [9]Department of Medical Oncology, Chris O'Brien Lifehouse, Sydney, Australia

Background: The De-Squamate study met its primary endpoint with a high rate of clinical or pathological complete response (cpCR) observed in 17 (63%) patients, comprised of a pathological complete response (pCR) in 4 (15%) and a clinical complete response (cCR) in 13 (48%) following pembrolizumab for resectable cutaneous squamous cell carcinoma. This allowed for omission of planned surgery and radiotherapy (total de-escalation) in 48% and omission of adjuvant radiotherapy (partial de-escalation) in 15%. The outcomes of patients achieving a cpCR are unknown.

Methods: We conducted a phase 2, multicenter study to evaluate pembrolizumab in patients with resectable stage II-IV(M0) cSCC. Patients received 4 cycles of 3 weekly pembrolizumab before proceeding to ^{18}F-FDG-PET assessment. Patients who achieved a cCR, defined as a complete metabolic response and negative mapping biopsies of the target site(s) had total de-escalation whilst those without a cCR underwent surgery with partial de-escalation if achieving a pCR. Maintenance pembrolizumab was given for a further 13 cycles in patients achieving a cpCR. Patients underwent CT/MRI scans and clinical review every 12 weeks during follow-up. The secondary endpoint of event-free survival (EFS) was defined as the time from first dose of pembrolizumab to recurrence, progressive disease during upfront pembrolizumab precluding surgical resection, or death.

Results: A total of 27 patients received pembrolizumab. At a median follow-up of 24 months, 7 events (26%) had occurred including 6 deaths. Three deaths occurred secondary to disease, 1 of which precluded surgery, while 3 resulted from serious adverse events unrelated to study drug. Recurrence following surgery was observed in 3 patients, none of whom had achieved a pCR, leading to death in 2 patients. There were no recurrences observed in patients who achieved a cpCR. The median EFS was not reached in this group, with a 24-month EFS rate of 94% (95% CI: 76%-99%). In those without a cpCR, the median EFS was 7 months (95% CI: 0-14.7) and the 24 months EFS rate 40% (95% CI: 15%-70%).

Conclusions: Patients achieving a cpCR following pembrolizumab maintained a high EFS rate demonstrating the potential for treatment de-escalation without compromising longer term outcomes.

Clinical trial identification: NCT05025813.

Legal entity responsible for the study: Metrosouth hospital and health services.

Funding: Merck Sharpe and Dohme.

Disclosure: R. Ladwa: Financial Interests, Institutional, Research Grant: MSD; Financial Interests, Institutional, Speaker, Consultant, Advisor: MSD, Sanofi, L'oreal; Non-Financial Interests, Personal, Other, Attendance for educational conference: Sanofi; Financial Interests, Personal, Steering Committee Member: MSD. B.G..M. Hughes: Financial Interests, Personal, Advisory Board: MSD, BMS, AstraZeneca, Roche, Pfizer, Regeneron, Sanofi, Amgen, Eisai. B. Panizza: Financial Interests, Personal, Invited Speaker: Sanofi; Financial Interests, Institutional, Research Funding: Decibel; Financial Interests, Personal, Stocks or ownership: Cochlear Ltd. S.V. Porceddu: Financial Interests, Personal, Steering Committee Member: Regeneron. J. Lee: Financial Interests, Personal, Research Grant: MSD; Financial Interests, Personal, Speaker, Consultant, Advisor: Boxer Capital, MSD, Dr Reddy, AstraZeneca; Financial Interests, Personal, Other, Attendance and travel for meetings: Pfizer, Novartis, MSD; Financial Interests, Personal, Advisory Board: MSD, Dr Reddy. All other authors have declared no conflicts of interest.

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1658P Neoadjuvant plus adjuvant treatment with cemiplimab in surgically resectable, high risk stage III/IV (M0) cutaneous squamous cell carcinoma (NEO-CESQ): Biomarker analysis

C. Trojaniello[1], L. Piccin[2], P. Bossi[3], M. Mandalà[4], P. Queirolo[5], F. Spagnolo[6], F. Bassetto[7], V. Rampinelli[8], F. Giovacchini[9], E. Pennacchioli[10], C. Caracò[11], G. Parrinello[12], D. Mallardo[1], D. Giannarelli[13], D. Massi[14], P.A. Ascierto[1]

[1]Melanoma, Cancer Immunotherapy & Developmental Therapeutics, Istituto Nazionale Tumori IRCCS - Fondazione G. Pascale, Naples, Italy; [2]DiSCOG Scienze Chirurgiche Oncologiche Gastroenterologiche, IOV - Istituto Oncologico Veneto IRCCS, Padua, Italy; [3]Medical Oncology Department, IRCCS Humanitas Research Hospital, Rozzano, Italy; [4]Medicine and Surgery Department, Università degli Studi di Perugia, Perugia, Italy; [5]Division of Medical Oncology for Melanoma, Sarcoma and Rare Tumors, IEO - Istituto Europeo di Oncologia IRCCS, Milan, Italy; [6]Medical Oncology, IRCCS Ospedale Policlinico San Martino, Genoa, Italy; [7]Plastic Surgery Clinic , Neuroscience Dept, Padua University Hospital, Padua, Italy; [8]Unit of Otorhinolaryngology -Head and Neck Surgery, Department of Medical, University of Brescia, Brescia, Italy; [9]Maxillofacial Surgery Unit, University of Perugia, Perugia, Italy; [10]Melanoma and Soft Tissue Sarcoma Surgery Dept., IEO - Istituto Europeo di Oncologia, Milan, Italy; [11]SC Chirurgia Melanoma e dei Tumori Cutanei, Istituto Nazionale Tumori IRCCS - Fondazione G. Pascale, Naples, Italy; [12]Otolaryngology Clinic, IRCCS Ospedale Policlinico San Martino, Genoa, Italy; [13]Facility of Epidemiology and Biostatistics, Fondazione Policlinico Universitario Agostino Gemelli IRCCS, Rome, Italy; [14]Department of Health Sciences, UniFI - Università degli Studi di Firenze, Florence, Italy

Background: The NEO-CESQ study (NCT04632433) investigates a perioperative immunotherapy approach combining neoadjuvant and adjuvant cemiplimab in patients with resectable high-risk cutaneous squamous cell carcinoma (CSCC). Here we report an update of clinical results and preliminary biomarkers analysis.

Methods: Between May 2021 and Oct 2022, 25 patients with high-risk resectable stage III/IV CSCC received 2 cycles of neoadjuvant cemiplimab (350 mg Q3W), followed by surgery and up to 1 year of adjuvant therapy. Primary endpoint was MPR by central pathology. Baseline FFPE gene profiling and serial IL-6 serum levels were assessed.

Results: Pathological complete response (pCR) was achieved in 9 patients (39%) and near-pCR in 2 (9%), while 1 patient showed partial response and 11 showed no response. As of January 31, 2025, one patient discontinued due to disease progression (no response), and two did not undergo surgery—one due to non-compliance and one due to unrelated death. Overall, 11 patients (48%) achieved MPR, and relapse-free survival (RFS) showed a trend toward benefit in this group. Patients with pCR had significantly lower IL-6 levels before adjuvant therapy (median 5.6 vs. 13.5 pg/mL; $p=0.003$), suggesting IL-6 as a potential biomarker. No significant differences were observed at other time points. Gene expression analysis revealed that MPR was associated with upregulated IFN-γ—signature and downregulation of WNT5A signaling, indicating a more immune-active and less proliferative tumor environment. High CD276 (B7-H3) expression correlated with significantly worse event-free survival ($p = 0.01$), supporting its role as a negative prognostic marker. No grade 3/4 adverse events were observed.

Conclusions: Neoadjuvant immunotherapy confirmed its important role in patients with resectable locally advanced CSCC. Post-surgery IL-6 levels emerged as a potential biomarker. Responders exhibited increased IFN-γ—related gene expression, while high CD276 (B7-H3) expression was associated with poorer event-free survival supporting its prognostic relevance. Further analyses are ongoing.

Clinical trial identification: NCT04632433.

Legal entity responsible for the study: Fondazione Melanoma Onlus.

Funding: Sanofi.

Disclosure: P. Bossi: Financial Interests, Personal, Advisory Board: MSD, Merck, Regeneron, SunPharma, GSK, Molteni, Angelini, Nestlé, Nutricia, Pfizer; Financial Interests, Institutional, Coordinating PI: MSD, Pfizer, BeiGene, Regeneron Pharmaceuticals; Financial Interests, Personal and Institutional, Coordinating PI: GSK; Non-Financial Interests, Leadership Role, National research group - Board of Directors: GONO; Non-Financial Interests, Leadership Role, Board of Directors: MASCC; Non-Financial Interests, Leadership Role: AIOCC. M. Mandalà: Financial Interests, Personal, Advisory Board: MSD, Novartis, Sanofi, BMS, Pierre Fabre; Financial Interests, Personal, Invited Speaker: Sun Pharma. D. Giannarelli: Financial Interests, Personal, Other, Educational Course in Biostatistics: Amgen; Financial Interests, Personal, Other, Course on GRADE system in Hematology: AstraZeneca; Financial Interests, Personal, Advisory Board, Advisory Board on a Sanofi sponsored study: Sanofi. P.A. Ascierto: Financial Interests, Personal, Other, Consultant and Advisory Role: BMS, Roche Genentech, Novartis, Merck Serono, Sun Pharma, Sanofi, Sandoz, Immunocore, Boehringer Ingelheim, Regeneron; Financial Interests, Personal, Other, Consultant, Advisory Role and Travel support: MSD, Pierre Fabre; Financial Interests, Personal, Other, Consultant and Advisory Role, Travel support: Pfizer/Array; Financial Interests, Personal, Other, Consultant Role: Italfarmaco; Financial Interests, Personal, Other, Consultant Role: Pfizer, Nouscom, Lunaphore; Financial Interests, Personal, Other, Consultant role: Medicenna; Financial Interests, Personal, Other, Consultant role and travel support: Bio-AI Health; Financial Interests, Personal, Advisory Board, Consultant and Advisory role: ValoTx; Financial Interests, Personal, Advisory Board, Consultant and Advisor role, Travel support: Replimmune; Financial Interests, Personal, Advisory Board, Advisor role: Bayer; Financial Interests, Personal, Other, Consultant and Advisory: Erasca; Financial Interests, Personal, Other, Consultant: Philogen, Incyte; Financial Interests, Personal, Other, Consultant and Advisory role: BioNTech, Genmab; Financial Interests, Personal, Advisory Board: Anaveon, Menarini; Financial Interests, Institutional, Funding, Clinical trial and translational research: BMS; Financial Interests, Institutional, Funding, Clinical Trial: Roche Genentech, Pfizer/Array, Sanofi; Non-Financial Interests, Leadership Role, President since 2010: Fondazione Melanoma Onlus Italy; Non-Financial Interests, Leadership Role, President since 2014: Campania Society of ImmunoTherapy of Cancer (SCITO) Italy; Non-Financial Interests, Other, Member of Steering Committee since 2016: Society for Melanoma Research (SMR); Non-Financial Interests, Member of Board of Directors, November 2017 - December 2021: Society for Immunotherapy of Cancer (SITC); Non-Financial Interests, Member: ASCO, SITC, EORTC Melanoma Cooperative Group, AIOM, SMR. All other authors have declared no conflicts of interest.

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1659P Updated pathologic responses from a pilot phase II trial of neoadjuvant immunotherapy for stage II-IV resectable cutaneous squamous cell carcinoma of the head and neck (CSCC-HN)

M. Amit[1], R. Ferrarotto[2], P. Nagarajan[3], G. Li[1], Y. Yuan[4], W. Morrison[5], D. Rosenthal[5], L. de Sousa[2], F. Johnson[2], E. Diaz[1], P. Gidley[1], C. Lewis[1], J. Myers[7], N. Gross[1]

[1]Head and Neck Surgery, University of Texas MD Anderson Cancer Center, Houston, United States of America; [2]Thoracic Head and Neck Medical Oncology, The University of Texas MD Anderson Cancer Center - Main Building, Houston, United States of America; [3]Pathology, MD Anderson Cancer Center, Houston, United States of America; [4]Biostatistics, University of Texas MD Anderson Cancer Center, Houston, United States of America; [5]Radiation Oncology, University of Texas MD Anderson Cancer Center, Houston, United States of America

Background: In a pilot phase II trial, we investigated the use of neoadjuvant cemiplimab to induce a pathologic response in patients with advanced, resectable cutaneous squamous cell carcinoma of the head and neck (CSCC-HN). Here, we report pathologic responses with an expansion cohort including patients with stage II disease. NCT03565783.

Methods: Patients with newly diagnosed or recurrent stage II-IV (M0) (AJCC 8th Ed) CSCC-HN were treated with 2 doses of cemiplimab 350 mg intravenously every 3 weeks prior to surgery. The primary endpoint was overall response rate (ORR) per RECIST v1.1. Secondary endpoints included safety, pathologic response, disease-free and overall survival.

Results: Forty-four patients (39M, 5F) enrolled with a median age 70 years (range: 42-90) and stage II ($n = 8$), stage III ($n = 22$) or IV ($n= 14$) disease. Ten (23%) patients presented with recurrent disease, including 2 (5%) patients with prior radiation. Six (14%) patients withdrew prior to treatment or refused surgery. Neoadjuvant immunotherapy was generally well-tolerated and there were no surgical delays among the remaining 38 patients. Adverse events (AEs) were observed in 18/42 (43%) patients; 1 (2%) grade 3 diarrhea, 6 (14%) $\leq$ grade 2 AEs. Major pathologic response (MPR, $\leq$10% viable tumor) was observed in 26/38 (68%) patients, including 19 (50%) patients with a pathologic complete response (pCR). Patients with a pCR did not receive radiotherapy after surgery. At a median follow up of 40.5 months (range: 2.8-81.4), none of the patients with a MPR have recurred.

Conclusions: MPR is frequently observed following neoadjuvant immunotherapy in patients with resectable stage II-IV CSCC-HN. These data further support the completion of a randomized phase III trial currently enrolling. NCT06568172.

Clinical trial identification: NCT03565783.

Legal entity responsible for the study: The authors.

Funding: Regeneron.

Disclosure: R. Ferrarotto: Financial Interests, Personal, Advisory Board: Regeneron-Sanofi, Ayala Pharmaceuticals, Bicara Therapeutics, Prelude Therapeutics, Guidepoint, Elevar Therapeutics, G1 Therapeutics, ExpertConnect, Remix, Eisai, Coherus, Rgenta, RAPT, LEK; Financial Interests, Personal, Invited Speaker: Merck; Financial Interests, Personal, Other, DSMC chair (BioAtla study): LabCorp; Financial Interests, Personal, Royalties: UptoDate; Financial Interests, Institutional, Coordinating PI: Merck, Genentech, Pfizer, Ayala, EMD Serono, Gilead; Financial Interests, Institutional, Other, Co-PI: Rakuten, Nanobiotix; Financial Interests, Institutional, Local PI: Prelude, ISA Therapeutics, Remix, Rgenta, Viracta, Mersana Therapeutics. Y. Yuan: Financial Interests, Personal, Other, Founder: Polaris Consulting. L. de Sousa: Financial Interests, Institutional, Research Funding: Regeneron, Agenus; Financial Interests, Personal, Advisory Board: Coherus Biosciences. P. Gidley: Financial Interests, Personal, Stocks/Shares: AbbVie, Amgen, Eli Lilly, GE Healthcare, Medtronic, Merck, Organon, Novartis, Pfizer, Roche, Sandoz Group Ag, Johnson and Johnson. N. Gross: Financial Interests, Institutional, Research Funding: Regeneron, Ascendis; Financial Interests, Personal, Advisory Board: PDS Biotechnology, Regeneron, Merck, GeoVax; Financial Interests, Personal, Royalties: Up-to-Date; Financial Interests, Personal, Speaker, Consultant, Advisor: AiCME, OncLive. All other authors have declared no conflicts of interest.

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1660P Adjuvant cemiplimab for high-risk cutaneous squamous cell carcinoma: Evaluating dosing intervals in a phase III trial

D. Rischin[1], S. Porceddu[2], F. Day[3], H. Christie[4], J.E. Jackson[5], B. Stein[6], Y.B. Su[7], R. Ladwa[8], N. Yamazaki[9], P. Bossi[10], A. Challapalli[11], A. Hauschild[12], F.A. Seebach[13], F. Cardozo[14], S-Y. Yoo[15], H. Ju[16], S. Bansal[17], P. Goncalves[18], M. Fury[19], J. Visich[20]

[1]Medical Oncology, Peter MacCallum Cancer Centre, Melbourne, Australia; [2]Department of Radiation Oncology, Peter MacCallum Cancer Centre, Melbourne, Australia; [3]Medical Oncology, Calvary Mater Newcastle, Waratah, Australia; [4]Medical Oncology, Cancer Care Centre Hervey Bay, Urraween, Australia; [5]Radiation Oncology, Radiation Oncology Centers, Gold Coast, Australia; [6]Medical Oncology, ICON Cancer Centre Adelaide (Kurralta Park), Adelaide, Australia; [7]Head and Neck Medical Oncology, Nebraska Cancer Specialists, Omaha, United States of America; [8]Medical Oncology, Princess Alexandra Hospital - Metro South Health, Woolloongabba, Australia; [9]Dermatologic Oncology, National Cancer Center - Tsukiji Campus, Chuo-ku, Japan; [10]Medical Oncology, IRCCS Humanitas Research Hospital, Rozzano, Italy; [11]Bristol Cancer Institute, University Hospitals Bristol & Weston NHS Foundation Trust, Bristol, United Kingdom; [12]Dermatology, Venereology and Allergology, UKSH - Universitätsklinikum Schleswig-Holstein - Campus Kiel, Kiel, Germany; [13]Oncology and Translational Sciences, Regeneron Pharmaceuticals, Inc., Tarrytown, United States of America; [14]Quantitative Pharmacology, Regeneron Pharmaceuticals, Inc., Tarrytown, United States of America; [15]Biostatistics Dept., Regeneron Pharmaceuticals, Inc., Tarrytown, United States of America; [16]Global Patient Safety, Regeneron Pharmaceuticals, Inc., Tarrytown, United States of America; [17]Clinical Oncology & Translational Sciences, Regeneron Pharmaceuticals, Inc., Tarrytown, United States of America; [18]Oncology, Regeneron Pharmaceuticals, Inc., Tarrytown, United States of America; [19]Oncology Clinical Sciences, Regeneron Pharmaceuticals, Inc., Tarrytown, United States of America; [20]Clinical Pharmacology, Regeneron Pharmaceuticals, Inc., Tarrytown, United States of America

Background: Cemiplimab is the first adjuvant systemic therapy to improve disease-free survival (DFS) compared with placebo for patients with high-risk cutaneous squamous cell carcinoma (CSCC; HR 0.32). Here we present pharmacokinetic (PK) and safety data for two regimens of cemiplimab in patients with high-risk CSCC.

Methods: In this double-blind, placebo-controlled study, the primary endpoint was DFS. Secondary endpoints included safety, steady-state PK and immunogenicity.

Results: A total of 209 patients were randomised to cemiplimab and 206 patients to placebo. DFS was improved compared with placebo in patients assigned to a regimen of every 3 weeks (Q3W; HR 0.44) and a regimen that includes dosing every 6 weeks (Q6W; HR 0.25). In the safety analysis set, 129 patients received Q3W only and 280 received Q3W start/Q6W switch. For cemiplimab-treated patients, observed sparse PK data and modelling indicate mean trough concentration at steady-state ($C_{trough,ss}$) was similar for both regimens. The model-predicted data indicates a modest reduction in mean $C_{trough,ss}$ (20.8%) and increase in mean maximum concentration at steady-state (51.3%) for Q3W start/Q6W switch regimen vs Q3W only (Table). Patients treated with Q3W start/Q6W switch had a lower incidence of grade $\geq$3 treatment-emergent adverse events (AEs), serious AEs, or AEs leading to treatment discontinuation compared with Q3W only.

Table: 1660P PK and safety of cemiplimab Q3W and Q3W start/Q6W switch regimens

	Cemiplimab 350 mg Q3W for 48 weeks	Cemiplimab 350 mg Q3W for 12 weeks/700 mg Q6W switch for 36 weeks
Pharmacokinetics, mean (SD)		
PK analysis set	n=59	n=112
Cycle 3 $C_{trough,ss}$,mg/L	56.5 (23.9)	49.0 (23.7)
MP $C_{trough, ss}$,mg/L	66.3 (30.7)	52.5 (31.4)
MP $C_{max, ss}$,mg/L	154 (43.1)	233 (64.9)
Safety, n (%)		
Safety analysis set	n=65	n=140
TEAEs, any grade	59 (90.8)	128 (91.4)
TEAEs, grade 3–5	23 (35.4)	26 (18.6)
TE-SAEs	17 (26.2)	19 (13.6)
TEAEs leading to treatment discontinuation	16 (24.6)	4 (2.9)
TEAEs leading to death	1 (1.5)	1 (0.7)

$C_{max\ ss}$, maximum concentration at steady-state; MP, model-predicted; TEAE, treatment-emergent adverse event; TE-SAE, treatment-emergent serious adverse event.

Conclusions: Efficacy, PK and immunogenicity were similar between the Q3W and Q3W/Q6W switch regimens. The safety profile of cemiplimab regardless of dosing regimen was consistent with the known safety profile of cemiplimab in advanced solid malignancies. These results support cemiplimab 350 mg Q3W/700mg Q6W switch as a more convenient regimen for patients and health care providers.

Clinical trial identification: NCT03969004.

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1661P Efficacy and safety of RP1 + nivolumab (nivo) in patients with non-melanoma skin cancer (NMSC)

D. Schadendorf[1], M.K. Wong[2], G.M. Beasley[3], A.C. Pavlick[4], K.J. Harrington[5], B. Chmielowski[6], J.J. Niu[7], A.M. VanderWalde[8], T.L. Bowles[9], K.K. Tsai[10], E.T. Hall[11], T.M. Wise-draper[12], J. Michels[13], A.M. Arance[14], M. Amini Adle[15], J. Zhu[16], M. Viana[17], J.W. Hou[17], M.M. Milhem[18]

[1]Department of Dermatology, West German Cancer Center, University Hospital Essen, Essen & National Center for Tumor Diseases (NCT-West), Campus Essen & University Alliance Ruhr, Research Alliance Ruhr, Research Center One Health, University Duisburg-Essen, Campus Essen, Essen, Germany; [2]Department of Medicine, Roswell Park Comprehensive Cancer Center, Buffalo, United States of America; [3]Department of Surgery, Duke Cancer Institute, Duke University, Durham, United States of America; [4]Division of Medical Oncology, Department of Medicine, Weill Cornell Medical College, New York, United States of America; [5]Division of Radiotherapy & Imaging, The Institute of Cancer Research/Royal Marsden NIHR Biomedical Research Centre, London, United Kingdom; [6]Division of Hematology and Medical Oncology, Jonsson Comprehensive Cancer Center, University of California Los Angeles, Los Angeles, United States of America; [7]Department of Medical Oncology, Banner MD Anderson Cancer Center, Gilbert, United States of America; [8]Department of Clinical Research, West Cancer Center and Research Institute, Germantown, United States of America; [9]Department of Surgical Oncology, Intermountain Medical Center, Murray, United States of America; [10]Division of Hematology/Oncology, Department of Medicine, Helen Diller Family Comprehensive Cancer Center, University of California San Francisco, San Francisco, United States of America; [11]Division of Oncology, Fred Hutchinson Cancer Center, University of Washington School of Medicine, Seattle, United States of America; [12]Division of Hematology/Oncology, University of Cincinnati Cancer Center, Cincinnati, United States of America; [13]Département de Médecine Oncologique, Gustave Roussy, Villejuif, France; [14]Medical Oncology Department, Hospital Clinic, Translational Genomics and Targeted Therapies in Solid Tumors, IDIBAPS, Barcelona, Spain; [15]Department of Onco-dermatology, Centre Léon Bérard, Lyon, France; [16]Biometrics, Replimune, Inc., Woburn, United States of America; [17]Clinical Development, Replimune, Inc., Woburn, United States of America; [18]Department of Internal Medicine, Medical Oncology, Holden Comprehensive Cancer Center, University of Iowa, Iowa City, United States of America

Background: Patients (pts) with NMSC that has progressed on an anti–PD-(L)1–containing therapy have poor clinical outcomes and limited treatment options. RP1 (vusolimogene oderparepvec) is an oncolytic immunotherapy that expresses human granulocyte-macrophage colony-stimulating factor and a fusogenic glycoprotein (GALV-GP-R^{-}). Here, we report the efficacy of RP1 + nivo from the NMSC cohort of the phase 1/2 IGNYTE trial (NCT03767348).

Methods: The trial enrolled pts with anti–PD-1–naïve and –failed NMSC, including Merkel cell carcinoma (MCC), basal cell carcinoma (BCC), angiosarcoma, and cutaneous squamous cell carcinoma (CSCC). RP1 was administered intratumorally, at 1×10^6 PFU/mL initially, then at 1×10^7 PFU/mL Q2W ($\leq$7 doses) with intravenous nivo. The objective response rate (ORR) was assessed by investigator assessment

Table: 1661P Confirmed response by NMSC type

BOR, n (%)	MCC		BCC		Angiosarcoma		CSCC PD-1 naïve			CSCC PD-1 failed		
	PD-1 naïve (n = 4)	PD-1 failed (n = 19)	PD-1 naïve (n = 3)	PD-1 failed (n = 10)	PD-1 naïve (n = 6)	PD-1 failed (n = 8)	LA (n = 5)	Met (n = 11)	Total (n = 16)	LA (n = 8)	Met (n = 25)	Total (n = 33)
CR	4 (100)	3 (15.8)	1 (33.3)	0	2 (33.3)	2 (25.0)	2 (40.0)	4 (36.4)	6 (37.5)	1 (12.5)	1 (4.0)	2 (6.1)
PR	0	2 (10.5)	0	3 (30.0)	2 (33.3)	1 (12.5)	1 (20.0)	2 (18.2)	3 (18.8)	1 (12.5)	2 (8.0)	3 (9.1)
ORR	4 (100)	5 (26.3)	1 (33.3)	3 (30.0)	4 (66.7)	3 (37.5)	3 (60.0)	6 (54.5)	9 (56.3)	2 (25.0)	3 (12.0)	5 (15.2)

BOR, best overall response.

using modified RECIST (mRECIST); the key modification was that progression had to be confirmed by further tumor increase to allow for the potential of pseudoprogression.

Results: A total of 99 pts with NMSC were evaluable for efficacy; 71% of pts overall had disease progression on prior anti—PD-1 therapy. Substantial responses to RP1 + nivo occurred across tumor types (MCC, BCC, angiosarcoma), with confirmed responses seen both in pts with anti—PD-1—naïve and —failed disease, as well as in locally advanced (LA) and metastatic (met) CSCC (Table). The most common treatment-related adverse events (TRAEs; ≥15%) were fatigue, chills, and pyrexia. The most common grade ≥3 TRAEs (≥2 events in pts with anti—PD-1—naïve or —failed disease) were fatigue, rash maculo-papular, abdominal pain, diarrhea, hyponatremia, and pyrexia.

Conclusions: RP1 + nivo induced responses across multiple NMSC tumor types, including anti—PD-1—failed disease, and represents a promising treatment approach for pts with advanced skin cancers, including those with disease progression on prior anti—PD-1 therapy.

Clinical trial identification: NCT03767348.

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Wong: Financial Interests, Personal, Advisory Board: Regeneron, Castle Biosciences, Replimune, Inc., Incyte, Sun Pharmaceuticals. G.M. Beasley: Financial Interests, Institutional, Research Funding: Replimune, Inc., Delcath, Philogen, Skyline; Financial Interests, Personal, Speaker, Consultant, Advisor, Payment/honoraria: Bristol Myers Squibb. A.C. Pavlick: Financial Interests, Personal, Speaker, Consultant, Advisor: Bristol Myers Squibb, Merck; Financial Interests, Personal, Advisory Board: BioNTech. K.J. Harrington: Financial Interests, Institutional, Advisory Board: Amgen, AstraZeneca, BMS, Boehringer Ingelheim, Merck, MSD, Pfizer, Replimune, Oncolys, Vyriad, Idera, Onchilles, BeiGene, GSK, Johnson and Johnson, QBiotics; Financial Interests, Institutional, Other, Honoraria for lectures: Amgen, AstraZeneca, BMS, Boehringer Ingelheim; Financial Interests, Institutional, Advisory Board, Advisory board for CD47 assets: Arch Oncology; Financial Interests, Institutional, Advisory Board, Development of DDR assets: ARTIOS; Financial Interests, Institutional, Advisory Board, Development of exosomal STING agonist: Codiak; Financial Interests, Institutional, Advisory Board, Development of oncolytic adenovirus: PsiVac; Financial Interests, Institutional, Advisory Board, Steering committee: ALX Oncology; Financial Interests, Institutional, Funding, Research: AstraZeneca, Boehringer Ingelheim, MSD; Financial Interests, Institutional, Funding, Development of oncolytic HSV platform: Replimune; Non-Financial Interests, Leadership Role, Chair of Steering Committee: ART NET; Non-Financial Interests, Other, Member of Global Steering Committee: MR - Linac. B. Chmielowski: Financial Interests, Institutional, Research Funding: Bristol Myers Squibb , Macrogenics, Infinity Pharmaceuticals, Advenchen Laboratories, Xencor, Compugen, Iovance, RAPT Therapeutics, IDEAYA Biosciences, Ascentage, Atreca, Replimune, Inc., InstilBio, Adagene, Trisalus Life Sciences, Kinnate , PTC Therapeutics, Xilio Therapeutics, Kezar Life Sciences, Immunocore, AskGene Pharma, Krystal Bio, Nested Therapeutics, Pierre Fabre, Georgiamune, Immatics; Financial Interests, Personal, Advisory Role, Data monitoring board: Servier, SpringWorks Therapeutics; Financial Interests, Personal, Advisory Board: Regeneron, Treeline Biosciences, Atreca. J.J. Niu: Financial Interests, Personal, Advisory Board: AstraZeneca, Bristol Myers Squibb, Daiichi Sankyo, Pfizer, Sanofi, EMD Serono. A.M. VanderWalde: Financial Interests, Personal, Full or part-time Employment: Caris Life Sciences; Financial Interests, Personal, Advisory Role: West Clinic, George Clinical; Financial Interests, Institutional, Research Funding: Amgen, Merck, Genentech/Roche, Millennium, AstraZeneca, Lilly, Bristol Myers Squibb, Replimune, Inc., Caris Life Sciences, EMD Serono, Immunomedics/Gilead. T.L. Bowles: Financial Interests, Institutional, Research Funding: Amgen, Genentech, Natera, Iovance; Non-Financial Interests, Personal, Leadership Role: Society of Surgical Oncology, Melanoma Research Foundation, AIM at Melanoma. K.K. Tsai: Financial Interests, Institutional, Research Funding: ABM Therapeutics, AstraZeneca, BioAtla, Bristol Myers Squibb, Genentech, Georgiamune, Ideaya, Innovent, OnKure, Pfizer, Regeneraon, Replimune, Inc.; Financial Interests, Personal, Speaker, Consultant, Advisor, Consulting fees: Bristol Myers Squibb. E.T. Hall: Financial Interests, Institutional, Research Funding: Replimune, Inc., Bristol Myers Squibb, NiKang Therapeutics, Neoleukin Therapeutics, Pfizer, ImCheck Therapeutics, AstraZeneca, Gilead, Immunocore, Cullinan Oncology, Kezar Life Sciences; Financial Interests, Personal, Speaker, Consultant, Advisor, Consulting fees: Eisai; Financial Interests, Personal, Advisory Board: Microbiotica. T.M. Wise-draper: Financial Interests, Personal, Advisory Board: Merck & Co, Adaptimmune, EMD Serono, Genmab, Replimune; Financial Interests, Personal, Other, Consultant for Molecular Tumor Board: Caris Life Sciences; Financial Interests, Personal, Other, Consultant: NEED INC; Financial Interests, Personal, Invited Speaker: Medscape; Financial Interests, Personal, Ownership Interest, Funded 10K as initial investment: High Enroll; Financial Interests, Personal, Research Grant: BMS, Merck & Co, Tesaro/GSK, AstraZeneca, Janssen; Financial Interests, Personal and Institutional, Local PI: BMS, EMD Serono, Replimune, AstraZeneca, Alkermes, Exicure, Shattuck Labs, Boston Medical, Debio Pharm, Eli Lilly, Epizyme, Iovance, Agenus, SQZbiotech, AdlaiNortye, Yingli, Vyriad, Ideaya, MINK, Sinogen, Takeda, duality Bio, Daiichi Sankyo, Coherus, Xencor, Bicara. J. Michels: Financial Interests, Personal, Advisory Board: GSK, Regeneron; Financial Interests, Personal, Invited Speaker: MSD; Financial Interests, Institutional, Research Grant: MSD, Replimune; Financial Interests, Institutional, Coordinating PI: MSD; Non-Financial Interests, Advisory Role: Replimune. A.M. Arance: Financial Interests, Personal, Advisory Board, Consulting or Advisory Role, Speakers' Bureau: Pierre- Fabre; Financial Interests, Personal, Advisory Board, Consulting or Advisory Role, Speakers' Bureau: BMS, MSD; Financial Interests, Personal, Advisory Board, Consulting or Advisory Role: BioNTech, Almirall, Genmab; Financial Interests, Institutional, Local PI, Research Funding: Novartis, BioNTech, BMS, MSD, Pierre Fabre, Replimune, IO Biotech, HUYABIO International, Iovance Biotherapeutics, Immunocore, Sairopa, Mural Oncology; Other, Travel, Accomodations, Expenses: MSD, Pierre Fabre. M. Amini Adle: Financial Interests, Institutional, Research Funding: Bristol Myers Squibb ; Financial Interests, Personal, Speaker, Consultant, Advisor: Bristol Myers Squibb, Sun Pharmaceuticals; Financial Interests, Personal, Speaker, Consultant, Advisor, Payment/honoraria: Pierre Fabre, Bristol Myers Squibb, Sun Pharmaceuticals, Merck Sharp & Dohme; Other, Personal, Other, Travel support: Pierre Fabre, Bristol Myers Squibb, Sun Pharmaceuticals, Merck Sharp & Dohme; Other, Personal, Other, Receipt of equipment, materials, drugs, medical writing, gifts or other services: Pierre Fabre. J. Zhu, J.W. Hou: Financial Interests, Personal, Full or part-time Employment: Replimune, Inc.; Financial Interests, Personal, Stocks/Shares: Replimune, Inc. M. Viana: Financial Interests, Personal, Full or part-time Employment: Replimune, Inc.; Financial Interests, Personal, Stocks/Shares: Replimune, Inc.; Other, Personal, Other, Patent: Zentalis Pharmaceuticals. M.M. 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1662P A phase II, open label study to evaluate the safety and clinical activity of balstilimab in patients with advanced/metastatic non-melanoma skin cancers (AGENONMELA)

I. Lugowska[1], A. Dawidowska[2], S. Jaczewska[2], M.A. Slowinska[3], A.M. Szumera-Cieckiewicz[4], M. Mikula[5], M. Rosinska[6], J. Dylewska-Rzeznik[2], A. Korycka[1], H. Badziak-Sterczewska[1], A. Janowska[1], I.M. Agnieszczak[1], P. Rutkowski[7]

[1]Centre for Excellence in Precision Oncology, Maria Sklodowska-Curie National Research Institute of Oncology, Warsaw, Poland; [2]Clinical Trials Support Center, Maria Sklodowska-Curie National Research Institute of Oncology, Warsaw, Poland; [3]Department of Dermatology, Military Institute of Medicine, Warsaw, Poland; [4]Department of Pathology, Maria Sklodowska-Curie National Research Institute of Oncology, Warsaw, Poland; [5]Laboratory of High Throughput Technologies, Department of Genetics, Maria Sklodowska-Curie National Research Institute of Oncology, Warsaw, Poland; [6]Computational Oncology Department, Maria Sklodowska-Curie National Research Institute of Oncology, Warsaw, Poland; [7]Department of Soft Tissue/Bone Sarcoma and Melanoma, Maria Skłodowska-Curie National Research Institute of Oncology, Warsaw, Poland

Background: Non-melanoma skin cancers (NMSCs) account for ∼30% of all cancer diagnoses. Systemic treatment for patients ineligible for surgery or radiotherapy remains challenging, especially in rare subtypes, but the role of immunotherapy (IO) is increasing. Balstilimab is a clinically validated anti-programmed cell death protein 1 (anti-PD-1) antibody demonstrating activity in refractory cancers.

Methods: AGENONMELA is a phase 2 single-center academic trial evaluating the anti-PD-1 therapy in pts with advanced/metastatic NMSC. Balstilimab was administered at 3 mg/kg IV every 2 weeks, for up to 24 months or until progression/toxicity. Palliative radiotherapy (RT) was allowed per physician discretion. Tumor response was assessed every 12 weeks using RECIST v1.1 and iRECIST. The primary endpoint was clinical benefit rate (CBR) at 12 weeks, defined as the proportion of pts with complete response, partial response (PR) or stable disease (SD), with a 30% futility threshold.

Results: Of 43 pts screened, 32 had at least one dose of balstilimab. Median age was 70.5 years (range 32—86), none had prior IO. CBR at 12 weeks was 64.5% (95% CI: 47.7—81.4%; $p < 0.0001$ vs. 30% futility threshold), with SD in 16 and PR in 4 pts (Table). 6 pts had RT during the study. Median time on treatment was 8.3 months (95% CI: 3.0—16.0). Of 32 pts, 24 (75%) discontinued treatment before completing the 24-month protocol: 16 (50.0%) due to progression, 6 (18.8%) due to toxicity, 2 (6.3%) for other reasons. 6 pts (18.8%) completed the protocol; 2 remained on therapy at data cutoff. Estimated 36-month overall survival was 72.3% (95% CI: 51.7—85.3%). Grade ≥3 treatment-related adverse events occured in 3 pts (9.4%), all reported as serious adverse events.

Table: 1662P			
		Baseline Pts characteristics N (%)	**CBR at 12 weeks N (%)**
Overall		32 (100)	20 (64.5)
Sex	Female	12 (37.5)	4 (33.3)
	Male	20 (62.5)	16 (84.2)
Histology	Basal cell carcinoma (BCC)	8 (25)	7 (87.5)
	Squamous cell carcinoma	4 (12.5)	1 (25)
	Adnexal adenocarcinoma	6 (18.8)	4 (66.7)
	Angiosarcoma	6 (18.8)	2 (33.3)
	Other	8 (25): Kaposi's sarcoma 3 (9.4) Gorlin-associated BCC 2 (6.3) Extramammary Paget 2 (6.3) Adenoid cystic carcinoma 1 (3.1)	6 (85.7)
Stage	Locally advanced	25 (78.1)	15 (62.5)
	Metastatic	7 (21.9)	5 (71.4)
Systemic treatment lines	0	14 (43.8)	7 (53.8)
	1	14 (43.8)	10 (71.4)
	2	3 (9.4)	2 (66.7)
	3	1 (3.1)	1 (100)

Conclusions: Balstilimab showed promising activity and an acceptable safety profile in advanced NMSC, supporting further investigation of PD-1 blockade in this heterogeneous population.

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1663P Proof of concept study of low-dose nivolumab in cancers with unresectable cutaneous squamous cell carcinoma

G. Thibault[1], M. Amini Adle[2], E.M. Neidhardt[1], R. Varnier[3], S. Aho[2], P-E. Heudel[1], J-Y. Blay[1], J. Fayette[2]

[1]Department of Medicine, Centre Léon Bérard, Lyon, France; [2]Medical Oncology Department, Centre Léon Bérard, Lyon, France; [3]Oncology Dept., Centre Léon Bérard, Lyon, France

Background: Immunotherapy is an innovative but expensive treatment with proven efficacy across various cancers. An Indian trial suggested low-dose nivolumab (20 mg q3w) could be as effective as standard dosing in head and neck cancers. Before cemiplimab reimbursement for cutaneous squamous cell carcinoma (cSCC) in France, our institution offered low-dose nivolumab to selected patients at its own expense, following multidisciplinary team approval and informed consent. Pharmacokinetic data suggest lower doses may preserve efficacy while reducing toxicity, which is particularly relevant in frail populations. We evaluated low-dose nivolumab in rare tumors lacking formal immunotherapy approval, using cSCC as a model.

Methods: This retrospective proof-of-concept study included patients with recurrent or unresectable cSCC who received at least two cycles of nivolumab 20 mg every 3 weeks (q3w) between December 2023 and April 2025. The primary endpoint was 6-month progression-free survival (PFS). Tumor responses were assessed every 8 weeks according to RECIST criteria. Safety was monitored throughout the treatment period.

Results: Thirteen patients (with lymphadenopathy: N=4, periauricular: N=3; temporal: N=2; vertex: N=1; other: N=3), with a mean age of 87.3 years, received a mean of 7.6 cycles of low-dose nivolumab. At the interim cut-off (April 2025), six patients had $\geq$6 months of follow-up. The 6-month PFS rate was 77% (90% CI: 0.57–1.00). Overall response rate (ORR) was 61% (2 complete and 6 partial responses; 90% CI: 35–83%), with 2 cases of stable disease and 3 progressions. Both stable disease cases had short follow-up; response with longer treatment remains possible. Treatment was well tolerated, with no grade 3–4 adverse events or treatment discontinuations.

Conclusions: Low-dose nivolumab (20 mg q3w) showed promising efficacy and good tolerability in recurrent/unresectable cSCC. The 6-month PFS of 77% appears favorable compared to the historical benchmark of approximately 55% observed with full-dose cemiplimab, supporting the hypothesis that reduced dosing retains benefit. Further validation and cost-effectiveness studies are warranted.

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1664P Long-term updates of the I-TACKLE trial: pembrolizumab (P) followed by cetuximab (C) in locally advanced/metastatic (LA/M) cutaneous squamous cell carcinomas (cSCC)

C. Resteghini[1], C. Bergamini[2], A. Alberti[3], S. Farinatti[4], S. Alfieri[5], S. Cavalieri[6], E. Colombo[7], A. Grammatica[8], C. Gurizzan[9], L.D. Locati[10], L. Lorini[11], A. Maurichi[12], R. Patuzzo[13], V. Rampinelli[14], M. Ravanelli[15], V. Tovazzi[16], M. Zamparini[17], P.A. Ascierto[18], L.F.L. Licitra[19], P. Bossi[20]

[1]Head & Neck Oncology Unit, Humanitas University, Pieve Emanuele, Italy; [2]Medical Oncology Department, Fondazione IRCCS - Istituto Nazionale dei Tumori, Milan, Italy; [3]Medical Oncology, Università degli Studi di Brescia - Facoltà di Medicina e Chirurgia, Brescia, Italy; [4]Oncology Dept., Humanitas University, Pieve Emanuele, Italy; [5]Department of Medical Oncology, Fondazione IRCCS - Istituto Nazionale dei Tumori, Milan, Italy; [6]Oncologia medica 3 - Tumori Testa - Collo, Istituto Nazionale dei Tumori di Milano - Fondazione IRCCS, Milan, Italy; [7]Medical Oncology Department, Fondazione IRCCS - Istituto Nazionale dei Tumori, Milan, Italy; [8]Unit of Otolaryngology and Head and Surgery, ASST Spedali Civili di Brescia, University of Brescia, Italy, Brescia, Italy; [9]Department of Oncology, IRCCS Humanitas Research Hospital, Rozzano, Italy; [10]Department of Internal Medicine and Therapeutics, University of Pavia, Pavia, Italy; [11]Dipartimento di Oncologia Medica, IRCCS Humanitas Research Hospital, Rozzano, Italy; [12]Surgery, Fondazione IRCCS - Istituto Nazionale dei Tumori, Milan, Italy; [13]Melanoma Surgery Unit, Fondazione IRCCS Istituto Nazionale Tumori, Milan, Italy; [14]Unit of Otorhinolaryngology -Head and Neck Surgery, Department of Medical, University of Brescia, Brescia, Italy; [15]Radiology Unit, Department of Medical and Surgical Specialties, Radiological Sciences, ASST Spedali Civili di Brescia, Brescia, Italy; [16]Dipartimento di Oncologia Medica, Azienda Ospedaliera Spedali Civili di Brescia, Brescia, Italy; [17]Center for Public Health Research, University of Milan-Bicocca, Monza, Italy; [18]Melanoma, Cancer Immunotherapy & Developmental Therapeutics, Istituto Nazionale Tumori IRCCS - Fondazione G. Pascale, Naples, Italy; [19]Head and Neck Medical Oncology Unit, Fondazione IRCCS - Istituto Nazionale dei Tumori, Milan, Italy; [20]Medical Oncology Department, IRCCS Humanitas Research Hospital, Rozzano, Italy

Background: The I-TACKLE trial investigated P and C in LA/M cSCC, obtaining a cumulative overall response rate (ORR) of 63% [95% CI 48−77] with 19/43 (44%) responding to P alone and 8/21 (38%) responding to P+C after resistance. Here we report on long-term survival and safety endpoints.

Methods: I-TACKLE is an open-label, nonrandomized, phase II trial conducted in 3 Italian centers. Eligible pts with LA/M cSCC not manageable with surgery / radiation received P 200 mg every 3 weeks. In case of partial (PR) or complete response (CR), pts continued to receive P alone. In case of stable disease (SD) or progression (PD), pts received C (400 mg/sm loading dose, then weekly 250 mg/sm) in addition to P until further PD. The primary endpoint was cumulative ORR by single agent or by combination strategy; safety, PFS (since the start of P and P+C), and OS were secondary endpoints.

Results: At a median follow up of 52 (range 36-NE) months, all pts discontinued study treatment. No new progressions were recorded in either treatment groups. A total of 28 deaths were recorded, 17 due to PD during treatment (60%). Median OS in the overall population were 27 months (95%CI 17-45). Median PFS with P alone and P+C were 12 (3-NE) and 7 months (4-NE), respectively. Overall, we recorded 3 additional grade 3-4 treatment-related adverse events (TRAE): warm autoimmune hemolytic anemia (G4), cutaneous rash (G3), in pts treated with P and bullous pemphigoid (G3) in pt treated with P+C. Four out of 43 (9%) pts discontinued treatment because of TRAE (3 P; 1 P-C).

Conclusions: Long term observations confirm the benefit of antiEGFR in reverting primary and acquired resistance to immunotherapy in R/M SCC. Long term follow-up is needed to identify and manage late TRAE.

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Locati: Financial Interests, Personal, Invited Speaker: Bayer, IPSEN, Novartis, Seagen, SunPharma; Financial Interests, Personal, Advisory Board: Eli Lilly, MSD, Roche, Johnson and Johnson, Eisai; Financial Interests, Institutional, Local PI: EISAI; Financial Interests, Institutional, Funding: Pfizer; Non-Financial Interests, Advisory Board: AIOM (Italian Association of Medical Oncology); Non-Financial Interests, Advisory Role: MSGS (Multidisciplinary Salivary Glands Society), AIOCC (Associazione Italiana Oncologia Cervico Cefalica); Non-Financial Interests, Advisory Role, Thyroid Task Force: EORTC. P.A. Ascierto: Financial Interests, Personal, Other, Consultant and Advisory Role: BMS, Roche Genentech, Novartis, Merck Serono, Sun Pharma, Sanofi, Sandoz, Immunocore, Boehringer Ingelheim, Regeneron; Financial Interests, Personal, Other, Consultant, Advisory Role and Travel support: MSD, Pierre Fabre; Financial Interests, Personal, Other, Consultant and Advisory Role, Travel support: Pfizer/Array; Financial Interests, Personal, Other, Consultant Role: Italfarmaco; Financial Interests, Personal, Other, Consultant Role: Pfizer, Nouscom, Lunaphore; Financial Interests, Personal, Other, Consultant role: Medicenna; Financial Interests, Personal, Other, Consultant role and travel support: Bio-AI Health; Financial Interests, Personal, Advisory Board, Consultant and Advisory role: ValoTx; Financial Interests, Personal, Advisory Board, Consultant and Advisor role, Travel support: Replimmune; Financial Interests, Personal, Advisory Board, Advisor role: Bayer; Financial Interests, Personal, Other, Consultant and Advisory: Erasca; Financial Interests, Personal, Other, Consultant: Philogen, Incyte; Financial Interests, Personal, Other, Consultant and Advisory role: BioNTech, Genmab; Financial Interests, Personal, Advisory Board: Anaveon, Menarini, ImCheck therapeutics; Financial Interests, Institutional, Funding, Clinical trial and translational research: BMS; Financial Interests, Institutional, Funding, Clinical Trial: Roche Genentech, Pfizer/Array, Sanofi; Non-Financial Interests, Leadership Role, President since 2010: Fondazione Melanoma Onlus Italy; Non-Financial Interests, Leadership Role, President since 2014: Campania Society of ImmunoTherapy of Cancer (SCITO) Italy; Non-Financial Interests, Other, Member of Steering Committee since 2016: Society for Melanoma Research (SMR); Non-Financial Interests, Member of Board of Directors, November 2017 - December 2021: Society for Immunotherapy of Cancer (SITC); Non-Financial Interests, Member: ASCO, SITC, EORTC Melanoma Cooperative Group, AIOM, SMR. L.F.L. Licitra: Financial Interests, Personal, Advisory Board, for expert opinion in advisory boards: MSD, Boehringer Ingelheim, Hoffmann-La Roche Ltd, Roche, GSK, Alentis; Financial Interests, Personal, Advisory Board: Medscape, EMD Serono, Seagen International GmbH, Genmab US, AbbVie, Purple Biotech Ltd, Aveo Pharmaceuticals, Alx Oncology Inc, Bicara Therapeutics Inc, MSD It, Merck Healtcare KGaA; Financial Interests, Institutional, Research Grant, Funds received by my institution for clinical studies and research activities in which I am involved: AstraZeneca, BMS, Boehringer Ingelheim, Celgene International, Eisai, Exelixis, Debiopharm International SA, Hoffmann-La roche ltd, Medpace, Merck−Serono, Merck Healthcare KGaA, MSD, Novartis, Pfizer, Roche, Adlai Nortye; Financial Interests, Institutional, Local PI, Funds received by my institution for clinical studies and research activities in which I am involved: Alentis; Financial Interests, Institutional, Research Grant: Eli Lilly, Isa Therapeutics, Kura Oncology, Nektar Therapeutics, Sanofi, Regeneron, Syneos Healthe Clinical, Sun Pharmaceutica, Incyte Biosciences International Srl, Gilead Sciences; Non-Financial Interests, Member: AIOM, ASCO, ESTRO, EORTC, ACC; Non-Financial Interests, Leadership Role: AIOCC. P. Bossi: Financial Interests, Personal, Advisory Board: MSD, Merck, Regeneron, SunPharma, GSK, Molteni, Angelini, Nestlè, Nutricia, Pfizer; Financial Interests, Institutional, Coordinating PI: MSD, Pfizer, BeiGene, Regeneron Pharmaceuticals; Financial Interests, Personal and Institutional, Coordinating PI: GSK; Non-Financial Interests, Leadership Role, National research group - Board of Directors: GONO; Non-Financial Interests, Leadership Role, Board of Directors: MASCC; Non-Financial Interests, Leadership Role: AIOCC. All other authors have declared no conflicts of interest.

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1665P Integrated immune and genomic liquid biopsy profiling in patients with advanced cutaneous squamous cell carcinoma (CSCC) treated with anti-PD-1 therapy

A. Boutros[1], I. Vanni[2], M. Croce[3], L. Pastorino[2], E. Allavena[3], S. Coco[4], S. Santamaria[1], A. Rosa[5], V. Fontana[6], A. Guadagno[7], S.E. Bellomo[8], C. Marchiò[8], E. Berrino[9], P. Ghiorzo[10], F. Spagnolo[4], E.T. Tanda[4]

[1]U.O. Clinica di Oncologia Medica, IRCCS Ospedale Policlinico San Martino, Genoa, Italy; [2]Genetica Oncologica, IRCCS Ospedale Policlinico San Martino, Genoa, Italy; [3]Bioterapie, IRCCS Ospedale Policlinico San Martino, Genoa, Italy; [4]U.O. Oncologia Medica 2, IRCCS Ospedale Policlinico San Martino, Genoa, Italy; [5]U.O. Epidemiologia Clinica, IRCCS Ospedale Policlinico San Martino, Genoa, Italy; [6]Department of Mathematics, University of Genoa, Genoa, Italy; [7]Pathology Unit, IRCCS Ospedale Policlinico San Martino, Genoa, Italy; [8]Pathology Unit, IRCCS Istituto di Candiolo - FPO, Candiolo, Italy; [9]Pathology Unit, Università degli Studi di Torino - Dipartimento di Oncologia - IRCCS Candiolo, Candiolo, Italy; [10]Department of Internal Medicine and Medical Specialties (DiMI), University of Genoa, Genoa, Italy

Background: PD-1 inhibitors have markedly improved outcomes in advanced CSCC and are now the standard of care. However, up to 50% of patients fail to respond or relapse after initial benefit, and validated predictive biomarkers are still lacking. We investigated soluble immune markers and ctDNA profiling from liquid biopsy to identify correlates of response to anti-PD-1.

Methods: This prospective translational study enrolled patients with advanced CSCC eligible for treatment with cemiplimab. Baseline and on-treatment serum and plasma samples were collected for integrated immunogenomic liquid biopsy analysis. immune-checkpoints (sB7-H1, sLAG-3, sTIM-3, sCTLA-4) and IFN-γ were quantified longitudinally using immunoassays. Baseline and on-treatment ctDNA was analysed with the next-generation sequencing TruSight Oncology 500 panel to assess somatic and germline alterations, tumor mutational burden (TMB), microsatellite instability, and copy number variations. Associations with clinical outcomes were explored using multivariable regression models.

Results: Forty-one patients with advanced CSCC were included. Each unit increase in baseline IFN-γ levels was significantly associated with non-response (RR = 1.21; 95% CI: 1.03−1.43) and disease progression (HR = 1.54; 95% CI: 1.14−2.08). A greater longitudinal decrease in IFN-γ levels during treatment was observed in non-responders. Baseline cfDNA and sCTLA-4 levels showed non-significant associations with worse outcomes. Median TMB was 14.7 Muts/Mb (range: 1.7−185.1). Among 28 patients with available baseline ctDNA, one responding patient displayed a deletion in the androgen receptor (AR) gene detectable both at baseline and during treatment. Germline confirmation of an MSH2 pathogenic variant enabled diagnosis of Lynch syndrome in a previously untested patient.

Conclusions: IFN-γ emerged as a promising prognostic and dynamic biomarker in advanced CSCC on anti-PD-1. ctDNA showed frequent alterations and high TMB heterogeneity. AR gene deletion may be a biomarker of response. Findings support liquid biopsy for stratification, pending validation in larger cohorts.

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1666P CemiplimAb-rwlc survivorship and epidemiology (CASE): A prospective, non-interventional study of the safety and effectiveness of cemiplimab in immunocompromised/ immunosuppressed (IC/IS) patients with advanced cutaneous squamous cell carcinoma (CSCC) at 2 years' follow-up

S. Park[1], G. Rabinowits[2], M. Kelley[3], E. Ruiz[4], F. Worden[5], C. Zhu[6], D. McIntyre[6], R. Quek[7], J. Wang[8], K. Tumaneng[9], S. Nia[10], J-F. Pouliot[11]

[1]Medicine, University of California San Diego - UCSD, La Jolla, United States of America; [2]Hematology and Oncology, Miami Cancer Institute/ Baptist Health South Florida, Miami, United States of America; [3]Department of Medicine, Durham VA Medical Center, Duke University School of Medicine, Durham, United States of America; [4]Dermatology, Brigham and Women's Hospital, Boston, United States of America; [5]Hematology and Oncology, University of Michigan, Ann Arbor, United States of America; [6]Biostatistics Medical Affairs, Regeneron Pharmaceuticals, Inc., Tarrytown, United States of America; [7]Health Economics and Outcomes Research, Regeneron Pharmaceuticals, Inc., Tarrytown, United States of America; [8]Clinical Oncology Research, Regeneron Pharmaceuticals, Inc., Tarrytown, United States of America; [9]Global Medical Affairs, Oncology, Regeneron Pharmaceuticals, Inc., Tarrytown, United States of America; [10]Oncology, Regeneron Pharmaceuticals, Inc., Tarrytown, United States of America; [11]Global Medical Affairs, Regeneron Pharmaceuticals, Inc - Corporate Headquarters, Tarrytown, United States of America

Background: Cemiplimab is a programmed death-1 (PD-1) inhibitor approved in patients with locally advanced or metastatic CSCC who are not candidates for curative surgery or radiation. Despite high prevalence of CSCC, IC/IS patients have been excluded from registration studies due to potentially reduced treatment efficacy.

Methods: CASE is a phase 4, multicentre, prospective, non-interventional study evaluating the effectiveness and safety of cemiplimab in patients with advanced CSCC (NCT03836105). Data were collected from 65 United States academic and community oncology centres.

Results: As of 24 March 2025, of the 254 patients with advanced CSCC who received $\geq$1 dose of intravenous cemiplimab 350 mg every 3 weeks, 42 were IC/IS. Most IC/IS patients were white (90.5%), male (76.2%), $\geq$65 years of age (88.1%) and had locally advanced disease (57.1%). The patients were divided into 3 categories: hematologic malignancies (45% - CLL, myeloproliferative disorders or leukemia), immunosuppressed (33% - immune disorders or HIV) and organ transplants (22% - kidney, liver or pancreas). Median duration of exposure was 43.9 weeks. Objective response rate was achieved in 45.2% of patients, with 23.8% having a complete response. Median progression-free survival was similar in IC/IS vs non-IC/IS patients, at 14.6 and 15.7 months, respectively. Treatment-related immune-related adverse events (AEs) occurred in 31.0% of patients, and treatment-related serious AEs occurred in 7.1% of patients. The proportion of IC/IS patients experiencing AEs was similar to that of the non-IC/IS population. In the 9 transplant patients, 7/9 had stable transplant status, and 2/9 had rejection episodes leading to treatment-related allograft loss. All 7 patients receiving an mTOR inhibitor (5) or tacrolimus (2)-based immunosuppression had stable transplant function.

Conclusions: This analysis suggests the safety and efficacy of cemiplimab in IC/IS CSCC patients are similar to those observed in non-IC/IS patients in this real-world study.

Clinical trial identification: NCT03836105.

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1667P Nivolumab (NIVO) +/- relatlimab (RELA) or ipilimumab (IPI) for patients (pts) w/ treatment-naïve or -refractory advanced basal cell carcinoma (aBCC)

G. Warrier[1], J.S. Deutsch[2], K.M. Schenk[3], M. Schollenberger[4], T. McMiller[5], W.H. Sharfman[6], A. Pons[7], E.M. Will[8], V. Madan[9], I. Morales[5], M. Wang[10], M. Considine[11], L. Cope[11], J. Hoang[12], D. lewis[13], A. Berger[5], H. Wang[11], J. Taube[9], S.L. Topalian[14], E.J. Lipson[15]

[1]Medical Oncology Department, Johns Hopkins University, Baltimore, United States of America; [2]Pathology Department, Johns Hopkins Hospital, Baltimore, United States of America; [3]Oncology Department, Johns Hopkins Medicine - The Sidney Kimmel Comprehensive Cancer Center, Baltimore, United States of America; [4]Medical Oncology, Johns Hopkins Medicine - The Sidney Kimmel Comprehensive Cancer Center, Baltimore, United States of America; [5]Surgery, Johns Hopkins University - School of Medicine, Baltimore, United States of America; [6]Medical Oncology Department, Johns Hopkins Kimmel Cancer Center at Greenspring Station, Lutherville, United States of America; [7]Oncology, Johns Hopkins University - School of Medicine, Baltimore, United States of America; [8]Pathology, Johns Hopkins Hospital, Baltimore, United States of America; [9]Pathology, Johns Hopkins University - School of Medicine, Baltimore, United States of America; [10]Surgery, Johns Hopkins University, Baltimore, United States of America; [11]Oncology - Biostatistics, Johns Hopkins University - School of Medicine, Baltimore, United States of America; [12]Radiology, Johns Hopkins University - School of Medicine, Baltimore, United States of America; [13]Dermatology, Johns Hopkins University - School of Medicine, Baltimore, United States of America; [14]Surgery, Johns Hopkins University - School of Medicine, Baltimore, United States of America; [15]Oncology, Johns Hopkins Medicine - The Sidney Kimmel Comprehensive Cancer Center, Baltimore, United States of America

Background: More effective therapies for pts with locally-advanced/metastatic BCC are sorely needed. Anti-PD-1 is standard of care in 2nd line (2L) after hedgehog inhibitors (HHi), mediating an overall response rate (ORR; CR+PR) of $\sim$22-31%. This phase 2 trial (NCT03521830) investigates if 1L NIVO (anti-PD-1) will improve ORR, and if NIVO+RELA (anti-LAG-3) or NIVO+IPI (anti-CTLA-4) can salvage pts w/ anti-PD-1 refractory (aPD1R) aBCC.

Methods: Adults w/ HHi-naïve or -exposed aBCC received NIVO alone for $\leq$48W. Primary endpoint was ORR per RECIST v1.1. Among 32 HHi-naïve pts, $\geq$14 responses (CR+PR) had 81% power to detect ORR 50% (null hypothesis 31%; $\alpha = 0.1$). Pts w/ aPD1R aBCC (stable disease $\geq$36W or progressive disease) received NIVO+RELA; those refractory to NIVO+RELA received NIVO+IPI. Pre- and on-NIVO tumor biopsies were assessed for expression of candidate immune markers with multiplex qRT-PCR and IHC.

Results: From 12/2018 – 4/2025, 35 pts received NIVO alone (Table). Among 28 evaluable HHi-naïve pts, ORR was 50% (14/28, 95% CI: 31 – 69%) meeting the endpoint; median follow-up 25.6 mo, median progression free survival 13.9 mo, median duration of response 17.3 mo. In pts w/ aPD1R aBCC, responses were seen with NIVO+RELA or NIVO+IPI, primarily in HHi-naïve pts. Safety profiles were consistent w/ prior experience. By IHC, pre-NIVO tumor biopsies exhibited a trend toward increased ratio of LAG-3+/CD3+ T cells in non-responders (NR; mean 0.40 v 0.20, 1-sided $p = 0.16$); CD3+ T cells increased from pre- to on-NIVO in R ($p = 0.03$) but not in NR. With gene expression profiling, R (but not NR) showed a significant shift from an immune-suppressive to -reactive tumor milieu after NIVO.

Table: 1667P Patient cohorts, treatments, and response to therapy

Patient population (Therapy)	Pts enrolled	ORR among evaluable pts
Anti-PD-1 naïve (NIVO 480 mg IV q4W x 12)		
All	35	47% (15/32) 1 unconfirmed PR
HHi-naïve	31	50% (14/28) 1 unconfirmed PR
HHi-exposed	4	25% (1/4)
Anti-PD-1 refractory (NIVO 480 mg + RELA 480-960 mg IV q4W x 12) Pts may have received anti-PD-1 monotherapy on or outside of this study		
All	20	21% (3/14)
HHi-naïve	13	38% (3/8)
HHi-exposed	7	0% (0/6)
Anti-PD-1 + anti-LAG-3 refractory (NIVO 240 mg + IPI 1 mg/kg IV q3W x 4, then NIVO 480 mg q4W x 7) Pts may have received anti-PD-1 monotherapy on or outside of this study		
All	9	60% (3/5)
HHi-naïve	5	100% (1/1)
HHi-exposed	4	50% (2/4*) *1 pt did not receive RELA

Conclusions: In aBCC, 1L NIVO improves ORR over that reported w/ 2L anti-PD-1 after HHi. In aPD1R aBCC, NIVO+RELA and NIVO+IPI can mediate tumor response.

Our findings support consideration of NIVO as 1L therapy for pts w/ aBCC. Further testing of NIVO+RELA and NIVO+IPI in aBCC is warranted.

Clinical trial identification: NCT03521830.

Legal entity responsible for the study: Johns Hopkins University School of Medicine Baltimore, MD 21287 USA PI: Evan J. Lipson, MD Associate Professor, Medical Oncology.

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1668P A phase II trial combining pulsed sonidegib (soni) and cemiplimab (cemi) in advanced basal cell carcinoma (aBCC)

E. Ramelyte[1], G. Restivo[1], Z. Balázs[2], A.L. Suter[3], X. Yang[2], P. Turko[1], F. Sella[1], M. Esposito[3], V. Hermes[3], A. Tastanova[3], M. Schmid[3], L. Pawlik[3], A. Boerlin[4], A. Ciernik[5], N.M. Roshardt Prieto[5], J. Narainsing[6], M. Krauthammer[7], M.P. Levesque[8], R. Dummer[9]

[1]Dermatology Department, University Hospital Zurich, Zurich, Switzerland; [2]Department of Quantitative Biomedicine, University Hospital Zurich, Zurich, Switzerland; [3]Dermatology USZ, University Hospital Zurich, Zurich, Switzerland; [4]Department of Dermatology, USZ - University Hospital Zurich, Zurich, Switzerland; [5]Dermatology, USZ - University Hospital Zurich, Zurich, Switzerland; [6]Dermatology, University Hospital Zurich, Zurich, Switzerland; [7]DQBM, UZH - Universitat Zurich Schlieren (WAA), Schlieren, Switzerland; [8]Dermatology Department, USZ - University Hospital Zurich, Zurich, Switzerland; [9]Dermatology Department, University Hospital Zurich, University Zurich, Kantonsspital Aarau, Zurich and Aarau, Switzerland

Background: Hedgehog inhibitors (HHI) are the preferred 1st line and the anti-PD1 antibody Cemi is the recommended 2nd line therapy in aBCC. A short term combo of Soni with Cemi was investigated in this clinical study.

Methods: 20 aBCC patients (pts) with various comorbidities were treated with combined pulsed Soni 200mg QD (2w on, 2w off) and Cemi 350mg Q3W for 26 weeks (NCT04679480). The primary endpoint was the best response by week 26. Secondary endpoints were safety, best overall response (OR). The trial included extensive translational investigations. We performed single-cell RNA sequencing (scRNAseq) on peripheral blood mononuclear cells (PMBC) (n=9 pts), measured serum cytokines by multiplex ELISA (n= 19 pts) longitudinally and analyzed baseline and on treatment tumor biopsies using spatial transcriptomics (n=3 pts) and multicolor immunohistochemistry (n=4 pts).

Results: Treatment-related adverse events all grades included muscle cramps (60%), dysgeusia (35%) and CPK elevations (60%). The incidence of hypothyroidism (35%) and rash (40%) was higher than expected suggesting enhanced immune activity by the combo. The OR at week 26 was >80% in 19 patients (1 not evaluable due to a treatment-unrelated death) with 4 CR, 12 PR, and 3 SD. OR further improved during follow up. scRNAseq after Soni treatment showed changes in the composition of PBMC and induction of IFNγ signalling in CD8 and CD4 T-cells and a TNFα response in monocytes. Multiplex ELISA demonstrated an increase in cytokines including IL-12, IL-18, MIG, IP-10, I-TAC and TNF-α after Cemi initiation. Spatial transcriptomics highlighted an increased transcription of antigen presentation machinery molecules after Soni treatment and the close proximity of lymphocytes and tumor cells during therapy.

Conclusions: Short term pulsed Soni and Cemi is feasible, shows promising clinical efficacy associated with improved antigen presentation accompanied by an IFNγ response. Unexpectedly, Soni alone already resulted in systemic immune activation.

Clinical trial identification: NTC04679480.

Legal entity responsible for the study: The authors.

Funding: Regeneron, Sanofi, Sun Pharma.

Disclosure: E. Ramelyte: Financial Interests, Personal, Advisory Board, and invited Speaker: BMS; Financial Interests, Personal, Speaker, Consultant, Advisor: Eli Lilly, MSD, Regeneron; Financial Interests, Personal, Other, Travel Grant: Galderma; Financial Interests, Personal, Advisory Board: Leo Pharma; Financial Interests, Personal, Advisory Board, and invited Speaker and other Travel support: Pierre Fabre; Financial Interests, Personal, Advisory Board, and invited Speaker and writing engagement and travel grant: Sanofi; Financial Interests, Personal, Other: Uptodate inc. Royalties; Non-Financial Interests, Personal and Institutional, Coordinating PI: Sanofi; Non-Financial Interests, Personal, Other: ASCO Member, SGDV Member. P. Turko: Financial Interests, Personal, Stocks/Shares: Oncobit. M. Krauthammer: Financial Interests, Personal, Other, Advisor: Oncobit AG; Financial Interests, Personal, Ownership Interest: Oncobit AG. M.P. Levesque: Financial Interests, Personal and Institutional, Steering Committee Member: Oncobit; Financial Interests, Personal and Institutional, Funding: Bacoba, Leica, Scailyte. R. Dummer: Financial Interests, Personal, Other, Consulting and/or advisory role: Novartis, Merck Sharp & Dohme (MSD), Bristol Myers Squibb (BMS), Roche, Amgen, Takeda, Pierre Fabre, Sun Pharma, Sanofi, Catalym, Second Genome, Regeneron, MaviVAX SA, T3 Pharma, Pfizer, Simcere; Financial Interests, Personal, Other, medical advisor: Oncobit. All other authors have declared no conflicts of interest.

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1669P Prognostic impact of organ-specific metastasis on survival in patients with Merkel cell carcinoma

K. Khaddour[1], T. Akaike[2], D. Hippe[3], A. Menon[2], K. Murchison[4], A. Camargo[5], F. Bahar[4], M. Lobo[6], J.A. DeCaprio[7], P. Nghiem[2], M. Thakuria[8], A. Silk[9]

[1]Medical Oncology, Dana-Farber Cancer Institute- Harvard Medical School, Boston, United States of America; [2]Department of Dermatology, University of Washington, Seattle, United States of America; [3]Department of Biostatistics, Fred Hutchinson Cancer Research Center, Seattle, United States of America; [4]Department of Medicine, Dana-Farber Cancer Institute, Boston, United States of America; [5]Cutaneous Oncology, Dana-Farber Cancer Institute, Boston, United States of America; [6]Department of Surgery, A.C. Camargo Cancer Center - Unidade Antonio Prudente, Sao Paulo, Brazil; [7]Department of Medicine, Dana-Farber Cancer Institute- Harvard Medical School, Boston, United States of America; [8]Department of Dermatology, Brigham and Women's Hospital- Harvard Medical School, Boston, United States of America; [9]Department of Medicine, Dana-Farber Cancer Institute- Harvard Medical School, Boston, United States of America

Background: Merkel cell carcinoma (MCC) is a rare aggressive skin neuroendocrine cancer with a high predilection for metastases. This study investigates survival of patients with metastatic MCC based on the site(s) involved at the time of diagnosis.

Methods: In this study conducted at Dana-Farber Cancer Institute/ Brigham and Women's Hospital and University of Washington, we included patients who developed metastatic MCC and were enrolled in observational research studies. Clinical and pathologic characteristics were collected. We analyzed associations of the sites of metastases with overall survival (OS) and disease-specific survival (DSS) using multivariable Cox and Fine-Gray regression models. We compare a new proposed M-stage to the AJCC 8th edition using the c-index and the Akaike information criterion (AIC).

Results: Among 687 patients, 126 had distant metastases at time of initial diagnosis, while 561 patients developed distant metastases following diagnosis of local or regional disease. The cohort was 73% male, and 22% immunosuppressed. The most common sites of metastases were lymph nodes (50%), skin/subcutaneous (24%), bone (24%), liver (24%), and lung (8%). In multivariable analyses, metastases in the liver (hazard ratio [HR] > 1.8, p< 0.001), bone (HR > 1.7, p < 0.001), and pancreas (HR > 1.9, p = 0.001) each had a significant and independent association with worse OS and DSS relative to metastases in the skin/subcutaneous tissue or lymph nodes only. We propose a new M-stage classification for MCC which includes the number of high-risk sites (liver, bone and pancreas) and predicted DSS better than AJCC 8th edition M-stage (c-index: 0.56 vs. 0.58, p = 0.002; ΔAIC: 26.9).

Table: 1669P

	Current AJCC 8th edition M-stage	New proposed M-stage
M1a	Lymph nodes & subcutaneous	Any non-high-risk organs
M1b	Lung	1 high-risk organ: liver, bone or pancreas
M1c	All other sites	2 or 3 high-risk organs

Conclusions: Liver, bone and pancreas metastases in patients with metastatic MCC are associated with worse survival. The number of high-risk sites can provide better prognostic information on clinical outcomes in MCC. (Table) AJCC 8^{th} edition compared to New proposed M-stage in MCC.

Legal entity responsible for the study: K. Khaddour.

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Disclosure: All authors have declared no conflicts of interest.

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1670P Merkel cell carcinoma: Trends in incidence, epidemiology and genomic landscape in the united states

L. Kamal

Department of Clinical Oncology, Menofia University Faculty of Medicine, Shebein Elkom, Egypt

Background: Merkel cell carcinoma (MCC) is a rare form of skin cancer that had been associated with poor prognosis. Studies that describe the epidemiology and genomics of MCC were often limited by the small sample size and short duration of follow-up. Additionally, it remains unclear how much changes in disease epidemiology have occurred as a result of advances in MCC management.

Methods: This was a retrospective study of cases diagnosed with MCC . The epidemiology analysis set included cases diagnosed between 2000 and 2022 in the Surveillance, Epidemiology, and End Results Program (SEER 17) database (Nov 2024 submission. The genomics analysis by AACR GENIE Cohort v.17.0.We used cbioportal to extract data on mutations, copy number alterations, and structural variants in patients with MCC. Only genes that were profiled in at least 50 samples were included in the analysis.

Results: In the epidemiology analysis set, 12,291 patients with MCC were diagnosed between 2000-2022. Most cases were males (n=7,857), white (n=11,662), and originating from the head and neck region (n=4319). The highest peak of MCC diagnoses was in 2021 with 716 diagnoses. The overall annual incidence rate was 0.7 per 100,000. MCC incidence rates increased with age with the highest incidence observed in patients over 85 ($p < .05$) . The Observed five- years survival rate was 48.70% (95% CI 47.5%-49.9 %). Higher survival was noted in patients diagnosed between 2017 and 2022 of 68.10% (95% CI 63.50%- 72.30%) The genomics analysis set included 442 patients. The most frequently mutated genes were TP53 (34.2%), RB1 (27.4%), KMT2D (20.2%), and NOTCH1 (20.1%). The most frequent copy number alterations were observed in RB1 (homdel in 4%), TRIP13 (amp in 3.8%), CYSLTR2 (homdel in 3.6%), and TNFRSF14 (homdel in 3.3%). The most common genes with structural variants included TP53 (0.7%), RB1 (0.5%), ESWR1 (0.9%), and PPARG (0.3%).

Conclusions: This is the largest study to date to report on epidemiological trends and genomic landscape of MCC. It demonstrates that most patients with MCC were white, male, and the diagnosis incidence increased with age. Improved survival was observed in patients diagnosed after 2017. Alterations in multiple genes were observed in MCC with possible therapeutic implications.

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1671P Treatment outcomes of cutaneous angiosarcoma of the head and neck in Japan: A multicenter retrospective study of 336 cases

E. Nakano[1], Y. Fujisawa[2], M. Masuzawa[3], S. Mori[4], T. Yanagi[5], H. Kato[6], M. Aoki[7], C. Watanabe[8], D. Yokoyama[9], Y. Kiniwa[10], Y. Nakamura[11], S. Ohe[12], K. Yamakawa[13], T. Suyama[14], T. Ishii[15], S. Kawashima[16], H. Kokubu[17], T. Takenouchi[18], T. Funakoshi[19], K. Namikawa[1]

[1]Dermatologic Oncology, National Cancer Center Hospital, Tokyo, Japan; [2]Dermatology, Ehime University, Ehime, Japan; [3]Dermatology, Kitasato University School of Medicine, Kanagawa, Japan; [4]Dermatology, Nagoya University Graduate School of Medicine, Nagoya, Japan; [5]Dermatology, University of the Ryukyus Graduate School of Medicine, Okinawa, Japan; [6]Geriatric and Environmental Dermatology, Nagoya City University Hospital, Nagoya, Japan; [7]Dermato-Oncology, NHO Kagoshima Medical Center, Kagoshima, Japan; [8]Dermatology and Plastic Surgery, Kumamoto University, Kumamoto, Japan; [9]Division of Dermatology, Department of Internal Related, Kobe University Graduate School of Medicine, Kobe, Japan; [10]Dermatology, Shinshu University, School of Medicine, Matsumoto, Japan; [11]Dermatology, University of Tsukuba, Tsukuba, Japan; [12]Dermatologic Oncology, Osaka International Cancer Institute, Osaka, Japan; [13]Environmental Immuno-Dermatology, Yokohama City University Hospital, Yokohama, Japan; [14]Dermatology, Dokkyo Medical University, Saitama Medical Center, Koshigaya, Japan; [15]Dermatology, Toyama Prefectural Central Hospital, Toyama, Japan; [16]Dermatology, Chiba University, Chiba, Japan; [17]Dermatology, Shiga University of Medical Science, Otsu, Japan; [18]Dermatology Dept, Niigata Cancer Center Hospital, Niigata, Japan; [19]Dermatology Department, Keio University School of Medicine, Shinjuku-ku, Japan

Background: Cutaneous angiosarcoma (cAS) is a rare and aggressive malignant vascular tumor that most commonly develops in the head and neck of elderly patients. Although surgery combined with postoperative radiotherapy is considered the standard treatment for cAS in Western countries, many patients in Japan present with large tumors, which are often deemed inoperable. Evidence regarding the efficacy of treatment strategies for patients with cAS in Japan remains limited. Therefore, we conducted a multicenter retrospective study to evaluate real-world treatment outcomes.

Methods: We retrospectively collected and analyzed clinical data of patients treated at 35 institutions in Japan between April 2018 and March 2023. Eligible patients received surgery, radiotherapy, or chemotherapy and had no regional or distant metastases at diagnosis.

Results: A total of 336 patients were included, with a median follow-up of 24.1 (range, 0.3—88.3) months. The median age was 79 (41—98) years, 244 patients (74.4%) were male, and 105 (31.3%) had a tumor over 15 cm. Twenty-five patients (7.6%) received a combination of surgery, radiotherapy, and chemotherapy, while 196 patients (59.8%) were treated with chemotherapy and radiotherapy. The median progression-free survival (PFS) and overall survival (OS) for all patients were 12.9 and 29.9 months, respectively. Among those treated with a combination of surgery, radiotherapy, and chemotherapy, the median PFS was 35.3 months, and the median OS was not reached. In patients treated with chemotherapy and radiotherapy, the median PFS and OS were 15.8 months and 34.5 months, respectively. In a landmark analysis, patients who continued chemotherapy for more than 6 months after achieving a complete response to combined chemotherapy and radiotherapy demonstrated significantly better OS than patients who discontinued within 6 months (median OS: not reached vs. 29.6 months; p=0.04).

Conclusions: Multimodal therapy for head and neck cAS in Japan was associated with favorable outcomes. Prolonged chemotherapy following a complete response may further improve survival in this aggressive disease.

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Disclosure: All authors have declared no conflicts of interest.

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1672P A phase II study of boron neutron capture therapy (BNCT) using CICS-1 and SPM-011 in patients with unresectable angiosarcoma

T. Kashihara[1], S. Nakamura[2], K. Namikawa[3], D. Ogata[3], E. Nakano[3], H. Endo[2], M. Takemori[2], T. Goka[4], N. Yamazaki[3], H. Igaki[1]

[1]Department of Radiation Oncology, NCCH - National Cancer Center Hospital-Tsukiji Campus, Chuo-ku, Japan; [2]Department of Medical Physics, NCCH - National Cancer Center Hospital-Tsukiji Campus, Chuo-ku, Japan; [3]Department of Dermatologic Oncology, National Cancer Center Hospital, Chuo-ku, Japan; [4]Department of Radiological Technology Radiological Oncology, NCCH - National Cancer Center Hospital-Tsukiji Campus, Chuo-ku, Japan

Background: Unresectable angiosarcoma is an aggressive malignancy with poor outcomes and limited treatment options. Boron neutron capture therapy (BNCT) is a radiotherapy that selectively delivers high linear energy transfer radiation to tumor cells by nuclear reaction between thermal neutrons and boron-10. We conducted a single-arm phase II trial to assess the efficacy and safety of BNCT using an accelerator-based neutron source (CICS-1) and the boron compound (SPM-011; Steboronine®) in patients with unresectable angiosarcoma.

Methods: Ten patients with unresectable locally advanced or recurrent angiosarcoma were enrolled, and treated with BNCT at a prescribed skin dose of 18 Gy-Eq, based on phase I safety data. Nine were patients with primary cutaneous angiosarcoma of the scalp, and one patient had radiation-induced cutaneous angiosarcoma after postoperative radiotherapy for breast cancer. The primary endpoint was objective response rate (ORR) within 90 days after BNCT, assessed by central review according to RECIST v1.1. The ORR of 18.5% in the ANGIOTAX trial was used as the threshold, with a significance level of 5% one-sided and a power of 80%, and the primary endpoint was considered to have been achieved if a response was confirmed in at least 5 out of 10 patients.

Results: All patients completed BNCT as planned. Two patients achieved a complete response and three had a partial response, resulting in a confirmed ORR of 50% (5/10), meeting the predefined efficacy threshold. Treatment was well tolerated; acute grade ≥2 adverse events were limited to asymptomatic grade 3 hyperamylasemia in all patients and grade 2 leukopenia in one patient, all resolved without sequelae. Skin toxicity was only grade 1 radiation dermatitis observed in two patients.

Conclusions: Accelerator-based BNCT using CICS-1 and SPM-011 demonstrated promising efficacy with a manageable safety profile in patients with unresectable angiosarcoma, significantly surpassing historical benchmarks. These encouraging results highlight BNCT's potential as a practice-changing therapeutic approach for this challenging malignancy, warranting further evaluation for difficult-to-treat disease.

Clinical trial identification: NCT05601232, 1 November 2022.

Legal entity responsible for the study: Cancer Intelligence Care Systems, Inc. and Stella Pharma Corporation.

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Disclosure: T. Kashihara: Financial Interests, Personal, Speaker's Bureau: HIMEDIC, Inc., AstraZeneca Co., Ltd., Eisai Co., Ltd., Guerbet Japan; Financial Interests, Personal, Advisory Role: Novartis Pharma K.K. S. Nakamura: Financial Interests, Personal, Research Grant: Cancer Intelligence Care Systems, Inc. H. Igaki: Financial Interests, Personal, Speaker's Bureau: HIMEDIC, Cancer Intelligence Care Systems, Inc.; Financial Interests, Personal, Research Grant: HIMEDIC, Cancer Intelligence Care Systems, Inc. All other authors have declared no conflicts of interest.

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1673P Phase II clinical trial of trastuzumab emtansine for HER2-positive advanced extramammary Paget's disease (TEMENOS-2)

S. Arakawa[1], I. Hirai[1], Y. Nakamura[1], H. Uchi[2], S. Ohe[3], T. Takenouchi[4], S. Mori[5], T. Takai[6], S. Yoshikawa[7], T. Fujimura[8], Y. Yamamoto[9], R. Takemura[10], T. Funakoshi[1]

[1]Department of Dermatology, Keio University School of Medicine, Shinjuku-ku, Japan; [2]Department of Dermato-Oncology, National Hospital Organization Kyushu Cancer Center, Fukuoka, Japan; [3]Department of Dermatologic Oncology, Osaka International Cancer Institute, Osaka, Japan; [4]Division of Dermatology, Niigata Cancer Center Hospital, Niigata, Japan; [5]Department of Dermatology, Nagoya University Graduate School of Medicine, Showa, Japan; [6]Department of Dermatology, Hyogo Cancer Center, Akashi, Japan; [7]Department of Dermatology, Shizuoka Cancer Center, Sunto-Gun, Japan; [8]Department of Dermatology, Tohoku University, Sendai, Japan; [9]Department of Dermatology, Wakayama Medical University, Wakayama, Japan; [10]Department of Preventive Medicine and Public Health, Biostatistics Unit at Clinical and Translational Research Center, Keio University School of Medicine, Shinjuku-ku, Japan

Background: Metastatic extramammary Paget's disease (EMPD) is a rare cutaneous adenocarcinoma lacking standard chemotherapy due to limited clinical evidence. Approximately 40% of EMPD patients overexpress human epidermal growth factor receptor 2 (HER2), indicating a potential therapeutic target. Although trastuzumab, a monoclonal antibody targeting HER2, has demonstrated some clinical benefit in HER2-positive EMPD, its efficacy remains limited. Trastuzumab emtansine (T-DM1), an antibody-drug conjugate that combines trastuzumab with the microtubule inhibitor emtansine (DM1) via a stable linker, may offer a promising treatment strategy for advanced HER2-positive EMPD.

Methods: This phase II, single-arm, investigator-initiated, multicenter trial enrolled 14 Japanese patients with centrally confirmed HER2-positive metastatic EMPD who had not received prior HER2-targeted therapy. HER2-positivity was defined as IHC 3+ or ISH amplification. T-DM1 (3.6 mg/kg) was administered intravenously every 3 weeks for up to two years. Imaging assessments were performed every six weeks. The primary endpoint was the objective response rate (ORR) by independent central review. Secondary endpoints included investigator-assessed ORR, progression-free survival, overall survival, tumor shrinkage, and safety, which was monitored throughout the study.

Results: The primary analysis population for efficacy included all 14 patients in the Full Analysis Set (FAS). All patients received the planned treatment. The median number of dosing cycles was 13, and the median dosing duration was 9.2 months. The investigator-assessed ORR, a secondary endpoint, was 85.7% (95% CI: 57.2—98.2 %), with the lower limit exceeding the predefined threshold of 5%. No unexpected or serious adverse events were reported. The primary endpoint, centrally reviewed ORR based on interim data, is expected to be available by the end of May 2025.

Conclusions: Although the final analysis is pending, T-DM1 appears to show promising clinical activity and manageable safety in patients with advanced HER2-positive EMPD. These findings support T-DM1 as a viable treatment option for this rare and potentially aggressive malignancy.

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1674P Real-world use of encorafenib + binimetinib in unresectable/metastatic melanoma: Results from the MelBase French database

C. Lebbe[1], L. Mortier[2], H. Montaudie[3], O. Dereure[4], S. Dalle[5], C. Dutriaux[6], G. Jeudy[7], E. Maubec[8], M. Amini Adle[9], M-T. Leccia[10], D. Legoupil[11], F. Brunet Possenti[12], C. Gaudy Marqueste[13], C. Pages[14], P. Stoebner[15], S. Aviles[16], M. Mouri[16], B. Oriano[1], W. Lefevre[17], L. Da Meda[18]

[1]Dermatology Dept., Hopital Saint Louis AP-HP, Paris, France; [2]Dermatology Dept., CHRU de Lille - Hôpital Roger Salengro, Lille, France; [3]Dermatology Dept., Hopital del l'Archet, Nice, France; [4]Dermatology Dept., CHU de Montpellier - Hopital Saint-Eloi, Montpellier, France; [5]Dermatology Dept., Lyon Sud Hospital Center, Pierre Benite, France; [6]Dermatology Dept., CHU Bordeaux - Hopital St. André, Bordeaux, France; [7]Dermatology, CHU Dijon, Dijon, France; [8]Dermatology Department, Hopital Avicenne AP-HP, Bobigny, France; [9]Dermatology Dept., Centre Léon Bérard, Lyon, France; [10]Dermatology Dept., CHU Grenoble Alpes - Site Nord (La Tronche), La Tronche, France; [11]Dermatology Dept., CHRU Brest - Hôpital Morvan, Brest, France; [12]Oncodermatology, Hopital Bichat - Claude-Bernard AP-HP, Paris, France; [13]Oncodermatology, AP-HM - CHU La Timone Enfants, Marseille, France; [14]Oncodermatology, IUCT - Institut Universitaire du Cancer de Toulouse - Oncopole, Toulouse, France; [15]Oncodermatology, CHU Nimes - Hopital Universitaire Carémeau, Nimes, France; [16]Industry, Pierre Fabre Medicament, Boulogne Billancourt, France; [17]Dermatology Dept., Assistance Publique - Hopitaux De Paris, Paris, France; [18]Oncodermatology, APHP Saint Louis, Paris, France

Background: Encorafenib + binimetinib (E+B) is the third anti-BRAF/anti-MEK combination approved in France for advanced melanoma. We assessed the real-world efficacy and safety of this combination.

Methods: Prospective data from MelBase database were analysed. Patients with BRAFV600-mutated unresectable/metastatic melanoma initiated E+B between Aug 2019—Dec 2024 and were followed until Mar 2025. OS, PFS, duration of treatment (DoT) and adverse events (AEs) were assessed.

Results: 220 patients were included (median age: 59). E+B was used in 18% of patients as first line (1L) and in 39% of patients as second line (2L), mostly after immunotherapy. At 10.4-mo median follow-up, 12-mo OS and PFS were 49.8% (95% CI 43.5—57.1) and 26.6% (95% CI 21.3—33.3) respectively; 24-mo OS and PFS were 33.8% (27.8—41.1) and 17.5% (13.0—23.6) respectively. Median DoT was 6.3 mo (5.1—7.9). In subgroup analyses, 24-mo OS for E+B in 1L vs 2L strict was 12.2 mo (4.7—31.6) and 50.4 mo (40.0—63.4) respectively. This could be explained by the fact that patients receiving E+B in 1L had poorer prognostic factors (Table). In patients with brain metastases, 24-mo OS was 19.4 mo (12.7—29.8) vs 45.9 mo (37.3—56.4) in patients without brain metastases. Out of the patients that received E+B in 1L (n=39), 9 patients switched early to immunotherapy before progression; of those, 44.4% had brain metastases, 55.5% had hepatic metastases and 55.5% had LDH > ULN. Among the early switched patients, 12-mo OS and PFS was 38.9% (16.2—93.1) and 14.8% (2.6—86.0) respectively. 19.1% of patients permanently discontinued and 20.0% of patients temporarily discontinued E+B. Grade ≥3 AEs occurred in 21.3%.

Table: 1674P Baseline characteristics

n patients (%)	E+B 1L (N=39)	E+B 2L (N=86)
LDH		
≤ULN	22 (56.4)	59 (68.6)
>ULN	17 (43.6)	27 (31.4)
Number of metastatic sites		
<3	15 (38.5)	47 (54.7)
≥3	24 (61.5)	39 (45.3)
Brain mets		
Yes	17 (43.6)	33 (38.4)
No	22 (56.4)	53 (61.6)
Liver mets		
Yes	21 (53.8)	27 (31.4)
No	18 (46.2)	59 (68.6)
ECOG PS		
0	18 (46.2)	46 (53.5)
1	7 (17.9)	25 (29.1)
2	7 (17.9)	8 (9.3)
≥3	7 (17.9)	7 (8.1)

Conclusions: Despite more heavily pretreated patients with worse prognosis than in phase III COLUMBUS study, our analysis shows real-world robust efficacy for E+B, especially in 2L. Some high-risk patients (brain mets, liver mets, high LDH) may benefit from the early switch. Safety was consistent with known profile.

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1675P PD-1 inhibitor therapy for cutaneous squamous cell carcinoma in patients with indolent lymphoma: A multi-center retrospective

S. Batchu[1], A. Burbano[2], F. Aya Moreno[3], G. Roccuzzo[4], C. Oh[5], C. Trojaniello[6], J. Sambale[7], L. Thompson[8], R. Falcone[9], I. Pires da Silva[10], L. Zimmer[11], G.K. In[12], M. Brandlmaier[13], V.T-L. Ma[14], S. Park[15], F. Spagnolo[16], W. Samlowski[17], U. Leiter-Stoppke[18], S. Cohen[19], A. Algazi[20]

[1]School of Medicine, UCSF - University of California San Francisco - Parnassus Campus, San Francisco, United States of America; [2]Surgery, Massachusetts General Hospital Cancer Center, Harvard Medical School, Boston, United States of America; [3]Medical Oncology Department, Hospital Clinic of Barcelona, Barcelona, Spain; [4]Department of Dermatology, University of Turin, AOU Città della Salute e della Scienza di Torino, Turin, Italy; [5]University of Miami, University of Miami, Sylvester Comprehensive Cancer Center, Deerfield Beach, United States of America; [6]Melanoma and immunotherapy, Istituto Nazionale Tumori IRCCS - Fondazione G. Pascale, Naples, Italy; [7]Department of Dermatology, Uniklinikum Erlangen, Friedrich-Alexander University Erlangen-Nürnberg (FAU), Erlangen, Germany; [8]School of Medicine, OHSU - Oregon Health Science University, Portland, United States of America; [9]Phase I Unit, Fondazione Policlinico Universitario Agostino Gemelli IRCCS, Rome, Italy; [10]Medical Oncology Department, Melanoma Institute Australia, Wollstonecraft, Australia; [11]Dermatology, Universitätsklinikum Essen, Essen, Germany; [12]Medicine Dept., USC Norris Comprehensive Cancer Center, Los Angeles, United States of America; [13]University Department of Dermatology and Allergology, Paracelsus Medizinische Privatuniversität, Salzburg, Austria; [14]Department of Internal Medicine, University of Wisconsin Carbone Cancer Center, Madison, United States of America; [15]Cutaneous Disease Program, University of California San Diego - UCSD, La Jolla, United States of America; [16]Medical Oncology, IRCCS Ospedale Policlinico San Martino, Genoa, Italy; [17]Medical Oncology, University of Nevada, Reno, United States of America; [18]Department of Dermatology, Universitaets-Hautklinik Tuebingen, Tuebingen, Germany; [19]Surgical Oncology, Massachusetts General Hospital Cancer Center, Harvard Medical School, Boston, United States of America; [20]Medicine: Hematology / Oncology Department, UCSF Medical Center at Mount Zion, San Francisco, United States of America

Background: Indolent CD5+ hematologic malignancies, including chronic lymphocytic leukemia (CLL), are associated with an increased risk for advanced cutaneous squamous cell carcinoma (cuSCC), but data regarding the clinical efficacy of PD-1 antibodies in these patients are limited. Here, we investigate cuSCC outcomes in this population.

Methods: Included patients had an indolent CD5+ malignancy, a diagnosis of cuSCC, and received at least 3 doses of PD-1 antibody monotherapy or discontinued due to toxicity. The primary endpoint was investigator-assessed objective response rate by RECIST v1.1 or, when not possible, clinical assessment. Secondary endpoints included PFS, OS, TTP and adverse events based on available data. CLL Rai staging changes during PD-1 antibody therapy and the impact of baseline Rai stage on cuSCC response rates were also described.

Results: A total of 50 patients (46 male) with a median age of 81 years (range 58-91) were included. 47 patients had CLL/SLL (baseline Rai Stage: 0 = 14; I = 9; II = 6; III = 7; IV = 9; Unknown = 2). There was 1 patient with Stage 1 cuSCC, 4 with Stage II, 13 with Stage III, and 32 with Stage IV. The median number of PD-1 antibody doses was 7.5 (range: 1-59). The overall response rate was 30/50 (60.0%) with 15 complete responses, 15 partial responses, 9 stable disease, and 11 progressive disease per RECIST v1.1 criteria or, in 11 cases, clinical assessment. The median PFS for the entire population was 18.5 months (95% CI 7.0 to 30.1), and the median OS was 35.0 months (95% CI 30.1 to 39.8). The median PFS in responding patients was 34.5 months (95% CI 20.2 to 48.7). cuSCC responding patients included 8/16 patients with Rai Stage III/IV CLL/SLL. Rai staging remained unchanged in all but 6 patients with evaluable data (39/45). Four grade 3 adverse events were reported: rash, abscess, acute renal failure, and hemolytic anemia. There were no grade 4 or 5 TEAEs described.

Conclusions: PD-1 antibody monotherapy for cuSCC in indolent CD5+ non-Hodgkin lymphoma patients induces frequent and sustained clinical responses despite the immune compromise conferred by the underlying hematologic malignancy.

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1676P Early discontinuation of immunotherapy in patients with objective responses: Clinical outcomes in real-world Danish patients with advanced melanoma

R.S. Jurlander[1], M. Donia[1], T.H. Borch[1], S.K. Petersen[2], L.M. Guldbrandt[3], R.B. Friis[3], A.A. Luczak[4], H. Schmidt[3], C.H. Ruhlmann[2], L. Bastholt[2], E. Ellebaek[1], I.M. Svane[1]

[1]Department of Oncology, National Center for Cancer Immune Therapy (CCIT-DK), Herlev Hospital, Herlev, Denmark; [2]Department of Oncology, OUH - Odense University Hospital, Odense, Denmark; [3]Oncology Department, Institute of Clinical Medicine, Aarhus University Hospital Skejby, Aarhus, Denmark; [4]Oncology Department, Aalborg University Hospital, Aalborg, Denmark

Background: Immune checkpoint inhibitors (ICI) have significantly improved outcomes in advanced melanoma, but optimal treatment duration remains unclear. Many patients continue treatment until two years of therapy, treatment limiting toxicity, or progression. Longer treatment is associated with higher cumulative risk of immune-related adverse events and increased resource utilization for both patients and healthcare systems. Since 2019, Danish guidelines have recommended discontinuation of ICI before the 2-year landmark upon confirmed complete response (CR) or confirmed and stable partial response (PR). In this study, we evaluated the real-world outcome (PFS/OS/MSS) of patients who discontinue ICI treatment early due to objective response.

Methods: We retrieved data from the Danish Metastatic Melanoma Database (DAMMED) on patients who initiated first-line ICI therapy (anti-PD1 or anti-PD1/anti-CTLA4) between January 2019 and January 2024. All included patients discontinued treatment before the 2-year landmark due to CR or PR while patients stopping due to toxicity, progression or other causes were excluded.

Results: A total of 167 patients were included (63% male, median age 71 [33-91]). Most had cutaneous melanoma (83%) and received single-drug anti-PD1 (79%). The majority had M1c disease (43%), but 10 (6%) had M1d disease. In total 90 and 77 patients discontinued due to CR and PR, respectively. The median and minimum treatment duration was 8 and 7 months (CR) and 12.4 and 9.6 months (PR). Median follow-up time was 48 months [IQR: 34.3-60.3]. At 12- and 36-months, PFS was 100% and 87% (CR), 96% and 81% (PR). At 12- and 36- months, OS was 99% and 98% (CR), 100% and 86% (PR). Melanoma-specific survival (MSS) at 12- and 36 months was 100% and 99% (CR), 100% and 92% (PR).

Conclusions: Early discontinuation due to an objective response is associated with MSS above 90% at 3-years for patients with PR after a minimum of nine months of treatment. These findings support a potential for shortening the treatment duration in patients with melanoma achieving a PR.

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1677P Recombinant human adenovirus type 5 (H101) combined with Anti-PD-1 therapy in advanced melanoma resistant to PD-1 inhibitors: Mechanistic insights from spatial tumor microenvironment analysis

J. Lin[1], Z. Lin[2], H. Zhang[2], L. Shi[1], D. Wang[2], Y. Huang[1], L. Chen[1], L. Chen[1], Y. Chen[1]

[1]Department of Medical Oncology, Clinical Oncology School of Fujian Medical University, Fujian Cancer Hospital, Fuzhou, China; [2]Department of Phase I Clinical Trial Ward, Clinical Oncology School of Fujian Medical University, Fujian Cancer Hospital, Fuzhou, China

Background: While PD-1 blockade shows clinical benefit in melanoma patients, 30-40% of those who develop resistance continue to pose a significant barrier to achieving durable responses. Rrecombinant human adenovirus type 5 (H101) may remodel the immune tumor microenvironment (TME) via viral mimicry and immunogenic cell death.This study examines the efficacy and safety of the H101 combined with anti-PD-1 therapy in patients with anti-PD-1 refractory advanced melanoma and particularly emphasizes the comprehensive omics characterization of the spatial TME.

Methods: This prospective single-arm study includes 10 patients with PD-1 resistant advanced melanoma. All patients completed at least two cycles of intratumoral administration of H101 in combination with intravenous anti-PD-1 therapy. Spatial transcriptomics (Xenium Prime 5K Human Pan Tissue & Pathways Panel) was used to map the dynamic TME during treatment. Cell type annotation was performed based on a predefined list of marker genes. Additionally, baseline cellular subtypes of treatment responders and non-responders were compared.

Results: The study enrolled 10 patients with melanoma. Tumor response assessment revealed CR (0), PR (2), SD (4) and PD (4), yielding a DCR of 60% (6/10). In the comparison of responders (n=6) and non-responders (n=4) at baseline, the proportion of monocyte(5.83%)and melanoma cell (72.85%) were higher in responders. Non-responders have higher proportion of endothelial cells(9.33%), fibroblasts (8.25%) and B cells(0.63%). Moreover, the dynamic TME analysis revealed that compared to baseline, in P004 and P009 (responder group), there was a significant increase in the proportion of T cells (+18.75% $p<0.001$) and macrophages (+5.55%, $p<0.001$) within 50μm of endothelial.

Conclusions: In conclusion, the spatial analysis of the immune microenvironment provides evidence for the activation of a systemic anti-tumor response by H101 combined with anti-PD-1 therapy. This combination therapy have been provide a new strategy for melanoma resistant to PD-1 inhibitors, thereby enabling precise patient selection to maximize therapeutic benefits.

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1678P Neoadjuvant immunotherapy for patients with resectable stage III/IV cutaneous melanoma: A Swedish retrospective real-world study (NEO-MEL)

A.F. Nelson[1], E. Krabbe[2], K. Björkström[3], B. Jovanovic[4], C.U. Blank[5], A. Carneiro[6], K. Isaksson[7], G. Ullenhag[8], H. Helgadottir[9], L. Ny[10], R. Olofsson Bagge[11]

[1]Oncology Dept., Sahlgrenska University Hospital - Jubileumskliniken, Göteborg, Sweden; [2]Surgery, Institute of Clinical Sciences, Sahlgrenska Academy, University of Gothenburg, Gothenburg, Sweden; [3]Department of Oncology, Karolinska University Hospital, Stockholm, Sweden; [4]Tema Cancer, HHLH, Karolinska University Hospital-Solna, Solna, Sweden; [5]Medical Oncology Dept., NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [6]Hematology, Oncology and Radiation Physics Department, Skane University Hospital CCC, Lund, Sweden; [7]Surgery, Lund University, Lund, Sweden; [8]Oncology Department, University Hospital Uppsala/Akademiska Sjukhuset, Uppsala, Sweden; [9]Oncology Department, Karolinska Institutet - Cancer Center Karolinska (CCK), Solna, Sweden; [10]Dept. of Oncology, Sahlgrenska University Hospital, Göteborg, Sweden; [11]Department of Surgery, University of Gothenburg - The Sahlgrenska Academy, Göteborg, Sweden

Background: Recent clinical trials have demonstrated that neoadjuvant treatment with PD-1 inhibitor, with or without CTLA-4 blockade, gives superior outcomes compared to adjuvant PD-1 monotherapy in patients with resectable stage III—IV cutaneous melanoma. However, the generalizability of these results to real-world, unselected patient populations remains unclear.

Methods: We conducted a retrospective study across four academic centers in Sweden, including patients with macroscopic, resectable cutaneous stage III-IV melanoma who received neoadjuvant immune checkpoint inhibitor (ICI) therapy between January 1, 2022, and September 30, 2024. Clinical and pathological data were extracted from medical records.

Results: A total of 172 patients were included. The median age was 71 years. Patients had lymph node metastases (67%), in-transit metastases (19%), combined lymph node and in-transit metastases (8%), and distant stage IV disease (6%). Patients received a median of two neoadjuvant ICI treatments, with 96% receiving PD-1 inhibitor monotherapy. Planned surgery was completed in 92% of patients, the remaining 8 % did not undergo surgery due to disease progression to inoperable regional disease (only one patient), stage IV or patient preference. Adjuvant PD-1 inhibitor was administered to 64% of patients, with a median of five treatment cycles, while 25% of patients received no adjuvant therapy. Among all patients initiating ICI treatment, response rates, EFS and RFS stratified by response, are presented in the table. Median follow-up time was 12 months. Grade ≥3 adverse events related to neoadjuvant ICI occurred in 8% of patients.

Table: 1678P

Characteristic	All	Lymph node metastases only	In-transit and/or distant metastases
	N = 171	*N* = 115	*N* = 56
Radiological/clincal responders, n (%)	84 (49%)	58 (50%)	26 (46%)
Pathological response, n(%)			
MPR, n (%)	72 (42%)	56 (49%)	16 (28%)
pPR	20 (12%)	15 (13%)	5 (9%)
pNR	53 (31%)	32 (28%)	21 (38%)
No surgery	18 (11%)	9 (8%)	9 (16%)
EFS at 12 m, % (95% CI):			
-All	73% (66-82)	78% (69-87)	63% (49-81)
RFS at 12 m, % (95% CI):			
-MPR	95% (89-100)	96% (89-100)	92% (79-100)
-pPR	93% (80-100)	91% (75-100)	NE
-pNR	58% (44-77)	63% (47-85)	51% (29-91)

Conclusions: In this population-based, real-world study, neoadjuvant ICI therapy for resectable stage III—IV cutaneous melanoma was feasible and efficacy outcomes were comparable to those reported in clinical trials. Patients with regional lymph node metatstases only appeared to have better outcomes.

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1679P Circulating metabolic profiling for predicting tebentafusp efficacy

E.F. Saldanha[1], R.C. Laister[2], B. Wang[3], S. Penny[4], D.M. Pinto[5], M.O. Butler[6], S. Saibil[7]

[1]Clinical Oncology Department, Princess Margaret Cancer Centre, University Health Network, Toronto, Canada; [2]Department of Medical Oncology and Hematology, UHN - University Health Network - Princess Margaret Cancer Centre, Toronto, Canada; [3]Immune Profiling Team, Princess Margaret Cancer Centre, Toronto, Canada; [4]Human Health Therapeutics, National Research Council Canada, Montreal, Ottawa, Canada; [5]Human Health Therapeutics, National Research Council Halifax, Halifax, Canada; [6]Medical Oncology and Hematology Department, UHN - University Health Network - Princess Margaret Cancer Center, Toronto, Canada; [7]Medical Oncology, UHN - University Health Network - Princess Margaret Cancer Center, Toronto, Canada

Background: Tumour metabolites drive T cell dysfunction and hamper immune checkpoint blockade efficacy. However, the impact of metabolic rewiring on the immune mobilizing T cell receptor against cancer (ImmTAC) therapy is unknown. Here, we interrogate the circulating metabolome of pts with metastatic uveal melanoma (MUM) treated with tebentafusp (TEBE), a bispecific ImmTAC, to identify metabolic parameters associated with TEBE benefit.

Methods: The relative abundance of 115 metabolites in baseline (BL) plasma of 18 HLA-A*02:01+ pts with MUM was assessed by targeted metabolomics. Pts treated at the Princess Margaret Cancer Centre (2020-2023) were included. Health records were reviewed to extract clinical data. Descriptive statistics were used to summarize the pts characteristics, treatment modalities, and outcomes. Time-to-event outcomes were analyzed using the Kaplan-Meier method. Overall survival (rwOS) was defined as the time from the first treatment dose to death from any cause. The response rate was assessed using the RECIST v1.1. Statistical significance was determined as a false discovery rate (FDR) <0.05 and p-value <0.05.

Results: 18 HLA-A*02:01+ pts with MUM and treated with TEBE were profiled. The median age was 66 (range: 34 – 91), and 52% were female. 71% received TEBE in the first line. Responding (CR/PR) pts exhibited significantly longer median rwOS not reached ([NR], 95%CI: 21 – NR) vs. 13 months (95%CI: 6 – NR) p = 0.034, for non-responding (SD/PD). Metabolomic analysis of BL plasma identified 8 metabolites whose abundance significantly differed between responding and non-responding pts. Specifically, the bile acid Chenodeoxyglycocholic acid (CDCA) was increased in the plasma of the non-responding pts. Whereas citrulline was increased in responding pts. The Elevated CDCA and the ratio of CDCA and citrulline were associated with a lack of TEBE benefit, ROC AUC = 0.925 (0.738 - 1), ROC AUC 0.963 (0.818 - 1), respectively.

Conclusions: Leveraging a metabolomics approach, we identified the bile acid CDCA as a potential metabolic biomarker hampering TEBE benefit. Future work will aim to validate this finding in a larger cohort and to understand the biological mechanism underpinning this correlation.

Legal entity responsible for the study: The authors.

Funding: Department of Medicine, University of Toronto.

Disclosure: M.O. Butler: Financial Interests, Personal, Advisory Board: Merck, Novartis, Bristol Myers Squibb, Novartis, Pfizer, Sanofi, Immunocore, Medison, IDEAYA, Bio, Regeneron , IOVANCE; Financial Interests, Personal, Funding: Merck, Takara Bio, Novartis ; Financial Interests, Personal, Other: Adaptimmune; Financial Interests, Personal, Invited Speaker: Merck, Bristol Myers Squibb, Novartis, Sanofi, Medison, Pfizer. S. Saibil: Financial Interests, Personal, Funding: Medison, Novartis, Pfizer. All other authors have declared no conflicts of interest.

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1680P A plasma proteomics-based model optimizes first-line treatment decisions for metastatic melanoma

R.J. Sullivan[1], Y. Elon[2], Y. Brody[3], M. Harel[4], N. Dahan[3], S. Raveh Shoval[5], G. Boland[6], A. Lawless[7], I. Puzanov[8], F-X. Danlos[9], A. Marabelle[10], C. Robert[11], R. Shapira-Frommer[12], A. Bacchiocchi[13], R. Halaban[14], M. Sznol[15], A.P. Dicker[16]

[1]Mass General Cancer Center, MGH - Massachusetts General Hospital and Harvard Medical School, Boston, United States of America; [2]Physics, OncoHost, BGiv'at Ada, Israel; [3]Diagnostics, OncoHost, Binyamina-Giv'at Ada, Israel; [4]R&D, OncoHost, Binyamina-Giv'at Ada, Israel; [5]Clinical Affairs, OncoHost, Binyamina-Giv'at Ada, Israel; [6]Surgical Oncology, Massachusetts General Hospital Cancer Center, Harvard Medical School, Boston, United States of America; [7]Medical Oncology, Massachusetts General Hospital Cancer Center, Harvard Medical School, Boston, United States of America; [8]Medicine Department, Roswell Park Comprehensive Cancer Center, Buffalo, United States of America; [9]Drug Development Department - LRTI, Institut Gustave Roussy, Villejuif, France; [10]Drug Development Department, Institut Gustave Roussy, Villejuif, France; [11]Dermatology Department, Institut Gustave Roussy, Villejuif, France; [12]Oncology Department, The Ella Institute for Treatment and Research of Melanoma and Skin Cancer - Sheba Medical Center, Ramat Gan, Israel; [13]Dermatology, Yale University School of Medicine - Yale Cancer Center, New Haven, United States of America; [14]Dermatology, Yale School of Medicine, New Haven, United States of America; [15]Cancer Center, Yale University School of Medicine - Yale Cancer Center, New Haven, United States of America; [16]Department of Radiation Oncology, Thomas Jefferson University, Philadelphia, United States of America

Background: Multiple immune checkpoint inhibitor (ICI)-based therapies are approved for metastatic melanoma, yet biomarkers for informing treatment decisions are lacking. PROphet is a plasma proteomic test previously shown to predict benefit from anti-PD(L)1-based therapy in patients with metastatic non-small cell lung cancer. Here, we evaluate the clinical utility of PROphet for treatment selection in melanoma.

Methods: Pre-treatment plasma samples were collected from 248 patients with metastatic melanoma treated with anti-PD1 monotherapy (ICI mono) or anti-PD1 + anti-CTLA4 combination (ICI combo) in a real-world, prospective, observational study (Table). Long-term survival aligned with trends reported in comparable large-scale studies. PROphet classified patients as PROphet-POSITIVE or -NEGATIVE based on plasma proteomic profiles. Overall survival (OS) was analyzed using Kaplan-Meier analysis.

Results: Significantly longer OS was observed in PROphet-POSITIVE patients (n=183, 74%) compared to PROphet-NEGATIVE patients (n=65, 26%); the subset of patients receiving ICI mono displayed the strongest effect, indicating that the model is predictive for anti-PD1 treatment (Table). Multivariate analysis showed significant interaction between PROphet output and treatment modality (HR=0.22, p=0.02). Notably, PROphet-POSITIVE patients displayed similar survival with ICI mono and ICI-combo (HR=0.98, p=0.93), suggesting that these patients could be treated with ICI mono, thus avoiding potential toxicity from ICI combo. Conversely, PROphet-NEGATIVE patients had superior OS with ICI combo over ICI mono (HR=0.57, p=0.08), suggesting that ICI combo is preferable for these patients.

Table: 1680P Treatment type, model output and OS

Patient treatment	Drug	n	Hazard ratio (HR) [95% CI]	p-value	PROphet-POSITIVE mOS	PROphet-NEGATIVE mOS
All (ICI mono, ICI combo)	Pembrolizumab, nivolumab, nivolumab + ipilimumab	248	0.50 [0.33 – 0.75]	0.0009	118.4 months	17.4 months
ICI mono	Pembrolizumab, nivolumab	120	0.36 [0.21-0.64]	0.0005	Not reached	10.2 months
ICI combo	Nivolumab + ipilimumab	128	0.65 [0.36-1.17]	0.1547	118.4 months	29.7 months

Conclusions: PROphet predicts OS benefit for melanoma patients, with high predictivity for PD1 inhibitors. With further validation, PROphet output may serve to guide the choice between ICI mono vs ICI combo.

Legal entity responsible for the study: The authors.

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A. Marabelle: Financial Interests, Personal, Advisory Board: Innate Pharma, Deka Biosciences, Hotspot Therapeutics, Johnson & Johnson, Bioline Rx, Grey Wolf Therapeutics, Lytix Biopharma, HiFiBiO Therapeutics, ImCheck Therapeutics, Marengo Therapeutics, Medicxi, Guidepoint, Pathios Therapeutics, Asgard Therapeutics, Adagene, Daiichi Sankyo, Endeavor Biomedicines, Vivaldi Therapeutics, QBiotics; Financial Interests, Personal, Other, Associate Editor: European Journal of Cancer; Financial Interests, Personal, Stocks/Shares: HiFiBiO Therapeutics, Deka Biosciences, Hot-Spot Therapeutics, Shattuck Labs, Marengo Tx, Phoenix, Centessa, Adagene, IMCheck, Grey Wolf Therapeutics; Financial Interests, Institutional, Research Grant: BMS, MSD, Boehringer Ingelheim, Unicancer, AstraZeneca, Sofinnova; Non-Financial Interests, Other, Associate Editor: IOTECH Journal; Non-Financial Interests, Leadership Role, Vice President and Co-Founder: French Society for Cancer Immunotherapy (FITC); Non-Financial Interests, Advisory Role, Annual Meeting Organizing Committee: SITC; Non-Financial Interests, Advisory Role, ESMO IO Annual Meeting co-chair: ESMO; Non-Financial Interests, Institutional, Product Samples: BMS, IDERA, MSD, Transgene; Non-Financial Interests, Leadership Role: Société Française d'Immunothérapie des Cancers; Non-Financial Interests, Member: American Association for Cancer Research, American Society for Clinical Oncology, European Society for Medical Oncology, Society for Immunotherapy of Cancer (SITC); Non-Financial Interests, Advisory Role: Ligue contre le Cancer. C. Robert: Financial Interests, Personal, Advisory Board, Consultancy fees: BMS, Roche, Pierre Fabre, Novartis, Sanofi, MSD, Pfizer, Sun Pharma, Merck, Ultimovacs, Regeneron, Egle, Philogen, Maat Pharma; Financial Interests, Personal, Other, Steering Committee: Novartis, Regeneron, Pfizer, IO Biotech; Financial Interests, Personal, Other, IDMC: Ultimovacs; Financial Interests, Personal, Invited Speaker: Pierre Fabre, Sanofi, BMS, MSD, Novartis. R. Shapira-Frommer: Financial Interests, Personal, Advisory Board: Novartis, BMS, MSD, Roche, AstraZeneca. A.P. Dicker: Financial Interests, Advisory Role: Janssen, OncoHost, Orana Med, IBA, CVS, Hengrui Pharmaceuticals, Onconova Therapeutics, SBR Biotechnologies, EmpiricaLab, Aptar Pharma, Blue Spark Technologies, Imagene AI. All other authors have declared no conflicts of interest.

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1681P Comparative analysis of clinicopathological features and treatment responses in cutaneous head and neck vs other cutaneous melanomas

A. Popovic[1], M. Situm[2], A. Jalovcic-Suljevic[3], J. Maric Brozic[4], D. Herceg[5], I. Marijanovic[6], I. Stojkovski[7], B. Dujović[8], I.Z. Petkovic[1], T.M.S. Amaral[9], U. Leiter-Stoppke[10], D. Stulhofer Buzina[11], J. Glisic[12], A. Stojkovic[1], E. Bicakicic Filipovic[13], I. Salatic[8], B.L. Nikolin[14], S.V. Matkovic[15], L. Kandolf Sekulovic[8]

[1]Oncology Department, Clinic of Oncology, University Clinical Center Nis, Nis, Serbia; [2]Dermatology Department, KBC Sestre Milosrdnice - University Hospital Center, Zagreb, Croatia; [3]Clinic of Oncology Dept., Clinical Centre University of Sarajevo, Sarajevo, Bosnia and Herzegovina; [4]Klinika za Onkologiju i Nuklearnu Medicinu, KBC Sestre Milosrdnice - University Hospital Center, Zagreb, Croatia; [5]Department Of Oncology, KBC - University Hospital Centre Zagreb, Zagreb, Croatia; [6]Oncology Dept., University Clinical Hospital Mostar, Mostar, Bosnia and Herzegovina; [7]Oncology, Ss. Cyril and Methodius University - Faculty of Medicine, Skopje, North Macedonia; [8]Dermatology and Venerology Department, Military Medical Academy VMA, Belgrade, Serbia; [9]Dermato-Oncology Department, Skin Cancer Clinical Trials Center, University of Tuebingen, Tübingen, Germany; [10]Dermato-Oncology Department, Universitaets-Hautklinik Tuebingen, Tuebingen, Germany; [11]Dermatology and Venerology Department, KBC - University Hospital Centre Zagreb, Zagreb, Croatia; [12]Oncology Dept, University Clinical Center Kragujevac, Kragujevac, Serbia; [13]Oncology, Clinical Centre University of Sarajevo, Sarajevo, Bosnia and Herzegovina; [14]Clinic of Medical Oncology, Oncology Institute of Vojvodina, Sremska Kamenica, Serbia; [15]DH Department, Institute for Oncology and Radiology of Serbia, Belgrade, Serbia

Background: Cutaneous head and neck melanomas (cHNM) exhibit distinct clinicopathological characteristics, distinguishing them from other region melanomas (cORM). We aim to assess and compare the clinicopathological characteristics of cHNM and corm within our study population and response to immune-checkpoint inhibitors (ICI) and targeted therapy (TT) in the first-line metastatic setting.

Methods: We conducted a retrospective analysis, including patients (pts) with cutaneous melanoma from tertiary centers in Serbia, Croatia, Bosnia and Herzegovina, and North Macedonia. Data were obtained from the Central South Eastern European Registry (CSEEREG), a part of the European Melanoma Registry (EUMelaReg), with a data cut-off date of March 2025.

Results: A total of 7869 pts were included, 1133 (14.4%) with cHNM, with a median follow-up of 21.18 months. These pts were more often male ($p < .0001$) and older ($p < .0001$), had a higher prevalence of lentigo maligna ($p < .0001$) and a lower prevalence of superficial spreading melanoma ($p < .0001$). cHNM were thicker ($p = .0004$), BRAF wild-type, had higher initial substage II ($p < .0001$), had less SLNB performed ($p < .0001$), and received adjuvant radiotherapy more frequently ($p = .0139$). The median local relapse-free survival ($p < .0001$) and distant metastasis-free survival ($p < .0001$) were shorter in cHNM subgroup. 752 pts received 1st line ICI, 19% with cHNM, 81% with cORM. No significant difference was observed in mPFS (19.70 vs 11.48 months; $p=0.11$) and mOS (NR vs 35.46; $p=0.10$). When only pts >40 years were included ($n = 717$), cHNM showed significantly better mPFS (20.0 vs 11.5 m; $p = 0.02$) and mOS (NR vs 34.9 m; $p = 0.04$). 633 pts received 1st line TT, 17% with cHNM and 83% with cORM. No statistically significant difference was observed in mPFS (7.53 vs 9.08 months; $p = 0.18$) and mOS (18.59 vs 23.62; $p = 0.15$) across all ages and in the >40-year subgroup.

Conclusions: Despite the clinical and pathological differences, pts with cHNM seem to have a similar benefit to ICI and TT as pts with cORM. Notably, cHNM pts >40 years exhibited a significantly better response to ICI. However, due to the shorter times to recurrence, cHNM may require an intensified follow-up after initial diagnosis.

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1682P Outcomes of first-, second- and third-line treatment with all available combinations of BRAF plus MEK inhibitors in advanced melanoma patients: A real-world study

A.M. Czarnecka[1], P. Blonski[2], B. Cybulska-Stopa[3], W. Bal[4], R. Dziura[5], M. Ciazynska[6], K. Ostaszewski[7], K. Krawczak[8], M. Wiśniewska[9], L. Galus[10], K. Kozak[11], N. Kempa-Kaminska[12], G. Kaminska-Winciorek[13], M. Pabianek[6], P. Rogala[11], T. Kubiatowski[8], M. Ziętek[14], J. Mackiewicz[10], P. Rutkowski[2]

[1]Department of Outpatient Chemotherapy, Maria Sklodowska-Curie National Research Institute of Oncology, Warsaw, Poland; [2]Department of Soft Tissue/Bone Sarcoma and Melanoma, Maria Sklodowska-Curie National Research Institute of Oncology, Warsaw, Poland; [3]Department of Clinical Oncology, Lower Silesian Oncology, Pulmonology and Hematology Center, Wrocław, Poland; [4]Outpatient Chemotherapy Department, National Oncology Institute Maria Sklodowskiej-Curie National Research Institute, Gliwice Branch, Gliwice, Poland; [5]Department of Clinical Oncology, Holy Cross Cancer Center, Kielce, Poland; [6]Chemotherapy Unit and One-Day Chemotherapy Unit, Specialist Oncology Hospital NU-MED, Tomaszow Mazowiecki, Poland; [7]Department of Outpatient Chemotherapy, National Research Institute of Oncology, Warsaw, Poland; [8]Department of Clinical Oncology with a Subunit of Gynecologic Oncology, Provincial Specialized Hospital in Rzeszów, Rzeszow, Poland; [9]Clinical Department of Oncology, Professor Franciszek Lukaszczyk Oncology Center in Bydgoszcz, Bydgoszcz, Poland; [10]Department of Medical and Experimental Oncology, Institute of Oncology, Poznan University of Medical Sciences, Poznan, Poland; [11]Soft Tissue/Bone Sarcoma and Melanoma Department, Maria Sklodowska-Curie National Research Institute of Oncology, Warsaw, Poland; [12]Department of Clinical Oncology, Lower Silesian Oncology, Pulmonology and Hematology Center, Wroclaw, Poland; [13]Department of Bone Marrow Transplantation and Oncohematology Skin Cancer and Melanoma Team, Maria Sklodowska-Curie National Research Institute of Oncology Gliwice Branch, Gliwice, Poland; [14]Soft Tissue, Skin and Melanoma Cancer Center, Lower Silesian Oncology, Pulmonology and Hematology Center, Wroclaw, Poland

Background: For melanoma patients with *BRAF*-mutated tumours, the BRAF+MEK inhibitors are a viable treatment option. Three combinations are used in clinical practice: vemurafenib + cobimetinib (V+C), dabrafenib + trametinib (D+T) and encorafenib + binimetinib (E+B). Direct head to head comparison of combination is not available.

Methods: We conducted retrospective, multicentre study enrolling patients treated with all three available combinations of BRAF+MEK inhibitors in a routine clinical practice. The aim of the study was to assess and compare treatment efficacy.

Results: We enrolled 886 patients. In the first-line treatment, 491 patients were treated with D+T, 146 with V+C and 111 with E+B. The ORR was 49.3% in patients treated with D+T, 58.2% in patients treated with V+C and 52.3% in patients treated with E+B. First-line D+T treatment resulted with the median PFS (mPFS) of 8.08 months (95% CI: 7.13 - 9.23), V+C - 8.67 months (95% CI: 6.93 - 11.01), while E+B - 11.04 months (95% CI: 7.79 - 13.8). The mOS in D+T treated patients was 14.36 months (95% CI: 12.62 - 15.67), in V+C patients12.02 months (95% CI: 10.74 - 15.77) and in E+B patients 14.32 months (95% CI: 11.24 - 23.85). In the second-line treatment, 60 patients were treated with D+T, 5 - with V+C and 46 - with E+B. For D +T treatment the ORR was 51.7% while for E+B - 32.6%. 4/5 patients treated with V +K had PR. The median PFS for D+T treatment was 9.82 months (95% CI: 6.93 - 15.18), while for E+B - 8.84 months (95% CI: 7.59 - 13.5). In the third-line treatment, 60 patients were treated with D+T, 46 with E+B, 1 with V+K. The mPFS for D+T treatment was 4.5 months (95% CI: 2.76 - 6.05) and for E+B - 3.4 months (95% CI: 2.76 - 7.92).

Conclusions: All three combinations had similar and significant clinical activity, manifested in high ORR and favourable survival outcomes in real-world setting.

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Financial Interests, Personal, Advisory Board: MSD, BMS, Pierre Fabre, Merck, Sanofi, Blueprint Medicines, Philogen, Medison Pharma, Genesis Pharma; Financial Interests, Personal, Invited Speaker: Merck, Sanofi, Novartis, AstraZeneca, Genesis Pharma; Financial Interests, Institutional, Research Grant, research grant for ISS: Pfizer; Financial Interests, Institutional, Funding, research grant for institution: BMS; Non-Financial Interests, Member of Board of Directors: Polish Society of Surgical Oncology; Non-Financial Interests, Member of Board of Directors, President: Polish Oncological Society. All other authors have declared no conflicts of interest.

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1683TiP OMIT trial: omitting therapeutic lymph node dissection in patients with stage IIIB/C melanoma and major pathological response in the index lymph node after neoadjuvant immunotherapy

A.W. Schurink[1], J.A. van den Broek[1], C. Verhoef[1], E.E.A.P. Mulder[1], L. Been[2], J. van Bastelaar[3], H. Bonenkamp[4], A. Dassen[5], S. Estourgie[6], A. Francken[7], A. Kuijpers[8], M. Anderegg[9], S. Van der Ploeg[10], J. Van Der Hage[11], K. Wevers[12], A. Witkamp[13], E. Zeestraten[14], A.A.M. van der Veldt[15], D. Gruenhagen[1]

[1]Surgical Oncology, Erasmus MC Cancer Institute, Rotterdam, Netherlands; [2]Surgical Oncology, UMCG - University Medical Center Groningen, Groningen, Netherlands; [3]Surgical Oncology, Zuyderland Medical Center, Heerlen, Netherlands; [4]Surgical Oncology, Radboud University Medical Center, Nijmegen, Netherlands; [5]Surgical Oncology, Hospital Medisch Spectrum Twente, Enschede, Netherlands; [6]Surgical Oncology, Medical Center Leeuwarden, Leeuwarden, Netherlands; [7]Surgical Oncology, Isala Hospital, Zwolle, Netherlands; [8]Division of Surgical Oncology, Netherlands Cancer Institute, Amsterdam, Netherlands; [9]Surgical Oncology, Amsterdam UMC, locatie VUmc, Amsterdam, Netherlands; [10]Surgical Oncology, Maxima Medisch Centrum -Veldhoven, Veldhoven, Netherlands; [11]Surgery, LUMC - Universitair Medisch Centrum, Leiden, Netherlands; [12]Surgical Oncology, Maastricht UMC+, Maastricht, Netherlands; [13]Surgical Oncology, Univeristy Medical Center Utrecht, Utrecht, Netherlands; [14]Surgical Oncology, Amphia Ziekenhuis-location Molengracht, Breda, Netherlands; [15]Medical Oncology and Radiology & Nuclear Medicine Department, Erasmus MC Cancer Institute, Rotterdam, Netherlands

Background: The current standard of care for resectable stage III melanoma includes neoadjuvant immunotherapy followed by therapeutic lymph node dissection (TLND). In the randomized NADINA trial, neoadjuvant ipilimumab plus nivolumab induced a major pathological response (MPR; <10% viable tumor cells) in 59% of patients with resectable stage III melanoma, with an 18-month distant metastasis-free survival (DMFS) of 96.9% (Blank et al, NEJM 2024). These results question the necessity of TLND, especially given its associated morbidity, such as lymphedema. In the phase II PRADO trial (Reijers et al., *Nat Med* 2022), TLND was omitted in patients with an MPR in the index node, defined as the lymph node harboring the largest metastasis, with 1-year and 3-year DMFS rates of 100% and 98%, respectively. However, the PRADO trial was underpowered to draw definitive oncological conclusions regarding the omission of TLND.

Trial design: The OMIT trial (NCT06754904) is a nationwide, multicenter, prospective, single-arm interventional study conducted across all designated melanoma centers in the Netherlands. A total of 213 patients with resectable stage III melanoma eligible for neoadjuvant treatment with ipilimumab plus nivolumab will be enrolled. Following neoadjuvant treatment, patients will undergo an index node procedure. If MPR is achieved in the index node, TLND will be omitted. The two co-primary endpoints are 2-year DMFS and 2-year in-basin locoregional recurrence-free survival (LRFS). Secondary endpoints include pathological response rates, post-surgical complications, melanoma-specific and overall survival, and health-related quality of life (HRQoL). All patients will undergo 5 years of follow-up according to standard surveillance protocols. Follow-up includes radiological imaging and HRQoL assessments using the EQ-5D-5L, FACT-Melanoma, Cancer Worry Scale, and Work Performance Questionnaire. The trial is currently open for inclusion.

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1684TiP Intismeran autogene (V940/mRNA-4157) + pembrolizumab versus pembrolizumab alone as first-line therapy for advanced melanoma: The phase II INTerpath-012 study

P.A. Ascierto[1], C. Robert[2], M.S. Carlino[3], A.I. Daud[4], J.J. Luke[5], C. Gebhardt[6], T. Posadas[7], L.S. Ojalvo[8], X. Fang[9], M. Kalabis[10], G.V. Long[11]

[1]Melanoma, Cancer Immunotherapy & Developmental Therapeutics, Istituto Nazionale Tumori IRCCS - Fondazione G. Pascale, Naples, Italy; [2]Dermatology Department, Institut Gustave Roussy, Villejuif, France; [3]Medical Oncology, Melanoma Institute Australia, The University of Sydney, and Westmead and Blacktown Hospitals, Sydney, Sydney, Australia; [4]Department of Medicine, University of California San Francisco, San Francisco, United States of America; [5]Immunotherapy and Drug Development Center, UPMC Hillman Cancer Center and University of Pittsburgh, Pittsburgh, United States of America; [6]Dermatology/Skin Cancer Center, University Medical Center Hamburg-Eppendorf, Hamburg, Germany; [7]Research and Development, Moderna, Inc., Cambridge, United States of America; [8]Oncology, Moderna, Inc., Cambridge, United States of America; [9]BARDS, Merck & Co., Inc., Rahway, NJ, United States of America; [10]Oncology Department, Merck & Co., Inc., North Wales, United States of America; [11]Faculty of Medicine and Health, The University of Sydney, and Mater and Royal North Shore Hospitals, Melanoma Institute Australia, University of Sydney, and Royal North Shore and Mater Hospitals, Sydney, Australia

Background: Immune checkpoint inhibitors are the standard of care for advanced melanoma, but there remains a need to improve outcomes in this setting. Intismeran autogene (intesmeran) is a novel mRNA-based individualized neoantigen therapy (INT) directed against up to 34 patient-specific neoantigens. In the phase 2b KEYNOTE-942 study, adjuvant intismeran + pembrolizumab was shown to have manageable safety and prolonged RFS vs pembrolizumab alone in participants with resected stage IIIB-IV melanoma. The benefit of INT in metastatic disease is unknown. Here, we describe the methodology for the randomized, double-blind, phase 2 INTerpath-012 study (NCT06961006) designed to evaluate intismeran + pembrolizumab vs placebo + pembrolizumab in participants with previously untreated, unresectable advanced melanoma.

Trial design: Eligible participants are aged ≥18 years, have unresectable histologically confirmed stage III or IV cutaneous melanoma per AJCC 8th ed criteria, have ≥1 measurable lesions per RECIST v1.1, an ECOG PS of 0 or 1, and a blood and tumor biopsy sample for intismeran generation. Participants must not have received prior treatment for unresectable melanoma; prior (neo)adjuvant targeted therapy or immunotherapy is allowed provided relapse has not occurred within 12 months of treatment discontinuation. Approximately 160 participants will be randomly assigned 1:1 to intismeran 1 mg IM Q3W or matching placebo for ≤9 doses + pembrolizumab 400 mg IV Q6W for ≤17 cycles. Randomization will be stratified by prior anti−PD-1−based therapy (yes vs no) and *BRAF* V600 status (mutated vs wild type or unknown). The primary end point is PFS per RECIST v1.1 by investigator. Secondary end points are ORR and DOR per RECIST v1.1 by investigator, OS, and safety and tolerability. Imaging by CT or MRI will be performed Q12W ≤week 96, and Q24W thereafter. Participants will be followed up for survival Q12W until death, withdrawal of consent, or end of study, whichever occurs first. AEs will be evaluated throughout the study and for 30 days after treatment end (90 days for serious AEs or 30 days if new anticancer therapy is initiated). AEs will be graded per NCI CTCAE v5.0. Recruitment is ongoing.

Clinical trial identification: NCT06961006.

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1685TiP SWE-NEO: Swedish neoadjuvant trial comparing anti-PD-1 monotherapy to combined anti-CTLA-4/anti-PD-1 blockade in resectable stage III melanoma

H. Helgadottir[1], G. Jönsson[2], J.A. Nilsson[3], K. Björkström.[4], A.L.T. Berglund[5], A. Carneiro[6], L. Ny[7], R. Olofsson Bagge[8]

[1]Oncology Department, Karolinska Institutet - Cancer Center Karolinska (CCK), Solna, Sweden; [2]Department of Oncology, Lund University, Lund, Sweden; [3]Surgery, University of Gothenburg - The Sahlgrenska Academy, Göteborg, Sweden; [4]Department of Oncology-Pathology, Karolinska Institute, Stockholm, Sweden; [5]Research, Epistat AB, Uppsala, Sweden; [6]Hematology, Oncology and Radiation Physics Department, Skane University Hospital CCC, Lund, Sweden; [7]Dept. of Oncology, Sahlgrenska University Hospital, Göteborg, Sweden; [8]Department of Surgery, University of Gothenburg - The Sahlgrenska Academy, Göteborg, Sweden

Background: Patients resected for clinically detected stage III melanoma have a high risk of recurrence and death from melanoma. At present two studies (SWOG S1801 and NADINA) have demonstrated superiority when using neoadjuvant immune checkpoint inhibitor (ICI) treatment compared to adjuvant ICI treatment only, but no larger studies have compared aPD-1 monotherapy to aPD-1/aCTLA-4 combination therapy in the neoadjuvnat setting. The SWE-NEO trial aims to compare these two regimens in patients with clinically detected stage III melanoma, where the aPD-1/aCTLA-4 combination is potentially more effective, but also associated with more side effects.

Trial design: SWE-NEO is a phase III randomized controlled multicenter open-label trial initiated by the Swedish Melanoma Study Group. Patients are included after being diagnosed with resectable stage III cutaneous, acral or unknown primary melanoma with lymph node metastases and/or up to 3 in-transit metastases. Patients are randomized to having either two courses of ipilimumab (80mg) and nivolumab (240 mg) combination therapy (n=64) or nivolumab (480 mg) monotherapy (n=64), before the pre-planned operation (index node resection). In both arms, theuapeutic lymphadenectomy and one year of adjuvant treatment with nivolumab, or with dabrafenib+trametinib in patients with a BRAF V600E mutation, is administrated only to patients with no major pathological response (MPR, $\leq$10% viable cells) in the operated tumor. Patients are followed for up until 5 years. The primary objective is to study event-free survival (EFS). In a power analysis, a sample size of 64 + 64 patients was found to be required to detect down to a 13% difference in EFS at two years (70% power, alpha 0.05). Secondary objectives include further efficacy outcomes (relapse-free survival (RFS), distant metastasis-free survival (DMFS), overall survival (OS), MPR and its correlation to RFS, DMFS and OS and radiological response and its correlation to MPR, RFS, DMFS and OS), and safety analysis (surgery conducted according to plan, surgical complications, and ICI related adverse events). Exploratory objectives include biomarker analyses from sequential blood and tumor samples.

Clinical trial identification: NCT06794775, EMA Clinical Trials Information System (CTIS) Trial ID: 2024-519593-39-00, protocol approved April 30th 2025.

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1686eP Local skin reactions due to topical 5-fluorouracil treatment: Does DPYD-genotype matter?

N. Heersche[1], M. de With[1], E. Oomen-de Hoop[2], S.F.A. Meier[2], F.S. Verhaar[2], M. Matic[3], K.C. Noz[4], T. Brinkhuizen[5], R.H.N. Van Schaik[6], R.H. Mathijssen[7], M.J. Deenen[8], M. Wakkee[9], S.L. Koolen[10]

[1]Medical Oncology and Clinical Chemistry, Erasmus MC Cancer Institute, Rotterdam, Netherlands; [2]Medical Oncology, Erasmus MC Cancer Institute, Rotterdam, Netherlands; [3]Clinical Chemistry, Erasmus MC University Medical Center, Rotterdam, Netherlands; [4]Dermatology, Adrz-Admiraal de Ruijter Hospital-Goes, Goes, Netherlands; [5]Dermatology, Catharina Hospital Eindhoven, Eindhoven, Netherlands; [6]Clinical Chemistry, Erasmus MC - University Medical Center, Rotterdam, Netherlands; [7]Medical Oncology, Erasmus MC Daniel den Hoed Cancer Center, Rotterdam, Netherlands; [8]Clinical Pharmacy, Catharina Hospital Eindhoven, Eindhoven, Netherlands; [9]Dermatology, Erasmus MC, Rotterdam, Netherlands; [10]Medical Oncology, Erasmus MC - University Medical Center, Rotterdam, Netherlands

Background: Fluoropyrimidines, such as 5-fluorouracil (5-FU), are mainly metabolised by the dihydropyrimidine dehydrogenase (DPD) enzyme. Genetic variants in the *DPYD* gene can reduce DPD activity and are a well-established cause of severe toxicity during systemic 5-FU treatment. Since 2020, the European Medicine Agency recommends *DPYD*-genotype guided dosing in systemic treatment of fluoropyrimidines. However, the role of *DPYD*-variants for topical 5-FU treatment has not yet been prospectively studied.

Methods: This multicentre, patient-partnered, prospective observational trial enrolled patients initiating topical 5-FU treatment for cutaneous (pre)malignancies. Genotyping was performed for clinically relevant *DPYD* variants (i.e. *DPYD**2A, *7, *13, c.2846C>T and HapB3 (c.1236G>A and c.1129-5923 C>G)), with both patients and investigators blinded to genotype. Participants submitted weekly photographs of the treated lesion over four weeks and again at three months. Local skin reactions (LSR) were assessed using the validated LSR Grading Scale (range 0 − 24) (Rosen *et al.* Dermatol Ther 2014). Study endpoints included comparison of maximum LSR scores between wild-types and variant carriers, as well as the incidence of severe LSR (score >13).

Results: Among 233 evaluable patients, 21 (9%) carried a *DPYD* variant. Severe LSRs occurred more frequently in variant carriers (14%) compared to patients wild-type for *DPYD* (3%; $P = 0.037$), with the highest incidence in *DPYD**2A carriers (2/4; 50%). The highest overall LSR score (i.e. 20), was also seen in a *DPYD**2A carrier. Median maximum LSR scores were similar (wild-type: 6 [IQR: 5-9] *vs.* variant: 6 [4.5-9]; $P = 0.48$). All skin reactions resolved without scarring. None of the patients was hospitalised after start of topical 5-FU treatment.

Conclusions: Severe local skin reactions were significantly more frequent in variant carriers, in particular those with *DPYD**2A. However, overall LSR scores did not differ between *DPYD* wild-types and variant carriers. While *DPYD* genotype may help guide treatment management in known variant carriers (e.g. imiquimod instead of 5-FU), current data do not support routine *DPYD* genotyping prior to topical 5-FU.

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Legal entity responsible for the study: Erasmus University Medical Centre, Rotterdam, the Netherlands.

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1688eP Minimal residual disease (MRD) predicts worse survival in high-risk melanoma: A meta-analysis

J.E.P. Conrado[1], P. Cotta Abrahão Reis[2], L.F.L. da Silva[3], M.M. Noronha[4], E.F. Saldanha[5], L.G. Santana[6]

[1]*Department of Surgery, Universidade Federal do Ceara, Fortaleza, Brazil;* [2]*Oncologia, UFRJ - Universidade Federal do Rio de Janeiro, Rio De Janeiro, Brazil;* [3]*Departamento de Medicina Clínica, UFF - Universidade Federal Fluminense - Faculdade de Medicina, Niteroi, Brazil;* [4]*Department of Pathology and Forensic Medicine, Universidade Federal do Ceara, Fortaleza, Brazil;* [5]*Clinical Oncology Department, UHN - University Health Network - Princess Margaret Cancer Centre, Toronto, Canada;* [6]*Department of Oncology, A.C. Camargo Cancer Center - Unidade Antonio Prudente, Sao Paulo, Brazil*

Background: High-risk cutaneous melanoma is a heterogeneous disease with a variable risk of recurrence. Despite adjuvant therapy, up to 50% of these pts will experience disease relapse. Hence, optimal recurrence risk assessment is key to improving outcomes. Here, we conducted a meta-analysis to determine whether ctDNA detection after curative-intent resection is associated with survival outcomes.

Methods: We systematically surveyed databases to identify studies reporting the association of ctDNA detection with relapse-free survival (RFS), distant metastasis-free survival (DMFS), and overall survival (OS), in patients with resected high-risk melanoma (IIB-IV). Timepoints of ctDNA assessment were classified as landmark (after curative resection) and follow-up (> 12 weeks after resection). We synthesized the detection rate, sensitivity and specificity of ctDNA-MRD in predicting recurrence. ctDNA was treated as a binary variable (present/absent) for the analysis. For efficacy outcomes, we pooled hazard ratios (HRs) and 95% confidence intervals (95%CI) for RFS, DMFS, and OS.

Results: Twelve studies met the inclusion criteria, including a total of 1354 patients. Distribution according to the staging was reported for 1146 patients, Stage II = 136 (11,8%), Stage III = 947 (82,6%), Stage IV = 63 (5,4%). The median detection rate at any timepoint was 17% (range: 6,8% - 88%). Pooled sensitivity of ctDNA for detecting MRD and predicting recurrence in resectable melanoma was 35.6% (95%CI: 19.9%−55.2%; $I^2 = 78.2\%$), while the pooled specificity was 95.0% (95%CI: 90.2%−98.4%; $I^2 = 70.2\%$). The detection of ctDNA-MRD at the landmark period was associated with worse RFS (HR 3.63, 95%CI: 2.63−5.03; $p < 0.001$; $I^2 = 45.6\%$), DMFS (HR 6.25, 95% CI: 2.45-15.93; $p < 0.001$; $I^2 = 64.4\%$), and OS (HR 3.45, 95%CI: 2.54-4.68; $p < 0.001$; $I^2 = 0\%$).

Conclusions: ctDNA assessment for MRD detection in pts with high-risk melanoma, at the landmark timepoint, demonstrated high specificity and was associated with worse survival outcomes. While leveraging ctDNA for MRD detection is promising, the low detection rate and the sensitivity warrant caution. Future prospective studies will shed light on the role of ctDNA for MRD detection.

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1689eP Demographic shifts in melanoma: Trends from 2000 to 2022

D.A. Hassan[1], B.H. Ali[2], M.H. Ali[3], S. Gharib[4]

[1]*Medicine, Suez Canal University Hospital, Ismailia, Egypt;* [2]*Medicine, Faculty of Medicine - Suez Canal University, Ismailia, Egypt;* [3]*Pharmacy, Sinai University, Ismailia, Egypt;* [4]*Internal medicine, Suez Canal University Hospital, Ismailia, Egypt*

Background: Melanoma is a potentially lethal skin malignancy with rising global incidence. Although several studies have explored melanoma risk factors, limited attention has been given to age- and race-specific trends across two decades. This study aims to assess age-adjusted incidence trends of melanoma focusing on racial, gender, and age-based disparities.

Methods: We extracted data of 448,790 patients diagnosed with melanoma between 2000 and 2022 from the SEER database. Age-adjusted incidence rates (per 100,000) were calculated using the 2000 U.S. standard population (19 age groups - Census P25-1130).Confidence intervals are 95% for rates (Tiwari mod) and trends. Trends were analyzed using the weighted least squares method. Annual percent change (APC) and percent change (PC) were calculated for each subgroup.

Results: The overall incidence rate of melanoma was 23, with a percent change (PC) of 39.7 and an annual percent change (APC) of 1.3. Stratified by race, the White population showed a PC of 38.4 and APC of 1.4, while the Black and Asian populations exhibited declining trends with PCs of −14.8 and −38, and APCs of −0.6 and −1.3, respectively. The American Indian population showed a mild increase, with a PC of 3.3 and APC of 2. Females and males showed comparable trends, with PCs of 39.3 and 38.3, and APCs of 1.3 and 1.2, respectively. By age, incidence rates increased among older adults aged 55 and above, while younger age groups showed decreasing trends, highlighting a shift in melanoma burden toward older populations.

Conclusions: Melanoma incidence continues to rise, particularly among White populations and older age groups. In contrast, a decline is observed among Black and Asian populations and younger individuals. These findings underscore the need for age- and race-specific prevention strategies and raise concern about the aging population's increasing burden of melanoma.

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1690eP Stage III and IV malignant melanoma in Sweden: A nationwide study on health care resource utilization, productivity loss and associated cost, prescription drug use and survival prognosis

A. Valachis[1], G. Ullenhag[2], H. Park[3], P. Carlqvist[3], T. Dibbern[3], A. Havell[4]

[1]*Oncology Department, Örebro University Hospital (Universitetssjukhuset Örebro), Örebro, Sweden;* [2]*Oncology Department, University Hospital Uppsala/Akademiska Sjukhuset, Uppsala, Sweden;* [3]*Health Economics and Outcome Research, Nordic Market Access NMA AB, Stockholm, Sweden;* [4]*Essential Medicines, Novartis Sverige AB, Kista, Sweden*

Background: Malignant melanoma (MM) is an aggressive cancer with rising incidence and high recurrence. This study describes healthcare resource utilisation, productivity loss and associated cost, prescription drug use and survival associated with primary and recurrent stage III and IV MM in Sweden.

Methods: Adults (≥18 years) with stage III or IV MM from Jan 2010 - Dec 2022 were identified in the Swedish Cancer Register (SCR). To identify recurrence, additional data were extracted from Jan 2000 - Dec 2009. Individuals were classified as primary stage III (n=653), primary stage IV (n=289), locoregional recurrent (n=606), or distant recurrent (n=2,595) MM. Data from SCR were linked with the National Patient Register, Prescribed Drug Register, and MIDAS to assess healthcare resource utilisation, prescriptions and mortality.

Results: Mean follow-up was 3.38 (primary stage III), 2.13 (primary stage IV), 3.80 (locoregional recurrent), and 1.65 (distant recurrent) years. Distant recurrent had highest inpatient care utilization despite shortest follow-up: 3 stays and 6 days/stay on average, totalling 858M SEK in costs. Outpatient visits were most frequent in primary stage III (24 visits/person on average). Long-term sickness absence was similar across groups (265−317 compensated net days/year; ∼110,000 SEK/person). Compensated short-term sickness absence was highest in primary stage IV and distant recurrent (177 net days/year respectively). Analgesics, sedatives and sleeping pills were the most dispensed drug groups. Kaplan-Meier analysis showed survival differences between groups ($p<0.0001$), with poorest survival observed in recurrent stage IV MM. Primary stage IV showed better but rapidly declining survival and recurrent stage III demonstrated better survival than primary stage III.

Conclusions: This nationwide study highlights the substantial healthcare and societal burden of stage III and IV MM in Sweden, with high resource utilisation, productivity loss and associated cost with poor survival outcomes, especially in stage IV MM. These findings underline the need for early detection and personalised treatment strategies to enhance patient outcomes.

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1693eP Diagnosis and follow-up of immune-related hypophysitis with magnetic resonance imaging (MRI) of pituitary gland

D.C. Ziogas, A. Angelousi, S. Asimakopoulou, C. Bourgioti, A. Anastasopoulou, S. Kazanas, P. Diamantopoulos, A. Karabela, E. Tasouli, L-A. Moulopoulos, H. Gogas

First Department of Medicine, National and Kapodistrian University of Athens - School of Medicine, Athens, Greece

Background: Immune checkpoint inhibitors(ICI) have transformed the prognosis of many solid malignancies. However, their use has been associated with distinct endocrine immune-related adverse events(irAEs), with ir-hypophysitis being among the most common, but not adequately studied.

Methods: A retrospective analysis of pituitary MRI scans was conducted in cancer patients treated with ICI-based regimens and diagnosed with biochemically documented anterior pituitary deficiency from January 2016 to September 2024. The 1^{st} MRI was performed at the time of diagnosis of ICI-induced hypophysitis, and the 2^{nd} during follow-up (FU) and were evaluated by a blinded central radiologist team.

Results: 66 ICI-treated cancer patients with anterior pituitary deficiency were included. 60 patients had received ICI for melanoma, 3 for lung cancer, 2 for hepatocellular carcinoma and 1 for colon cancer. Initial MRI was available in 60 patients and performed at a median time of 2 weeks post-diagnosis of ir-hypophysitis. MRI abnormalities were found in 32 patients-53.3%, including enlargement of the pituitary gland-25%,reduced enhancement-10%, empty sella turcica-8.3%, heterogeneous enhancement of the pituitary gland-5%,reduced pituitary size-3.3% and slight deviation of the pituitary stalk-1.7%. An MRI assessment, after a median follow-up of 1.6 years of ir-hypophysitis diagnosis, was available in 37 patients (31 had also an initial MRI); in 14 of 31 with a 2ndMRI-45%, imaging changes were detected including a partially empty sella turcica, a reduced size or an enlargement of the pituitary gland. No specific ICI-based monotherapy nor combinations were associated with a particular imaging pattern or abnormal findings at FU. Patients with multiple axis deficiencies had more frequent MRI abnormalities than those with isolated corticotrope deficiency at both time points(46.3% vs. 68.4% initially; 56.4% vs.71.3% at FU).

Conclusions: MRI pituitary abnormalities were detected in ∼50% of patients with ICI-induced hypophysitis (45% of them changed on FU); these were not specific to the underlying malignancy or the administered ICI. The other half presented normal pituitary MRI in both assessments, keeping the challenge of imaging for this irAE.

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1694eP Real-world induction targeted therapy (TT) with immune checkpoint inhibitor (ICI) therapy switch in advanced BRAF-mutant melanoma: A multi-institutional study

J. McKenzie[1], T. Davidson[2], S.R. Karri[3], A. Lomax[4], E. Sneath[4], Y-J. Kuo[5], N. Harel[1], S. Portela[1], K. Cao[1], G. McArthur[6], G. Au-Yeung[1], A.M. Menzies[5], M.S. Carlino[3], H.A. Tawbi[2], S.K. Sandhu[1], L. Spain[1]

[1]Medical Oncology Department, Peter MacCallum Cancer Centre, Melbourne, Australia; [2]Department of Melanoma Medical Oncology, The University of Texas MD Anderson Cancer Center, Houston, United States of America; [3]Medical Oncology Department, Westmead and Blacktown Hospitals, Westmead, Australia; [4]Medical Oncology Department, Barwon Health Andrew Love Cancer Centre, Geelong, Australia; [5]Medical Oncology Department, Melanoma Institute Australia, Wollstonecraft, Australia; [6]Victorian Comprehensive Cancer Center, Peter MacCallum Cancer Centre, Melbourne, Australia

Background: Sequential treatment with induction BRAF and MEK inhibitor TT and a planned ICI switch prior to progression as evaluated in the SECOMBIT trial may benefit some patients (pts) with advanced melanoma and adverse prognostic factors, who are typically not represented in clinical trial populations. In clinical practice, this strategy is typically reserved for symptomatic pts with critical sites of disease. We report real-world outcomes in this population.

Methods: This multi-institutional study retrospectively reviewed clinical outcome data from pts with advanced *BRAF*-mutant melanoma that had induction TT with planned ICI switch within 16 weeks from Aug 2020-Dec 2024. Primary endpoint was overall survival (OS).

Results: 42 pts were identified (median age 57 years, range 29-81) with a median baseline ECOG of 1 (range 0 – 3). Of these, 26 (62%) had brain metastases (81% symptomatic), 17 (40%) had liver metastases and 20 (of 36 known; 55%) had elevated baseline LDH. At a median TT-ICI interval of 52 days (IQR 35-72), 36 (86%) pts underwent planned ICI switch, most commonly with combination ipilimumab/nivolumab (97%). Of 24 (57%) requiring corticosteroids (CS) for initial symptom relief, 19 received ICI and of these 18 ceased CS prior to switch. At a median follow up of 21.1mo (95%CI 16.1-47.8), median PFS was 4.77mo (3.3-7.8) and median OS 17.3mo (12.3-NR). 12-, 24- and 36-mth PFS rates were 35.5%, 22.6% and 22.6%, while OS rates were 66.5%, 42.9% and 30%. Several factors predicted poor survival (Table). On multivariate analysis, only elevated LDH at end of TT independently predicted OS (Table).

Table: 1694eP Variables associated with overall survival

Variable	Univariate analysis		Multivariate analysis	
	HR (95% CI)	p-value	HR (95% CI)	p-value
ECOG ≥1	3.6 (1.0-12.4)	**0.04**	1.50 (0.24-9.39)	0.7
BRAF V600 non-E	2.32 (0.90-6.01)	0.08		
Metastatic sites ≥3	2.88 (0.84-9.95)	0.09	2.56 (0.45-14.7)	0.3
Brain metastases	3.4 (1.2-9.4)	**0.02**	1.62 (0.32-8.09)	0.6
Required CS at baseline	3.25 (1.3-8.1)	**0.01**	2.07 (0.38-11.3)	0.4
Baseline LDH ≥ ULN	1.67 (0.66-4.24)	0.28		
LDH ≥ ULN at end of TT	3.77 (1.2-12)	**0.02**	4.64 (1.18-18.2)	**0.015**
Any grade irAE	0.30 (0.11-0.81)	**0.02**		

CS – corticosteroid; irAE – immune-related adverse event; LDH – lactate dehydrogenase; TT – targeted therapy; ULN – upper limit of normal range

Conclusions: This real-world cohort demonstrates meaningful OS with a TT-ICI switch strategy for poor prognosis pts with advanced *BRAF*-mutant melanoma. Elevated LDH at end of TT independently predicted poor OS.

Legal entity responsible for the study: The authors.

Funding: Has not received any funding.

Disclosure: G. McArthur: Financial Interests, Institutional, Coordinating PI, Financial interest is reimbursed costs for activities on clinical trials: Genentech Roche; Non-Financial Interests, Other, Advisory Board Member: Canthera; Non-Financial Interests, Institutional, Product Samples, IMP for investigator initiated and funded trial: Pierre-Fabre, Pfizer; Non-Financial Interests, Other, Director: Melanoma World Society; Non-Financial Interests, Other, Board of Directors: National Breast Cancer Foundation (Australia); Non-Financial Interests, Leadership Role, Board of Directors: Children's Cancer CoLab. G. Au-Yeung: Financial Interests, Institutional, Advisory Board: Incyclix Bio; Financial Interests, Institutional, Coordinating PI: AstraZeneca, Roche–Genentech. A.M. Menzies: Financial Interests, Personal, Advisory Board, advisory board: BMS, MSD, Novartis, Roche, Pierre-Fabre, QBiotics. M.S. Carlino: Financial Interests, Personal, Advisory Board, Consultant Advisor: MSD, BMS, Novartis, Amgen, Oncosec, Merck, Sanofi, Ideaya, Pierre Fabre, Eisai, Nektar, Regeneron, Stand Therapeutics. H.A. Tawbi: Financial Interests, Personal, Advisory Board: Bristol Myers Squibb, Merck, Novartis, Genentech, Eisai, Karyopharm, Iovance, Pfizer, Jazz Pharmaceuticals, Regeneron Pharmaceuticals, IO Biotech, Immunocore, Corcept, Krystal Biotech; Financial Interests, Institutional, Trial Chair: Bristol Myers Squibb; Financial Interests, Institutional, Local PI: Merck, RAPT Pharmaceuticals; Financial Interests, Institutional, Steering Committee Member: Novartis, Genentech; Financial Interests, Institutional, Funding: GSK, Eisai, Syntrix; Financial Interests, Institutional, Coordinating PI: Regeneron Pharmaceuticals. S.K. Sandhu: Financial Interests, Institutional, Advisory Board, I have served on advisory boards for MSD. The e contracts for my role are set up with the institution and the funds go to a research fund at Peter MacCallum Cancer Centre: Merck Sharp and Dohme; Financial Interests, Institutional, Advisory Board, I have served on advisory boards for AstraZeneca. The contracts for my role as an advisor are set up with the institution and the funds go to a research fund at Peter MacCallum Cancer Centre: AstraZeneca; Financial Interests, Institutional, Advisory Board, I have served on advisory boards for Novartis. The contracts for my role as an advisor are set up with the institution and the funds go to a research fund at Peter MacCallum Cancer Centre: Novartis; Financial Interests, Institutional, Advisory Board, I have served on an advisory board for Merck Serono. The contracts for my service are set up with the institution and funds go into a research fund at the Peter MacCallum Cancer Centre: Merck Serono; Financial Interests, Institutional, Advisory Board, I have served on an advisory board for SkylineDx. The contracts for my service are set up with the institution and funds go into a research fund at the Peter MacCallum Cancer Centre: SkylineDx; Financial Interests, Institutional, Advisory Board, I have served on an advisory board for AbbVie. The contracts for my role are with the institution and funds go into a research fund at the Peter MacCallum Cancer Centre: AbbVie; Financial Interests, Institutional, Invited Speaker, I have received honoraria for invited speaker role only which is paid to my institution. Funds go into a research fund at the Peter MacCallum Cancer Centre: Janssen; Financial Interests, Personal, Other, Equity (stock): AdvanCell; Financial Interests, Personal, Stocks/Shares: AdvanCell; Financial Interests, Institutional, Research Grant, My institution receives grant funding to run an investigator initiated trial that I lead: Novartis, Genentech, Amgen, AstraZeneca, Merck Serono, Merck Sharp and Dohme; Financial Interests, Institutional, Research Grant, Pfizer are providing funding to my institution for the conduct of an investigator initiated clinical trial: Pfizer; Financial Interests, Institutional, Research Grant, Senwha are providing funding to my institution for the conduct of an investigator initiated clinical trial: Senwha; Non-Financial Interests, Other, I serve on the Independent Safety and Data Monitoring committee for 2 of Novartis sponsored studies and dont receive any compensation for this: Novartis; Non-Financial Interests, Other, I serve on the steering committee for several Janssen sponsored trials and i do not receive compensation for this: Janssen; Non-Financial Interests, Other, I serve on the steering committee for one AstraZeneca sponsored trial and I do not receive compensation for this: AstraZeneca; Non-Financial Interests, Other, I serve on the steering committee for one Genentech sponsored trials and I do not receive compensation for this: Genentech; Non-Financial Interests, Other, I serve on the steering committee for a Bristol Myer Squibb sponsored trial and I do not receive renumeration for this: Bristol Myer Squibb; Non-Financial Interests, Principal Investigator, I am a principal investigator on several of Janssen sponsored studies and don't receive any renumeration for this: Janssen; Non-Financial Interests, Principal Investigator, I am a principal investigator on several Novartis sponsored studies and dont receive any renumeration for this: Novartis; Non-Financial Interests, Principal Investigator, I am a principal investigator on several of Genentech sponsored studies and don't receive any renumeration for this: Genentech; Non-Financial Interests, Principal Investigator, I am a principal investigator on several of Bristol Myer Squibb sponsored studies and don't receive any renumeration for this: Bristol Myer Squibb; Non-Financial Interests, Principal Investigator, I am the Principal Investigator for several AstraZeneca sponsored studies and am not remunerated for this: AstraZeneca; Non-Financial Interests, Principal Investigator, I am a principal investigator on a MacroGenics sponsored study and don't receive any renumeration for this: MacroGenics; Non-Financial Interests, Principal Investigator, I am a principal investigator on a Regeneron sponsored study and do not receive any renumeration for this: Regeneron; Non-Financial Interests, Principal Investigator, I am the Principal Investigator on several Merck Sharp and Dohme studies and I do not receive any renumeration for this: Merck Sharp and Dohme; Non-Financial Interests, Principal Investigator, I am a principal investigator on an IO Biotech study and I do not receive any renumeration for this: IO Biotech; Non-Financial Interests, Principal Investigator, I am principal investigator on an Evaxion sponsored study and I do not receive any renumeration for this: Evaxion. L. Spain: Financial Interests, Institutional, Advisory Board, Renal Cell Ad Board: Ipsen; Financial Interests, Personal, Invited Speaker, Renal Cell dinner meeting 08/2023: BMS; Financial Interests, Personal, Other, Engagement in steering committee of non-interventional study (ongoing from 2022); also invited chair at local meeting 08/2024: MSD; Financial Interests, Personal, Stocks/Shares, Stock in Impedimed: Impedimed; Financial Interests, Institutional, Local PI, PI for TILVANCE-301 trial: IOVANCE; Financial Interests, Institutional, Local PI, PI for PRISM-MEL study at Peter Mac: Immunocore. All other authors have declared no conflicts of interest.

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1695eP Systematic evaluation of polygenic risk scores (PRS) for autoimmune thyroid disease and the risk of thyroiditis following checkpoint immunotherapy (ICI) of melanoma in a North American patient (pt) population

A.A. Tarhini[1], Z. Chen[2], S.J. Lee[3], F.S. Hodi[4], T. Li[2], H. Streicher[5], V.K. Sondak[6], J.M. Kirkwood[7], X. Wang[8], P.A. Kanetsky[9]

[1]Cutaneous Oncology and Immunology, Moffitt Cancer Center, Tampa, United States of America; [2]Biostatistics and Bioinformatics, Moffitt Cancer Center, Tampa, United States of America; [3]Biostatistics, Dana-Farber Cancer Institute, Boston, United States of America; [4]Melanoma, Dana-Farber Cancer Institute - Longwood Center, Boston, United States of America; [5]DCTD Department, NIH - National Institutes of Health, Bethesda, United States of America; [6]Cutaneous Oncology, Moffitt Cancer Center, Tampa, United States of America; [7]Melanoma Program Department, University of Pittsburgh Medical Center Hillman Cancer Cente, Pittsburgh, United States of America; [8]Biostatistics and Bioinformatics, H. Lee Moffitt Cancer Center, Tampa, United States of America; [9]Epidemiology, Moffitt Cancer Center, Tampa, United States of America

Background: Predicting the risk of immune related adverse events (irAEs) including thyroiditis following exposure to ICI continues to be an urgent need.

Methods: We conducted genome-wide genotyping on samples from 744 consenting pts enrolled in ECOG-ACRIN E1609 trial (NCT01274338) that tested ipilimumab in melanoma in the U.S. and Canada. We used Illumina Infinium Global Screening Array v.3.0 + Multi-Disease BeadChip with 730059 designed loci. Genotypes were called using Illumina Genome Studio 2.0 and IAAP CLI v1.1.0, quality controlled, and harmonized with the 1000 Genomes Phase 3 (1KGP3) references. We used Michigan Imputation Server 2, pipeline v2.0.6, to impute the genotypes against the hrc-r1.1 reference panel and calculate all PRS listed in PGS-Catalog v20240318. Here, we report on the relationship between the distribution of PRSs for hypothyroidism and the occurrence of endocrine irAEs in E1609 pts, treating irAE grades as categorical (Kruskal-Wallis test) and as dichotomous (Wilcoxon rank-sum test) variables. IrAEs were graded using CTCAE v.4.

Results: We examined 18 distinct PRSs associated with hypothyroidism. For most, there was a significant difference in distribution of the PRS across grade of reported endocrine irAEs (Kruskal-Wallis $p<0.05$) or when irAE grade was dichotomized at 0 vs. other (Wilcoxon $p<0.05$). The most significant association was found with PRS PGS code PGS000761 (*Khan. Nat Commun. 2021*), which was trained in a non-melanoma European population with genetic similarity to our study sample based on our prior ancestry analysis (*Tarhini. AACR 2025*) and for which all markers in the PRS were available in our dataset. Interestingly, in multivariable survival analysis adjusting for primary status (known, unknown) and sex (male, female), no significant associations were noted.

Conclusions: Our analysis in the context of a U.S. Intergroup phase 3 trial supports the use of PRS to predict the risk of thyroiditis, selected based on inherited genetic similarity with the population used to train the score and regardless of pt tumor histology. Such use may have implications to better inform pt risk and follow up in the clinic.

Clinical trial identification: NCT01274338.

Legal entity responsible for the study: ECOG-ACRIN Cancer Research Group.

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1696eP Impact of metastasectomies on survival outcomes in melanoma patients treated with immunotherapy

J. Lyon[1], G. Rogers[2], A. Giobbie-Hurder[3], E. Lilley[4], C. Yoon[5], J.V. Cohen[2], P.A. Ott[2], R. Haq[2], D. Liu[6], M. Insco[2], F.S. Hodi[7], M. Manos[8], E. Buchbinder[2]

[1]Medicine, BIDMC - Beth Israel Deaconess Medical Centerl, Boston, United States of America; [2]Medical Oncology, Dana-Farber Cancer Institute, Boston, United States of America; [3]Statistics, Dana-Farber Cancer Institute, Boston, United States of America; [4]Surgery, Brigham and Womens Hospital, and Harvard Medical School, Boston, United States of America; [5]Surgery/Surgical Oncology Department, Brigham and Women's Hospital, Boston, United States of America; [6]Medical Oncology, Dana-Farber Cancer Institute, Harvard Medical School, Boston, United States of America; [7]Melanoma, Dana-Farber Cancer Institute - Longwood Center, Boston, United States of America; [8]Melanoma, Dana-Farber Cancer Institute, Boston, United States of America

Background: With the introduction of immunotherapy as first-line treatment for patients with advanced melanoma, overall and progression-free survival have improved considerably. Surgical resection of metastatic lesions is often performed on patients who receive immunotherapy, but the benefit is not known. This study investigated survival outcomes and patterns of utilization related to surgical metastatectomies in melanoma patients being treated with immunotherapy.

Methods: We identified melanoma patients who underwent surgery during immunotherapy treatment at Dana-Farber Cancer Institute. Patient characteristics, including age, gender, melanoma stage, treatments, reason for surgery, surgical history and survival status, were collected. Univariate and multivariate associations between overall survival (OS), time to progression (TTP) and clinical or treatment-related variables were assessed using Cox proportional hazards models.

Results: Between April 2014 and October 2023, 115 melanoma patients who underwent surgery while on immunotherapy were identified. The study population was predominantly male (63%) with an average age of 59 years, with 92% having stage IV melanoma. Surgery resulted in no evidence of disease (NED) in 26.1% of patients, with progressive metastasis removal being the most common surgical indication (41%). On multivariate analysis the study found statistically significant relationships between overall survival and patient age, ECOG status, time between treatment initiation and surgery and NED status. The greatest benefit on survival was observed for patients who underwent metastatectomy greater than 6 months from initiation of immunotherapy (HR for <6 months 2.67, 95% CI 1.37 − 5.20, $p = 0.004$). Survival outcomes were not significantly associated with site of surgery nor surgical intent, though surgeries done for isolated or progressive metastases showed a trend towards improved survival (HR 0.62, 95%. CI 0.28 − 1.36, $p = 0.06$).

Conclusions: This study sought to determine the benefit of surgical metastatecomies in melanoma patients on immunotherapy. Our findings suggest there are potential improvements in survival outcomes in certain melanoma patients, though future prospective studies are needed.

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Funding: Dana-Farber Cancer Institute.

Disclosure: F.S. Hodi: Financial Interests, Personal, Advisory Board: Checkpoint, Bioentre, Gossamer, Immunocore, Iovance, Trillium, Catalym, Kairos, BMS, Eisai, Corner therapeutics; Financial Interests, Personal, Member of Board of Directors: Bicara, Apricity. E. Buchbinder: Financial Interests, Personal, Advisory Board: BMS, Iovance, Merck, Zola, Obsidian, Anaveon, Werewolf, Immunocore; Financial Interests, Institutional, Coordinating PI: Genentech, Partners therapeutics; Other, Spouse employment: Takeda. All other authors have declared no conflicts of interest.

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1697eP Grand SLAM: A prospective randomized international multicenter study of shortened versus standard duration adjuvant immunotherapy for stage IIb-c, III and IV cutaneous malignant melanoma (CMM)

P. Ejder[1], H. Jespersen[2], M. Hernberg[3], L. Ny[4], A. Carneiro[5], S. Ellegård[6], K. Mattila[7], R. Olofsson Bagge[8], G. Fountoukidis[9], Y. Brandberg[10], M. Mandalà[11], B. O'Carrigan[12], B. Schilling[13], A.M. Arance[14], O. Hamid[15], A.L.T. Berglund[16], H. Helgadottir[17], G. Ullenhag[1]

[1]Oncology Department, Uppsala University Hospital, Uppsala, Sweden; [2]Oncology Department, Oslo University Hospital - The Norwegian Radium Hospital, Oslo, Norway; [3]Comprehensive Cancer Center, HUS - Helsinki University Hospital - Syopatautien klinikka, Helsinki, Finland; [4]Dept. of Oncology, Sahlgrenska University Hospital, Göteborg, Sweden; [5]Hematology, Oncology and Radiation Physics Department, Skane University Hospital CCC, Lund, Sweden; [6]Oncology Dept., Linköping University - Department of Health, Medicine and Caring Sciences (HMV), Norrköping, Sweden; [7]Oncology Department, Turku University Hospital (Tyks), Turku, Finland; [8]Department of Surgery, University of Gothenburg - The Sahlgrenska Academy, Göteborg, Sweden; [9]Oncology, Örebro University Hospital (Universitetssjukhuset Örebro), Örebro, Sweden; [10]Oncology Department, Karolinska Institutet - Cancer Center Karolinska (CCK) - OnkPat, K7, Solna, Sweden; [11]Medicine and Surgery Department, Università degli Studi di Perugia, Perugia, Italy; [12]Cancer Research UK Cambridge Institute, Cambridge University Hospitals NHS Foundation Trust, Cambridge, United Kingdom; [13]Department of Dermatology, UKW - University Hospital Würzburg, Würzburg, Germany; [14]Medical Oncology Department, Hospital Clinic y Provincial de Barcelona, Barcelona, Spain; [15]Research Department, The Angeles Clinic and Research Institute - West Los Angeles, Los Angeles, United States of America; [16]Research, Epistat AB, Uppsala, Sweden; [17]Oncology Department, Karolinska Institutet - Cancer Center Karolinska (CCK), Solna, Sweden

Background: Adjuvant treatment with PD-1 inhibitors for 12 months is the established standard of care for resected stage IIB-IV cutaneous malignant melanoma (CMM) patients. More recently neoadjuvant immunotherapy for macroscopic stage III and stage IV disease has been introduced. There is no clear biologic rationale for the chosen duration, and no studies have compared shorter adjuvant durations. In other solid tumours (breast, colorectal), shorter duration of adjuvant treatment has been shown to be non-inferior with improved toxicity profile. A reduced duration of adjuvant therapy could lead to less toxicity from reduced drug exposure, patients returning to normal life sooner, and significantly lower drug costs and healthcare resource utilization. There remains significant interest from patients and clinicians to address this important question.

Methods: Grand SLAM is a prospective phase III randomized, controlled international multi-center non-inferiority study. The primary objective is to investigate if short (6 months) has equal efficacy as long (12 months) duration of peri-operative immunotherapy (adjuvant ± neoadjuvant) in relation to distant metastasis free survival (DMFS) and relapse free survival (RFS) at landmark analysis of 2 years. After radical surgery of stage IIB-C, III or IV CMM, patients are randomly assigned 1:1 to short (6 months) or long (12 months) treatment with single agent approved anti-PD-1 regimens. Neoadjuvant patients with major pathological response (CR or near CR) are not eligible. The sample size of 1880 patients is based on a non-inferiority margin of 4%, $\alpha = 0.045$ and 80% power. An interim analysis will be conducted at 2/3 accrual. Biomarkers and the role of food supplements for relapse (MelKo) will be investigated in sub-studies.

Results: As of May 2025, the study is recruiting patients in the Nordic countries. Centers in other countries will open shortly.

Conclusions: This is the first randomized study to assess a shorter duration of (neo)-adjuvant immunotherapy (6 vs 12 months) in CMM patients.

Clinical trial identification: NCT06488482.

Legal entity responsible for the study: Uppsala University Hospital.

Funding: The Swedish Cancer Society.

Disclosure: All authors have declared no conflicts of interest.

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1698eP Incidence of chronic immune related adverse events (irAEs) in patients with advanced melanoma treated with immune checkpoint inhibitors in British Columbia

S. Yeo[1], A. Angeles[2], X. Ye[1], P.T. Nguyen[3], G. Bahl[4], V. Bernstein[5], E. Sonke[1], V.I. Poon[1], A. Weppler[1], K.J. Savage[1]

[1]Medical Oncology, BC Cancer Agency - Vancouver, Vancouver, Canada; [2]Medical Oncology, BC Cancer Agency - Surrey, Surrey, Canada; [3]Medical Oncology Dept., ARHCC - Abbotsford Regional Hospital and Cancer Centre, Abbotsford, Canada; [4]Radiation Oncology, BC Cancer - Abbotsford, Abbotsford, Canada; [5]Medical Oncology, BC Cancer Agency - Victoria, Victoria, Canada

Background: Immune checkpoint inhibitors (ICIs) are the preferred treatment for patients with locally advanced or metastatic melanoma (MM). However, long-term immune-related adverse events (irAEs), which may affect quality of life and overall long-term health, are often not captured in clinical trials due to limited follow-up.

Methods: Patients with locally advanced/MM who received ≥ 1 cycles of ICIs at BC Cancer from 2012-2022 were identified. irAEs were assessed using the CTCAEv5. Per SITC guidelines, chronic irAEs were defined as persisting > 3m after ICI discontinuation, chronic + active if ongoing inflammation requiring immunosuppression and chronic + inactive if absence of ongoing inflammation and not requiring immunosuppression.

Results: Among 530 patients, 136 (26%) received ipilimumab/nivolumab (ipi/nivo), 259 (49%) received a PD1 inhibitor and 135 (25%) received ipi as their first ICI. Using reverse censoring, median follow-up was 7.0 y (95% CI: 6.6-7.4 y). Acute irAEs occurred in 298 (56%) patients: 107 received ipi/nivo, 144 PD1and 47 ipi. Chronic irAEs occurred in 146 (28%) patients, of which 39/146 (27%) were chronic active and 107 (73%) were chronic inactive. 126 (24%) and 99 (19%) patients had chronic irAEs persisting at 6m and 12m, respectively. Patients treated with ipi/nivo had a greater frequency of chronic irAEs >3m (54/136=40%) compared to those who received PD1 (75/259=29%, p=0.03), and were more likely to develop chronic active irAEs (22/136=16% vs 13/259=5%, p<0.001). The most frequently affected systems were endocrine (37%) and dermatologic (36%, vitiligo 22%) but GI (11%), rheumatologic (9%), respiratory (6%), neurologic (1.5%), ophthalmologic (1%), cardiac (1%) and renal (.5%) toxicities were also observed. At maximal severity since onset of irAE, 48% were grade 1, 36% were grade 2 and 16% were grade 3/4. Following irAE onset, 99 (49%) of chronic irAEs required a subspecialist and 18 (9%) required hospitalization.

Conclusions: More than one in four patients with advanced melanoma treated with ICIs developed chronic irAEs, highlighting the importance of counselling on the risk of long-term toxicities and the need for extended follow-up for some patients.

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1699eP Treatment patterns using encorafenib plus binimetinib in BRAF mutated melanoma patients

P. Mohr[1], M. Weichenthal[2], N. Asher[3], I. Gavrilova[4], E. Ellebaek[5], I. von Wasielewski[6], J. Utikal[7], P. Terheyden[8], R. López Castro[9], J. Medina Martinez[10], R. Dummer[11], S. Levari[12], A. Jalovcic-Suljevic[13], I.M. Svane[5], P. Rutkowski[14], P.A. Ascierto[15], H. Gogas[16], J. Mangana[17], L. Bastholt[18], D. Schadendorf[19]

[1]Department of Dermatology, Elbe Hospital Buxtehude, Buxtehude, Germany; [2]Dermatology Department, Christian-Albrechts-University Kiel, Kiel, Germany; [3]Department of Oncology, Skin Cancer and Melanoma Center at Davidoff Cancer Center, Rabin, Israel; [4]Oncodermatology department, USBAL Oncology EAD Sofia, Sofia, Bulgaria; [5]Department of Oncology, National Center for Cancer Immune Therapy (CCIT-DK), Herlev Hospital, Herlev, Denmark; [6]Dermatology, Allergologie and Venereology, MHH - Medizinische Hochschule Hannover, Hannover, Germany; [7]Skin Cancer Unit, DKFZ - German Cancer Research Center, Heidelberg, Germany; [8]Dermatology, UKSH - Universitätsklinikum Schleswig-Holstein - Campus Lübeck, Lübeck, Germany; [9]Medical Oncology Department, Hospital Clínico Universitario de Valladolid, Valladolid, Spain; [10]Medical Oncology Department, Hospital Virgen de la Salud, Toledo, Spain; [11]Dermatology Department, University Hospital Zurich, Zurich, Switzerland; [12]Melanoma and Skin Cancer, Sheba Tel Hashomer Medical Center, Ramat Gan, Israel; [13]Clinic of Oncology Dept., Clinical Centre University of Sarajevo, Sarajevo, Bosnia and Herzegovina; [14]Department of Soft Tissue/Bone Sarcoma and Melanoma, Maria Skłodowska-Curie National Research Institute of Oncology, Warsaw, Poland; [15]Melanoma, Cancer Immunotherapy & Developmental Therapeutics, Istituto Nazionale Tumori IRCCS Fondazione Pascale, Naples, Italy; [16]First Department of Medicine, National and Kapodistrian University of Athens - School of Medicine, Athens, Greece; [17]Dermatologie department, USZ - Universitätsspital Zürich - Klinik für Dermatologie, Zurich, Switzerland; [18]Dept. of Oncology, OUH - Odense University Hospital, Odense, Denmark; [19]Department of Dermatology, Venereology and Allergology, University Hospital Essen, Essen, Germany

Background: This study aims at evaluating the treatment patterns, clinical characteristics, and outcomes in adult patients with unresectable or metastatic melanoma with a $BRAF^{V600}$ mutation who were treated with encorafenib plus binimetinib (E/B) in the real-life setting.

Methods: This retrospective study evaluated patients from the EUMelaReg registry with non-resectable or metastatic cutaneous melanoma including melanoma of unknown primary, who received E/B between Sep 2018 and Jan 2024. Patients were stratified into groups based on the line of treatment, and the class of preceding systemic treatments. Patterns of E/B treatment were the focus of the analysis. The main outcomes were response rates, progression-free survival (PFS) and overall survival (OS) from start of E/B treatment.

Results: The median age of the analyzed population (n=854) at start of E/B was 62 years and 61.6% were male. Stage IV M1c was diagnosed in 39.1% and M1d in 34.4% at the start of E/B. 52.7% of patients had an elevated lactate dehydrogenase and 48.2% had ≥ 3 metastatic sites at baseline. Among the different treatment patterns 79% ($n = 675$) of cases followed four main patterns, with the most common being: 1L E/B in treatment naive patients (36.3%); 2L+ E/B following 1L immune checkpoint inhibitors (ICI) without adjuvant therapy (35.4%); 1L E/B following adjuvant ICI (21.2%); and 2L+ E/B following both adjuvant and non-adjuvant ICI (7.1%). The remaining minority of cases received various other pretreatments. The overall response rate was 55.3% of the total population and varied from 54% to 64%, with no statistically significant difference between the four treatment patterns. Median PFS showed significant heterogeneity between the groups (p=0.004) with the shortest being 7.2 months in treatment naive patients receiving 1L E/B and the longest being 10.5 months in patients receiving 1L E/B following adjuvant ICI. Median OS from initiation of E/B treatment ranged from 15.2 to 20.5 months, without significant variation observed among the groups.

Conclusions: This study reflects the different treatment patterns of E/B use in real-world setting and informs on patient profiles and related outcome variables. Overall, E/B shows to be a valuable option for different treatment settings.

Legal entity responsible for the study: EuMelaReg gGmbH.

Funding: Has not received any funding.

Disclosure: P. Mohr: Financial Interests, Personal, Advisory Board: Bristol Myers Squibb, Merck Sharp & Dohme, Novartis, GSK, Pierre Fabre, Sanofi, Delcath, Regeneron, Immunocore, Merck Serono; Financial Interests, Personal, Invited Speaker: Bristol Myers Squibb, Merck Sharp & Dohme, Novartis, Pierre Fabre, Sanofi, Sun Pharma, Delcath; Financial Interests, Institutional, Funding: Bristol Myers Squibb, Merck Sharp & Dohme, Novartis; Non-Financial Interests, Principal Investigator: Bristol Myers Squibb, Merck Sharp & Dohme, Regeneron, Sanofi, Novartis, Sun Pharma. M. Weichenthal: Financial Interests, Personal, Advisory Board: Merck Sharp & Dohme, Bristol Myers and Squibb, Sanofi, Sun Pharma, Pierre Fabre; Financial Interests, Institutional, Local PI: Millennium, Merck Sharp & Dohme, 4SC, Innate Pharma; Financial Interests, Institutional, Research Grant: Bristol Myers and Squibb, Novartis; Non-Financial Interests, Local PI: EuMelaReg; Non-Financial Interests, Member of Board of Directors: DeCOG. I. Gavrilova: Financial Interests, Advisory Role: Merck Sharp & Dohme, Bristol Myers and Squibb, Pierre Fabre, Medison; Financial Interests, Other, Honoraria: Roche, Novartis, Merck Sharp & Dohme, Bristol Myers and Squibb, Sanofi, Merck, Pierre Fabre, Medison; Financial Interests, Speaker's Bureau: Roche, Merck Sharp & Dohme, Bristol Myers and Squibb, Novartis, Sanofi, Merck, Pierre Fabre, Medison; Financial Interests, Other, Travel/accommodations/expenses: Merck Sharp & Dohme, Bristol Myers and Squibb, Pierre Fabre. E. Ellebaek: Financial Interests, Personal, Invited Speaker: Pierre Fabre, BMS, Novartis, MSD; Other, Travel and conference expenses: MSD, Pierre Fabre. I. von Wasielewski: Financial Interests, Advisory Role: BMS, MSD, Sanofi; Financial Interests, Other, Honoraria payment: Novartis, BMS, MSD, Sanofi, Stemline, Kyowa Kirin; Financial Interests, Other, Travel/accommodations/expenses: Novartis, BMS, MSD, Sanofi, Stemline, Kyowa Kirin. J. Utikal: Financial Interests, Personal, Advisory Board: Amgen, Bristol Myers Squibb, Immunocore, Merck Sharp and Dohme, Novartis, Pierre Fabre, Rheacell, Sanofi. P. Terheyden: Financial Interests, Personal, Advisory Board: Almirall, Biotest, BMS, Sanofi; Financial Interests, Personal, Invited Speaker: Almirall, Bristol Myers Squibb, Sanofi, Kyowa Kirin, Biofrontera, 4SC; Financial Interests, Personal, Other, Travel support: BMS, Pierre-Fabre. J. Medina Martinez: Financial Interests, Personal, Other, Honoraria: BMS, MSD, Novartis, Pierre Fabre, Roche. R. Dummer: Financial Interests, Personal, Other, Consulting and/or advisory role: Novartis, Merck Sharp & Dohme (MSD), Bristol Myers Squibb (BMS), Roche, Amgen, Takeda, Pierre Fabre, Sun Pharma, Sanofi, Catalym, Second Genome, Regeneron, MaviVAX SA, T3 Pharma, Pfizer, Simcere; Financial Interests, Personal, Other, medical advisor: Oncobit. S. Levari: Financial Interests, Other, Travel/accommodations/expenses: BMS, MSD; Financial Interests, Other, Travel/Accommodation/Expenses: Medison; Financial Interests, Research Funding: BMS. I.M. Svane: Financial Interests, Institutional, Other, Conference participation: MSD; Financial Interests, Personal, Invited Speaker: BMS, Janssen; Financial Interests, Personal, Advisory Board: Mendus, Genmab; Financial Interests, Personal, Stocks/Shares, Cofounder and Founder warrents: IO Biotech; Financial Interests, Institutional, Research Grant: Adaptimmune, Enara Bio, Lytix Biopharma, TILT Biotherapeutics, Asgard Therapeutics, IO Biotech; Financial Interests, Institutional, Funding: Evaxion; Financial Interests, Institutional, Other, drug for investigator driven trial: BMS; Non-Financial Interests, Principal Investigator: BMS, Roche, TILT Biotherapeutics, Lytix Biopharma, Novartis, Immunocore, MSD. P. Rutkowski: Financial Interests, Personal, Invited Speaker, honoraria for lectures: MSD, BMS, Pierre Fabre; Financial Interests, Personal, Advisory Board: MSD, BMS, Pierre Fabre, Merck, Sanofi, Blueprint Medicines, Philogen, Medison Pharma, Genesis Pharma; Financial Interests, Personal, Invited Speaker: Merck, Sanofi, Novartis, AstraZeneca, Genesis Pharma; Financial Interests, Institutional, Research Grant, research grant for ISS: Pfizer; Financial Interests, Institutional, Funding, research grant for institution: BMS; Non-Financial Interests, Member of Board of Directors: Polish Society of Surgical Oncology; Non-Financial Interests, Member of Board of Directors, President: Polish Oncological Society. P.A. Ascierto: Financial Interests, Personal, Other, Consultant and Advisory Role: BMS, Roche Genentech, Novartis, Merck Serono, Sun Pharma, Sanofi, Sandoz, Immunocore, Boehringer Ingelheim, Regeneron; Financial Interests, Personal, Other, Consultant, Advisory Role and Travel support: MSD, Pierre Fabre; Financial Interests, Personal, Other, Consultant and Advisory Role, Travel support: Pfizer/Array; Financial Interests, Personal, Other, Consultant Role: Italfarmaco; Financial Interests, Personal, Other, Consultant Role: Pfizer, Nouscom, Lunaphore; Financial Interests, Personal, Other, Consultant role: Medicenna; Financial Interests, Personal, Other, Consultant role and travel support: Bio-AI Health; Financial Interests, Personal, Advisory Board, Consultant and Advisory role: ValoTx; Financial Interests, Personal, Advisory Board, Consultant and Advisor role, Travel support: Replimmune; Financial Interests, Personal, Advisory Board, Advisor role: Bayer; Financial Interests, Personal, Other, Consultant and Advisory: Erasca; Financial Interests, Personal, Other, Consultant: Philogen, Incyte; Financial Interests, Personal, Other, Consultant and Advisory role: BionTech, Genmab; Financial Interests, Personal, Advisory Board: Anaveon, Menarini, ImCheck therapeutics; Financial Interests, Institutional, Funding, Clinical trial and translational research: BMS; Financial Interests, Institutional, Funding, Clinical Trial: Roche Genentech, Pfizer/Array, Sanofi; Non-Financial Interests, Leadership Role, President since 2010: Fondazione Melanoma Onlus Italy; Non-Financial Interests, Leadership Role, President since 2014: Campania Society of ImmunoTherapy of Cancer (SCITO) Italy; Non-Financial Interests, Other, Member of Steering Committee since 2016: Society for Melanoma Research (SMR); Non-Financial Interests, Member of Board of Directors, November 2017 - December 2021: Society for Immunotherapy of Cancer (SITC); Non-Financial Interests, Member: ASCO, SITC, EORTC Melanoma Cooperative Group, AIOM, SMR. H. Gogas: Financial Interests, Personal, Advisory Board: MSD, BMS, Pierre Fabre, Sanofi; Financial Interests, Personal, Invited Speaker: MSD, BMS, Novartis, Pierre Fabre, Sanofi; Financial Interests, Steering Committee Member: Amgen, Replimmune; Financial Interests, Institutional, Local PI: Amgen, MSD, BMS, Replimmune, Iovance, Bayer; Financial Interests, Institutional, Research Grant: BMS, Pfizer, Lilly, Pierre Fabre, AstraZeneca. J. Mangana: Financial Interests, Personal, Advisory Board: Merck Sharp & Dohme; Financial Interests, Personal, Advisory Board, immune-related Adverse Events Management (ESMO): Bristol Myers and Squibb; Financial Interests, Personal, Other, Travel grant: Ultrasun, L'oreal, Merck Sharp & Dohme, Pierre Fabre; Financial Interests, Personal, Invited Speaker: Johnson and Johnson; Financial Interests, Personal, Other, Travel Grant: Sun pharmaceutical; Financial Interests, Institutional, Research Grant, The melanoma registry database of the Department of Dermatology has been partially supported by an unrestricted grant to the University of Zurich from Amgen, Novartis, BMS, MSD and Pierre Fabre: Novartis, BMS, Pierre Fabre, Amgen, MSD; Non-Financial Interests, Leadership Role, President WG Melanoma Group: SAKK; Non-Financial Interests, Leadership Role, president WG Melanoma: SAKK; Other, immune-related AEs podcast /paid personal: Bristol Myers Squibb International. L. Bastholt: Non-Financial Interests, Advisory Role, Scientific committee under Danish Medicines Agency regarding new treatments of melanoma, skin cancer and thyroid cancer: Danish Medicines Agency. D. Schadendorf: Financial Interests, Personal, Invited Speaker: BMS, MSD, Merck Serono, Regeneron; Financial Interests, Personal, Advisory Board: BMS, Novartis, MSD, Immunocore, Pierre Fabre, Regeneron, Philogen, BioAlta, Daiichi Sankyo, BioNTech, Immatics, CureVac, Formycon, Replimune, SunPharma, Skyline Dx, IOVance, Ipsen, IOBioTech, Boehringer-Ingelheim; Financial Interests, Institutional, Coordinating PI: Novartis, BMS, MSD, Pierre Fabre, Immatics; Financial Interests, Institutional, Research Grant: BMS, MSD; Financial Interests, Personal, Steering Committee Member: MSD; Financial Interests, Institutional, Local PI: Philogen, Immunocore, Regeneron, Biontech Rna Pharmacuticals GMBH, Replimune, IO Biotech; Non-Financial Interests, Member of Board of Directors: EORTC-MG; Non-Financial Interests, Leadership Role, Founding member and SC chair: European Melanoma Registry (EuMelaReg). All other authors have declared no conflicts of interest.

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1700eTiP SCOPE phase II clinical trial program evaluating off-the-shelf DNA plasmid vaccines, SCIB1 and iSCIB1+ administered in combination with checkpoint inhibitors in advanced unresectable melanoma

N. Varawalla[1], H.M. Shaw[2], P.G. Corrie[3], P. Patel[4], M. Payne[5], I. Karydis[6], K. Young[7], C. Barlow[8], S. Danson[9], S. Kumar[10], R.J. Lee[11], M.S. Highley[12], M. Marples[13], R.E. Board[14], S.J. Paston[1], J. Chadwick[1], J. Thornton[1], R. Miller[1], G.R. Goodhew[1], L. Durrant[1]

[1]Research & Development, Scancell Ltd., Oxford, United Kingdom; [2]Department of Medical Oncology, UCLH - University College London Hospitals NHS Foundation Trust, London, United Kingdom; [3]Oncology Department, Cambridge University Hospitals NHS Foundation Trust, Cambridge, United Kingdom; [4]Clinical Oncology, Nottingham Hospitals University Trust, Nottingham, United Kingdom; [5]Oncology Dept., Churchill Hospital, Oxford, United Kingdom; [6]Cancer Care Department, Southampton General Hospital, Southampton, United Kingdom; [7]Renal/Melanoma Department, The Royal Marsden Hospital - Chelsea, London, United Kingdom; [8]Oncology Dept., Musgrove Park Hospital - Taunton and Somerset NHS Foundation Trust, Taunton, United Kingdom; [9]Oncology and Metabolism Department, Weston Park Cancer Centre - Sheffield Teaching Hospitals NHS Foundation Trust, Sheffield, United Kingdom; [10]Department of Medical Oncology, South West Wales Cancer Institute Singleton Hospital, Sketty, United Kingdom; [11]Medical oncology department, The University of Manchester, Manchester, United Kingdom; [12]Oncology Dept., Derriford Hospital Plymouth Hospitals NHS Trust, Plymouth, United Kingdom; [13]Department of Medical Oncology, St. James's University Hospital, Leeds, United Kingdom; [14]Department of Medical Oncology, Royal Preston Hospital, Preston, United Kingdom

Background: SCIB1 and iSCIB1+ are DNA plasmid off-the shelf melanoma vaccines encoding two clusters of CD8 epitopes, TRP-2 and gp100. When administered by needle-free injection these are taken up by Antigen Presenting Cells which directly present the epitopes on MHC molecules to T cells that amplify the anti-tumour immune response, further boosted by dendritic cell cross presentation. Monotherapy SCIB1 induced dose dependent T cell responses in 88% of advanced melanoma patients with minimal side effects (Patel et al, 2018).

In patients with unresectable melanoma, immunotherapy with anti-PD-1 and anti-CTLA-4 checkpoint inhibitors reports an overall response rate (ORR) of 50%, with substantial adverse events (Lebbe et al, 2019). The addition of SCIB1 / iSCIB1+ could overcome the inadequate tumour recognition that causes immunotherapy to fail.

Trial design: The SCOPE phase 2 open label single arm clinical program evaluates within defined cohorts the safety and efficacy of SCIB1 and iSCIB1+ in over 140 patients with unresectable melanoma receiving immunotherapy with ipilimumab and nivolumab at 16 UK trial sites. The primary endpoint is ORR, as measured by RECIST 1.1. Secondary endpoints include duration of response, iRECIST determined ORR, Progression Free Survival and Overall Survival. T cell responses are assessed with a cultured ELISpot assays and adverse events are recorded.

Cohort 1 includes 43 patients with the target HLA haplotype (A2 positive) receiving plus SCIB1 administered intra-muscularly via a needle free injector. In Cohort 2 only,10 such patients, received pembrolizumab.

iSCIB1+ designed to increase eligible HLA haplotypes and avidity is similarly evaluated in Cohort 3, within two sub-groups of 31 and 17, without and with HLA restriction, respectively.

Intra-dermal administration enhances T cell responses in pre-clinical studies and accelerated dosing would prevent disruption of the regimen by checkpoint adverse events. These changes will be evaluated in 40 cohort 4 patients.

Collectively these cohorts will inform optimal product, patient characteristics and dosing regimen for the follow-on randomised clinical trial.

Clinical trial identification: NCT04079166.

Legal entity responsible for the study: Scancell Ltd.

Funding: Scancell Ltd.

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Corrie: Financial Interests, Personal, Advisory Board: Merck Sharpe and Dohme, Scancell, Astellas, SUN Pharma, Viatris, Iovance Biotherapeutics; Financial Interests, Personal, Invited Speaker: Pierre Fabre, Merck Sharpe & Dohme, Novartis; Financial Interests, Personal, Other, Consultancy: Microbiotica; Financial Interests, Institutional, Local PI: Bristol Myers Squibb, Novartis, AstraZeneca, Merck Sharpe and Dohme, Pfizer, Immunocore, Scancell; Financial Interests, Institutional, Research Grant: Pierre Fabre; Financial Interests, Institutional, Coordinating PI, Free drug supply for academic multi centre trial: Merck Sharpe & Dohme; Financial Interests, Institutional, Coordinating PI: Microbiotica. M. Payne: Financial Interests, Personal, Stocks or ownership, speaker, consultant, advisor: Genesis Care UK. I. Karydis: Financial Interests, Institutional, Local PI, Site for Company sponsored study: Genentech, BioNTech AG, Achilles Therapeutics, Replimmune Inc, ModernaTX, INc, ScanCell Ltd; Financial Interests, Institutional, Local PI, Site PI in a study co-funded by an unrestricted educational grant from Immunocore Ltd: Immunocore LTd; Financial Interests, Institutional, Local PI, Site for company sponsored study: T3 Pharmaceuticals AG. K. Young: Financial Interests, Personal, Invited Speaker, Speaker at educational events: BMS, Ipsen; Financial Interests, Personal, Invited Speaker, Speaker and Chair at educational events: Eisai; Financial Interests, Personal, Other, Consultancy project, Funded to attend ESMO: Merck Serono; Financial Interests, Institutional, Local PI, PI for SCOPE study: Scancell; Financial Interests, Institutional, Local PI, PI for Ultimovacs study: Ultimovacs; Financial Interests, Institutional, Local PI, PI for TiNivo2 study: Aveo Pharmaceuticals; Financial Interests, Institutional, Local PI, PI for SEACRAFT study: Erasca; Non-Financial Interests, Member of Board of Directors, Chair of New Consultants' Group and Rep for UKAOS: Association of Cancer Physicians; Non-Financial Interests, Member of Board of Directors: UKAOS; Non-Financial Interests, Member of Board of Directors, Trustee and Leader of Clinical Support Pillar: IOCN. C. Barlow: Non-Financial Interests, Personal, Speaker, Consultant, Advisor: Bristol Myers Squibb, Merck Sharpe and Dohme, Amgen, Roche, Pierre Fabre. S. Danson: Financial Interests, Personal, Other, Consultancy: Greywolf Therapeutics, Orion; Financial Interests, Personal, Advisory Board, MTH1 development advisory board: Oxcia; Financial Interests, Personal, Full or part-time Employment, Co-Clinical Lead for the UK Vaccine Innovation Pathway: NIHR; Financial Interests, Institutional, Coordinating PI, Dante lead: NIHR; Financial Interests, Institutional, Coordinating PI, Fortitude co-ordinating PI for the UK: Amgen; Financial Interests, Institutional, Research Grant, Ecmc lead: Cancer Research UK; Financial Interests, Institutional, Local PI, Local PI for the MSD-Anti-PD1 study: MSD; Financial Interests, Institutional, Local PI, Local PI for the TEADES study: Orion; Non-Financial Interests, Leadership Role, Lion steering group chair: NIHR; Non-Financial Interests, Personal, Proprietary Information, Have received prescribing information for immunotherapy to inform a project proposal: BMS. R.J. Lee: Financial Interests, Personal, Invited Speaker: AstraZeneca; Financial Interests, Personal, Advisory Board: Pierre Fabre; Financial Interests, Personal, Invited Speaker, Talk at Melanoma Focus: Pierre Fabre; Financial Interests, Institutional, Research Grant: Bristol Myers Squibb, AstraZeneca; Financial Interests, Institutional, Research Grant, Funding for translational research: Pierre Fabre; Financial Interests, Institutional, Other, Advisory role - payment going to research account in melanoma team: Lumisphere Technology. M. Marples: Financial Interests, Personal, Funding: Scancell Ltd; Financial Interests, Personal, Invited Speaker: BMS. S.J. Paston, J. Chadwick: Financial Interests, Personal, Full or part-time Employment: Scancell Ltd. J. Thornton: Financial Interests, Personal, Full or part-time Employment: Scancell Ltd; Financial Interests, Personal, Stocks/Shares: GSK, Merck Sharpe and Dohme, Moderna, Atai Life Sciences. R. Miller: Financial Interests, Personal, Full or part-time Employment: Scancell Ltd, Clarity Pharmaceuticals. G.R. Goodhew: Financial Interests, Personal, Full or part-time Employment: Scancell Ltd. L. Durrant: Financial Interests, Personal, Member of Board of Directors, shareholder and inventor of patent: Scancell Ltd. All other authors have declared no conflicts of interest.

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1701eTiP NRG-HN014 (C-PRE): A randomized phase III study of neoadjuvant immunotherapy with response-adapted treatment versus standard-of-care treatment for patients with resectable stage III-IV (M0) cutaneous squamous cell carcinoma (cSCC)

N. Gross[1], K. Atwood[2], B. Ashford[3], C. Barker[4], Y. Chen[5], P. Nagarajan[6], D. Rischin[7], N. Schmitt[8], S. Tam[9], T.M. Wise-Draper[10], M. Ward[11], X. Wu[12], S.S. Yom[13]

[1]Head and Neck Surgery, The University of Texas MD Anderson Cancer Center, Houston, United States of America; [2]Statistics and Data Management, NRG Oncology Statistics and Data Management Center, Philadelphia, United States of America; [3]Surgery, The Wollongong Hospital, Wollongong, Australia; [4]Radiation Oncology, Memorial Sloan Kettering Cancer Center, Commack, United States of America; [5]Radiology, Stephenson Cancer Center/University of Oklahoma, Oklahoma City, United States of America; [6]Pathology, MD Anderson Cancer Center, Houston, United States of America; [7]Medical Oncology Department, Peter MacCallum Cancer Centre, Melbourne, Australia; [8]Otolaryngology Head and Neck Surgery, Emory University, Atlanta, United States of America; [9]Otolaryngology Head and Neck Surgery, Henry Ford Cancer Institute - Henry Ford Health System, Detroit, United States of America; [10]Hematology Oncology Dept, Vontz Center for Molecular Studies, Cincinnati, United States of America; [11]Radiation Oncology, Levine Cancer Institute, Charlotte, United States of America; [12]Radiology, UCSF - University of California San Francisco, San Francisco, United States of America; [13]Radiation Oncology, UCSF - University of California San Francisco - Parnassus Campus, San Francisco, United States of America

Background: Cutaneous squamous cell carcinoma (CSCC) is associated with immune system dysfunction, and prior research has demonstrated durable pathologic responses and favorable survival using neoadjuvant anti-programmed death-1 (PD-1) therapy. NRG-HN014 (C-PRE) is an NCI-sponsored, open-label, randomized, phase 3 trial (NCT06568172) evaluating the efficacy of neoadjuvant cemiplimab with response-adapted treatment for advanced stage, resectable cSCC.

Trial design: Patients with histologically confirmed resectable stage III-IV (M0) CSCC will be enrolled, including select immunosuppressed patients. A total of 420 patients in the US, Canada and Australia will be randomly assigned 1:1 to Arm 1 (standard-of-care) or Arm 2 (neoadjuvant cemiplimab with response-adapted treatment). Patients in Arm 1 will proceed directly to surgery and receive adjuvant radiation as indicated by pathology. Patients in Arm 2 will receive cemiplimab 350 mg IV Q3W (up to 4 doses) followed by response-adapted oncologic surgery. Patients in Arm 2 with a complete pathologic response (pCR) will receive no further treatment. The remaining patients in Arm 2 with $<$pCR will receive adjuvant radiation as indicated by pathology followed by cemiplimab 700 mg IV Q6W (4 doses). Imaging will be performed at screening, during and after neoadjuvant cemiplimab (Arm 2), after adjuvant cemiplimab (Arm 2 as applicable), then Q3M for 2 years, Q6M for 1 year and annually thereafter. The primary end point is event-free survival by investigator assessment. Secondary end points include disease-specific and overall survival, safety, pCR (Arm 2), and patient-reported outcomes.

Clinical trial identification: NCT06568172, Activated 12/2024.

Legal entity responsible for the study: NIH/NCI.

Funding: Regeneron.

Disclosure: N. Gross: Financial Interests, Institutional, Research Funding: Ascendis, Regeneron; Financial Interests, Personal, Speaker, Consultant, Advisor: Regeneron, Merck, GeoVax, AiCME, OncLive; Financial Interests, Personal, Advisory Board: PDS Biotechnology; Financial Interests, Personal, Royalties: Up-to-Date. C. Barker: Financial Interests, Institutional, Research Funding: Regeneron, AlphaTau Medical, EMD Serono, Physical Sciences Inc, Amgen, Elekta; Financial Interests, Personal, Advisory Board: Regeneron; Financial Interests, Personal, Speaker, Consultant, Advisor: American Journal of Managed Care. Y. Chen: Financial Interests, Personal, Speaker, Consultant, Advisor: Mevion Medical Systems. D. Rischin: Financial Interests, Institutional, Research Funding: Merck, Regeneron, ALX Oncology, GSK, Bicara Therapeutics; Financial Interests, Personal, Advisory Board: Merck, Regneron, Eisai, Bicara Therapeutics, GSK. N. Schmitt: Financial Interests, Institutional, Research Funding: Taiho Oncology; Financial Interests, Personal, Speaker, Consultant, Advisor: Regeneron, GeoVax, Johnson and Johnson, Sensorion, Aspargo Labs, Synergy Research. S. Tam: Financial Interests, Institutional, Research Funding: Genentech, Pfizer. T.M. Wise-draper: Financial Interests, Institutional, Research Funding: IDEAYA, Merck, Lovance Biotherapeutics, Eli Lilly, Replimmune, Daiichi Sankyo, Duality Bio, Xencor, Bicara Therapeutics, Georgiamune, Coherus Biosciences; Financial Interests, Personal, Research Funding: Merck, Janssen, BMS, Tesaro/GSK; Financial Interests, Personal, Speaker, Consultant, Advisor: Merck, Replimmune, Genmab, EMD Serono, Caris Life Sciences, NEED Inc. M. Ward: Financial Interests, Personal, Speaker, Consultant, Advisor: Blue Cross Blue Shield NC, InformAI; Financial Interests, Personal, Royalties: Demos Medical Publishing. X. Wu: Financial Interests, Personal, Financially compensated role: Bayer. S.S. Yom: Financial Interests, Institutional, Research Funding: Bristol Myers Squibb, Merck, EMD Serono, Johnson & Johnson; Financial Interests, Personal, Royalties: Up-to-Date, Springer; Financial Interests, Personal, Financially compensated role: Elsevier. All other authors have declared no conflicts of interest.

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1702eTiP A phase II trial of quisinostat for high-risk uveal melanoma

J. Lutzky[1], L. Hernandez-Aya[2], L. Feun[1], J. King[1], S. Chandra[3], S. Rapisuwon[4], I. Mehmi[5], S. Chandrasekaran[6], P. Funchain[7], R.D. Carvajal[8], A.K.S. Salama[9], K. Montazeri[10], R.J. Sullivan[11], Z. Correa[12], J.W. Harbour[13]

[1]Cutaneous Oncology, University of Miami Sylvester Comprehensive Cancer Center, Miami, United States of America; [2]Medicine, University of Miami, Miami, United States of America; [3]Division of Hematology Oncology, Robert. H. Lurie Cancer Center of Northwestern University, Chicago, United States of America; [4]Medicine, Georgetown University School of Medicine, Washington, United States of America; [5]Medical Oncology, The Angeles Clinic & Research Institute, Santa Monica, United States of America; [6]Medicine, UTSW - University of Texas Southwestern Medical Center, Dallas, United States of America; [7]Medicine - Med/Oncology, Stanford Cancer Center Palo Alto, Palo Alto, United States of America; [8]Hemato-Oncology Dept., Northwell Health Cancer Institute, Lake Success, United States of America; [9]Medicine, Duke University, Durham, United States of America; [10]Department of Haematology and Oncology, Massachusetts General Hospital Cancer Center, Harvard Medical School, Boston, United States of America; [11]Medicine, MGH - Massachusetts General Hospital, Boston, United States of America; [12]Ophthalmology, University of Miami Bascom Palmer Eye Institute, Miami, United States of America; [13]Ophthalmology, UTSW - University of Texas Southwestern Medical Center, Dallas, United States of America

Background: Uveal melanoma (UM) comprises 5% of melanomas. Metastases occur in up to 50% of patients, for which here is no curative treatment. UM tumors can be classified into low (class 1) and high (class 2) metastatic risk categories based on their gene expression profiling (GEP). UM with Castle DecisionDx-UM GEP Class 2 harbor inactivating mutations in the tumor suppressor BAP1, which epigenetically regulates a wide range of genes involved in development and other functions. A novel epigenetic high-throughput drug screen for therapeutic targets that restore epigenetic regulation downstream of BAP1 identified quisinostat as the lead compound capable of reversing phenotypic effects of BAP1 loss in vitro and in vivo with minimal toxicity. Quisinostat is a histone deacetylase (HDAC) inhibitor with distinctive activity against HDAC4, which we showed to be deregulated with BAP1 loss. Based on these data, we initiated a clinical trial to evaluate the efficacy of adjuvant quisinostat treatment in patients with high-risk UM.

Trial design: Open-label, multicenter, investigator-initiated adjuvant phase 2 study with a synthetic control arm to evaluate the effect of adjuvant quisinostat treatment in high-risk UM patients on distant metastases-free survival(DMFS) at 36 months. A one-sided, one-sample log-rank test from a sample of 60 patients achieves an 85.1% statistical power at a 5% significance level to detect a significant change in 36-month DMFS of 75% in the treatment group when the 36-month DMFS in the historical control group is 60%. Patients are treated with 12 mg quisinostat orally (PO) 3 times a week, taken on a Monday, Wednesday, and Friday regimen for a total period of 51 weeks. Patients are evaluated by routine blood work and physical examination every 3 weeks while they are receiving quisinostat. Imaging studies are performed once at baseline and once every 12 weeks ($\pm$ 7 days) for the first 24 months and then once every 24 weeks ($\pm$ 7 days) thereafter. Main inclusion criteria are Castle Decision-UM GEP Class 2, largest basal diameter $>$ 12mm, untreated up to 183 days from definitive primary site therapy. Exploratory endpoints include PBMC histone protein and histone acetylation levels and circulating tumor prior to treatment, every 3 months and at relapse.

Clinical trial identification: NCT06932757.

Legal entity responsible for the study: University of Miami.

Funding: Mandell Family Foundation, Viriom, Inc (drug only).

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1703eTiP Phase I/IIa dose-finding study of triplet regimen of relatlimab (R) ipilimumab (IPI) and nivolumab (N) in first-line therapy of metastatic melanoma (MM) (TRINITY)

E. Burton[1], K. McVay[2], D. Milton[3], T. Davidson[4], R. Amaria[5], I.C. Glitza[6], J. Mcquade[7], T. Nguyen[8], A. Ikeguchi[8], W. Peng[9], M. Davies[2], H.A. Tawbi[10]

[1]Genomic Medicine & Melanoma Medical Oncology Dept., The MD Anderson Cancer Center, Houston, United States of America; [2]Melanoma Medical Oncology, The University of Texas MD Anderson Cancer Center, Houston, United States of America; [3]Biostatistics, The University of Texas MD Anderson Cancer Center, Houston, United States of America; [4]Department of Melanoma Medical Oncology, University of Texas MD Anderson Cancer Center, Houston, United States of America; [5]Melanoma Medical Oncology Department, The MD Anderson Cancer Center, Houston, United States of America; [6]Melanoma Medical Oncology Department, The University of Texas MD Anderson Cancer Center - Main Building, Houston, United States of America; [7]Melanoma Medical Oncology Department, The University of Texas MD Anderson Cancer Center, Houston, United States of America; [8]Melanoma Medical Oncology, University of Texas MD Anderson Cancer Center, Houston, United States of America; [9]Biology and Biochemistry, University of Houston, Houston, United States of America; [10]Melanoma Medical Oncology department, The University of Texas MD Anderson Cancer Center, Houston, United States of America

Background: Nivolumab (anti-PD1)(N) alone or in combination with IPI (anti-CTLA-4) (I) or RELA (anti-LAG3)(R) are approved immune checkpoint blockade (ICB) agents for the treatment (tx) of patients (pts) with advanced, unresectable MM. Doublet combinations induce higher rates of durable disease control and modestly improved survival vs single agent. Although there is no established dose-response relationship for N or R, higher doses of IPI induce higher objective response rates (ORR) but also increased gr $\geq$3 immune-related adverse events. Insights describing IPI's role in expanding TCR repertoire, modulating suppressive T cell populations, while N+R regulates exhaustion of activated T cells, suggest potential synergy with triplet ICB. RELATIVITY-048 (N 480 mg Q4W + R 160 mg Q4W + IPI 1 mg/kg Q8W) demonstrated impressive efficacy (ORR 59%) and improved progression free and overall survival (PFS, OS) over previously reported doublet tx. However, IPI was evaluated at a dose lower than the approved regimen and the study did not include dose escalation (DE). Our team seeks to optimize the dose and schedule of IPI to combine with N+R, identify the recommended phase II dose (RP2D) for triplet ICB, maximizing clinical benefit and maintaining a toxicity profile comparable to approved regimens.
Trial design: In this single center, PhI/IIa study evaluating triplet ICB (NCT06683755), untreated, unresectable MM pts will receive N 480mg + R 160mg IV Q4W (FDA approved dose). IPI DE will begin at 0.5mg/kg Q4W, incrementally escalating to 2mg/kg Q4W x4 induction doses, followed by maintenance tx with N+R Q4W at approved dose. Bayesian optimal interval design will be used to identify the maximum tolerated dose and RP2D (primary obj), accruing 12-18 pts. The PhII portion will enroll 12-18 additional pts at RP2D to determine the ORR (primary obj) by RECIST 1.1. Secondary objectives include PFS, OS, and correlatives analyses. Non IPI containing prior adjuvant or neo-adjuvant tx is allowed if >6 mos since last dose. Asymptomatic brain metastasis pts on $\leq$10mg corticosteroids are allowed. Safely biopsiable lesions are required for pts enrolled in the PhII portion. This study accruing at MD Anderson Cancer Center.

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1704eTiP SUPRAME: A phase III trial evaluating IMA203 T-cell receptor (TCR) T-cell therapy vs investigator's choice in previously treated advanced cutaneous melanoma

D. Schadendorf[1], W. Alsdorf[2], R. Amaria[3], A. Betof Warner[4], T. Holderried[5], A. Busse[6], C. Gebhardt[7], J.B.A.G. Haanen[8], J.C. Hassel[9], S. Klobuch[10], C. Robert[11], M. Wermke[12], S.C. DeVane[13], S. Hengler[14], D. Krishna[15], A. Mayer-Mokler[16], D. Pankov[15], C. Britten[17], J.J. Luke[18]

[1]Department of Dermatology - Hautklinik, WTZ - Westdeutsches Tumorzentrum Essen, Essen, Germany; [2]Medical Clinic/Oncology Study Center, University Medical Center Hamburg-Eppendorf, Hamburg, Germany; [3]Melanoma Medical Oncology Department, The MD Anderson Cancer Center, Houston, United States of America; [4]Medicine Department, Stanford University School of Medicine, Stanford, United States of America; [5]Department of Hematology, Oncology, Immuno-Oncology, Stem Cell Transplantation, University Hospital Bonn, Bonn, Germany; [6]Medizinische Klinik III, Charité - Universitätsmedizin Berlin, Berlin, Germany; [7]Dermatology and Venereology, University Medical Center Hamburg-Eppendorf, Hamburg, Germany; [8]Department of Medical Oncology, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [9]Section of DermatoOncology, NCT - Nationales Zentrum für Tumorerkrankungen, Heidelberg, Germany; [10]Medical oncology, NKI-AVL - Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, Netherlands; [11]Dermatology Department, Institut Gustave Roussy, Villejuif, France; [12]Early Clinical Trial Dept., Dresden University, Dresden, Germany; [13]Clinical Sciences, Immatics US., Inc., Houston, United States of America; [14]Translational Science, Immatics Biotechnologies GmbH, Munich, Germany; [15]Clinical Development, Immatics US., Inc., Houston, United States of America; [16]Clinical Development Department, Immatics Biotechnologies GmbH, Tuebingen, Germany; [17]Medical, Immatics Biotechnologies GmbH, Munich, Germany; [18]Malignant Hematology and Medical Oncology, University of Pittsburgh Cancer Institute, Pittsburgh, United States of America

Background: Outcomes in metastatic melanoma after failure of ICI are poor with limited long-term survival. There is an urgent need for therapies delivering more durable survival. IMA203 is an autologous TCR T-cell therapy engineered to recognize intracellular PRAME-derived peptides presented by HLA and initiate a potent and specific anti-tumor response. In the Ph 1 trial (NCT03686124), IMA203 exhibited favorable tolerability and encouraging anti-tumor activity in pts with melanoma. Confirmed ORR was 56% (18/32), mDOR was 12.1 mo, mPFS was 6.1 mo, and mOS was 15.9 mo (Wermke, ASCO 2025). The most frequent TEAEs were lymphodepletion (LD)-related cytopenias in 97%. CRS was mostly lower grade (82% G1/2; 18% G3; no $\geq$G4) and ICANS was infrequent (12% G1-3). SUPRAME will evaluate IMA203 in advanced cutaneous melanoma after ICI.
Trial design: SUPRAME (NCT06743126) is a prospective, multicenter, open-label, randomized, actively controlled, Ph 3 trial evaluating the efficacy and safety of IMA203 vs investigator's choice in previously treated pts with unresectable or metastatic cutaneous melanoma (including acral melanoma). Eligible pts are $\geq$18 y, HLA-A*02:01+, measurable disease (RECIST v1.1), ECOG PS 0-1 and disease progression on or after $\geq$1 PD-1 inhibitor. Pts with BRAF mutation should have received 1 prior BRAF-directed therapy ($\pm$ MEK inhibitor) at investigator discretion. Pts with active brain metastases, leptomeningeal metastases or with primary mucosal, uveal melanoma and melanoma of unknown primary are excluded. Those with asymptomatic stable brain metastasis will be assessed for eligibility. The study will randomize 360 pts 1:1, who will undergo leukapheresis. Pts in the IMA203 arm will undergo LD with Cy (500 mg/m^2 x 4 days) and Flu (30 mg/m^2 x 4 days), subsequent infusion of 1-10 x10^9 IMA203 PRAME-directed TCR T cells, followed by low-dose IL-2 (1 M IU QD x5 days, BID x5 days, SC). Pts in the control arm will receive nivolumab/relatlimab, nivolumab, ipilimumab, pembrolizumab, lifileucel (US), or chemotherapy. The primary endpoint is PFS. Secondary endpoints include OS, ORR, safety and PROs (EORTC QLQ-C30, EQ-5D-5L). The trial plans to enroll patients in the US and Europe.

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Financial Interests, Institutional, Local PI: Roche; Financial Interests, Institutional, Trial Chair: Obsidian. A. Betof Warner: Financial Interests, Personal, Advisory Board: Bristol Myers Squibb, Immatics, Instil Bio, Iovance Biotherapeutics, Lyell Immunopharma, Novartis, Merck, Pfizer, cTRL Therapeutics, Genmab, IO Biotech, Replimune; Financial Interests, Personal, Other, Consultant: Adaptimmune; Financial Interests, Institutional, Coordinating PI: Bristol Myers Squibb International; Financial Interests, Institutional, Local PI: Lyell Immunopharma, Obsidian Therapeutics, Replimune; Financial Interests, Institutional, Steering Committee Member: Replimune; Non-Financial Interests, Member: ASCO, AACR; Non-Financial Interests, Leadership Role: SITC. T. Holderried: Financial Interests, Personal, Advisory Role: Otsuka, Pfizer, Sanofi, Novartis, Kite/Gilead, Jazz Pharmaceuticals, GSK, BMS, Amgen; Financial Interests, Personal, Other, Travel expenses: AbbVie, Janssen, Astellas Pharma, Neovii, Immatics, Sobi. C. Gebhardt: Financial Interests, Personal, Advisory Role: Almirall, Amgen, Beirersdorf, BioNTech, Bristol Myers Squibb, Decalth, Immunocore, Janssen, MSD, Novartis, Pierre-Fabre, Regeneron, Sanofi, Genzyme, Sciomics, SUN Pharma, Sysmex/Inostix; Financial Interests, Personal, Ownership Interest: Dermagnostix, Dermagnostix R&D. J.B.A.G. Haanen: Financial Interests, Personal, Stocks or ownership: Neogene Therapeutics; Financial Interests, Institutional, Advisory Role: MSD Oncology, Pfizer, Bristol Myers Squibb, Novartis, Roche/Genentech, Ipsen, Achilles Therapeutics, Immunocore, Sanofi, Third Rock Ventures, Neogene Therapeutics, Molecular Partners, BioNTech, T-Knife, POEM Health, Instill Bio, Iovance Biotherapeutics, Cure Vac; Financial Interests, Institutional, Research Funding: Bristol Myers Squibb, Novartis, Neon Therapeutics, BioNTech. J.C. Hassel: Financial Interests, Personal, Invited Speaker: BMS, Novartis, sanofi, MSD, Sunpharma, Pierre Fabre, Immunocore, Delcath; Financial Interests, Institutional, Advisory Board: MSD, Pierre Fabre, Novartis, BMS, Immunocore, Philogen, Sanofi; Financial Interests, Personal, Advisory Board: Sunpharma, Onkowissen; Financial Interests, Institutional, Local PI: Philogen, Genentech, 4SC, BioNTech, Iovance, Pierre Fabre, Regeneron, Sanofi, Replimune, Immatics, MSD, Pfizer; Financial Interests, Institutional, Coordinating PI: BMS, Immunocore, Novartis, Genmab, Seagan; Financial Interests, Personal, Steering Committee Member: Immunocore, IO Biotech; Financial Interests, Institutional, Research Grant: Sanofi; Non-Financial Interests, Leadership Role: DeCOG; Non-Financial Interests, Member: ASCO. S. Klobuch: Financial Interests, Personal, Other, Travel expenses: Immatics, BioNTech. C. Robert: Financial Interests, Personal, Stocks or ownership: RiboNexus; Financial Interests, Personal, Advisory Role: Bristol Myers Squibb, Roche, Novartis, Pierre Fabre, MSD, Sanofi, AstraZeneca, Pfizer, Sun Pharma; Financial Interests, Institutional, Research Funding: Novartis, Phio Pharmaceuticals. M. Wermke: Financial Interests, Personal, Advisory Board: Novartis, Boehringer Ingelheim, Bayer, Bristol Myers Squibb, ImCheck Therapeutics, Immatics, Pfizer, AstraZeneca, Tacalyx GmbH, Regeneron, Zymeworks, Daiichi Sankyo, PharmaMar, Iovance Biotherapeutics, Tknife, Genentech; Financial Interests, Personal, Invited Speaker: Boehringer-Ingelheim, Synlab GmbH, Janssen, EMD Merck Serono, GWT TUD GmbH, Amgen, Novartis, MJH Life Sciences, Bristol Myers Squibb International; Financial Interests, Personal, Other, Member of DSMC: ISA Therapeutics; Financial Interests, Institutional, Funding: Roche; Financial Interests, Institutional, Research Grant: Iovance Biotherapeutics; Non-Financial Interests, Personal, Other, Congress Travel Cost Support: AstraZeneca, EMD Merck Serono, Janssen, Boehringer Ingelheim, Immatics, Daiichi Sankyo, Iovance. S.C. DeVane: Financial Interests, Personal, Full or part-time Employment: Immatics; Financial Interests, Personal, Stocks or ownership: Immatics; Financial Interests, Personal, Funding: Immatics; Financial Interests, Personal, Other, Travel expenses: Immatics. S. Hengler: Financial Interests, Personal, Full or part-time Employment: Immatics; Financial Interests, Personal, Stocks or ownership: Immatics. D. Krishna: Financial Interests, Personal, Full or part-time Employment: Immatics. A. Mayer-Mokler: Financial Interests, Personal, Full or part-time Employment: Immatics; Financial Interests, Personal, Stocks or ownership: Immatics; Financial Interests, Personal, Royalties: Immatics. D. Pankov: Financial Interests, Personal, Full or part-time Employment: Immatics, Westchester Medical Center; Financial Interests, Personal, Stocks or ownership: OradellBio. C. Britten: Financial Interests, Personal, Full or part-time Employment: Immatics; Financial Interests, Personal, Leadership Role: Immatics; Financial Interests, Personal, Stocks or ownership: Immatics. J.J. Luke: Financial Interests, Personal, Advisory Role: AbbVie, Agenus, AstraZeneca, Bayer, BMS, Clasp, Curadev, Eisai, EMD Serono, Geneos, Gilead, HotSpot, Krustal, Janssen, Ikena, Immatics, Incyte, IO Biotech, iTeos, LegoChem, Lyvgen, Merck, Mersana, Novartis, Pfizer, Pioneering Medicines, Regeneron, Replimmune, Storm, Sumoitomo, Synlogic, Teva; Financial Interests, Personal, Advisory Board: BioCytics, Bright Peak, RefleXion, Xilio, Actym, Duke Street, Elipacience, Kanaph, NeoTx, OncAI, OncoNano, Pyxis, Saros, Tempest, Zola. All other authors have declared no conflicts of interest.

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